GEK-83872



GE Drive Systems

Instructions

Gate-Turn-Off Thyristor Inverter Induction Motor Drive

GATE-TURN-OFF THYRISTOR INVERTER INDUCTION MOTOR DRIVE

GEK-83872

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These instructions do not purport to cover all details or variations in equipment, nor to provide for every possible contingency to be met during installation, operation, and maintenance. Should further information be desired or should particular problems arise that are not covered sufficiently for the purchaser's purpose, the matter should be referred to General Electric Company, Drive Systems Operations, Salem, Virginia, U.S.A.

Safety Symbol Legend

WARNING Commands attention to an operating procedure, practice, condition or statement, which, if not strictly observed, could result in personal injury or death.

CAUTION Commands attention to an operating procedure, practice, condition or statement, which, if not strictly observed, could result in damage to or destruction of equipment.

NOTE Commands attention to an essential operating of maintenance procedure, condition, or statement, that must be highlighted.

WARNING

THIS EQUIPMENT CONTAINS A POTENTIAL HAZARD OF ELECTRICAL SHOCK OR BURN. ONLY PERSONNEL WHO ARE ADEQUATELY TRAINED AND THOROUGHLY FAMILIAR WITH THE EQUIPMENT AND THE INSTRUCTIONS SHOULD INSTALL, OPERATE, OR MAINTAIN THIS EQUIPMENT.

ISOLATION OF TEST EQUIPMENT FROM THE EQUIPMENT UNDER TEST PRESENTS POTENTIAL ELECTRICAL HAZARDS. IF THE TEST EQUIPMENT CANNOT BE GROUNDED TO THE EQUIPMENT UNDER TEST, THE TEST EQUIPMENT'S CASE MUST BE SHIELDED TO PREVENT CONTACT BY PERSONNEL.

TO MINIMIZE HAZARD OF ELECTRICAL SHOCK OR BURN, APPROVED GROUNDING PRACTICES AND PROCEDURES MUST BE FOLLOWED.

CAUTION

THIS EQUIPMENT IS DESIGNED TO WORK IN A NORMAL INDUSTRIAL ATMOSPHERE. OPERATING RADIO TRANSMITTERS NEARBY MAY CAUSE THE CONTROL EQUIPMENT TO MALFUNCTION. IT IS RECOMMENDED THAT RADIO TRANSMITTERS NOT BE OPERATED WITHIN THE VICINITY OF THE CONTROL EQUIPMENT WHEN THE PANEL DOORS ARE OPEN, NOR AT ANY TIME WITHIN THE CONFINES OF A POWER CONTROL ROOM.

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CHAPTER 1
SYSTEM DESCRIPTION

1-1. BASIC DRIVE CONTROL

The Gate-Turn-Off thyristor (GTO) inverter Induction Motor Drive (IMD) is a static, adjustable frequency drive system which controls an induction motor. Depending on size of the load capacitors, C_1 but typically from 60-80% to 100% speed, the load inverter is load commutated by leading kvar load. From 0% to 50-70% speed the load inverter is forced commutated by turning off the GTO's. A simplified one-line diagram of a single channel IMD is given in Figure 1-1.

The transformer provides isolation from the ac system bus and provides the correct voltage at the terminals of the rectifier. Also, the internal impedance of the transformer limits the magnitude of any downstream bus faults.

The rectifier is a thyristor bridge whose gating is controlled to produce a variable dc voltage at its output. The output of the rectifier is fed through the dc link reactor, whose function is to smooth the current and keep it continuous over the operating range of the system.

The dc link reactor output is then fed into the GTO inverter bridge which produces variable frequency ac at the stator terminals of the motor. The inverter bridge and the rectifier bridge use different power hardware and are controlled by microprocessor-based electronics. Although the bridges are labelled "rectifier" and "inverter" it is possible for their roles and hence power flow to reverse. In this case the motor would be braked by pumping its energy back into the ac line. A more general nomenclature is to call the line side bridge the source converter, and to call the motor side bridge the load converter.

The pulse train firing signals for the thyristor bridge legs in the source converter(s) emanate from source NSFC card(s), which are ribbon cabled to the XDCE distribution card, then to the HPTK firing card. The HPTK output drives a series connected pulse transformer primary winding common to all the pulse transformers whose secondaries drive the gates of the series thyristors in a particular bridge leg..

In addition to the source converter(s) and load inverter firing signals there are a number of other interfaces between the signal electronic module (SEM) and the power conversion module (PCM) via the XDCD/XDCE distribution cards, principally source and load voltage and current feedbacks, 400 Hz inverter voltage status, also potential transformer (PT) and motor current feedbacks used for applications requring synchronizing and transfer of motor between inverter and 50/60 Hz lines.

The inverter load capacitor, typically sized at 2/3 the motor kVA, is a low pass filter, which provides near sine wave motor voltage and current, such that the motor need not be thermally derated for inverter output waveshape. Many applications of this drive are for fans, pumps, compressors, where load torque, thus also torque producing component of motor current, is proportional to speed squared. If constant motor flux, (volts per Hertz), is assumed over the motor speed range, (not the case in actual operation) then the flux producing component of motor current is constant and the load capacitor current is proportional to speed squared. The flux producing component of motor current is the

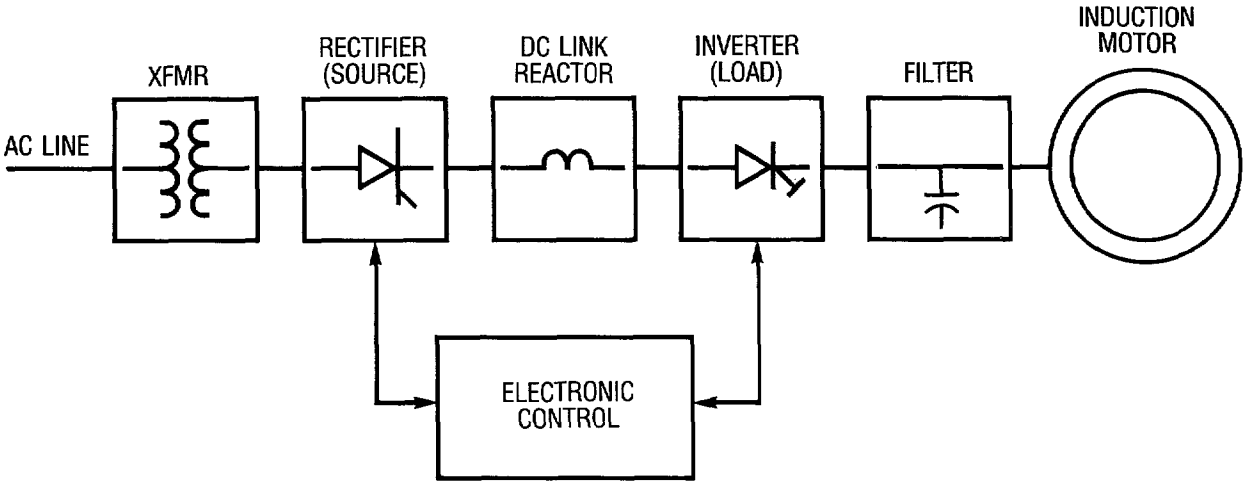


Figure 1-1. Simplified IMD One-Line Diagram

algebraic sum of the capacitor current and the flux producing component of inverter current. This is illustrated in Fig 1-2. At low speed, the capacitor current is less than the required flux producing component of motor current so the inverter must help provide motor excitation current. At high speed, the capacitor current exceeds the required motor excitation component of current, thus providing leading kvars so that the drive may operate in the more efficient self-commutating mode (rather than forced commutated), and the inverter must absorb the excess portion of capacitor current to prevent motor overvoltage.

A more detailed diagram of a single channel IMD is shown in Figure 1-3. As shown in this figure, the electronic control receives the following signal inputs:

1. Attenuated line and load bus voltage signals.
2. Line and load current signals from current transformers.
3. Speed reference signal.
4. Process commands such as stop, start, etc.

The attenuated bus voltage signals are used by the control to synchronize thyristor firing with the line and load voltages, to provide voltage sensing across the thyristor cells, and to provide electronic over/under voltage detection. The current signals are used for regulator current feedback, electronic overcurrent detection, and software-implemented fault detection.

The electronic control evaluates the process commands and internal status signals to determine whether the drive should be in a stopped, started, alarmed, or faulted condition. If in a started condition, the electronics provides gate signals to control the thyristor bridges. These signals are low level and are conditioned in the power bridge circuitry to provide the necessary isolation and power level.

The power bridges are of the six pulse, double way type and the thyristor bridge legs "fire" in the order that they are numbered in Figure 1-3 and as shown in Figure 1-4.

The source converter current is successfully transferred from one leg to the next by the ac line voltages. Similarly, the inverter current is successfully transferred by load's leading kvar (>75% SPD).

The process of switching bridge current from one leg to the next is illustrated in Figure 1-5A. The principles illustrated here apply to a rectifier bridge as well as to an inverter bridge (when load has leading kvars). This phase-controlled switching is accomplished by using the following two thyristor characteristics: (1) When the voltage across the thyristor is positive, it can be triggered into conduction, and (2) it will not permit current flow in the reverse direction - thus, in an alternating voltage circuit, thyristor conduction will cease and blocking voltage will begin to appear when the current becomes zero.

Current transfer must be completed before voltage cross-over, with a margin angle γ, see Figure 1-5B . This angle must be long enough to allow the previously conducting leg thyristors to recover to their blocking state before voltage is applied. This is why the fundamental component of current must lead the voltage, from an inverter/load viewpoint. From the rectifier/source viewpoint, the fundamental component of current will always lag the voltage; but for successful commutation, the angle $\gamma = 180° - \beta = 180° - \mu - \gamma$ must always be less than 180° ; a practical limit is 155°. Correspondingly, a practical minimum value for β , for the inverter bridge, is 25°.

1-2. GENERAL OPERATIONAL THEORY

a. In Figures 1-6A and 1-6C, vector diagrams illustrate drive operation. The load angle alpha γ is the angle between inverter output voltage and inverter output current. The diagram is divided into four quadrants shown in Table 1.

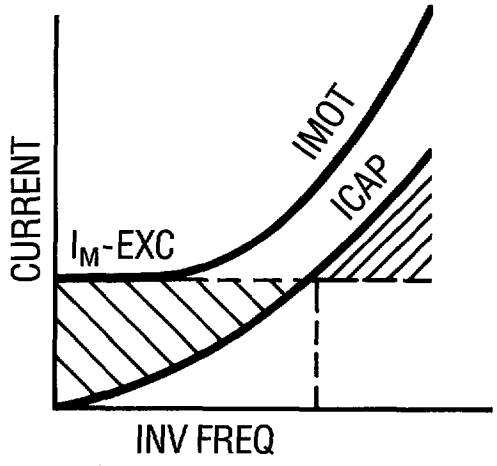


Figure 1-2.

Thus at alpha = 0 degree, the drive is producing maximum generating torque, that is, the motor is acting as a generator feeding power back into the ac lines through the inverter and source converter. At alpha = 90 degree, the inverter is at the maximum flux "eating" angle, that is working its hardest to reduce flux and making no torque. At alpha = 180 degree,the inverter is trying to make its maximum motor torque and no flux. At alpha = 270 degree, the inverter is trying to make maximum flux and no torque. In a later section, the alpha load angle control will be described.

The capacitor sizing and the flux versus speed profile setpoint setup in the control is

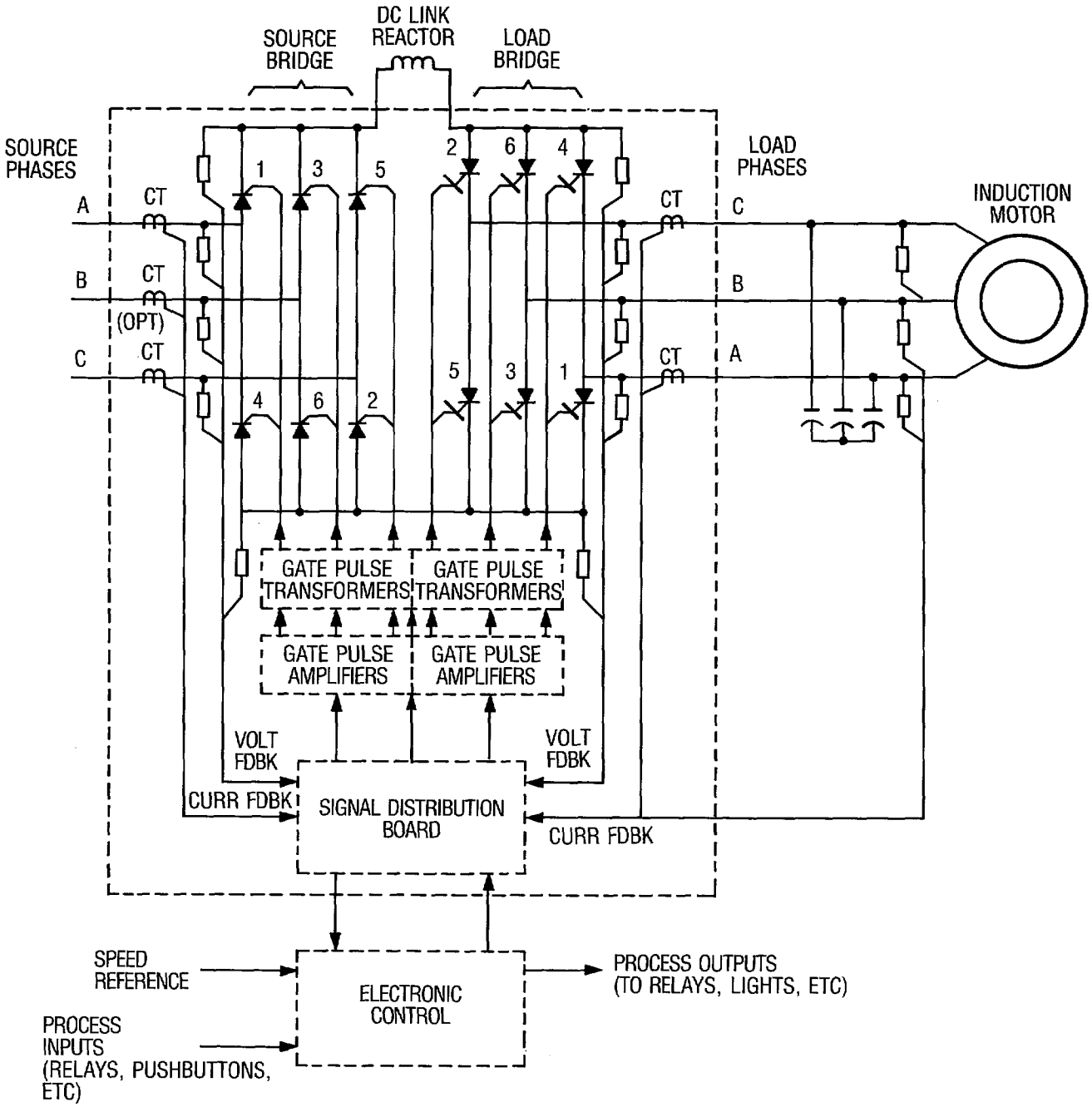


Figure 1-3. Single Channel IMD Drive

TABLE 1

QUADRANT	ALPHA ANGLE	DESCRIPTION
I	0 < alpha < 90	High speed generating
II	90 < alpha < 180	High speed motoring
III	180 < alpha < 270	Low speed motoring
IV	270 < alpha < 360	Low speed generating

such that for fan, pump, and compressor applications, in the speed range from .80 to 1.0 per unit speed, the capacitor provides nearly all the required motor excitation current. Therefore the inverter only needs to provide the motor kW, not its kvar and this in turn maximizes the power factor that the drive reflects to the source line voltage at high speed, high torque operation.

Thus the load capacitor's two main advantages are that it minimizes motor heating and maximizes source input power factor. The disadvantage of the load capacitor is that it creates a resonance determined by the value of the load capacitor and motor leakage inductance. Typically this resonance is in the range of 180-220 Hz. While this is above the operating frequency of the drive, harmonics of the inverter fundamental frequency can excite this resonance. For example, if the resonance is 200 Hz, at 40 Hz inverter frequency, the 5'th harmonic can excite the load resonance. This can cause undesirable torque harmonics. To avoid this problem, for any given inverter frequency, the harmonic of inverter current that would excite this load resonance is eliminated by inverter current waveform notching techniques.

This is made possible by virtue of the use of GTO's in the power circuit which allow near instantaneous commutation of current from one inverter phase to another. This notching technique is implemented in software and is illustrated in Figure 1-7. Since the inverter frequency is known and the load resonant frequency is known, the harmonics to be reduced or known, the harmonics to be reduced for a particular inverter frequency ranges are predetermined and the inverter firing angles to accomplish the desired harmonic reduced firing pattern can be precalculated and the results stored for table lookup by the computer control.

The control is presently designed to eliminate 0, 1, or 2 harmonics. Each notch per quarter cycle eliminates one harmonic. The notch width and position in the quarter cycle determines which harmonic (s) are eliminated. The particular harmonics eliminated versus fundamental frequency range is adjustable by EEPROM but a typical setup may be as shown in Table II.

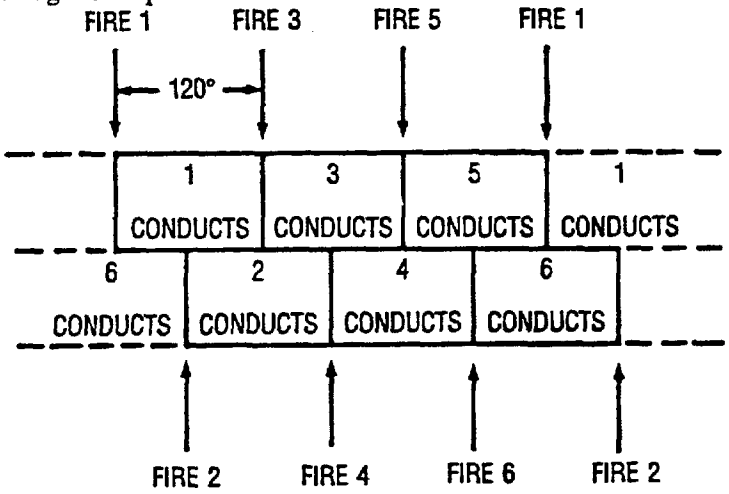


Figure 1-4. Thyristor Bridge Leg Conduction Sequence

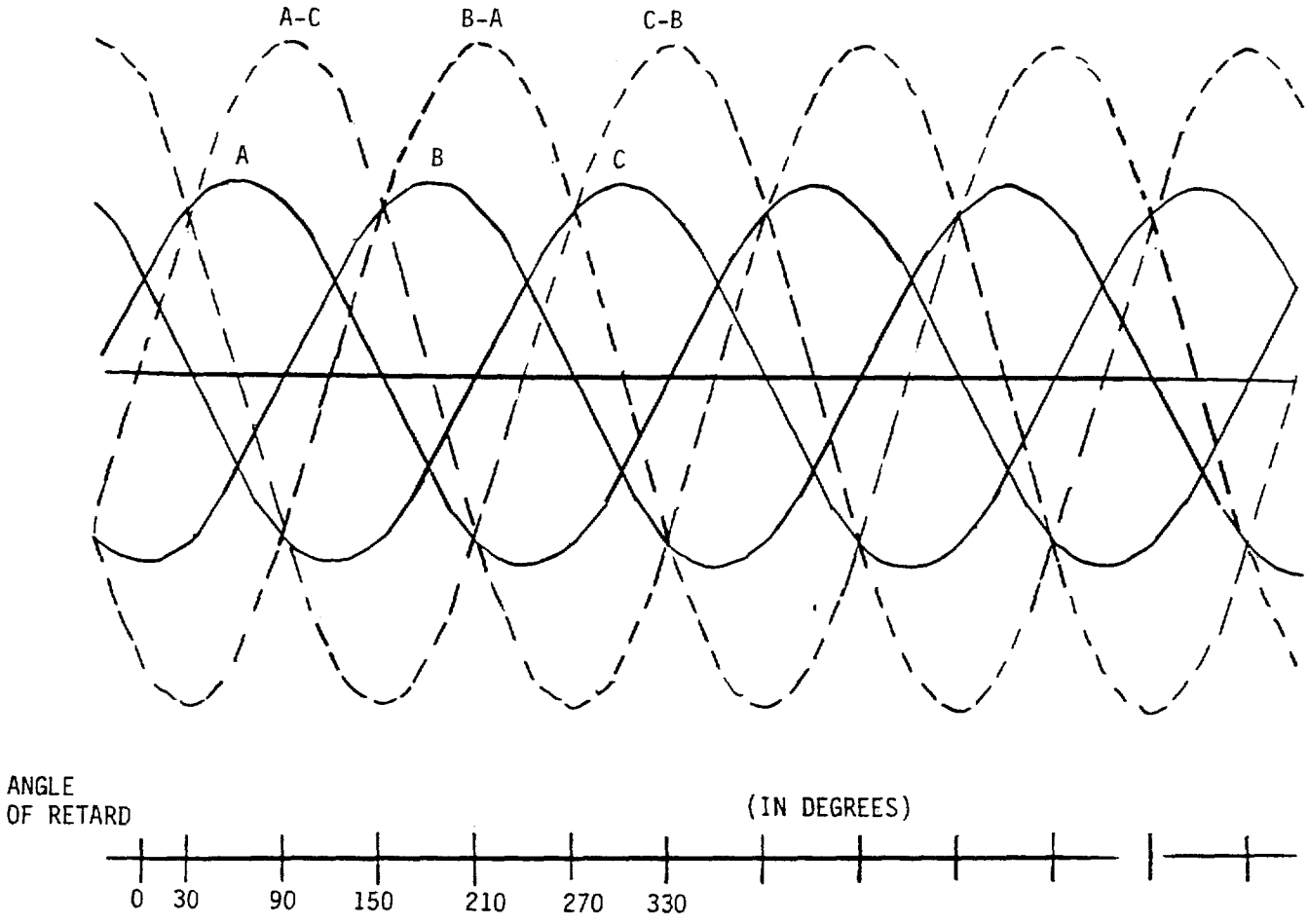


Figure 1-5A.. Source Voltage and Current

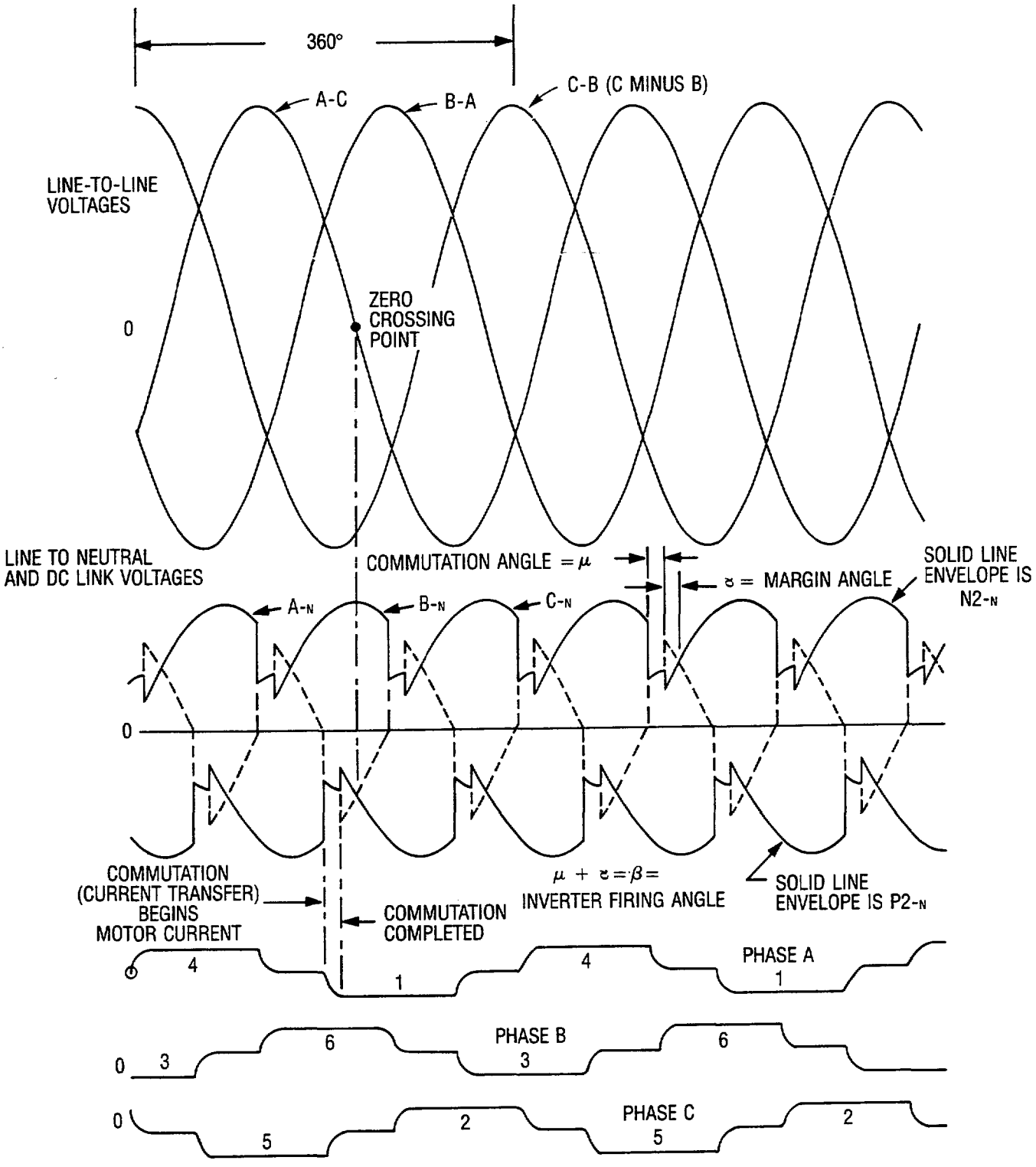
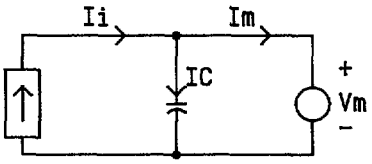


Figure 1-5B. Load Converter Voltage and Current (self commutation mode)



PER PHASE EQUIV. CIRCUIT

Ii = Ic + Im

Im = MOTOR CURRENT
Ic = LOAD CAPACITOR CURRENT
Ii = INVERTER LOAD CURRENT
Vm = MOTOR = CAPACITOR VOLTAGE

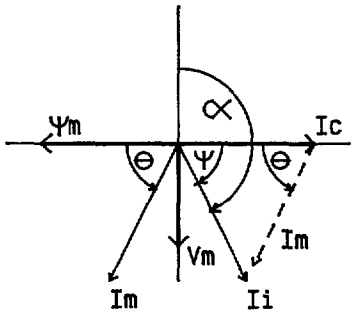


FIGURE 1-6A

$90° < \alpha \leq 180°$
HIGH SPEED MOTORING
LOAD CAPACITOR COMMUTATION

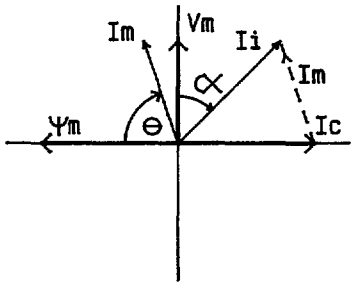


FIGURE 1-6B

$0° < \alpha \leq 90°$
TRANSIENT HIGH SPEED REGENERATION
LOAD CAPCAITOR COMMUTATED

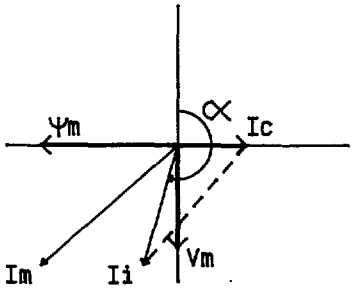


FIGURE 1-6C

$180° < \alpha \leq 270°$
LOW SPEED MOTORING
FORCE (GTO) COMMUTATED

Figures 1-6A, 1-6B and 1-6C. Vector Diagrams

TABLE II

Fun Freq. Range	Harmonic (s) Eliminated
8 - 20 Hz	11 - 13
20 - 30 Hz	7 - 11
30 - 40 Hz	5 - 7
40 - 50 Hz	5
50 - 66 Hz	0

Typical one half cycle waveforms of the inverter output current are shown in Figure 1-7 for chopping patterns to eliminate 0, 1 and 2 harmonics respectively. Figure 1-7 also illustrates the GTO switching frequency for the various chopping modes. The maximum switching frequency occurs at 40 Hz and is 5 * 40 or 200 Hz. At this frequency, the motor voltage for a fan load is 2/3 rated and load torque is 4/9 rated. At high speed, high load, the GTO switching frequency is fundamental frequency. Thus the switching losses are minimal for this chopping technique.

b. Startup modes and assumptions for IMD approach:

1. Commutation by turning the GTO converter on/off: (typically 0 to 45 Hz).
   a. Initially there is no motor flux, so at 0.25 PU IDC the OPEN_LOOP frequency is ramped from 1 to 5 Hz.
   b. At 5 Hz, motor flux has been established - thus the load PLL (phase-lock-loop), speed regulator, flux regulator, and load angle regulator are activated. (CLOSED-LOOP mode)
   c. Now in CLOSED-LOOP mode, continue to ramp up motor speed to 45 Hz.
   d. GTO commutation is maintained as long as the commanded load thyrister firing angle (alpha) is greater than (180 degrees - margin) i.e. about 165 degrees. For succesful transfer of current from GTO to GTO, the current must be low enough for the Gate Driver to drain. The DC link voltage waveshape is illustrated in Figure 1-7A for various inverter firing angles. For regenerative braking, the firing angle range is alpha between 270 to 90 degrees. As a means of maintaining speed regulator control down to light load-low speed conditions, when the commanded firing angle exceeds the max angle limit, link current is commanded to current-lower-limit and the angle spillover is used to stretch the GTO conduction interval - reducing torque and flux and increasing harmonic content in the inverter output. This all serves to make the control task of satisfying the speed regulator while ensuring successful load thyrister commutation in this low speed region very difficult - particularly because this is also the region of resonance of the load capacitor and motor leakage inductance with the 5th and 7th harmonics of the fundamental freguency. To reduce this "self-excitation", the current in the 30-40 Hz region is "chopped" to reduce the 5th and 7th harmonics and "chopped" in the 40-50 Hz region to reduce the 5th harmonic.

2. Load Commutation Mode: (typically 80-100% speed)
   For (0 degrees < commanded firing angle (alpha) < 165 deg) the load thyristers are commutated by the load capacitor. In fact, at high speed the load inverter must 'eat' some of the excess leading kvar to prevent motor overvoltage. In this region the GTO forced commutation

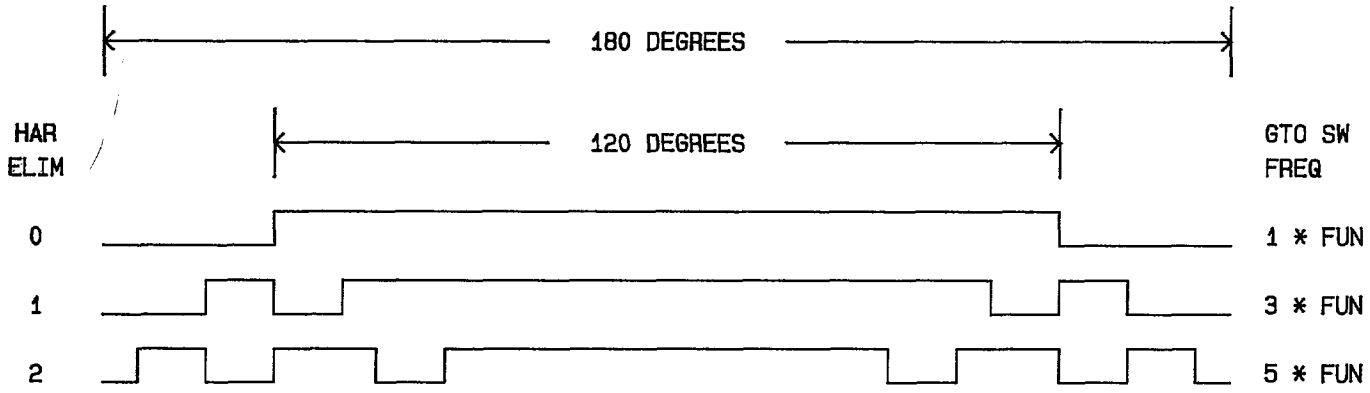
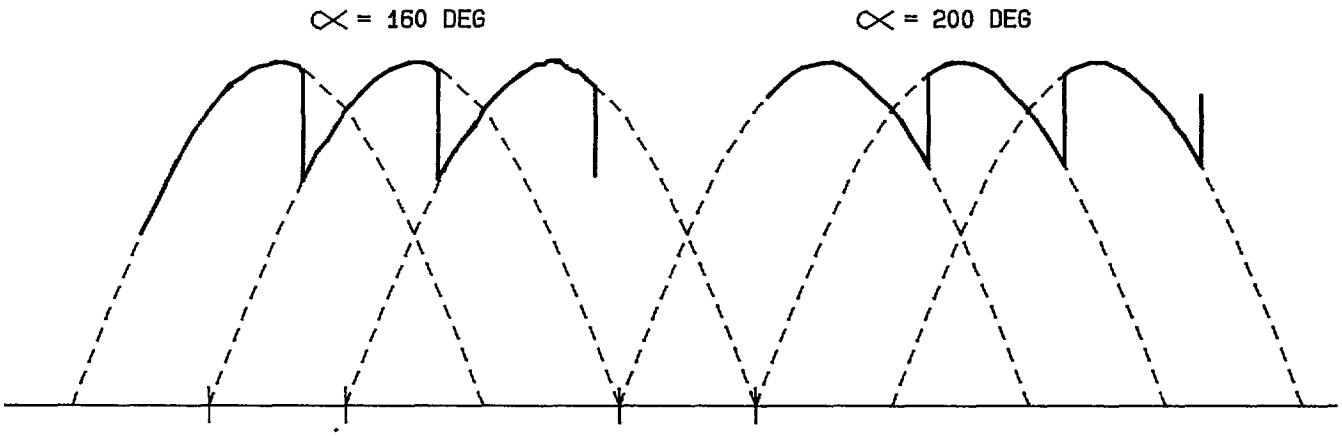


Figure 1-7. Typical Waveform of Inverter Output Current



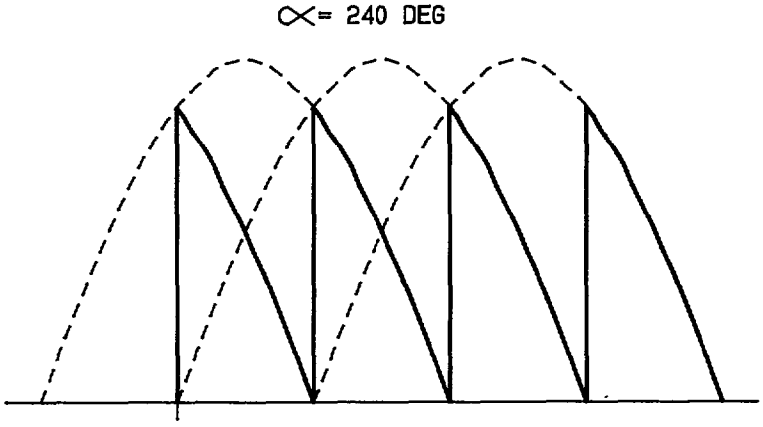


Figure 1-7A. DC Voltage Link Waveforms Various Firing Angles

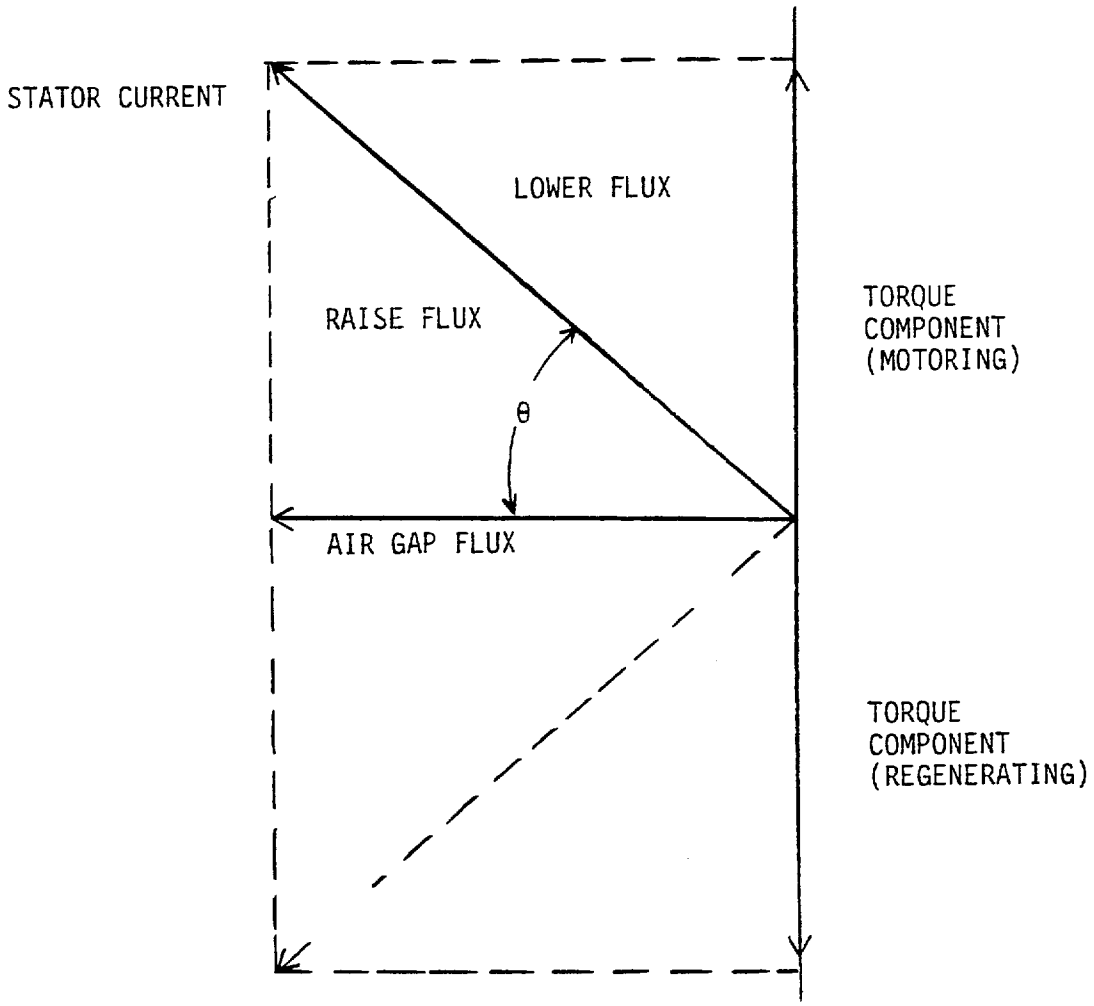


Figure 1-8. Motor Angle

mode does not need to operate. The relationship between motor, inverter, and load capacitor currents as a function of inverter firing angle is illustrated in Figure 1-6. At high speed motoring the firing angle (alpha) range is 90 degree < alpha < 180 degree. For transient high speed regeneration the alpha range is 0 degree < alpha < 90 degree. For low speed motoring operation the alpha range is 180 degree < alpha < 270 degree. These three cases are illustrated in vector diagrams in Figures 1-6A, 1-6B, 1-6C respectively.

3. Assumptions in Power Circuit Design:
   a. Fan load, IQ = K1*(SPD**2), IX = K2, IDC = K3*SQRT((IX**2)+(IQ**2))
   b. Nominal motor volts = rated voltage * actual speed / rated speed
   c. No Synchronization-and-Transfer-to-Bus feature. (Special option available)
   d. Does have Start-on-the-Fly feature.
   e. The GTO design uses either two, three or four series GTOs - depending on the voltage and horsepower requirements.

1-3. AC INDUCTION MOTOR

AC motor torque is produced by a force or pull between two magnetic fields in the air gap. One being produced by current flowing in the conductors of the stator or stationary member and the other by current flowing in the conductors of the rotor or rotating member.

In the controlled inverter system, only the stator circuit is accessible therefore both magnetic fields in the air gap must be controlled by controlling the stator parameters (or stator current, volts, and frequency).

Torque is proportional to the "torque producing component" of stator current, ($I_S \sin \Theta$) and motor flux,(ψ)

$$T_{AC} = K_4 I_S \sin \Theta \, (\psi)$$
$$= K_5 I_S \sin \Theta \, (I_S \cos \Theta)$$
$$= K_6 I_S^{\,2} \sin \Theta \cos \Theta$$

AC motor flux is produced by the flux producing component of stator current ($I_S \cos \Theta$). The motor angle Θ is described as the angle between the stator current and the motor flux produced in the air gap by the stator current. See Figure 1-8.

In the induction motor, constant torque can be maintained by controlling the torque producing component and the flux producing component of I_S which is done by regulating constant stator current and motor angle Θ.

When the motor takes on additional load, or torque is required to change, the speed drops back and both Θ and motor slip increase which, in turn increases the rotor current and draws corresponding torque producing stator current to maintain load.

Figure 1-9 shows the speed-torque characteristic of a typical induction motor. With a constant voltage applied and a frequency equal to B applied, the motor would accelerate to speed A where the torque curve equals the load torque at that speed. As the motor accelerates, the developed torque equals the characteristic shown. As long as the applied frequency, voltage and load remain constant, the drive will run at speed A and a slip equal to B-A. Under these conditions of constant volts/hertz and constant slip-stator current, motor angle Θ and flux are constant. To regenerate power back into the line, the motor speed exceeds the applied frequency B, the torque becomes negative, and the motor acts as a generator.

Constant torque is controlled by holding constant stator current and motor angle. Motor angle is increased (Figure 1-8) to reduce flux (and torque) for constant HP operation. Stator current and slip also change during this region.

In addition to torque control, motor speed can be controlled by being able to go from motoring to regenerating as required to hold the set speed. This can be done without changing speed, by changing the input frequency. See Figure 1-9. Assume the motor is running at speed A. The load changes to require a regenerating torque equal to E to maintain speed A. The control would change the input frequency to D (motor angle Θ and slip change polarities).

To control motor torque and speed, the inverter current and inverter voltage, signals are fed back to the control. These components are used to calculate flux, torque, capacitor current and motor current. The converter circuit is controlled to regulate current. The inverter circuit is controlled to regulate motor speed and volts.

At high load on the motor, the apparent power factor "seen" by the power source increases. This is because the source converter firing advances (is reduced) to obtain more current. The harmonics in the current and the resultant harmonics in the voltage, due to commutation "notching", will decrease.

The fundamental control strategy is: To increase motor current in response to a load torque increase, the dc link voltage, on the source side, is increased, thereby raising motor current so as to keep motor speed constant.

1-4. SERIES TWELVE-PULSE OPERATION (OPTION)

Series twelve-pulse operation consists of two identical, separate, six-pulse bridges operating from a common source at the same current with the voltage and firing reference angles shifted 30 degrees by an incoming isolation transformer. But the two drives are combined into one by connecting the "P(+)" bus of one to the "N(-)" bus of the other .

The two bridges deliver equal power and the transformer design makes the source-side converter voltage

equal, but also 30 degrees apart between the two channels. It is necessary that source-side and load-side voltages be phase-shifted. The transformer windings are shifted to reduce the harmonic distortion on the power system, to raise the harmonic frequencies, and to increase the power flow.

Normally the source master voltages (WYE) LEAD the source slave voltages (DELTA) .

1-5. BYPASS (ACROSS THE LINE) OPERATION- OPTION -

On drives with a bypass mode, the motor can be connected directly to the line - "bypassing" the IMD drive. This feature gives "back-up" capability to the system, i.e. the motor can still be run if the drive is faulted.

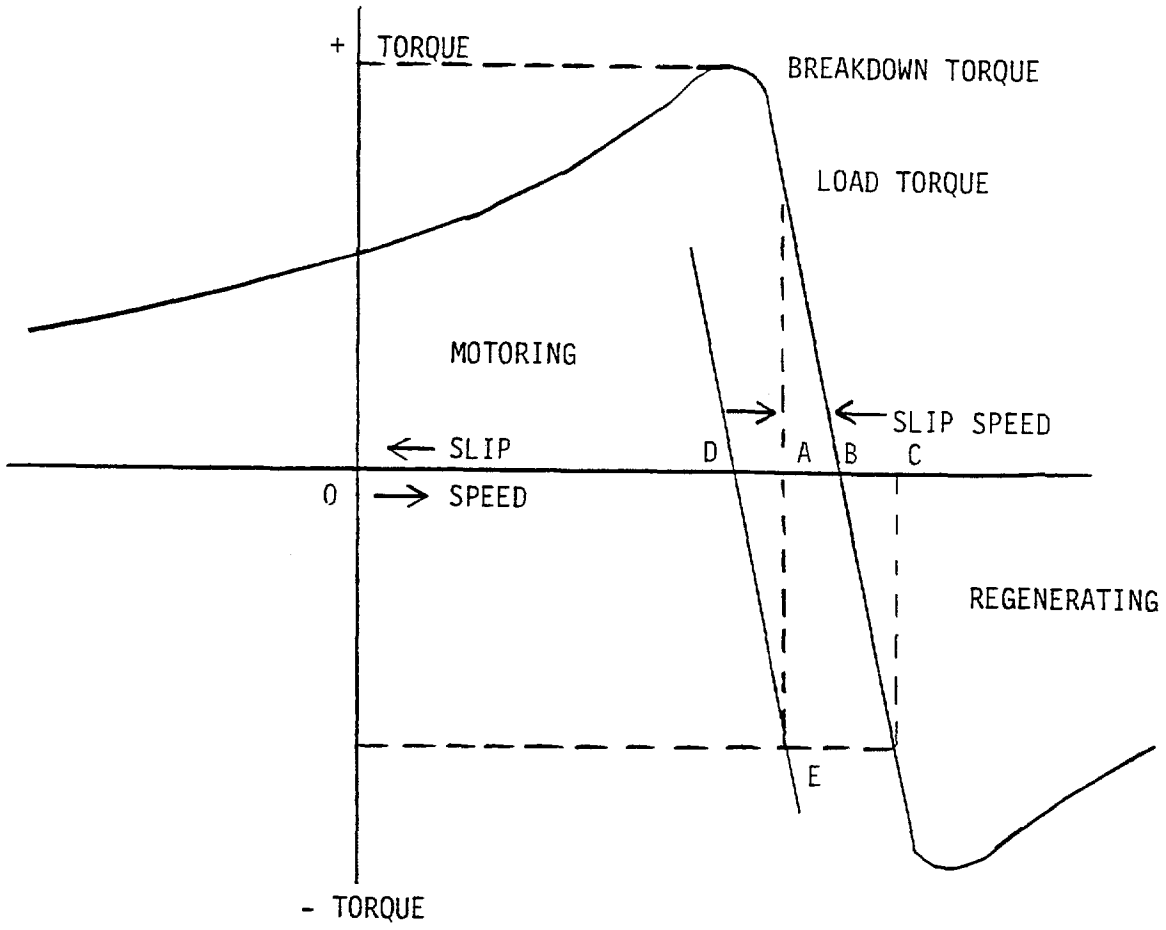


Figure 1-9. Speed Torque Characteristics of AC Induction Motor

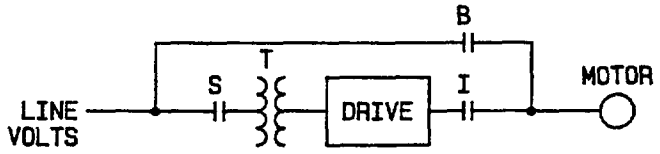
1-6. SYNCHRONIZING/DE-SYNCHRONIZING FUNCTION - [OPTION]

1-6.1. Overview

The synchronizing option on the IMD-GTO is a function which, when enabled, will match the motor voltage phase and magnitude to the source line voltage phase and magnitude and then, in a bumpless [bumpless is defined as < 3 p.u. current] fashion, transfer the motor to across-the-line operation. The IMD-GTO drive may be completely de-energized at this point (both high voltage and control power) if desired. The synchronizing function also provides the means to return the motor to IMD-GTO drive operation, again in a bumpless fashion.

1-6.2. System Configuration

1-6.2.1. Contactor/Disconnects. When the synchronizing function is required, a bypass contactor is required. The bypass contactor provides the means for connecting the motor directly to the ac line. The IMD-GTO drive is provided with an inverter contactor for connecting the motor to the load inverter section of the drive and a source contactor or disconnect swith for connecting the ac line to the source converter section of the drive. Refer to Figure 1-1



S = source contactor/disconnect switch
I = inverter contactor
B = bypass contactor
T = source isolation transformer

Figure 1-10. Typical System Configuration with Synchronizing Option

When the motor is running on the IMD-GTO drive, the source and inverter contactors are both closed and the bypass contactor is open. When the motor is running across-the-line (bypassed operation) the source and inverter contactors are open and the bypass contactor is closed.

1-6.2.2. Source Isolation Transformer. The source isolation transformer is recommended system option. In addition to providing additional system impedance to aid in limiting current in a fault condition, it provides the necessary ohmic isolation which will allow th IMD-GTO drive to operate normally even while the bypass contactor is closed. This allows the IMD-GTO drive to maintain uniterrupted synchronism of the motor to ac line and shut down only after the bypass contactor has closed in. Without the isolation transformer, the IMD-GTO drive must shut down before the bypass contactor closes, leaving a window where the motor, not being provided with any torque producing current, will tend to slip out of phase with the ac line. The amount of phase slip will depend upon the total motor and driven inertial, the motor loading, and the duration of the window. These matters will be discussed in more detail in a later section.

1-6.2.3. DC-Link Reactor. All IMD-GTO drives with the synchronizing option should be provided with air core dc link reactors. This will provide an unsaturable inductance to limit the magnitude of possible fault currents. This is especially important in systems without source isolation transformers where fault scenarios exist in which full source line to line voltage may be impressed across the dc link inductance, which cause saturation in iron core reactors.

1-6.3. SYNCHRONIZING SCENARIO WITH NO INPUT TRANSFORMER

The following scenario outlines the sequence of events culminating in the bumpless transfer of the motor to across-the-line operation. This scenario assumes there is not input isolation transformer.

- IF speed setpoint exceeds variable threshold

then

- speed is ramped to 50/60hz setpoint
- IF speed ramp is done and speed error is zero

then

- adjust speed setpoint to match motor and source voltage phase
- adjust flux setpoint to match motor and source voltage magnitude
- Check motor and source voltage potential transformer match
- IF all checks say synchronized

then

- adjust phase setpoint to make motor lead source voltage 20-30 degrees

then

- IF phase and magnitude errors are in tolerance then
- command bypass contactor to close
- start the bypass contactor operation timer
- IF the bypass contactor acknowledges closure within the maximum allowable closure time

then

- complete shutdown and trip inverter contactor

else
- trip bypass contactor
- IF sync fail fault enabled

then
- complete shutdown and trip inverter contactor
- declare sync fail fault which will cause bypass to reclose through undervoltage path

else
- restart drive
- declare sync fail alarm

1-6.4. De-Synchronizing Scenario

The following scenario outlines lthe sequence of events culminating in the bumpless transfer of the motor from across-the-line operation to IMDE-GTO drive operation. This scenario is the same whether there is an input isolation transformer or not.

- IF speed setpoint falls below variable threshold

then
- close the inverter contactor
- turn on the load phase lock loop
- IF the load phase lock loop is locked

then
- command the bypass contactor to trip
- IF the bypass contactor acknowledges opening within the maximum allowable opening time

then
- clear invlims to release the current regulator
- enable the flux/speed regulators and run motor on IMD-GTO drive

else
- shutdown and trip the inverter contactor
- declare de-synchronizing fail fault
- leave command to trip bypass contactor active

CHAPTER 2

2-1. EQUIPMENT OPTIONS

Functionally there are two main types of bridge for the power conversion equipment, namely the source bridge and load bridge. The source bridge is made from SCR stacks and the load bridge is made from GTO stacks.

2-2. Source Power Converter (bridge) Compartment.

The typical arrangement of the power converter SCR stack assemblies is shown in figure 2-1, 2-2 and 2-3. There is a diagram inside the bridge compartment to show which type is used.

These figures and the bridge elementary diagram show the major power components. Figure 2-3 is normally used for a series 12 - pulse source converter bridge. The SCR's are stacked horizontally; from three to six in series, between heat sinks. Refer to bridge elementary for number of SCR's in series per leg.

2-2.1. For Configurations as Shown in Figure 2-1, 2-2

The blower is mounted at the top of the bridge compartment, also at the top of the bridge are the source side CTs. The RC snubber network is located behind the cell assembly and the gate drive transformer.

2-2.2. For Configurations as Shown in Figure 2-3

There is a separate blower compartment which contains the cooling blowers, ac filter and dc filter network. MOVs and perhaps the auxiliary current transformer(CT) for source side. The hinged base in this compartment contains the blower starters, the 70V gating power supply and bottom blower selection relay. The CT for this type of bridge is located at the top of the bridge. Inside the bridge, each power converter leg module consists of three thyristors arranged in series in a heatsink assembly. The leg module is mounted in an insulated box with associated printed wiring boards (PWB) in front is shown on Fig. 2-3. An RC - Snubber network is also contained in the leg module box, above the cells.

2-2.3. For all Configuration

The stack assemblies are oriented horizontally in the bridge, so that air flow over the heatsinks is from front to back of the bridge enclosure.

2-2.4. Cell Assemblies

The cells are selected, matched, and assembled at the factory under the correct pressure. For this reason, it is recommended that the entire stack be replaced if a cell failure occurs. The defective stack may then be sent to the factory for repair.

2-2.5. Detecting Shorted Cells

Each cell in the module is paralleled by a voltage equalizing resistor, by a neon lamp and dropping resistor network which indicates when voltage is present across the cell. These lights are visible through the front doors of the equipment and can be used to detect a shorted cell.

2-2.6. Gate Driver Transformers

The cell gates are driven by current transformers which are wired so that all their primary windings are in series for each leg. The primary winding is simply a conductor through the center hole of the current transformers. Current pulses are applied in this conductor by the gate drive circuitry. This circuit insures that all thyristor in the string are gated simultaneously and it further provides voltage isolation between the thyristor gates and the electronic gate pulse generation circuitry.

2-2.7. Gate Driver Card (HPTK)

Gate pulses of proper duration and timing are developed in an electronic module located in a control compartment at the left end of the source converter bridge. These signals do not possess the necessary drive and isolation to interface directly with the gate driver current transformers. This interface is provided by the HPTK gate driver card.

The HPTK receives the low "level" signals and, after conditioning, passes them through a pulse transformer to provide isolation. The pulse transformer output drives a pulse amplifier circuit which is powered by a 70Vdc gating power supply. For configuration 2-1 and 2-2, the power supply is located inside the control compartment, for configuration 2-3, the power supply is located inside the blower compartment.) The pulse amplifier output drives all gate pulse transformer in the leg. A red LED on the board glows to indicate when gate pulses are present.

A second function of the HPTK card is to detect an undervoltage condition of the gating power supply. In the event that an undervoltage is detected, the HPTK outputs a logic signal to the electronic module, indicating the fault. The electronic control then takes appropriate action to shut down the drive. Since there is an undervoltage detector on each HPTK, wiring defects in the gating power will also be detected.

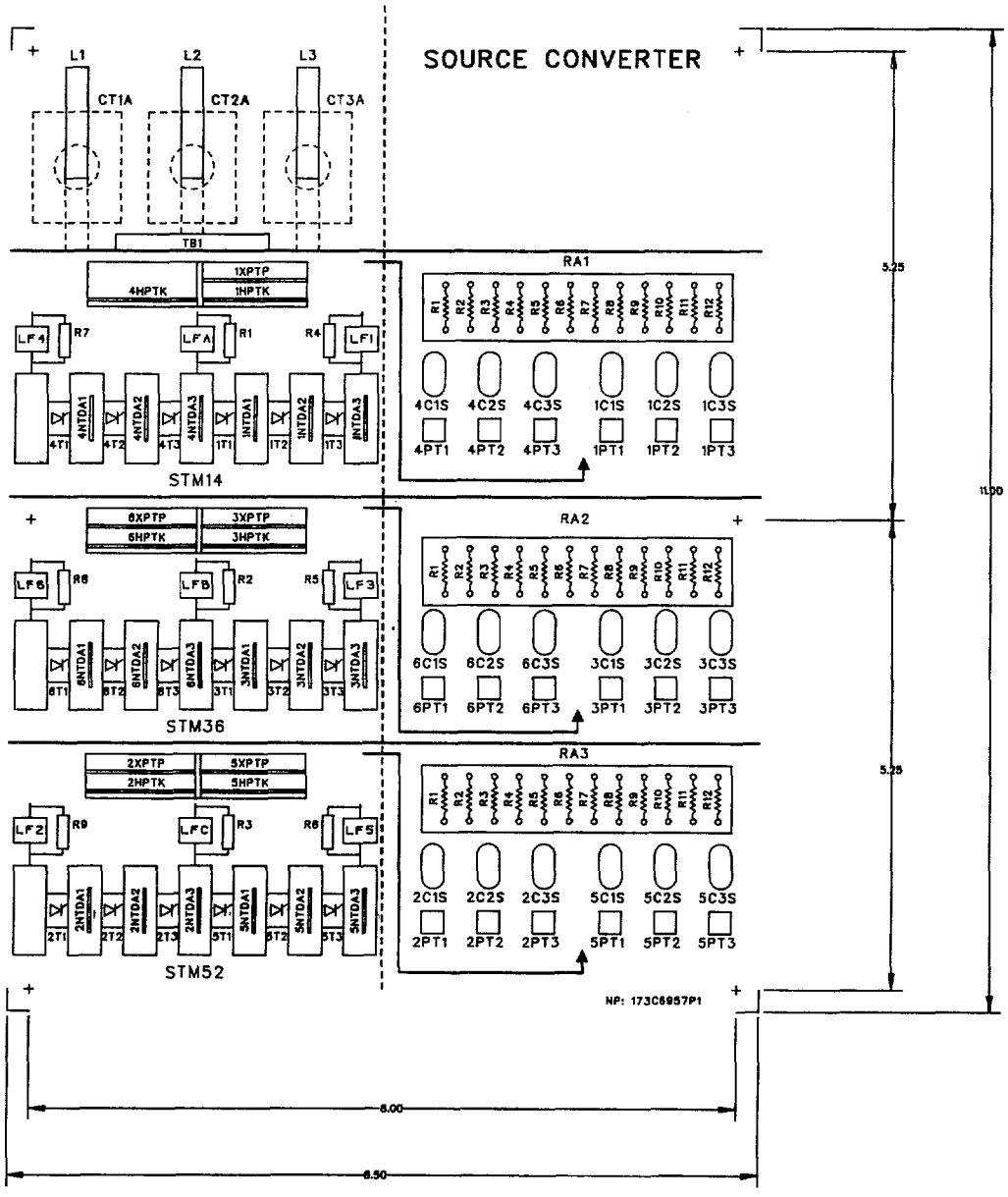


Figure 2-1. Front View of Power Bridge of 2kV Source and Load Assembly Compact Drives, 2300V 1500 Hp Maximum

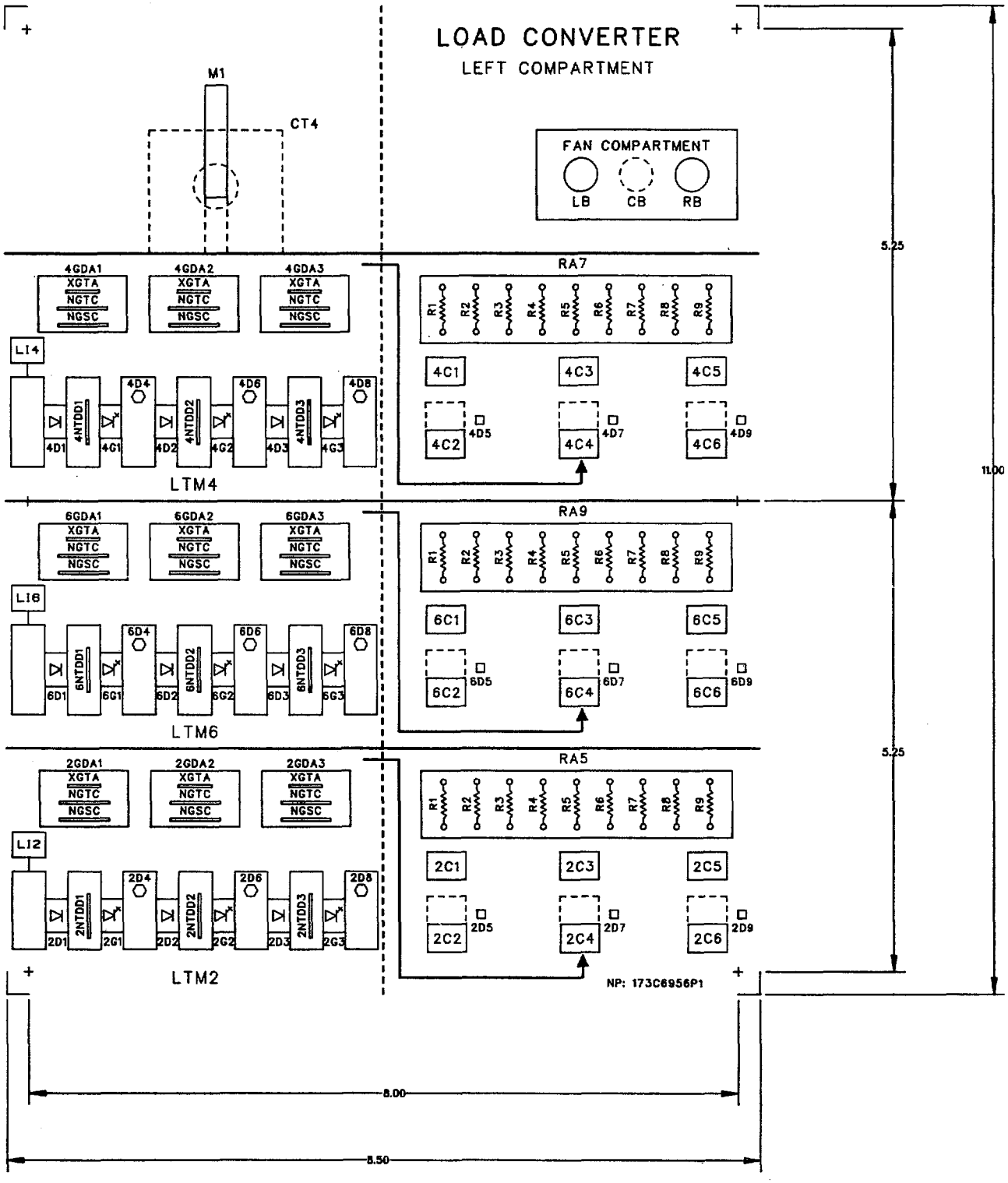


Figure 2-1A. Front view of Power Bridge of 2kV Source and Load Assembly Compact Drives, 2300V 1500Hp Maximum

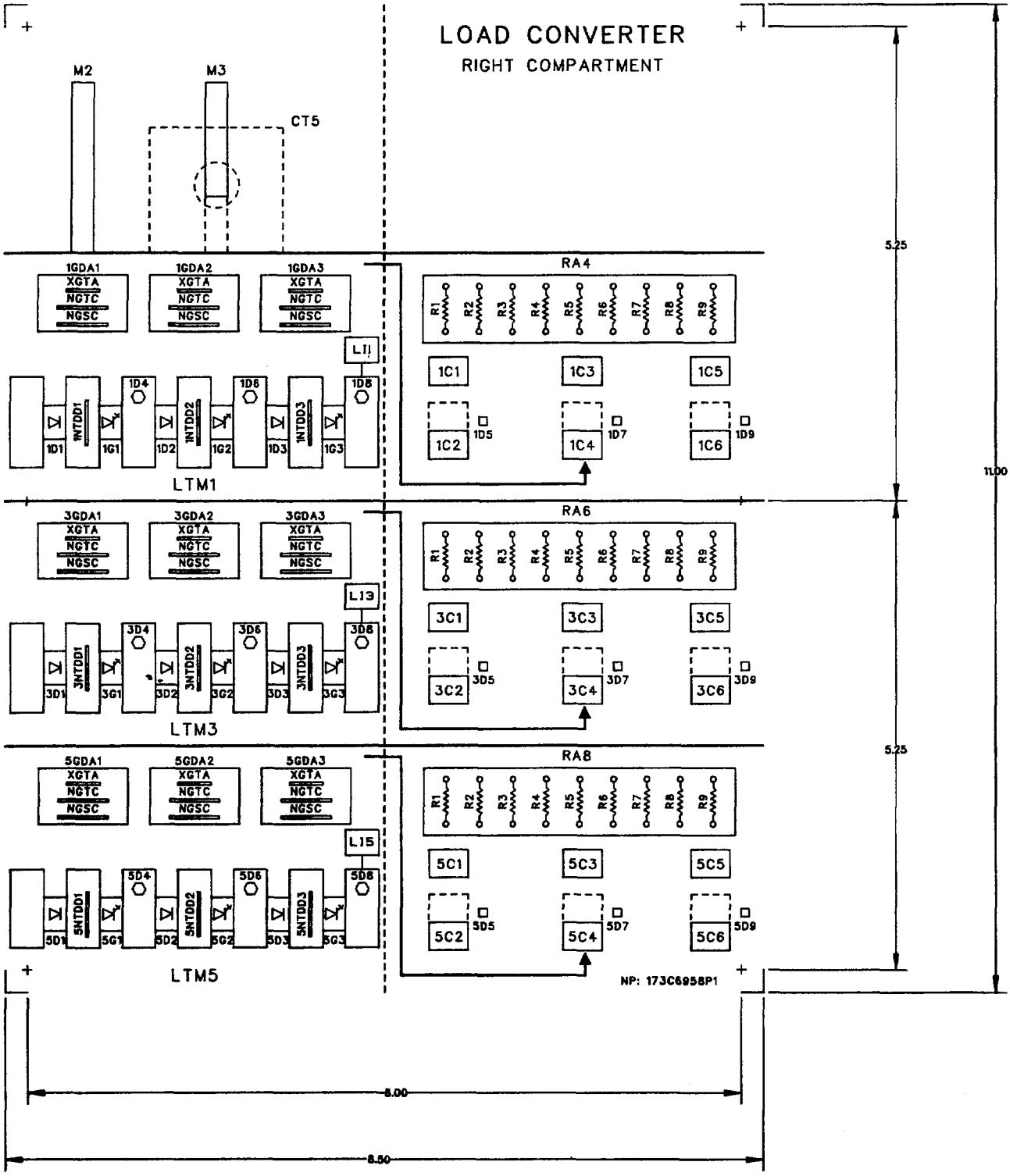


Figure 2-1B. Front View of Power Bridge of 2kV Source and Load Assembly Compact Drives, 2300V, 1500Hp Maximum

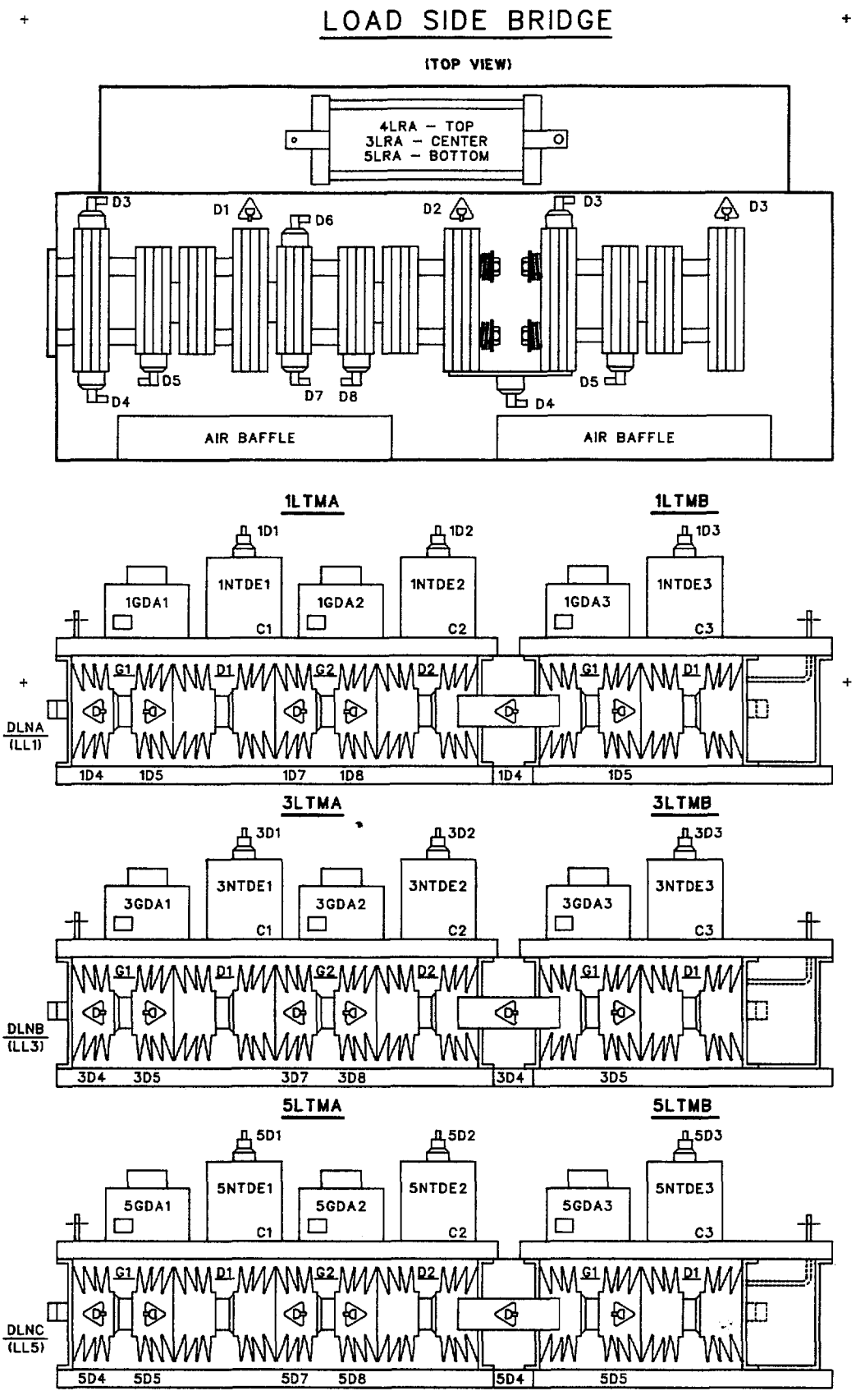


Figure 2-2A1. Front View of Power Bridge of 3kV Source and Load Assembly Drives Greater Than 1500Hp or Greater Than 2300V

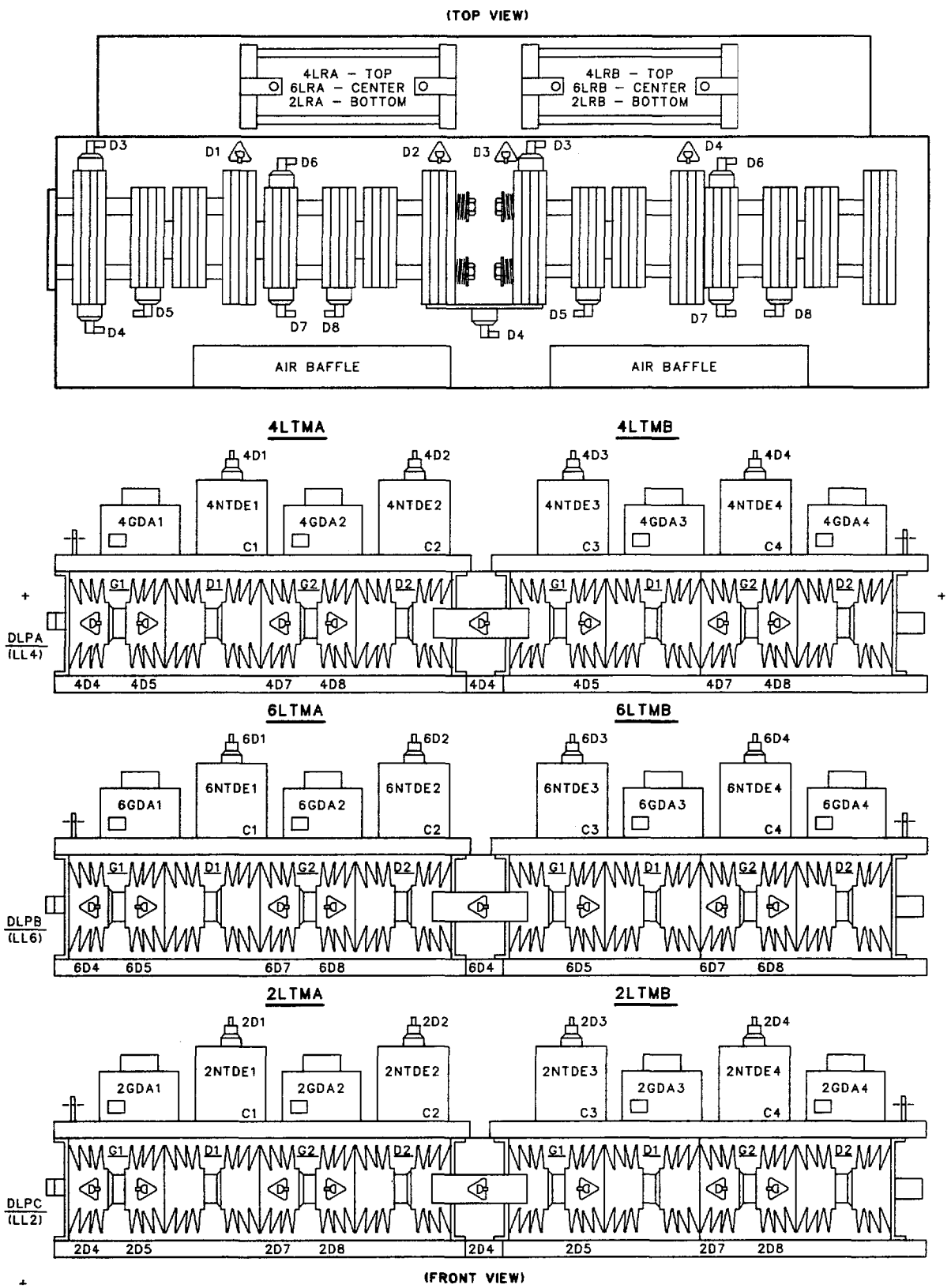


Figure 2-2A2. Front View of Power Bridge of 3kV Source and Load Assembly Drives Greater Than 1500Hp or Greater Than 2300V

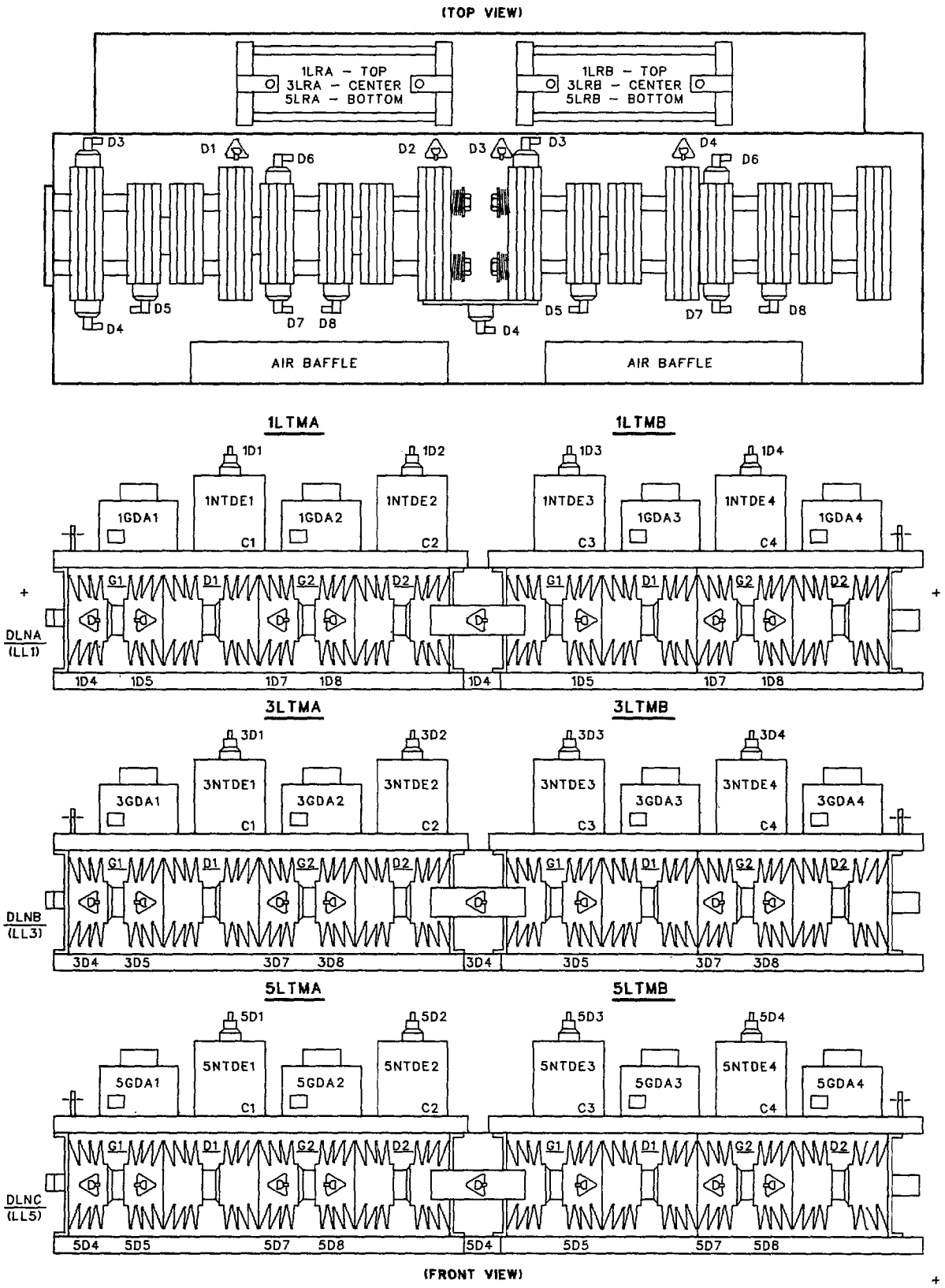


Figure 2-2B. Front View of Power Bridge of 4kV Source and Load Assembly Drive Greater Than 1500Hp or Greater Than 2300V

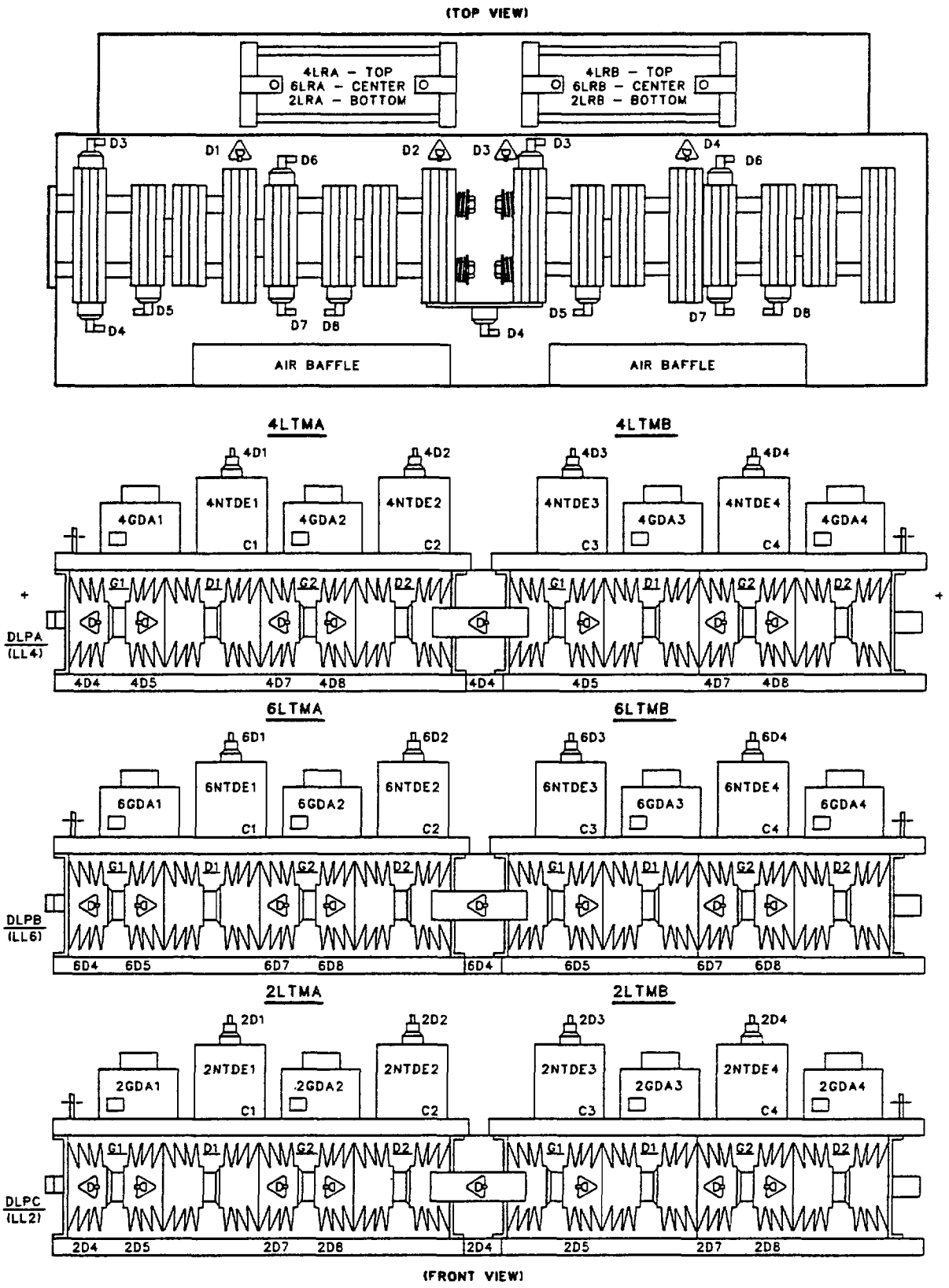
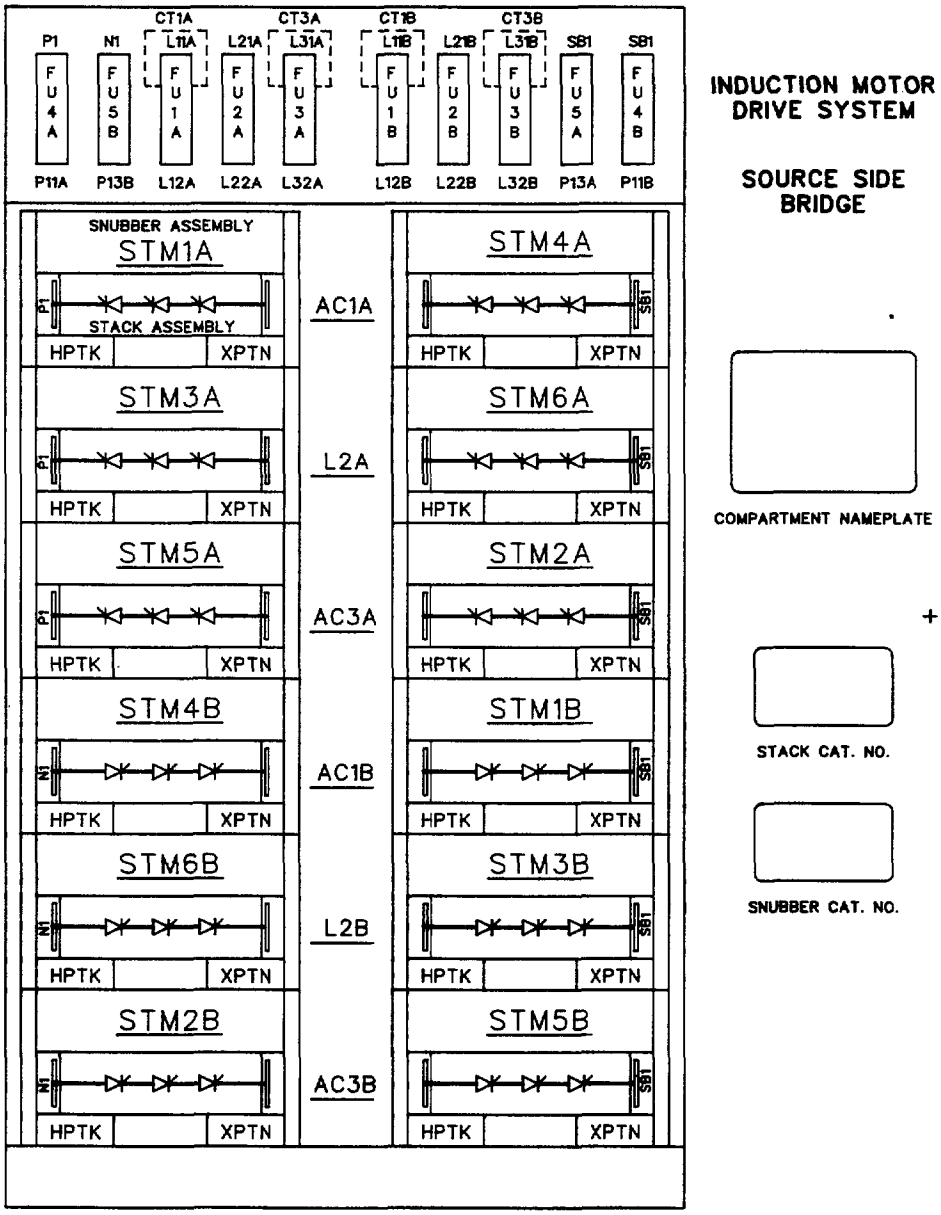


Figure 2-2B1. Front View of Power Bridge of kV Source Load Assembly Drive Greater Than 1500Hp or Greater Than 2300V



NP: 237B6890P1

Figure 2-3. Front View of Power Bridge of Series 12P, 2kV Source and 4kV Load Assembly Drives Greater Than 4000Hp with 77mm GTO's

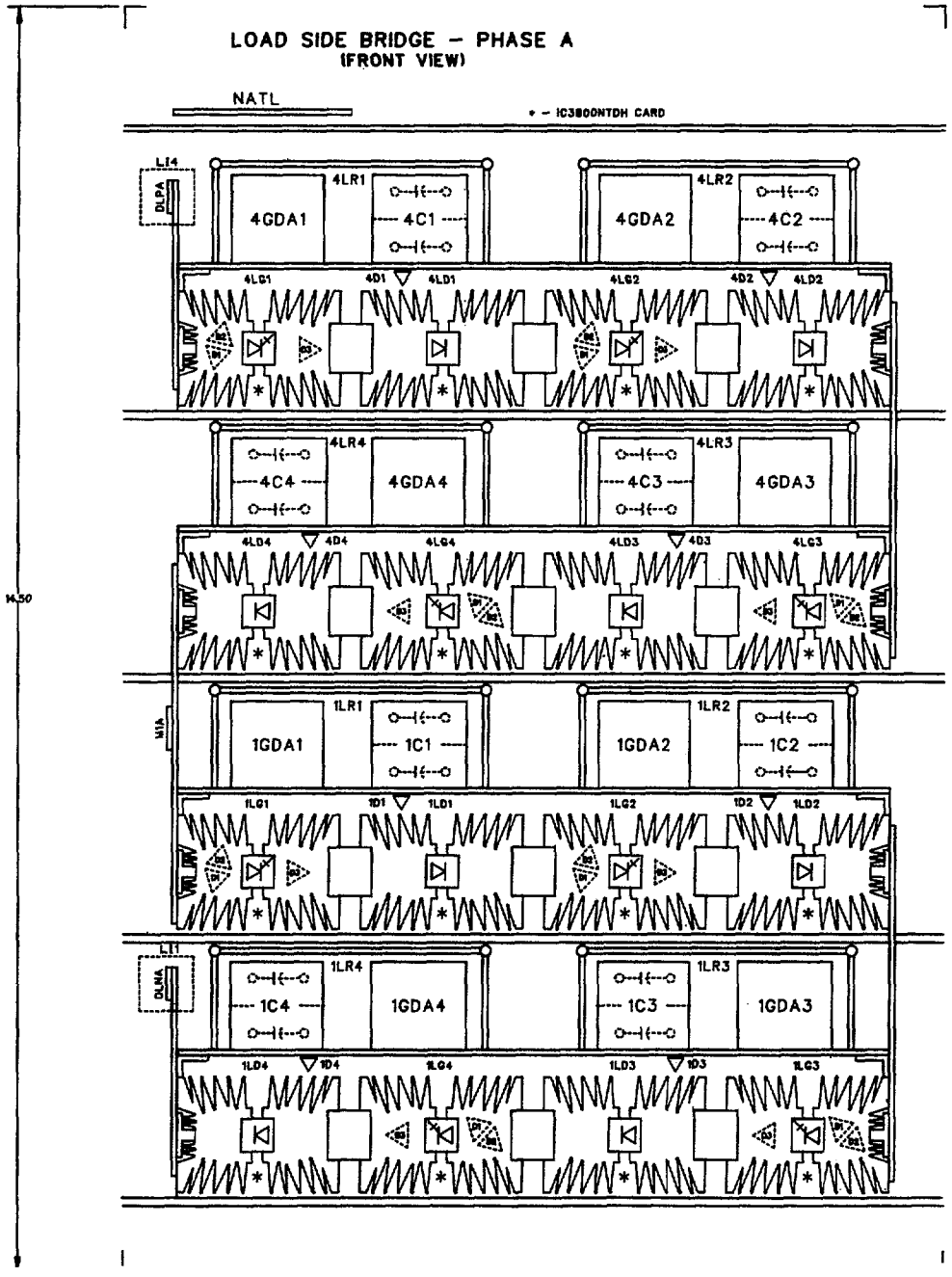


Figure 2-3A. Front View of Power Bridge of Series 12P 2kV Source and 4kV Load Assembly Drives Greater Than 4000Hp with 77mm GTO's

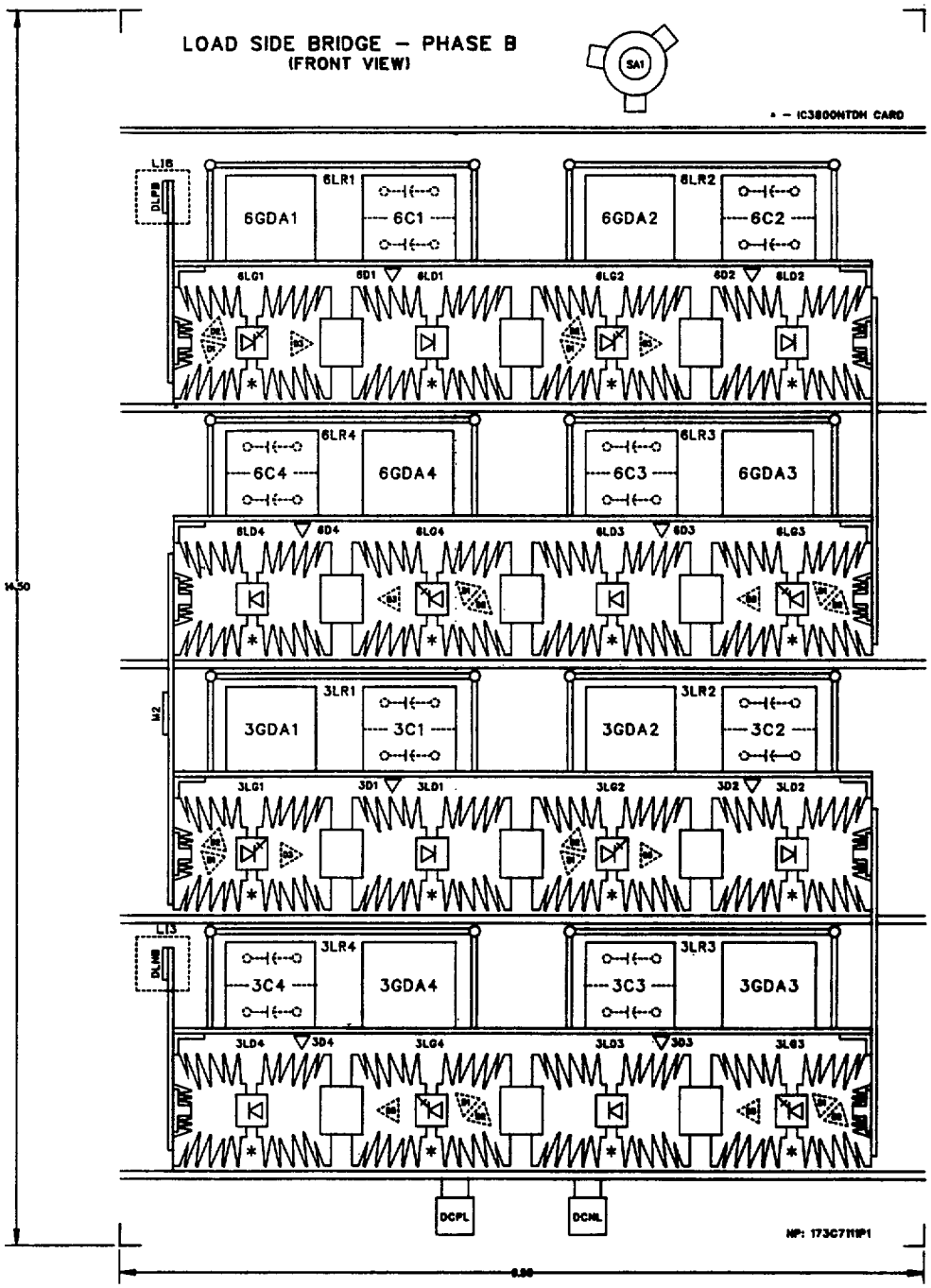


Figure 2-3B. Front View of Power Bridge of Series 12P 2kV Source and 4kV Load Assembly Drives Greater than 4000Hp with 77mm GTO's

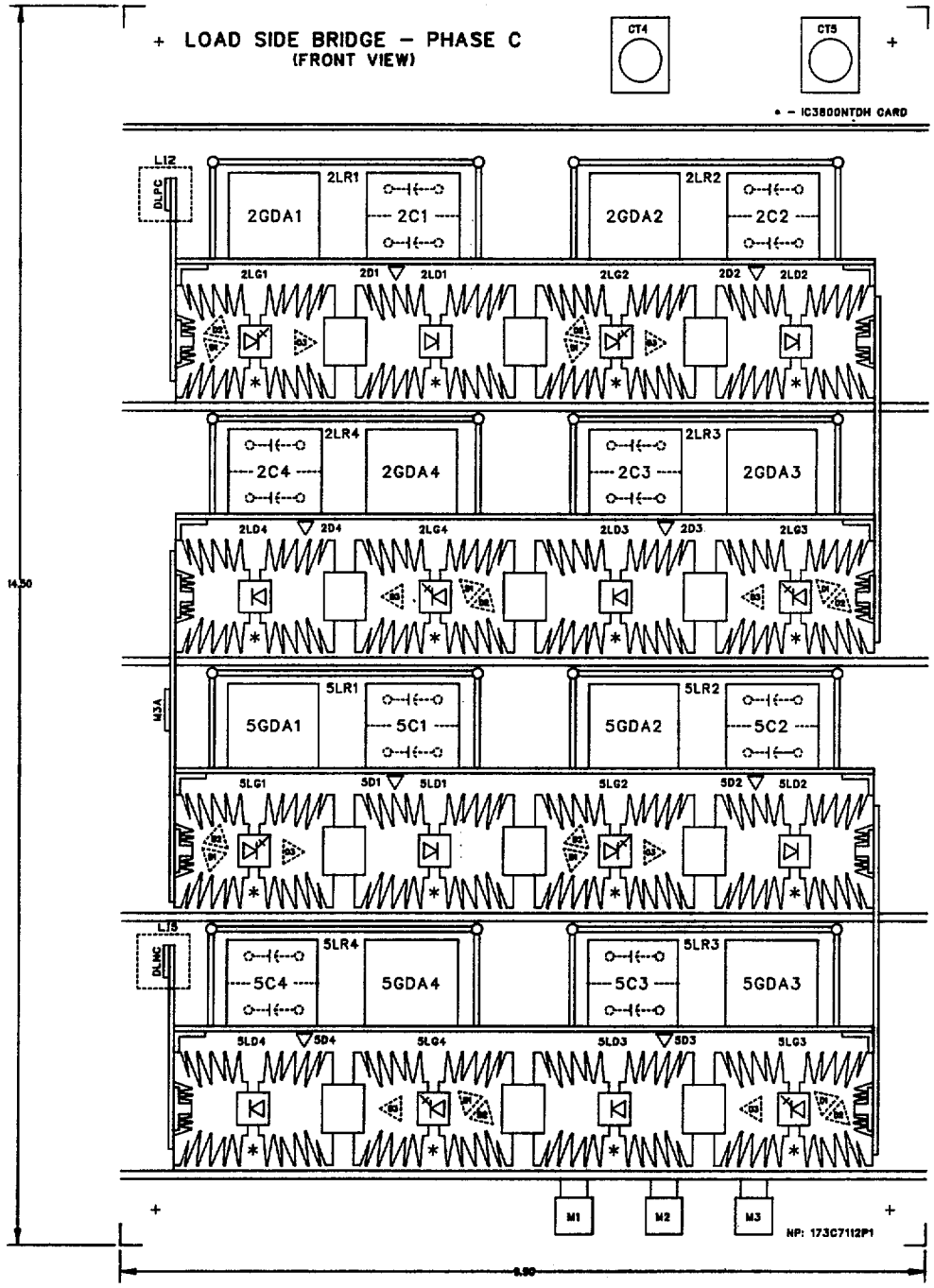


Figure 2-3C. Front View of Power Bridge of Series 12P 2kV Source and 4kV Load Assembly Drives Greater than 4000Hp- with 77mm GTO's

2-2.8. Voltage Feedback Alternator Card (XPTN)

In order to properly control the gating signals to the bridge, the electronics in the control compartment must have available analogs of all ac and dc bus voltages. The XPTN alternator cards in the thyristor modules provide bus-to-ground voltage signals which are combined in the electronic module to produce bus-to-bus signals for each converter bridge. The attenuator is variable via a berg jumper (BJ) for various system voltages: 1.75kV -STO; 2kV - LO; 4kV - HI.

2-3. LOAD POWER CONVERTER (BRIDGE) COMPARTMENT

The typical arrangement of the power converter GTO stack assemblies is shown in figure 2-1, 2-2 and 2-3. There is a diagram inside the bridge compartment to indicate the exact arrangement of the bridge. These figures and the bridge elementary diagram show the major power components. Figure: 2-3 is usually used for high horsepower drives.

The individual legs of the bridge are more complex than those of the source bridge because they are implemented with thyristors of the GTO (Gate Turn Off) type.

The function of the load bridge is identical to the source side bridge except for additional freedom in the allowed gating time of the thyristors. The source side thyristors can be gated over a gating delay angle, alpha, of approximately 10 degrees to 155 degrees. The reason for this restriction is that the source thyristors will not turn off at the end of a desired conduction period if the gating is outside the allowed range of values of alpha. The load side thyristors are of the GTO type and will turn off whenever gated off, so no restrictions on gating angle apply.

2-4. DETAILED CONFIGURATION OF THE LEG

Each leg of the load side bridge consists of several GTO's in series (two, three or four depending on the model). If specified, there is at least one more GTO in each series leg than required to withstand the voltage. Thus, on drives so equipped, the load side will continue to operate with one GTO in each leg failed.

2-4.1. GTO Auxiliaries

Each of the GTO's has several auxiliary devices associated with it to allow operation under all conditions. Figure 2-7A shows a typical GTO with its normal auxiliary devices: three diodes, a resistor-capacitor snubber, and equalizing resister, a cell light card and a gate driver. Figure 2-3 shows a typical mechanical arrangement of the same elements.

Diode D1 is in series with the GTO to block reverse voltage which will be applied in normal operation. The source side thyristors will block voltage in both directions but the load side GTO's will block forward voltage

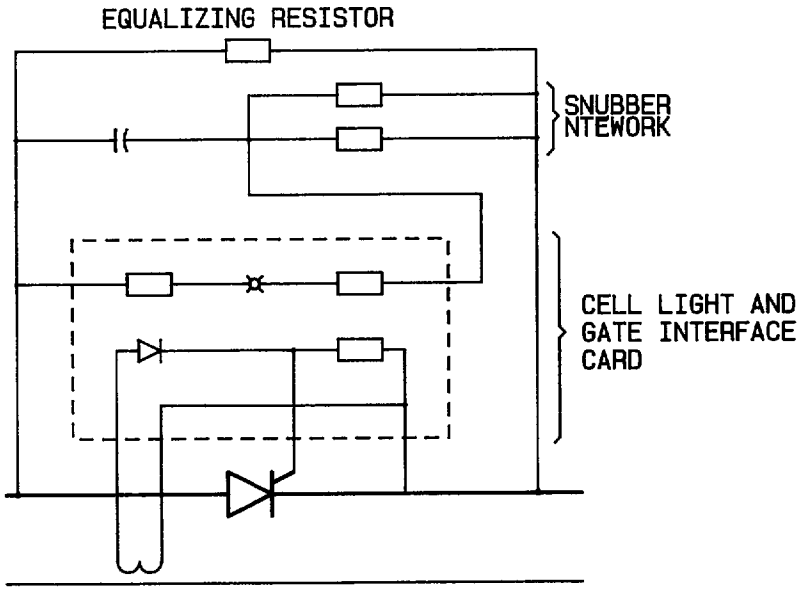


Figure 2-4. Cell With Gating, Snubber and Voltage Equalizing Circuits (typical)

only, therefore a series diode is needed. This diode is mounted under the same clamp with the GTO, in series mechanically as well as electrically.

A small diode Da is in parallel with the GTO with polarity reverse to the GTO. This diode is to assure that reverse voltage is never applied to the GTO. This diode is a stud type or flat base type diode with a tip flag terminal. It is attached directly to the GTO heat sink on the front side. In the case of 4500V GTO's this diode will be two series diodes each rated 2500V. Both are mounted on the front of the heatsink.

Another small diode Db is provided in parallel with the resistor which forms the snubber. This diode bypasses the snubber resistor in one polarity to make the capacitor more effective when the GTO turns off. This diode is mounted on the back side of the GTO heatsink, or in some cases on a bracket near the capacitor. It is of the same general type as the diode Da. When the voltage requires two devices in series for Da then there will also be two for Db. All these auxiliary diodes are similar mechanically but not necessarily interchangeable. Always replace a failed diode with one marked identically.

The capacitor Cs consists of one or more oil-filled metal can capacitors mounted behind or below the GTO heatsink.

The resistor Rs is of a flat-woven non-inductive type, mounted as an assembly behind the GTO heatsink stacks. The resistors associated with two or more GTO's may be located in a single assembly.

The equalizing resistor Re is connected across each GTO to force voltage sharing. This resistor consists of two power resistors mounted on the cell light card where possible. When its power rating is too high for card mounting it is a part of the resistor assembly.

The cell light card contains neon lamps which indicate the presence of voltage across the GTO and the series diode D1. These lights should be examined regularly to see if there are any failed components in the load side bridge. There is very little voltage across the diodes in some modes of operation, so the lights on all diodes may be dim. A failure is indicated only when one light is out while other similar lights are illuminated.

Failed GTO's should always be indicated by the red LED on the gate driver module any time control power is on. (High voltage is not required for this indication). If the gate driver indicated that the GTO is good and the GTO neon indicates that one GTO is not blocking voltage when other GTO neons are illuminated, then the failure is likely to be reverse diode Da which parallels the GTO.

2-5. WIRE DRESS

The type of wire and exact location of the wire are critical in the load side leg assemblies to prevent corona and to provide proper operation. If it is necessary to disconnect any of the wires on a load side assembly, then restore them to the original configuration as to wire type, quantity in parallel, routing and tying. Using another leg assembly as an example if necessary. Note that there are "right hand" and "left hand" stack assemblies and select one of the same configuration as the one disassembled to use as an example when rewiring.

2-5.1. For Configurations as shown in Figure 2-4, 2-5

The blowers are mounted at the top of the bridge compartment, also the top of the bridge houses the load

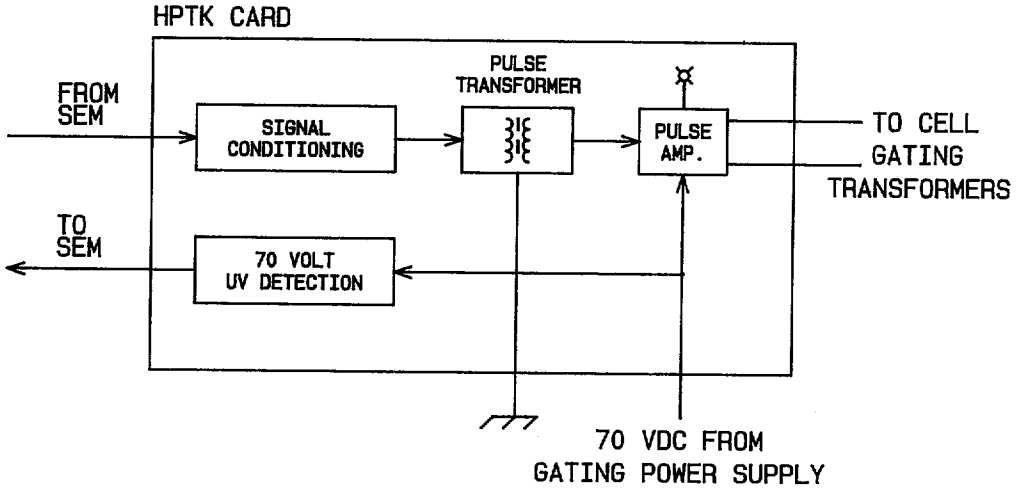


Figure 2-5. HPTK Block Diagram

side CTs. The RC Snubber Network is located behind the cell assembly.

2-5.2. For Configurations as shown in Figure 2-6

There is a separate blower compartment similar to that described in 2-2.2 except that there is no P70V power supply.

2-5.3. For all Configurations

The stack assemblies are oriented horizontally in the bridge so that air flow over the heatsinks is from front to back of the bridge enclosure.

2-5.4. For all Configurations

The gating Chopper Power Supply is located inside the load converter compartment.

2-5.5. Source Power Converter VS Load Converter

Different from the Source Power Converter, which has only one gating driver for each leg, the load converter has one gating driver for each GTO.

2-5.6. GTO Gate Driver Module

CAUTION

The gate driver module, including the signal level cards and enclosing sheet metal is connected directly to the high voltage circuit. Remove high voltage before maintaining.

The GTO Gate driver module enclosure is located in close physical proximity to the GTO and is electrically connected to the GTO cathode. The module should be considered a LIVE circuit component as long as it is mounted near the GTO in the power converter compartment. The gate driver module DS3820GGD provides drive to the GTO to turn it on and off. One driver is mounted directly above each GTO.

The signal that turns on a GTO is a current pulse into the GTO gate. In order to maintain conduction of the GTO, once the initial turn on is accomplished, a steady current must be kept in the GTO gate as long as the device is intended to remain in the conductive state. The turn-off of the GTO is accomplished by drawing current from the gate terminal with respect to the cathode. To maintain the GTO safely in the off-state, a negative voltage bias (not to exceed 15V) must be applied to the device, keeping the gate terminal negative with respect to the cathode. The tasks of the GTO driver module is to provide the means for controlling the GTO conduction at a maximum repetition rate of

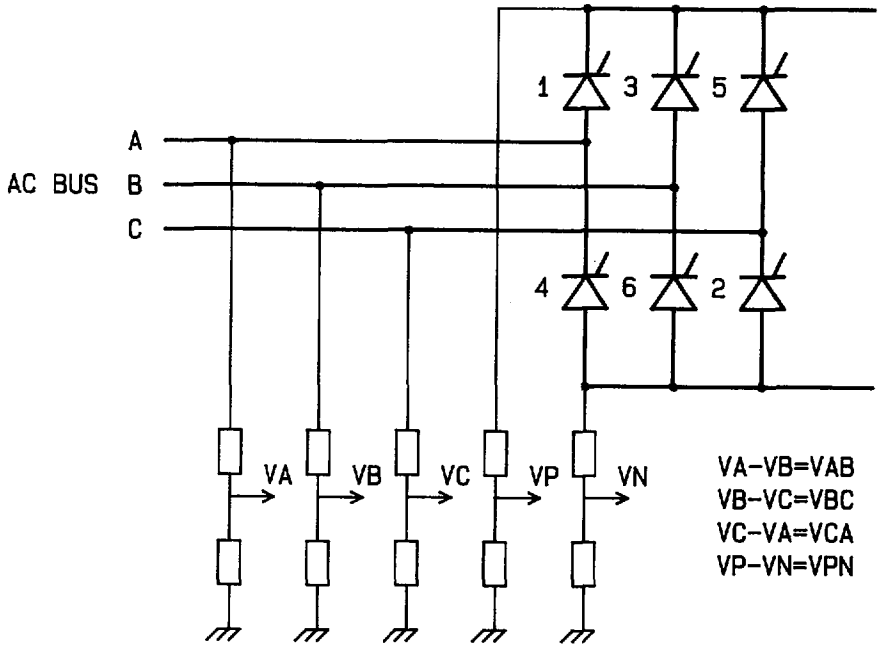


Figure 2-6. Bridge Attenuator Circuits

800 cycles per second. The control must be capable of pacing the ON and OFF intervals received into the input commands, so that the GTO can complete the turn-off phase before an ON-DRIVE signal is allowed. Also, the control must assure a minimum ON-DRIVE pulse duration so that the turn-on is completed.

The gate drive receives control power through an isolation transformer mounted at the rear of the driver module. It receives a signal to turn on or off the GTO by way of an optical fiber with a blue termination plugged into the blue receiver. It sends a diagnostic signal back to the control via the optical fiber with the grey termination plugged into the grey optical transmitter. The connections to the GTO gate are made by two screw terminals in the center of the upper card in the gate driver. The GTO leads are of controlled length and construction and should not be altered.

The optical fibers will be damaged if bent at a radius less than 35mm (1.38 inches), or if pinched or tightly cord tied.

The LED's on the gate drive modules indicate the status of the driver and GTO. The meaning of the LED's is as follows:

Left LED	Amber	Gate driver module OK when LED on
Left Center LED	Red	GTO failed (or gate driver output shorted)when led is on
Right Center LED	Amber	Gate driver receiving ON pulses via optical fiber
Right LED	Amber	400Hz power present when LED is on

If control power is present and the 400Hz power to the gate drivers is on, then the two outer amber LED's-should always be lit. The remaining amber LED will be lit only if the drive is running and will flicker at low speeds. The red LED may flash once when control power is first applied but should never be on in normal operation unless there is a failure. The diagnostic information fed back via the fiber optic cable to the control panel is the summation of the indications at the gate driver module. Thus any transient indication of malfunction at a gate driver will produce an alarm at the control and a latched LED lit on the fiber optic distribution board, XDCD, in the control panel.

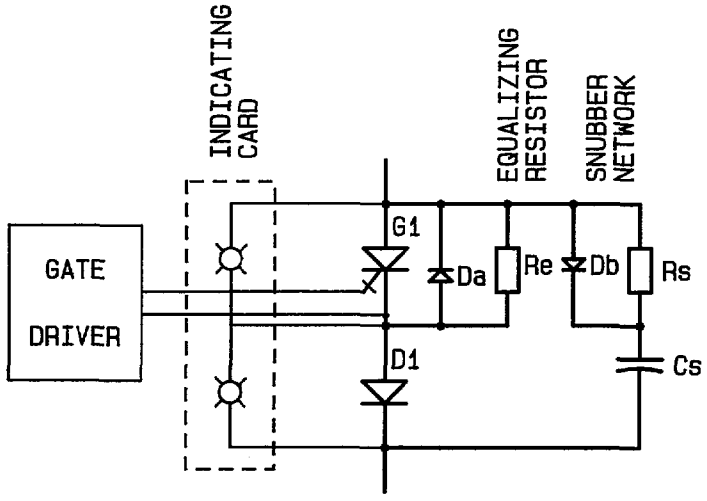


Figure 2-7A. GTO Thyristor with Auxiliaries

2-6. 400 HERTZ CONTROL POWER SUPPLY

The 115Vac 400 Hz control power for the gate drivers is provided by module DS3820CPSC mounted below or above the GTO leg shelves. Input is 115Vac 60Hz, output is 115Vac 400Hz squarewave.

2-7. THE LOAD SIDE SURGE SUPPRESSOR

The load side surge supressor SA1 is mounted in the bottom of the load side bridge cabinet. It serves to absorb the energy in the dc link inductor if all the load side GTO's are turned off while the current is flowing.

2-8. LOAD SIDE ATTENUATORS - NATL

The attenuator board NATL is located in the load bridge cabinet to sense voltages at the three motor terminals and the positive and negative dc link on the load side of the dc link inductor. The function is as illustrated in Figure 2-8. The signals are attenuated by a resistive divider consisting of the 10 meg ohms on NATL and a terminating resistor in the control panel. The surge supressors limit this signal to less than 75 volts to chassis in case the path to the terminating resistor is opened.The signals are passed to the control panel via a ribbon cable on connector JV. Signals from the remote attenuator which is located at the motor capacitors and senses the voltage at their neutral. These signals may be brought into the NATL board via connector JH for inclusion in the ribbon cable to the control panel.

2-9. LEADS TO MOTOR CAPACITORS

The capacitor connections must be made directly from the PCM to the capacitors (never connect the capacitors to the motor terminals or to the motor switchgear terminals). The capacitor cables must be bundled into a three phase set, tightly tied together. The three phases must never be separated into different bundles or conduits, and they should not be run in armored single phase cable. If it is impossible to meet the limits on length in the table which follows, then two or more bundles can be run in parallel. Each bundle should contain all three phases. When two bundles are used, length limitations are doubled.

The selection of capacitor leads and lead dress is to keep the inductance per phase, less than 2.0 uHy (.754 milliohms reactance on a 60 Hz basis). See sheets 2-19 and 2-20 from the General Electric Industrial Power Systems data book. Use these tables to calculate any installation. From table 1 the length limits for meeting a 2.0 uHy limit are as follows:

Wire Size AWG	Allowed Length of Three Conductor Cable (non -shielded 5kV cable in magnetic duct)
2	17 feet
1/0	18 feet
3/0	19 feet

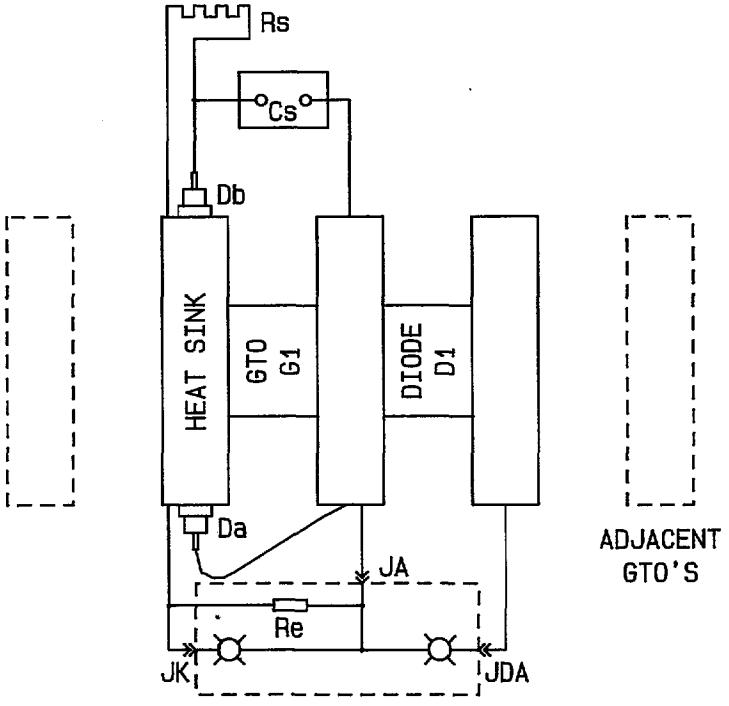


Figure 2-7B. Mechanical Arrangement, GTO and Auxiliaries, Top View (Gate Driver Not Shown)

As can be seen from the tables on sheets 2-19 and 2-20, shielded cable is a little worse, non-magnetic duct is a little better. As you can see from table 2, single conductors (as opposed to three conductor cable) are a little worse.This can be overcome by cord-tying the three cables together every three feet or so before pulling them into the conduit. If these limits seem too short, note that the allowed length for smaller wire is almost the same as for larger.Thus there is a big advantage to using two or more bundles of small wire in parallel, for providing two bundles doubles the allowed length and three bundles triples the length. Each bundle must contain all three phases.

These are upper limits. Shorter connections are to be preferred.

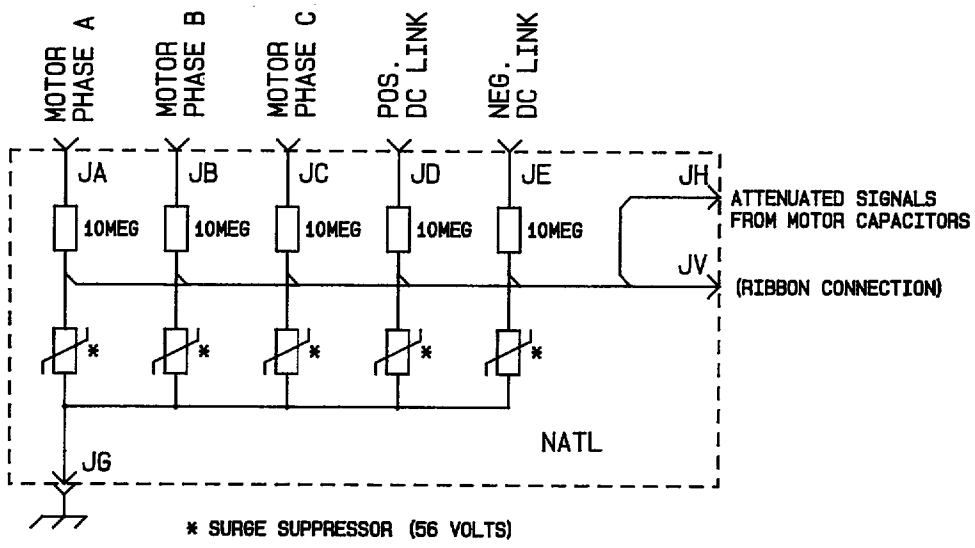


Figure 2-8. Attenuator NATL Interface

Conductor Constants

CABLE

Close approximations of resistance and inductive-reactance values for cable circuits can be made from the material in Section .6. The ready-reference material in Tables 1 and 2 below is less accurate but is satisfactory for preliminary analyses or rough checking, particularly when details of cable type and method of installation are not known. When conductor spacing (or insulation thickness) is known, closer approximations of reactance can be derived from Fig. 1 provided the conductors are not installed in magnetic duct or armor.

The resistance values are based on tinned Class B stranded-copper conductors (unless otherwise specified) at 60 cps and at 75 C.

The inductive reactance values are at 60 cps and are either positive- or negative-sequence values.

Ratios of Z_0/Z_1 for various arrangements of cable circuits can span a range of about 1 to 50 or more, depending on the path of the returning L-G fault current. Aluminum conduit shows a ratio of about 2, with or without an internal ground conductor. Steel conduit shows ratios from about 4 to 14 with the higher values associated with lower currents and larger conductors. When a full-size internal ground wire is used in steel conduit, the ratio will not exceed a value of about 4. Cable with magnetic armor shows higher ratios from about 15 to 30.

TABLE 1—Three-conductor Cables—Impedances L-N in Ohms/1000 Feet

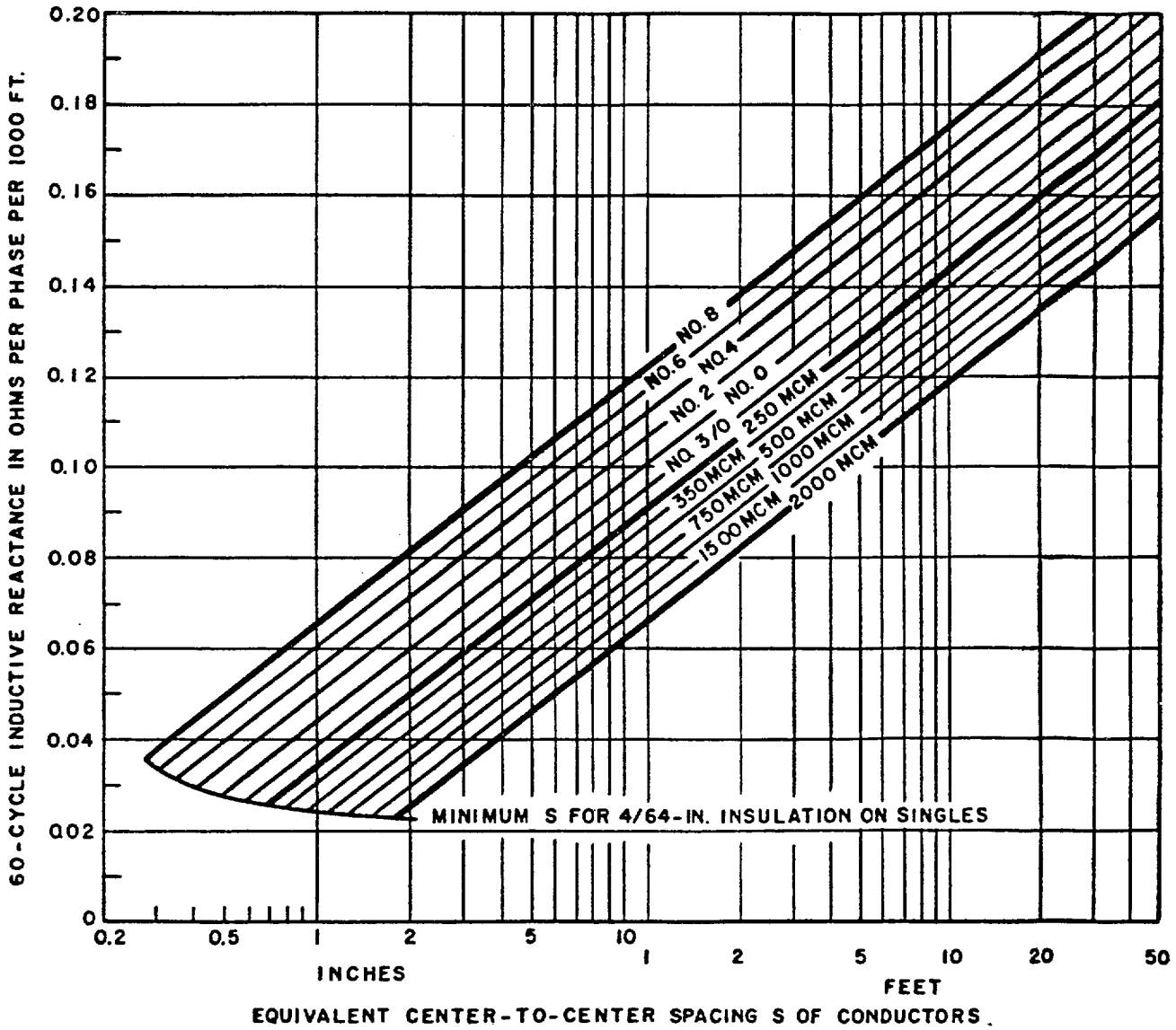
AWG or MCM	In Magnetic Duct or Armor						In Nonmagnetic Duct or Armor*					
	Nonshielded 600 V and 5 Kv			Shielded 5 Kv and 15 Kv			Nonshielded 600 V and 5 Kv			Shielded 5 Kv and 15 Kv		
	R	X	Z	R	X	Z	R	X	Z	R	X	Z
8	.811	.0577	.813	.811	.0658	.814	.811	.0503	.812	.811	.0574	.813
8 (solid)	.786	.0577	.788	.786	.0658	.789	.786	.0503	.787	.786	.0574	.788
6	.510	.0525	.513	.510	.0610	.514	.510	.0457	.512	.510	.0531	.513
6 (solid)	.496	.0525	.499	.496	.0610	.500	.496	.0457	.498	.496	.0531	.499
4	.321	.0483	.325	.321	.0568	.326	.321	.0422	.324	.321	.0495	.325
4 (solid)	.312	.0483	.316	.312	.0508	.317	.312	.0422	.315	.312	.0495	.316
2	.202	.0448	.207	.202	.0524	.209	.202	.0390	.206	.202	.0457	.207
1	.160	.0436	.166	.160	.0516	.168	.160	.0380	.164	.160	.0450	.166
1/0	.128	.0414	.135	.128	.0486	.137	.127	.0360	.132	.128	.0423	.135
2/0	.102	.0407	.110	.103	.0482	.114	.101	.0355	.107	.102	.0420	.110
3/0	.0805	.0397	.0898	.0814	.0463	.0936	.0766	.0346	.0841	.0805	.0403	.090
4/0	.0640	.0381	.0745	.0650	.0446	.0788	.0633	.0332	.0715	.0640	.0389	.0749
250	.0552	.0379	.0670	.0557	.0436	.0707	.0541	.0330	.0634	.0547	.0380	.0666
300	.0464	.0377	.0598	.0473	.0431	.0640	.0451	.0329	.0559	.0460	.0376	.0596
350	.0378	.0373	.0539	.0386	.0427	.0576	.0368	.0328	.0492	.0375	.0375	.0530
400	.0356	.0371	.0514	.0362	.0415	.0551	.0342	.0327	.0475	.0348	.0366	.0505
450	.0322	.0361	.0484	.0328	.0404	.0520	.0304	.0320	.0441	.0312	.0359	.0476
500	.0294	.0349	.0456	.0300	.0394	.0495	.0276	.0311	.0416	.0284	.0351	.0453
600	.0257	.0343	.0429	.0264	.0382	.0464	.0237	.0309	.0389	.0246	.0344	.0422
750	.0216	.0326	.0391	.0223	.0364	.0427	.0197	.0297	.0355	.0203	.0332	.0389

TABLE 2—Three Single Conductors—Impedances L-N in Ohms/1000 Feet

AWG or MCM	In Magnetic Duct						In Nonmagnetic Duct or Armor*					
	Nonshielded 600 V and 5 Kv			Shielded 5 Kv and 15 Kv			Nonshielded 600 V and 5 Kv			Shielded 5 Kv and 15 Kv		
	R	X	Z	R	X	Z	R	X	Z	R	X	Z
8	.811	.0754	.814	.811	.0860	.816	.811	.0603	.813	.811	.0688	.814
8 (solid)	.786	.0754	.790	.786	.0860	.791	.786	.0603	.788	.786	.0688	.789
6	.510	.0685	.515	.510	.0796	.516	.510	.0548	.513	.510	.0636	.514
6 (solid)	.496	.0685	.501	.496	.0796	.502	.496	.0548	.499	.496	.0636	.500
4	.321	.0632	.327	.321	.0742	.329	.321	.0506	.325	.321	.0594	.326
4 (solid)	.312	.0632	.318	.312	.0742	.321	.312	.0506	.316	.312	.0594	.318
2	.202	.0585	.210	.202	.0685	.214	.202	.0467	.207	.202	.0547	.209
1	.160	.0570	.170	.160	.0675	.174	.160	.0456	.166	.160	.0540	.169
1/0	.128	.0540	.139	.128	.0635	.143	.127	.0432	.134	.128	.0507	.138
2/0	.102	.0533	.115	.103	.0630	.121	.101	.0426	.110	.102	.0504	.114
3/0	.0805	.0519	.0958	.0814	.0605	.101	.0766	.0415	.0871	.0805	.0484	.0939
4/0	.0640	.0497	.0810	.0650	.0583	.0929	.0633	.0398	.0748	.0640	.0466	.0792
250	.0552	.0495	.0742	.0557	.0570	.0797	.0541	.0396	.0670	.0547	.0456	.0712
300	.0464	.0493	.0677	.0473	.0564	.0736	.0451	.0394	.0599	.0460	.0451	.0644
350	.0378	.0491	.0617	.0386	.0562	.0681	.0368	.0393	.0536	.0375	.0450	.0586
400	.0356	.0490	.0606	.0362	.0548	.0657	.0342	.0392	.0520	.0348	.0438	.0559
450	.0322	.0480	.0578	.0328	.0538	.0630	.0304	.0384	.0490	.0312	.0430	.0531
500	.0294	.0466	.0551	.0300	.0526	.0605	.0276	.0373	.0464	.0284	.0421	.0508
600	.0257	.0463	.0530	.0264	.0516	.0580	.0237	.0371	.0440	.0246	.0412	.0479
750	.0216	.0445	.0495	.0223	.0497	.0545	.0194	.0356	.0405	.0203	.0396	.0445

* These values may be used for preassembled aerial cable.

Conductor Constants



(Photo 1177701)

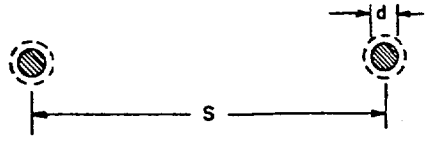
Fig. 1. Calculated positive- or negative-sequence inductive reactance for parallel conductors with standard stranding where values are per conductor for two-wire, single-phase circuits and line-to-neutral for three-phase circuits

OVERHEAD LINES

Practical transmission lines are often assumed to have a 60-cps positive- or negative-sequence reactance as high as 0.8 ohms/mile (or 0.15 ohms/1000 feet) line-to-neutral. Closer values can be obtained from Figure 1 if the conductor spacing is known. The values in Figure 1 were calculated from the equation

$$X_L = 10^{-6}\omega\left(15.2 + 140.4 \log \frac{2S}{d}\right)$$

with dimensions according to the following illustration where S and d are in the same units:



For an unsymmetrical arrangement of three conductors, an equivalent value of S for use in Fig. 1 can be derived from the relation

$$S = \sqrt[3]{(S_1)(S_2)(S_3)}.$$

Z_0/Z_1 ratios for overhead lines may vary from 2.25 to 6 or more depending on the number of circuits, and the number and disposition of ground wires. Approximations for actual arrangements can be obtained from "Circuit Analysis of A-C Power Systems"—Vol. I by E. Clarke.

CHAPTER 3
CONTROL ELECTRONICS PANEL

3-1. INTRODUCTION

The control electronics panel is made-up mainly by various modules interconnected thru ribbon cables and harnessed wiring. The main modules are:

DS3820IMF Main Control electronics

DS3820IOM Inputs/Outputs (I/O) Module

DS3820LDM Display Module

The modules are typically configured as in Figure 3-0. Following are brief descriptions of the functions of the module. For more specific information the corresponding modules elementary and documentation should be consulted.

3-2. ELECTRONICS MODULE (LCG)

The control electronics consists of cards in 2-rows x 6-slots module. Its primary purpose is to control the firing of the power converters SCR's and GTO's for driving a motor. It also provides various protective functions for the motor and bridges. Interface between the IMF to the customer is thru the IOM .

See Figure 3-1 for the physical layout of the module. Connection is made to the electronics module (SEM) through the plugs located at the top, and also through card front connectors.

From a functional standpoint, the electronics module processes signals as represented in Figure 3-2.

The signal interfaces for the SEM are as follows:

- Attenuator signals - Attenuated bus to ground signals from each converter.
- Line/load current feedback - Signals from line/load bus current transformer burden resistors.
- Power supply - The module requires +5, ±15, and +28 volts which are provided by a power supply.
- Thyristor gate signals - These are the electronic signals which drive the gate pulse circuits in the thyristor bridges.

The hardware of the electronics module is represented from a block diagram standpoint in Figure 3-3.

There is one microcomputer controlling the firing of both of the power converters.

The microcomputer establishes a local bus (L-bus) structure on which the input/output for it is interfaced. In addition, it communicates to an 8k byte read/write (RAM) memory and a 2k byte electrically alterable memory (EEPROM). This memory combines features

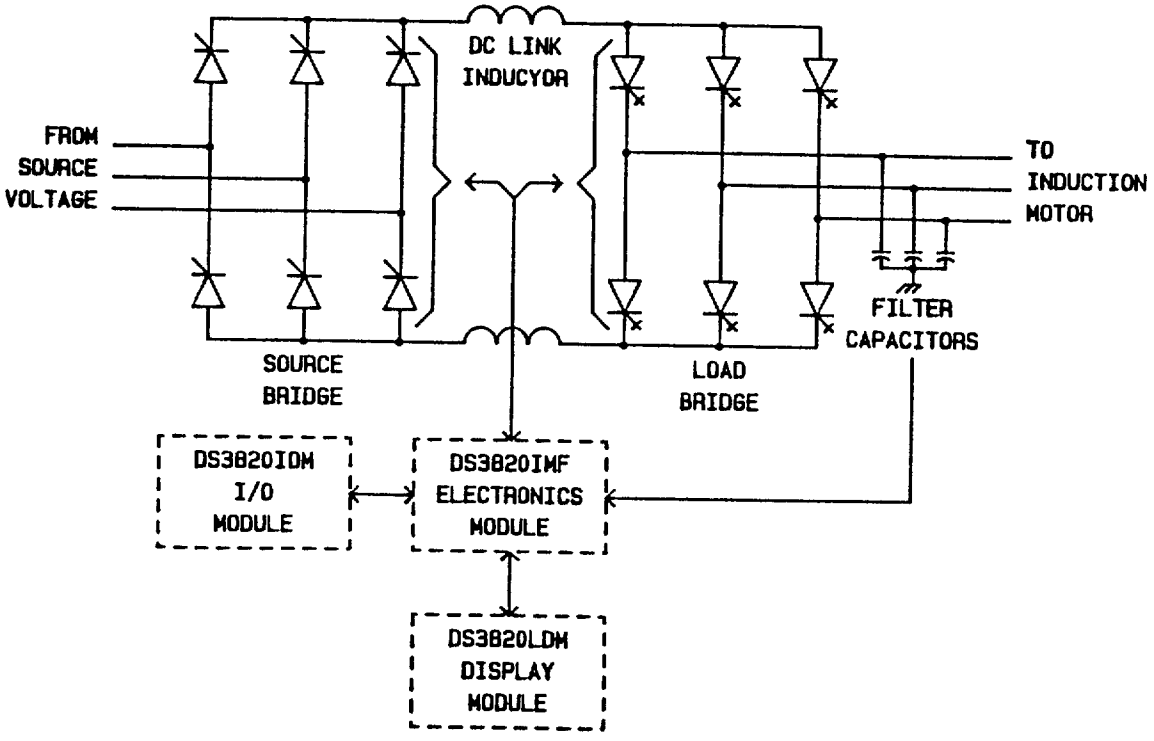


Figure 3-0. Control Electronic Panel

of read/write memory (RAM) and read only memory (ROM) in that it is electrically alterable (although slow) and at the same time will not lose its contents on power loss. The primary use of this EE
PROM memory is to store parameters which must be tailored to each different system. These parameters are field adjustable for drive tuneup.

The local bus has a timer and interrupt card which provides the necessary hardware for control of interrupt priority and timing for the microcomputer.

There are input/output ports on the local bus which interface static logic signals to/from the microcomputer. As an example, thyristor gating commands are written out to one of these ports (on each side) and sent to an analog card to produce the gating signals to the converter bridge.The electronic control contains a number of analog cards which process the voltage feedback, current feedback, meter signals, etc.

Analog to digital converter cards interface the speed reference, voltage feedback, current feedback signals, etc. to the microcomputer bus on each side.

I/O cards interface process command and status signals to the microcomputer. These inputs are combined logically by a sequencing program in the microcomputer to control the relays in the IOM. These outputs are used to control system devices such as panel lights, contactors, etc..

3-3. CONTROL OPERATIONAL DESCRIPTION

3-3.1. Basic Principles

Refer to the control system block diagram in Figure 3-4. The speed reference from the process controller is compared to a speed feedback derived from the integrated motor voltage to develop a speed error input to the speed regulator. The output of the speed regulator is a torque command.

The torque command signal is applied to both the source and load side control. As illustrated in Figures 1-6 and 1-7, motor torque is a function of voltage, current, and the angle between them. Therefore, torque can be controlled by adjusting current magnitude at a fixed angle or maintaining a constant current and

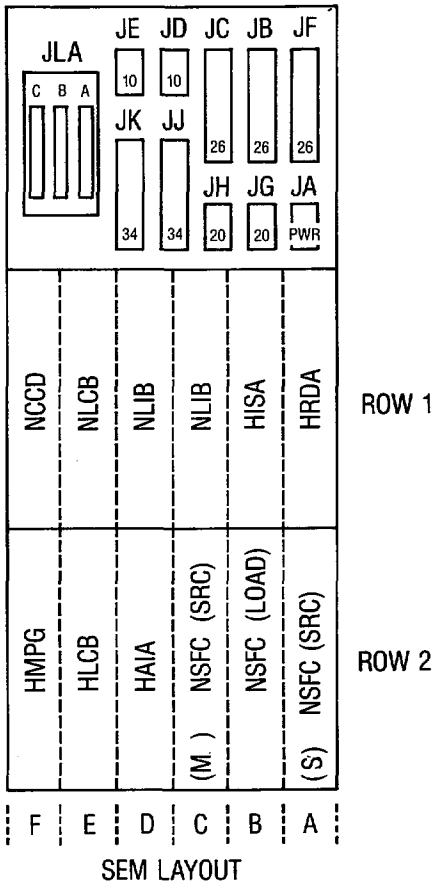


Figure 3-1. Small Electronics Module (SEM) Card and Connector Layout

varying the displacement angle θ. The current is controlled from the source side , whereas the angle is varied from the load side. At any instant in time, the torque is actively controlled by only one of these means.

The torque command to the source side control is applied to a maximum and minimum current limiter. The minimum current level is set to maintain continuous current in the dc link.

The current level control also affects the load side angle control. When the voltage spillover is not activated, the angle control is in one of its limits. If current is adjusted to control voltage, the angle control gain is decreased and displacement angle θ is increased, which reduces developed torque to that required by the load.

The current command is compared with current feedback, I_{DC}, and the error is applied to the current regulator. The current regulator controls the firing of the thyristors in the source side converter (rectifier). Thus, the source side control adjusts the dc link voltage as required (within its limits) to produce the motor current needed to drive the load.

The load side control is designed to adjust firing angle so as to maintain sufficient flux by controlling the angle between voltage and current.

Referring to Figure 1-5A and 1-5B, it is shown that the angle of advance, β is the sum of the commutation angle, μ and the margin angle, γ. For maximum power factor, it is desired to keep the margin angle as small as possible without encountering commutation failure. The objective is to make the displacement between voltage and current as small as possible while maintaining torque and flux.

When the drive requires regeneration, the load-side thyristors are fired full advance (point "X" in Figure 3-5), and the source side controls current by reversing the dc voltage to match the rectified motor voltage.

3-3.2. Phase Locked Loop Operation

Since the load side thyristors are fired in synchronism with the integrated motor stator voltages (flux), a phase locked loop (PLL) is used to maintain correspondence between the integrated voltage wave zero crossings (Figure 3-5) and the firing pulses.

When the PLL output is precisely in synchronism with the "flux" wave zero crossings, the loop is said to be "locked". If the output is lagging (slower than) the flux crossings, the PLL controller acts to increase the frequency (by decreasing PLLC LOAD). Similarly, if the frequency is leading (faster than) the flux crossings, the controller acts to decrease frequency (by increasing PLLC LOAD).

Although the output of a phase locked loop, when "locked", will have the same frequency as the reference signal, there will be a phase difference between the two due to the integrating action of the loop low-pass filters; thus, the name phase locked- loop.

3-3.3. Electronics Module (IMF) Printed Wiring Board (PWB) Descriptions

Following is a list of cards used in the electronics module. The function and condensed theory of operation of these cards follows the list. It is not the intent to describe in great detail the inner workings of each card,

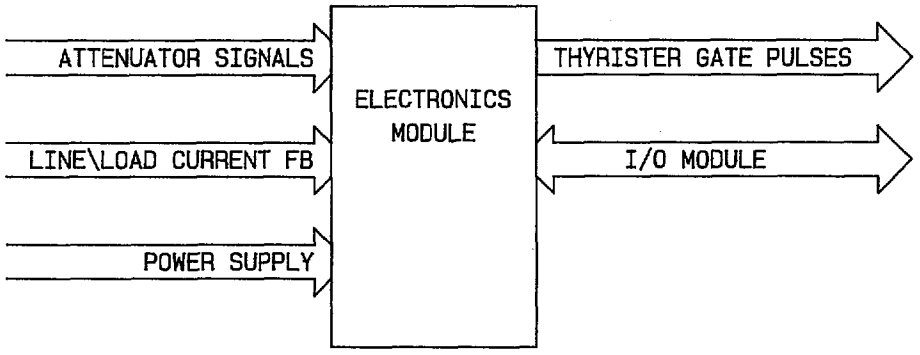


Figure 3-2. SEM Input/Output

but to relate their function in the overall system. For more detail, refer to the appropriate DS3800 drawing, Section 2, for operating notes and Section 4 for elementary diagram. It is also generally helpful to refer to the card symbols in the DS3820IMF elementary diagram.

3-3.4. Microprocessor - HMPG/DMPB

The DS3800HMPG is a single board computer with the following features.

1. 80286 microprocessor
2. Programmable communications interface
3. Programmable interval timer
4. On-board RAM memory (16K bytes)
5. On-board PROM memory (64K bytes)
6. Mapped I/O
7. Bus hog timer
8. Stall timer

The configuration chosen gives a powerful 16 bit processor.

A 50 pin ribbon connector is included on the HMPG to interface the EEPROM memory (DMPB).

A 25 pin male connector is present to serve as an RS232C communications port. (See Tables 3-1A and 3-1B.) A "null modem" is supplied to allow a standard data terminal interface cable to be used with either communications port.

3-3.5. Electrically-alterable ROM-DS3800DMPB

This is a non-volatile memory board containing 2k bytes of memory used to store drive tuneup parameters which may be customized on a per drive basis. Physically, the board is configured as a daughter board and mounted on the HMPG.

Constants stored in the board's memory are normally changed by first enabling board write enable with switch ""WREN" and then supplying the data through the drive monitor.

In the Drive Systems documentation procedures, DMPB and the two PROMS in sockets U2 and U3 comprise module DS3815PLH3.

3-3.6. Processor Extender Board - DS3800HLCB

The HLCB board extends the functions provided by HMPG to include those required for the system. The following functions are provided.

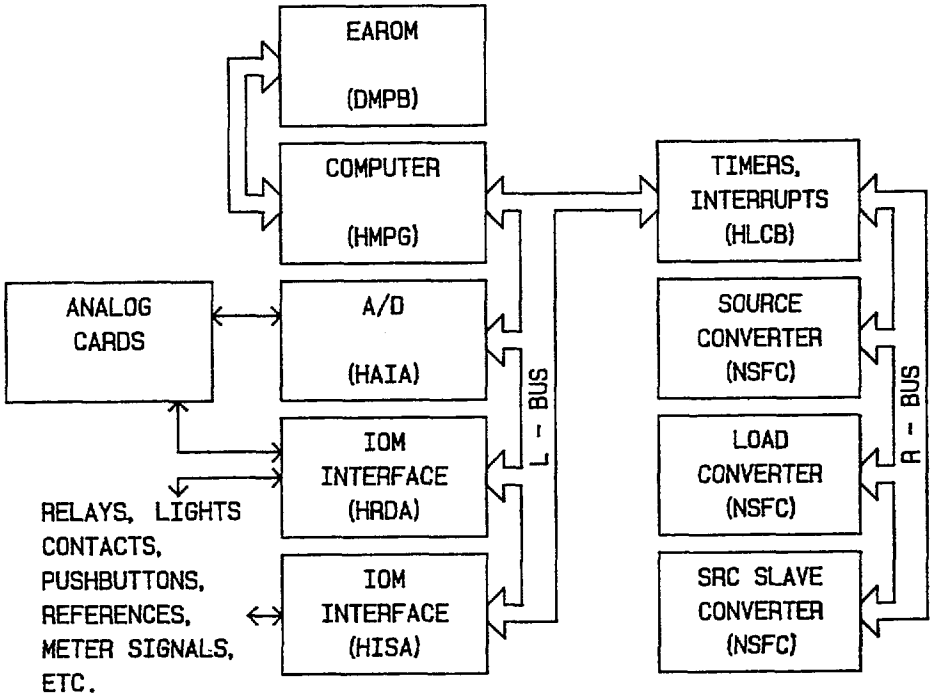


Figure 3-3. Electronics Module Block Diagram

1. Address decoding of on-board functions, NSFC
2. Interrupt Controller
   - Source crossover
   - Load crossover
   - Source angle control
   - Load angle control
3. 5 bit general purpose input port used for:
   - Ground fault
   - I/O module input ribbon cable continuity
   - I/O module signal cable continuity
   - I/O module output ribbon continuity
   - Door module ribbon cable continuity
4. Zero crossing detection. Circuity is included to derive source and load crossover interrupts from square wave information originating on the NLIB boards. A single clock pulse interrupt is obtained at each crossover.
5. Two programmable timers provide the functions of
   - PLL clock (source and load)
   - Time counter (source and load)
   - Speed calculation
   - Clock for serial communications port
6. RS-232 compatible port. Only the transmit function implemented at the port is intended for driving a printer located in the door module. The control line, DSR (data set ready), is also monitored to determine the status of the remote printer.
7. Digital to analog converters. Four 8-bit D/A converters are supplied. D/A 1 and D/A 2 are 8 bit non-committed outputs which are available for transmission to the user. D/A 3 is committed to putting out the drive horsepower signal. The 12 bit D/A 4 is committed to supplying the speed feedback signal for driving meters etc. System level documentation will define its use (if any) in other systems.

3-3.7. Analog/Digital Interface Card - DS3800HAIA

The HAIA board accepts 16 analog input signals, multiplexes them to one line and coverts to a two byte digital word. The A/D provides 11 bits of magnitude resolution. Also provided are three, 8-bit digital input ports, three 8-bit output ports and one 4-bit digital port. (See Table 3-3)

3-3.8. DC Link Commutator Control Card - DS3800NCCD

This card contains the auxiliary circuits used for source series 12 pulse, synchronizing, and for GTO protectives. It also receives the status (OK) of the GTO's. This information is available for use by the control program via the L-Bus.

These functions include:

1. Source slave bridge's current conditioning.
2. A fault trip comparator of source slave current vs load current.
3. A shorted motor filter capacitor detector.
4. Motor current calculation circuitry which consists of the filter capacitor current and subtracting it from the inverter current
5. Synchronizing function circuitry which includes PT match comparator (motor volts vs source (bypass) volts) and motor current as read from the bypass or motor isolator.
6. DC link overvoltage detection
7. Chopper power supply okay.
8. Ground fault detector based on the source drive current (zero sequence) for use where the source line is a grounded system.

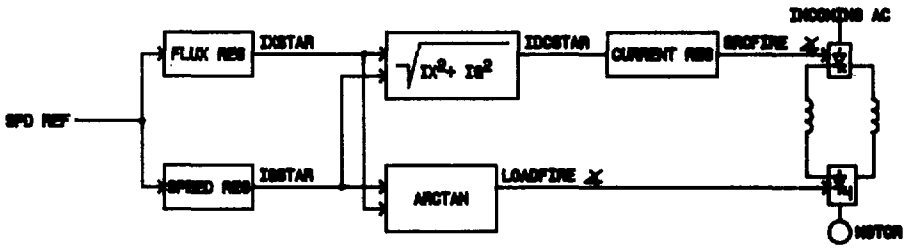


Figure 3-4. System Digital Regulator Block Diagram

9. Two 0-10v to 4-20mA converters for supplying signal indication to user if needed.
10. Source or load instantaneous overcurrent
11. Freeze load firing if having an instantaneous overcurrent or the chopper power supply is bad.

Test points and lights include:

(NCCD)

TP1	OIA2	SRC slave phase A current
TP2	OIB2	SRC slave phase B current
TP3	OIC2	SRC slave phase C current
TP4	1IFB2	SRC slave bridge current 5V=1PU
TP5	IERR	(SRC slave current)-(load current)
TP6	CIAB	Capacitor phase A-B current
TP7	CICA	Capacitor phase C-A current
TP8	MIAB	Motor phase A-B current
TP9	MIBC	Motor phase B-C current
TP10	MICA	Motor phase C-A current
TP11	MIFB	Motor current 5V=1PU
TP12	PTER	Motor PT(B) - Source PT(A)
TP14	CAPFLT	Shorted Capacitor detect
TP17	V/I2	Auxiliary voltage to current converter output 2
TP18	V/I1	Auxiliary voltage to current converter output 1
TP19	OIB1	Phase 2 of SRC current (for zero sequence GNDF detector)
TP23	GNDF2	Ground Fault based on current zero sequence
TP24	CPSOK	Chopper power supply OK
TP25	BIAS	Bias level for 4-20mA drivers

DAUGHTER BOARD(DCCD)

TP1/R1	CIB	Adjustment of current unbalance (IERR) trip point
TP2/R2	PTCOMP	Adjustment of PT match point
TP3/R3	CAPFLT	Adjustment of shorted capacitor detection point
TP4/R4	MOTG	Gain adjustment of contactor motor current for metering. 5V = 1PU
TP5/R5	VBOL	Adjustment of dc link overvoltage detection point
TP6/R6	GNDFLT	Ground fault point
TP7/R7	OVRI	Adjustment of source or load instantaneous overcurrent point

3-3.9. Flux Integrators - DS3800NLIB

The NLIB contains various analog signal conditioning and protective functions. Two identical boards are used, one for the source side and one for the loadside. The following functions are provided:

1. Voltage integrators
2. Line/Motor current transformer receivers
3. Voltmeter driver
4. Overvoltage protection
5. Undervoltage sensing
6. Suppression
7. Variable reference comparator
8. Current meter driver

3-3.9.1. Voltage Integrators. A signal proportional to each line-to-line voltage is obtained from the NSFC Board. This signal is applied to an analog integrator to obtain a replica of motor flux according to the equation:

$$V = K * d(Phi)/dt$$

or

$$Phi = \int V * dt$$

Components of line currents are used to compensate for drops due to motor resistance and commutating impedance so that the flux represents the motor internal counter-emf or the so called voltage behind the commutating reactance. The analog sine wave corresponding to flux is then applied to a zero crossing comparator where the digitized output is then available to the HLCB for phasing information. The analog flux signals are also read by the microprocessor based software for use in the control algorithms.

3-3.9.2. Line/Motor Current Transformer Receivers. Two signal conditioning differential receivers are included for line currents A and C. Line current B is calculated from A and C assuming a balanced load. Delta currents are also determined for use in the flux reconstruction.

3-3.9.3. Voltmeter Driver. The magnitude of the line-to-line voltage is determined by a 3 phase absolute value network. The absolute value is then scaled by an operational amplifier for use as a line voltage signal.

3-3.9.4. Overvoltage Protection.. The absolute value of line/motor volts obtained above is compared with an adjustable reference for a time determined by daughter board component, C37, then the overvoltage latch is set. The signal is wired to an I/O port for fault logging by the control software.

3-3.9.5. Undervoltage Sensing.. The absolute value of line volts is compared with an adjustable reference. If the voltage is less than the reference for time delay set by daughter board component, C17, then signal "1LV" becomes true. This signal is used by the software system sequencing to determine low source volts condition. The function is not used on the load side.

3-3.9.6. Suppression.. The absolute value sum of the line currents is taken and applied to a comparator. If the rectified current exceeds the reference threshold for 26ms, a fault latch is set. The time delay is included since the primary protection is in software which includes an on-line recovery. The function implemented here is strictly backup.

3-3.9.7. Current Meter Driver.. The absolute value of current is scaled and available from either source or load for meter driving. Scaling is 5v = 1pu/ (See Table 3-4)

3-3.10. Gating Reference Board - DS3800NSFC

The NSFC's primary task is to convert information from the microprocessor based control to cell pair and timing for firing 6 thyristors configured into a 3 phase - double way bridge. In performing this task, the following functions are used:

Table 3-1A. Microprocessor Board Features - RS232C Connector Signals (HMPG)

Pin#	Name	Signal	Direction	Function
		RS232C CONNECTOR SIGNALS		
JB-2	BA	TXD	output	transmit data
-3	BB	RXD	input	receive data
-4	CA	RTS	output	request to send
-5	CB	CTS	input	clear to send
-6	CL	DSR	input	data set only
-7	AB	DCOM	input	signal ground
-9	CD	DTR	output	data terminal ready

Table 3-1B. Microprocessor - Berg Jumpers, Testpoints and LED's (HMPG)

Nomenclature	Function	Position
	BERG JUMPERS	
J1	Selects type of PROM	8/16
J2	Selects size of PROM	16k
J3	Selects bootup method (on/off bd.)	DGN(not)
J4	Selects 8 or 16 address segments on L bus	8
J5	ground for RS232	GND
	TESTPOINTS	
RST	Reset jumper to common to reset processor	
DCOM	DCOM – Digital common	
I/S	Inhibit stall – Jumper to common to activate	
TNMI	Test nonmaskable interrupt – Jumper to common to activate	
	LED'S	
BHOG	Bushog – On when non-existent address is called by software	
TST	Software controlled and defined – On during initialization self test	

Table 3-1C. Electrically-Alterable ROM - LEDs and Switches (DMPB)

Nomenclature	Function
	LED'S
IMOK	Software controlled as a function of self diagnostics – On normally, will be off if selftest is not passed
WREN	Write enable. On when write capabiltiy to board is enabled with switch, "WREN"
	SWITCHES
WREN	Activates the board to allow alteration of the drive tuneup parameters.

1. R-bus interface
2. Firing mask latch
3. Gate drivers
4. Pulse train generation
5. Attenuator buffers
6. Line-to-line voltage signal conditioners
7. Cell state sensors
8. D-C voltmeter driver

Quantity 3 NSFCs are used, assigned as follows:

1. Source converter
2. Load converter
3. Slave source converter

3-3.10.1. R-bus Interface. Control logic is provided for decoding read and write strobes. Eight data lines (either upper or lower byte) are connected. All information exchange with host processor takes place through the R-bus interface.

3-3.10.2. Firing Mask Latch. The particular cell pair to be fired is selected in software and communicated to the NSFC over the R-bus. The mask information is temporarily stored in the primary firing mask latch. Also obtained from the software is time delay corresponding to the number of degrees the cell pair firing is to be delayed. This information is routed to a timer. At the end of the delay an interrupt is generated and on the load side the cell mask is clocked into the final latch. On the source side, the generated interrupt calls an "execute firing" program which (if all permissives are met) signals the NSFC to move the firing mask to the final latch.

3-3.10.3. Gate Drivers. Six Darlington transistors shift the gating information from logic level to analog levels sufficient to drive the remotely mounted gate pulse amplifiers. These drivers operate at 28 Vdc levels.

3-3.10.4. Pulse Train Generation. The gate pulse reference driver signals consist of a "0" true pulse train nominally 120 degrees long during the conduction period. The reference pulse train is generated from a crystal clock and shaped for a duty cycle of 15 microseconds on and 30 microseconds off by a timer and control logic. Only the source NSFC use pulse train firing. The load side uses "0" or "28Vdc" in offstate.

3-3.10.5. Buffers. Attenuator buffers are provided to receive each attenuator string connected to phase voltages 1,2,3 and also the positive and negative bus. These buffered signals are then used for analog line-to-line voltage signals as well as cell state sensor inputs.

3-3.10.6. Line-to-line Voltage Signal Conditioners

The buffered phase to neutral converter voltage signals are differentially combined to recreate line-to-line voltages. These are used after further processing (on NLIB) as information for the software control algorithms.

3-3.10.7. Cell State Sensors From a cursory check of the standard three phase double way bridge configuration, it can be seen that the voltage across each cell is a combination of one phase voltage and one dc bus voltage. Since each of these voltages is available from an attenuator, the voltage across each cell may be uniquely determined. Circuitry is provided for reading the "state" of all cells at a particular instant and for making the information available to the microprocessor via the L bus.

3-3.10.8. D-C Link Volts. The positive and negative dc link bus voltages are combined differentially to

obtain a dc link volts signal suitable for driving a remote analog meter with approximately 10 Vdc maximum scale (See Table 3-5).Figure 3-8 – Gating Reference - Berg Jumpers, Adjustments, LEDs and Testpoints.

3-3.11. Analog Board = DS3800NLCB

The NLCB provides various miscellaneous signal conditioning and test functions required for the drive. These features include:

1. Speed feedback generation from motor voltage zero crossing signals with overspeed protection and indication
2. External reference signal conditioning
3. Test speed reference
4. Torque reference signal conditioning
5. Selftest features
6. 4-20 MA signal drivers
7. Ground fault detector
8. Power applied reset
9. Fault relay drivers
10. Horsepower meter driver
11. Continuity detection circuits
12. Test switch
13. Restart upon power loss

3-3.11.1. Speed Feedback. A pulse rate signal proportional to drive speed is obtained from the zero crossing detection circuitry on the HLCB. A pulse rate to analog converter then produces a corresponding dc level proportional to speed suitable for driving a meter and protective functions.

The overspeed function is implemented by comparing an overspeed setting reference to the analog speed voltage attained above. A latch insures that transient conditions will be sealed. A red LED indicates the overspeed condition until reset.

3-3.11.2. External Reference Signal Conditioning

A differential amplifier committed to accepting the remote speed reference is supplied. Berg jumpers are used to configure the circuit to accept a 0 to 10 volt signal or a 4-20 ma signal. In either case, the output voltage is transmitted to a channel of the HAIA analog to digital converter for reading by the system processor.

3-3.11.3. Test Speed Reference. A differential amplifier is included to condition a voltage source signal from a potentiometer located on the I/O module. This is also known as the local speed reference.

Table 3-2. Extender - LED's and Testpoints (HLCB)

Nomenclature	Function
	LED's
IMOK	Software controlled as a function of selftest. On normally, will be off if selftest is not passed.
	TESTPOINTS
ACOM	Analog common.
DA1	Digital to analog output 1. Normally uncommitted.
DA2	Digital to analog output 2. Normally uncommitted.
DA3	Digital to analog output 3. Normally used for horsepower output.
DA4	Digital to analog output 4. Normally used for torque command to other channel (if supplied).
OTESTIN0	
OTESTIN1	
SFT1	Execution flag for source crossover and angle control routine.
SFT2	Execution flag for load crossover and angle control routine.
SFT3	Execution flag for field angle control interrupt.
SFT4	Execution flag for background program.
VA1	Phase A signal for source power.
VA2	Phase B signal for motor voltage.
DCOM	Digital common.

3-3.11.4. Torque Reference Signal Conditioning

A differential amplifier for receiving the master torque reference from the other six pulse drive (in a 12 pulse system) is also supplied.

3-3.11.5. Selftest Features. An array of static switches and voltage sources is included for use in testing the analog circuitry of this and other boards in the module during the module selftest mode. A sinewave oscillator is used to simulate line and motor voltage during "gate test mode". This is to allow the power converter electronics to be tested without the necessity of applying high voltage to the associated bridges.

3-3.11.6. 4 - 20 mA Signal Drivers . Two voltage to current converters provide capability of remote driving system information to customer interfaces. The signals selected for transmission are motor current and motor speed. In each case, 4ma corresponds to 0 volts.

3-3.11.7. Ground Fault Detector. The ground fault detector sums line-to-neutral voltages Va, Vb, and Vc. It operates on the principal that when all impedances from phase to neutral are balanced, the summation of the phase-to-neutral voltages will be zero. Any unbalanced impedance will cause an error to be detected and a ground fault noted by the system processor.

3-3.11.8. Power Applied Reset . A power applied reset is generated here and distributed throughout the electronics module. It is used to prevent false operations of relay drivers and fault latches during the process of powering up the module.

3-3.11.9. Fault Relay Drivers . Relay driver no. 1 is generally used as a method of tripping the source side drive switchgear from both hardware and software sensed fault conditions. It is inhibited from false operations during the power up phase by the power applied reset.

Table 3-3. Analog/Digital Input Signals - Berg jumpers and Testpoints (HAIA)

Nomenclature	Function	Position
	Berg Jumpers	
J1	Selects either a fixed +7.5v input to channel 15 or a variable input thru pin 12. The variable input is selected.	A
J2	Selects either a fixed +7.5v input to channel 14 or a variable input thru pin 12. The variable input is selected.	A
J3-J8	Selects L bus address for board.	F
J9	Selects filter for analog side of A/Dconverter. Filter is selected.	O
	Testpoints	
ACOM	Analog common.	
FOG	Test input. Tie to common to latch internalcard data bus.	
AN	Analog signal at input to A/D conversion	
DB	Data buffer direction. High transmits to microprocessor and low receives.	
DV	"O" data valid signal	
BSY	"1" busy signal	
ES	"1" enable switch	
CLK	card clock - 1.152 mhz	

Table 3-4. Flux Integrators - Jumpers, Adjustments, LED 's and Testpoints (NLIB)

Nomenclature	Function	Position
	BERG JUMPERS	
BJ1-BJ3	Selects voltage integrator scaling for 60Hz or 120Hz	60Hz or 120Hz (var)
	Adjustments	
FAB, FBC, FCA	Scale voltage integrators which calculate flux. Scaling is 4.75 volts RMS corresponds to rated volts.	
OV	Set overvoltage protection trip point.	
UV	Set undervoltage protection trip point.	
SUPP	Set trip point for overcurrent hardware fault.	
COMP		
VDC	Not used.	
	LED's	
OV	On when overvoltage fault latch set.	
SUPP	On when overcurrent fault latch set.	
IMOK	On if microprocessor based self test is passed and no operational amplifier saturation conditions are present.	
	Testpoints	
OVR	Analog dc voltage proportional to setpoint of over-voltage sensor.	
UVR	Analog dc voltage proportional to setpoint of source undervoltage sensor.	
SUPP	Analog dc voltage proportional to setpoint of overcurrent .	
FAB,FAC FCA	Analog voltage proportional to motor/line flux reconstructed. See fig. 3-6 for typical waveforms.	
FBAR, FCBR, FACR	Analog voltage proportional to motor/line flux reconstructed for IX drop. See fig. 3-6 for typical waveforms.	
OIA,OIB OIC	Analog waveform representative of inverted line current. See fig. 3-6 for typical waveforms.	

Relay driver no. 2 is generally used as a method of tripping the motor side isolator (if used) from both hardware and software sensed fault conditions. It is also inhibited from false operations during the power up hase.

3-3.11.10. Horsepower Meter Driver . An adjustable meter driver for powering a door mounted meter indicating horsepower is also included. The horsepower signal is calculated in software and converted to analog on the HLCB board.

3-3.11.11. Continuity Detection Circuits . Four circuits are included to monitor the continuity of various ribbon cables connected to the electronics module when it is integrated into a system control panel. These signals are only monitored during the panel selftest. An appropriate diagnostic message is printed out if a cable is not in the correct location.

3-3.11.12. Test Switch . The test switch on the board front edge is used to initiate a long selftest. When the switch is operated, the electronics module can execute a lonf selftest routine which extends to other components of the control panel. Upon completion of the selftest, the control will enter gate-test-mode using the three phase test oscillator as a source of voltage waveforms.

3-3.11.13. Restart Upon Power Loss . If the drive has been previously running and the control power is lost for a time (determined by setting of RTD potentiometer), then the drive will be automatically restarted when power returns by a pulse which appears at signal 1RSRT. Note this is an option and the I/O module must be properly configured. (See Table 3-7)

3-3.12. Relay Driver Board - DS3800HRDA

The HRDA is relay driver board with 16 open-collector outputs. Output voltages can range from 5 to 50 volts. In the system, it is applied at a nominal voltage of 28 volt dc. The board may be connected to the computer R-bus for accessibility by the system control.

The relay driver board is used principally as an interface between the system software control and the power devices located in the I/O module. (See Table 3-7)

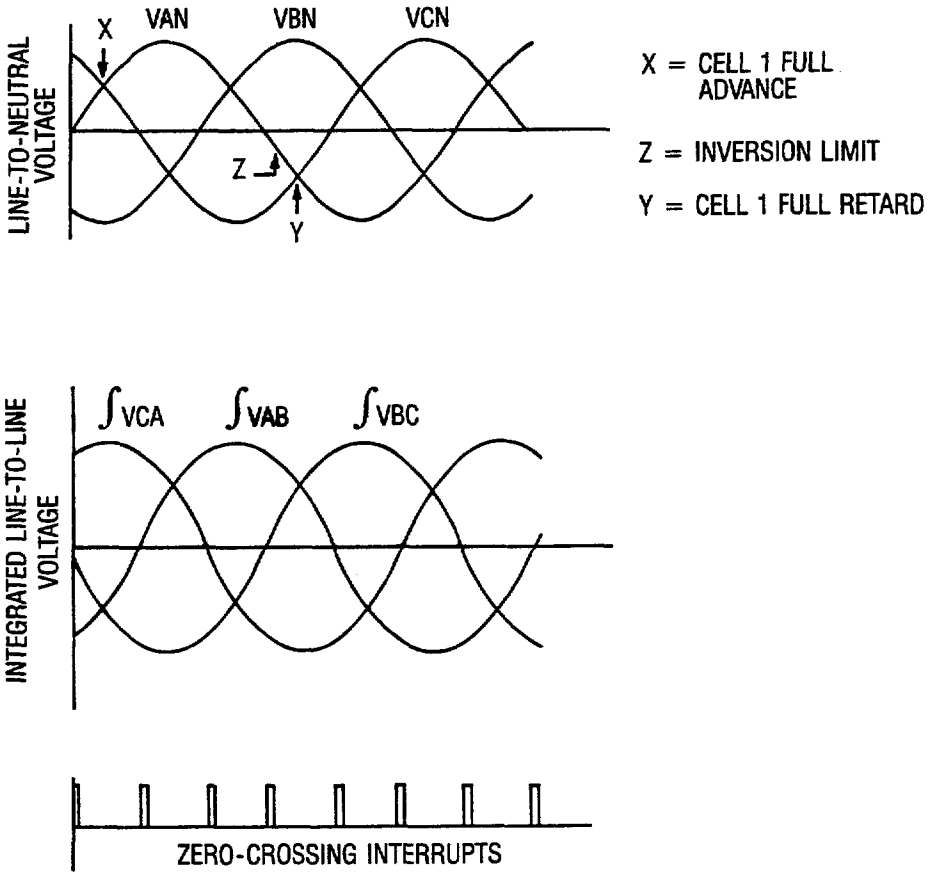


Figure 3-5. Flux Wave Zero-Crossings

3-3.13. Input Signal Conditioning - DS3800HISA

HISA is designed to work in conjunction with HRIA (in I/O module) to condition thirty high level (105Vdc) signals to logic levels and allow access to them through the processor R-bus. Each signal is separately isolated by a photocoupler and indicated by a bar LED. Two relay drivers are also included which are only utilized in the hardware selftest of the I/O module. (See Table 3-8)

3-4. Display Module (LDM)

The DS3820LDM is the display module for the control panel. The display provides fault, alarm and diagnostic information. The display module comes in two basic layouts.

Layout Option 1 provides the following features:

- 3 indicator lights (RUN, ALARM and FAULT)
- 2 push buttons (RESET AND DRIVE STATUS)
- 3 1/2 digit motor speed meter (RPM)
- 24 column printer with real-time clock

Layout Option 2 provides the following features:

- 4 push-to-test indicator lights (RUN, ALARM, FAULT and READY)
- 2 push buttons (RESET and DRIVE STATUS)
- 4 DB40 type meters (motor speed, motor current, motor voltage and horsepower)

All lights and push buttons are on a 28V circuit. The functions and color codes for each light are:

- Red for the motor RUN light
- Red for a system FAULT light
- Amber for a system ALARM light
- Green for the system READY light

The push buttons are connected to a 28 volt circuit and direct a 28 volt signal to the output when the button is depressed. The RESET push button is used to reset latched faults in either software or hardware (NLIB and NLCB latched faults - overspeed, suppression, overvoltage, etc.). The DRIVE STATUS push button; when depressed, provides information on the motor voltage, current, speed, etc. via the printer.

The digital meter is a dc voltage measurement panel meter. The conversion factor for the meter is 2 volts correspond to 1999 counts. The panel meter is powered by the 5 volt bus via the 26-pin ribbon cable.

Table 3-5. Gating Reference - Berg Jumpers, Adjustments, LED's and Testpoints

Nomenclature	Function	Position
	Berg jumpers	
J1	Determines if new mask should be clocked into final latch at termination of count down timer.	NINT(load) INT(source)
	Adjustments	
VDC	Sets remote meter full scale to corresponding D-C link volts.	
	LED's	
IMOK	Board hardware status indication. On normally. Off if Op amp saturation detector detects a board hardware failure.	
	Testpoints	
SIG 1-SIG 6	Gate pulse command. "0" true. Steady state length of 120 degrees relative to operating frequency. Each gate pulse is shifted 60 degrees relative to lower numbered ones. SIG numbers correspond to thyristor leg being gated.	
VDC	Output of D-C link voltage meter driver.	
VAC,VCB	Analog output proportional to motor/source line-to-line voltage.	
ACOM	Analog common.	

Table 3-6. Analog Board - Berg Jumpers, Adjustments, Testpoints and LEDs (NLCB)

Nomenclature	Function	Position
	BERG JUMPERS	
LOAD	Selects remote speed reference for either current input or voltage input. Should be coordinated with GAIN.	variable
GAIN	Selects gain necessary for remote speed when used as current input or voltage input.	variable
	Adjustments	
SPD	Adjusts analog speed feedback circuit for 10 volts at 60 Hz.	
SPM	Adjusts door mounted speed meter for proper fullscale reading	
OSP	Hardware overspeed setting.	
RTD	Adjusts allowable time for auto restart after loss of control power.	
	Testpoints	
SPD	Output of analog speed feedback calculation.	
OSP	Reference to hardware overspeed detection circuit. Scaled same as speed i.e. 10 volts = 60Hz.	
IOA	Output of channel A current.	
IOB	Output of channel B current.	
REF	Output of remote speed reference. With a voltage source input, will be 0 to 10 volts. With a current source input, will be 2 to 10 volts.	
TREF	Output of local test reference. Varies from 0 to 8.3 volts.	
12P	Output of torque reference obtained from other channel.	
NTL	Neutral. Summation of line to neutral voltages. Normally near 0 volts.	
RTD	Run time delay. Sets allowable time for loss of control power while still allowing an auto restart. Scaled 0 to 10 volts.	
PHA	Test oscillator, phase A. Shows a 55Hz. sine wave.	
PHB	Test oscillator, phase B. Shows a 55Hz. sine wave, lagging phase A by 120 degrees.	
	LED's	
OSP	Red LED indicating that the hardware overspeed latch is set.	
RSRT	Green LED which is pulsed when the auto restart circuit is activated.	
FT1	Green LED indicating the status of fault relay driver No. 1. Will be on when the relay driver is energized.	
FT2	Green LED indicating the status of fault relay driver No. 2. Will be on when the relay driver is energized.	
TEST	Green LED indicating when the test switches is closed and the system software indicates gate test mode.	
IMOK	Yellow indicating light will be on normally. An off condition indicates a possible hardware problem with the board. It can be turned off either locally or from the software as a function of selftest.	

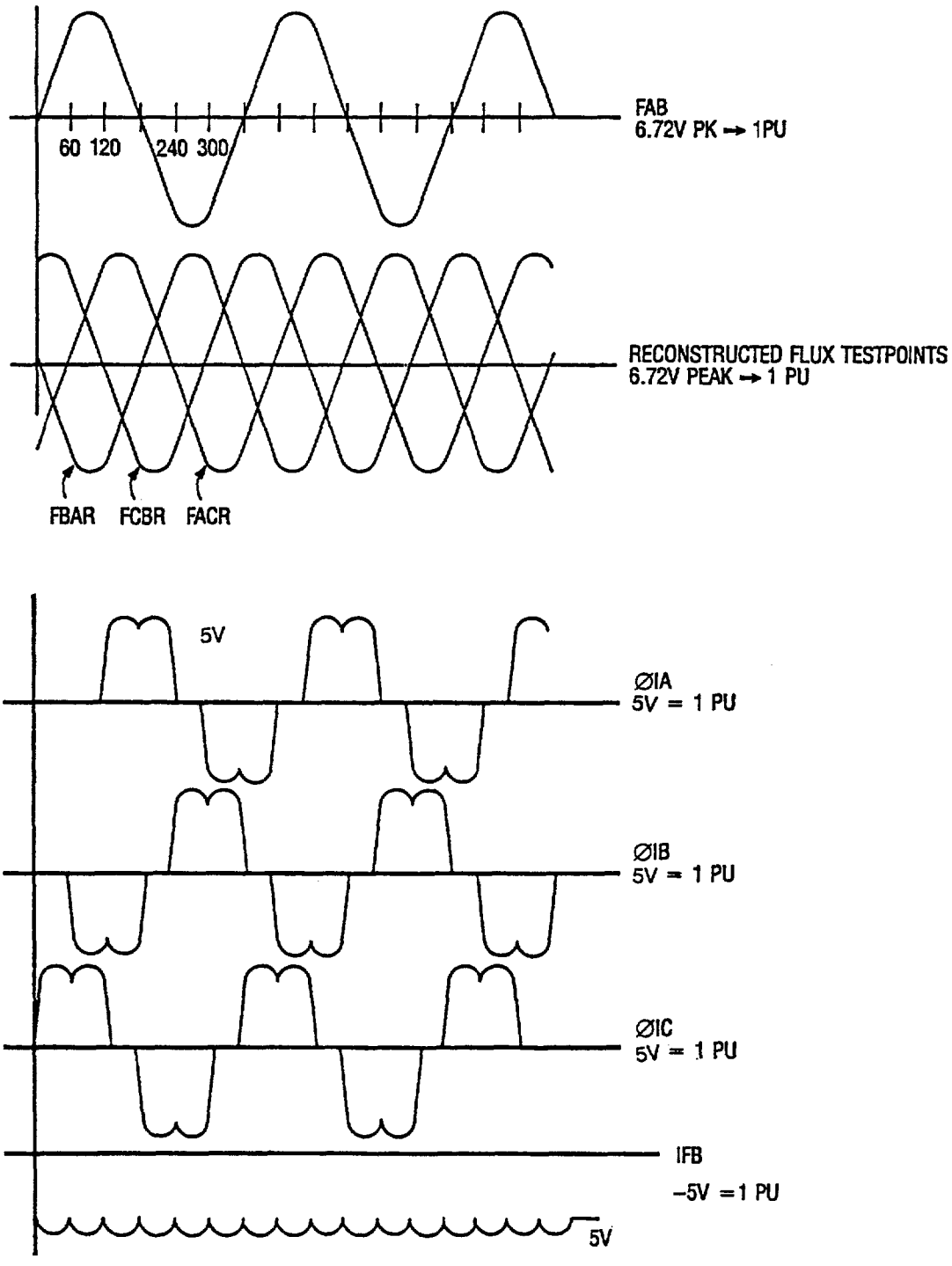


Figure 3-6. Analog Typical Waveforms

The four DB40 type analog meters are 10 volt movement.

The printer included in the display module is a Digitec Model 6610 Panel mount printer. The 24 column printer inputs data and control signals via a serial RS232 connection and uses an impact print head mechanism to output the data. The printer also supplies a real-time clock that will be used to log the time of status information and system failures.

The LDM interfaces to the IMF thru a 26 pin ribbon cable JK.

3-5. I/O Module (IOM)

The purpose of the DS3820IOMA (I/O) module is to provide signal conditioning and connection for interpanel and purchaser wiring in the control panel.The following functions are provided.

1. Signal conditioning and isolation for interface wiring at the 105Vdc level.
2. Isolated signal interlocks (form C) to remote locations as a function of programmed system requirements. Rated 0.25A, 115Vac (Resistive).
3. Fused 105Vdc power supply to provide a source of high level voltage for interface.
4. Power relay control for interface with a motor-side isolator as well as interface with a line-side switchgear device. Rated 10A, 600Vdc (Inductive).
5. Local operator's station for running the drive during maintenance modes. (Channel selector, local/remote, start/stop, local speed reference).
6. Terminal boards for terminating interconnecting wiring.
7. Diagnostic functions.

The IOMA module is designed to be intergrated into a system diagnostic test function residing in the system electronics module. Therefore, no local diagnostic messages are displayed. However, the necessary circuits and feedbacks are provided to allow the system function resident in the electronics module to isolate failed components to the replaceable board level.

Table 3-7. Relay Driver - Testpoints and LED's

(HRDA)

Nomenclature	Function
	TESTPOINT
TP1	Reset. Connecting testpoint 1 to testpoint 2 resets only this board. This function isn't useful in the LCI system.
TP2	DCOM
	LED's
0 - F	A red LED for each of the 16 relay drivers indicates the energized status of each circuit.
ADR	Yellow LED. Turns on when the board is addressed; stays on if the board is addressed at least every 200 microseconds.
WPOK	Parity check. Not used in this system. Will always be on.

Table 3-8. Input Signal Conditioning - Berg Jumpers, Adjustments, Testpoints and LEDs

HISA

Nomen.	Function
	LEDs
I0O - I1D	Green bar LEDs. Each indicates the status of the associated input, luminous when the input is energized.
IMOK	Yellow LED. On during normal operation. will be off only as the result of module selftest.

CHAPTER 4
FIELD ADJUSTABLE PARAMETERS AND PUBLIC VARIABLE PARAMETERS

Section A - EEPROM (Field) Variables

EEPROM Memory is a non-volatile memory space used to hold critical drive constants that may be system dependent. The use of EEPROM Memory allows the drive constants to be changed in the field in order to fine tune the system performance. EEPROM Memory space resides on the DMPB card (HMPG daughter board) and is configured to contain 1024 bytes (1K x 8).

Default values for the drive constants are burned into the ROMs residing on the HMPG (the ROMs also contain the executable software to run the drive). These default values are intended to provide a reasonable starting base for tuning the drive. They may be loaded into the EEPROM Memory by turning the DMPB write enable switch "on", jumpering testpoint TINO on the HLCB pwb to DCOM and then executing a hardware SEM reset (Touch HMPG TP "reset" to DCOM). In order to change a value in the EEPROM memory, turn the DMPB write enable switch "on" and use the monitor commands: (MODIFY and SAVE).

After power up or hardware reset, the microprocessor on the HMPG makes a local copy (shadow) in RAM of the EEPROM Memory. During normal running, the microprocessor accesses the drive constants from this local RAM shadow.

Table 4-1. Description of Variables

C R G A I N

Units - gain
Resolution - 1.0
Range - 10 to 1000

Definition - Current regulator gain. The relation to system parameters is:

CRGAIN = (758.5*Im*Wc*L)/Vll

Im = rms rated inverter current
Wc = desired crossover (180 rad)
L = commutating inductance
= reactor + 2*(trans comm reactance)
Vll = rated rms line-line source volts
(*2 for series 12-pulse)

The related software variable definition is
IREG G2 E = 16*CRGAIN/(6*DRSFRQ)

C R L I M P
Units - pu impedance
Resolution - 0.01
Range - 0 to 0.25 pu

Definition - Variable controlling the source side retard limit as a function of current based on source impedance.

CRLIMP = per unit source commutating impedance

The related software variable definition is

LINE IMP E = 63.75 * CRLIMP

C R M A R G
Units - degrees
Resolution - 1.0
Range - 0 to 30

Definition - Variable controlling the fixed source side margin when inverting. Normal margin is 20 degrees.

CRMARG = desired fixed margin in degrees

The related software variable definition is

RTARD MARG SRC E = 255[1+COS(180-CRMARG)]

C R M A X
Units - pu dc link current
Resolution - 0.01
Range - 0.25 to 1.50

Definition - Variable controlling the maximum clamp level for dc link current command. Usually picked on inverter rating and drive loading requirements. Normal setting is 1.15 pu.

CRMAX = max pu dc link current

The related software variable definition is

ISTAR CLMP E = 1024 * CRMAX

C R M I N
Units - pu dc link curren
Resolution - 0.01
Range - 0.01 to 0.5

Definition - Variable controlling the minimum clamp level for dc link current command. Usually set to maintain continuous current. The link reactor is usually picked to maintain continuous current at 0.12 pu.

CRMIN = min pu dc link current

The related software variable definition is

CURLL SRC E = 1024 * CRMIN

C R T C
Units - seconds
Resolution - 0.001
Range - 0.001 to 0.5

Definition - Time constant used to stabilize the current regulator. Typical value is 0.02 seconds.

CRTC = time constant (seconds)

The related software variable definition is

IREG G1 E = 4 * CRGAIN * CRTC

C R V D C
Units - gain
Resolution - 0.01
Range - 0.8 to 1.2

Definition - Gain applied to the feedforward section of the current regulator. It is intended to compensate for differentrated source line and motor voltages.

CRVDC = Vllm / Vlls

Vllm = rated rms line-to-line motor voltage

Vlls = rated rms line-to-line source voltage

The related software variable definition is

VDC GAIN E = 16 * CRVDC

C T A D V L
Units - pu source Vdc
Resolution - 0.01
Range - 0.5 to 1.00

Definition - Maximum output clamp on the current regulator output, source dc link volts command. Normal setting is 1 pu.

CTADVL = desired max pu dc link volts command

The related software variable definition is

ADVANC LINE E = 255 * CTADVL

C T G A I N
Units - gain
Resolution - 1.0
Range - 5 to 100

Definition - Cross-tie regulator gain. The relation to system parameters is:

CTGAIN = 4 * Wc Wc = desired crossover (9 rad)

The related software variable definition is

CRSST INT GAIN E = 64*CTGAIN/(2*DRFRQS)

C T M A R G
Units - pu source Vdc
Resolution - 0.01
Range - 0.0 to 0.5

Definition - Defines the minimum margin allowed between the actual current regulator output and the absolute maximum clamp on current regulator output, CTADVL. If current regulator output gets closer to the max clamp than is allowed by the margin CTMARG, then the crosstie becomes active to maintain the margin by reducing motor flux. Normal margin is 0.06 pu.

CTMARG = desired pu margin

The related software variable definition is

CROSS TIE MARG E = 255 * CTMARG

C T M A X
Units - pu flux
Resolution - 0.01
Range - 0.0 to 0.5

Definition - Maximum clamp on the output of the cross-tie regulator. It defines the maximum reduction that can be applied to the flux command by the cross-tie regulator. Normal setting is 0.4 pu

CTMAX = max pu cross-tie flux reduction

The related software variable definition is

CR$ST MAX LMT E = 8192 * CTMAX

C T T C
Units - seconds
Resolution - 0.001
Range - 0.001 to 0.5

Definition - Time constant used to stablilize the cross-tie regulator. Normal setting is 0.03 seconds

CTTC = time constant (seconds)

The related software variable definition is

CRSST PRP GAIN E = 8 * CTGAIN * CTTC

D R H P P U
Units - horsepower
Resolution - 1.0
Range - not clamped

Definition - Defines the drive's 1 pu horsepower rating.

DRHPPU = rated drive horsepower

There is no related software variable.

D R I P U

Units - amps
Resolution - 1.0
Range - not clamped

Definition - Defines the drive's 1 pu rms current rating.

DRIPU = rated drive rms current

There is no related software variable.

D R M C
Units - pu capacitance
Resolution - 0.01
Range - .5 to 3.0

Definition - Defines the size of the motor filter capacitors.

DRMC = pu motor filter capacitance

(per phase on motor base)

The related software variable definition is

MOT CAP E = 16 / (DRZRAT * DRMC)

D R M F R Q
Units - hertz
Resolution - 1.0
Range - 45 to 60

Definition - Defines the motor base frequency. Normally set to 60 hz.

DRMFRQ = motor base frequency (hz)

There is no related software variable.

D R M R P M
Units - rpm
Resolution - 1.0
Range - not clamped

Definition - Defines synchronous speed at 60 hz.

DRMRPM = motor 60 hz synchronous speed (rpm)

There is no related software variable.

D R M V P U

Units - volts
Resolution - 1.0
Range - not clamped

Definition - Defines rated 1 pu rms line-to-line motor voltage.

DRMVPU = 1pu rms motor voltage

There is no related software variable.

D R S F R Q
Units - hertz
Resolution - 1.0
Range - 50 to 60

Definition - Defines the source line frequency.

DRSFRQ = line frequency (hz)

There is no related software variable.

D R S V P U
Units - volts
Resolution - 1.0
Range - not clamped

Definition - Defines rated 1 pu rms source line to line voltage.

DRSVPU = 1 pu rms line voltage

There is no related software variable.

D R V C F G
Units - bit coded
Resolution - n/a
Range - not clamped

Definition - Control configurations and optional features are selected with this bit coded variable. The related software variables are assigned based on the bit patterns in DRVCFG:

Bit		Related Software Variable
0	__	SRC CONFIG E
1		
2		
3	__	XDCD CONFIG E
4		
5	__	START OPTION E
6		
7	__	SYNC EN P
8	__	ICAP FIL ENABLE P
9	__	FLX ERROR EN P
A	__	TORQ VS SPD EN P
B	__	unused
C	__	unused
D	__	unused
E	__	unused
F	__	unused

The related software variable bit pattern definitions are:

SRC CONFIG E: selects the source bridge configuration
00 - 6 pulse channel A
01 - 6 pulse channel B (factory test only)
10 - 12 pulse channel B lags
11 - 12 pulse channel B leads

XDCD CONFIG E: selects the number of series GTOs in an inverter leg
000 - invalid
001 - 1 cell
010 - 2 series cells
011 - 3 series cells
100 - 4 series cells
101 - 5 series cells
110 - 6 series cells
111 - invalid

START OPTION E: selects startup options
00 - disable KNOWN & SCAN start
01 - KNOWN start enable
10 - SCAN start enable
11 - KNOWN & SCAN start enable

SYNC EN P: synchronizing function
0 - disable
1 - enable

ICAP FIL ENABLE P: low pass filter on calculated capacitor current
(normally ENABLED) 0 - disable
1 - enable

FLX ERR FIL EN P: low pass filter on
the flux error signal
(normally DISABLED)
0 - disable
1 - enable

TORQ VS SPD EN P: function to limit torque
command as a function of speed
0 - disable
1 - enable

D R V T S T
Units - n/a
Resolution - n/a
Range - not clamped

Definition - Control test modes are selected with this variable. The related software variables are assigned based on the value of DRVTST.

Value	Related test variable set true
0	none
1	IO TEST MODE E - background is disabled, i/o can be exercised from monitor
2	CROWBAR TEST MODE E - diametrics are gated on the inverter while current is regulated at value specified by SUMTCH
3	MANUAL FIX FIRE E - inverter is gated at frequency specified by SUMFRQ while current is regulated at value specified by SUMTCH

D R Z R A T
Units - ratio
Resolution - 0.01
Range - 0.5 to 1.5

Definition - Defines the ratio between the motor base impedance and the inverter base impedance.

DRZRAT = Zbmot / Zbinv

Zbmot = motor base impedance

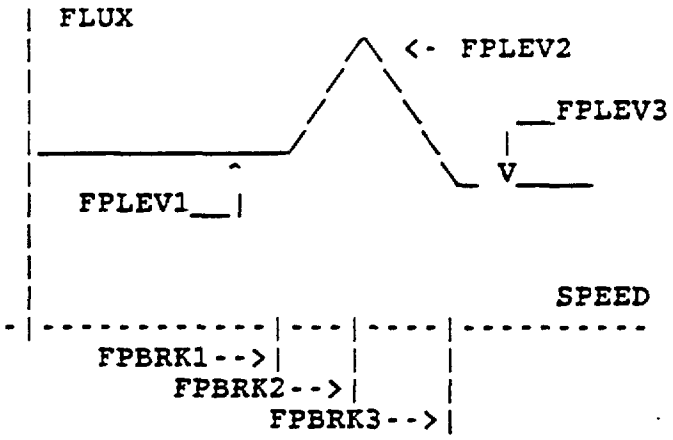
Zbinv = inverter base impedance

There is no related software variable.

F P B R K 1, 2, 3
Units - pu motor speed
Resolution - 0.01
Range - 0.0 to 1.5

Definition - Group of variables used in the "Flux Setpoint vs Speed' function that define the speed breakpoints.

FPBRK? = desired pu speed



Typical settings are:

FLX BRK 1 = 0.50
FLX BRK 2 = 0.85
FLX BRK 3 = 1.10

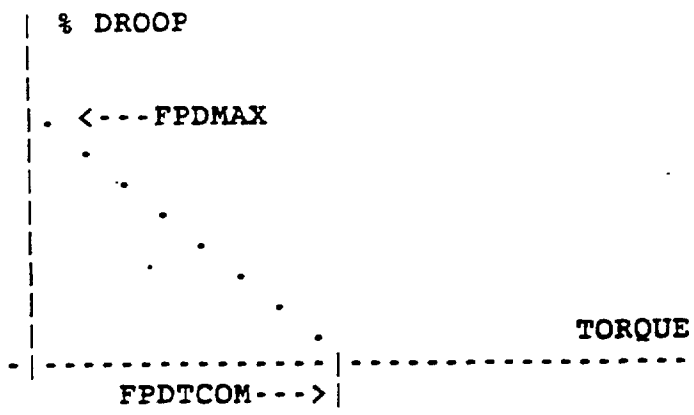
The related software variable definition is

FLX BRK? E = 1024 * FPBRK? * (DRFRQM/60)

F P D M A X
Units - percent
Resolution - 0.01
Range - 0.0 to 0.85

Definition - Used to program the "Flux Droop" function. It defines the percentage that flux command will be reduced (drooped) when the drive is at no-load. Typical setting is 0.65

FPDMAX = max flux droop at no-load (%)



The related software variable definition is

FLX DROOP MAX E = 512 * FPDMAX

F P D T C O M
Units - pu torque
Resolution - 0.01
Range - 0.0 to 0.6

Definition - Defines the torque level below which the "Flux Droop" function becomes active. See FPDMAX. Typical setting is 0.6 pu

FPDTCOM = desired pu torque

The related software variable definition is

FLX DROOP TCOM E = 1024 *FPDTCOM

F P L E V 1, 2, 3
Units - pu flux
Resolution - 0.01
Range - 0.5 to 1.45

Definition - Group of variables used in the "Flux Setpoint vs Speed" function that define the flux setpoint levels. See FPBRK?.

FPLEV? = desired pu flux

Typical settings are:

FLX LEV 1 = 1.00
FLX LEV 2 = 1.15
FLX LEV 3 = 0.90

The related software variable definition is

FLX LEV? E = 1024 * (FPLEV-FPX1/2)

F P X 1
Units - pu impedance
Resolution - 0.01
Range - 0.0 to 0.30

Definition - Defines motor parameter X1 per unit. It is used to compensate for the voltage drop across X1 in the motor volts calculation and in the flux profile levels.

FPX1 = motor parameter X1 pu

The related software variable definition is

XM1 E = 128 * FPX1

F R F E T C
Units - seconds
Resolution - 0.001
Range - 0.018 to 0.26

Definition - Time constant of the low pass filter applied to flux error. The filter can be disabled through DRVCFG. Typically set for 0.018 seconds and disable filter using DRVCFG.

FRFETC = filter time constant (sec)

The related software variable definition is

FLX ERR FIL GAIN E = 16 / (FRFETC*DRSFRQ)

F R G A I N
Units - gain
Resolution - 0.1
Range - 2.0 to 100.0

Definition - Gain of the flux regulator. The relation to system parameters is:

FRGAIN = Wc

Wc = desired crossover (10-15 radians)

The related software variable definition is

FLX INT GAIN E = 256 * FRGAIN / (2*DRSFRQ)

F R I C T C
Units - seconds
Resolution - 0.001
Range - 0.018 to 0.26

Definition - Time constant of the low pass filter applied to calculated capacitor current. The filter can be disabled through DRVCFG. Typical setting is for 0.083 seconds and enable filter using DRVCFG.

FRICTC = filter time constant (sec)

The related software variable definition is

ICAP FLTR GAIN E = 16 / (FRICTC*DRSFRQ)

F R L L I
Units - pu dc link current
Resolution - 0.01
Range - 0.0 to 0.50

Definition - Defines the threshold level for the test to determine light load operating condition based on level of commanded dc link current. Below this level light load operation is flagged. Typical setting is 0.17 pu.

FRLLIH = desired light load threshold
(pu dc link current)

The related software variable definition is

FLX LL IDC E = 1024 * FRLLIH

F R L L I H
Units - pu dc link current
Resolution - 0.01
Range - 0.0 to 0.2

Definition - Defines the hysterisis for the test to determine light load operating condition based on level of commanded dc link current. The hysterisis is centered about the threshold level defined by FRLLI. Typical setting is 0.09 pu.

FRLLIH = desired hysterisis
(pu dc link current)

The related software variable definition is

FLX LL IDC HYST E = 1024 * FRLLIH

F R L L S
Units - hertz
Resolution - 1.0
Range - 15 to 90

Definition - Defines the inverter fundamental frequency above which light load operation is exited regardless of level of commanded dc link current. This is to prevent overvoltage conditions on the motor. Typical settings are for approximately 5hz above the self- excitation frequency:

FRLLS = 5 + 1/(2*pi*sqr(Lm*Cm))

Lm = per phase motor magnetizing inductance (henries)
Cm = per phase motor filter capacitance (farads)

The related software variable definition is

FLX LL SPD E = 1024 * FRLLS / 60

F R T C
Units - seconds
Resolution - 0.001
Range - 0.2 to 2.0

Definition - Time constant used to stablilize the flux regulator. It is calculated to cancel one of the poles in the motor transfer function.

FRTC = (Lm + L2) / R2

Lm = magnetizing inductance
L2 = referred rotor leakage inductance
R2 = referred effective rotor resistance

The related software variable definition is

FLX PRP GAIN E = 4 * FRTC * FRGAIN

F R X M
Units - pu magnetizing inductance
Resolution - 0.01
Range - 0.5 to 6.0

Definition - Defines the pu motor magnetizing inductance.

FRXM = Xmpu

Xmpu = pu motor magnetizing inductance

The related software variable definition is

XM GAP E = 4 * FRXM * DRZRAT

I F H S 0 1
Units - hertz
Resolution - 1.0
Range - 0 to 75

Definition - Defines the transition freqeuncy threshold (Centered with ±2 Hz. hysterisis) between none and 5th harmonic elimination. The transition frequencies are based on the system resonant frequency which is defined

Fres = 1/(2*pi*sqr(Cm*Lst)

Cm = per phase motor filter capacitance

Lst = per phase motor starting inductance

The transition frequency is given by

IFHS01 = (0.2 * Fres) + 5 hertz

The related software variable definition is

SPD BRK HARM01 E = 1024 * IFHS01 / 60

I F H S 0 4
Units - hertz
Resolution - 1.0
Range - 0 to 40

Definition - Defines the transition freqeuncy threshold (centered with ±2 Hz. hysterisis) between 11-13th and none harmonic elimination. The transition frequency is given by

IFHS01 = (0.076923 * Fres) - 5 hertz

Fres = system resonant freq (see IFHS01)

The related software variable definition is

SPD BRK HARM04 E = 1024 * IFHS04 / 60

I F H S 1 2
Units - hertz
Resolution - 1.0
Range - 0 to 40

Definition - Defines the transition freqeuncy threshold between 5th and 5-7th harmonic elimination. The transition frequency is given by

IFHS12 = (0.171429 * Fres) hertz

Fres = system resonant freq (see IFHS01)

The related software variable definition is

SPD BRK HARM12 E = 1024 * IFHS12 / 60

I F H S 2 3
Units - hertz
Resolution - 1.0
Range - 0 to 40

Definition - Defines the transition freqeuncy threshold between 5-7th and 7-11th harmonic elimination. The transition frequency is given by

IFHS23 = (0.11688 * Fres) hertz

Fres = system resonant freq (see IFHS01)

The related software variable definition is

SPD BRK HARM23 E = 1024 * IFHS23 / 60

I F H S 3 4
Units - hertz
Resolution - 1.0
Range - 0 to 40

Definition - Defines the transition freqeuncy threshold between 7-11th and 11-13th harmonic elimination. The transition frequency is given by

IFHS34 = (0.083916 * Fres) hertz

Fres = system resonant freq (see IFHS01)

The related software variable definition is

SPD BRK HARM34 E = 1024 * IFHS34 / 60

I F S R L
Units - degrees per firing
Resolution - 1.0
Range - 1 to 30

Definition - Defines the rate limit applied to the inverter firing angle command when the firing angle quadrant changes. Typical setting is 5.

IFSRL desired rate limit (deg/fire)

The related software variable definition is

LD SLOW RATE LIM E = 1024 * IFSRL / 360

P O 1, 2, 3, 4, 5, 6
Units - n/a
Resolution - n/a
Range - 0 to 19

Definition - This collection of variables selects the six items to be displayed by the data logger on the door module printer. The number of the information to be printed is entered as data into each of the variables. If no data for a particular entry is required, then it should be set to zero. The selections are

Setting	Data
00	no print out
01	Motor Speed
02	Speed Command
03	Motor Current
04	Motor Voltage
05	Drive Horsepower
06	Drive Kilowatts
07	Link Voltage
08	Link Current
09	Source Current
10	Source Voltage
11	Source Firing Angle
12	Inverter Firing Angle
13	Inverter Power Factor
14	Source Power Factor
15	Source KVA
16	Source KVAR
17	Torque Limit
18	Current Limit
19	Torq Lim, Crosstie, Turning Gear Status

There is no related software variable.

P O E
Units - logical
Resolution - n/a
Range - T or F

Definition - Variable used to enable the "auto log" feature of the data logger. In this mode the printer prints the selected information at time intervals selected by the printer time interval setting.

POE = "T"rue enables auto log
= "F"alse disables auto log

There is no related software variable.

P T L O C
Units - pu dc link current
Resolution - 0.01
Range - 0.5 to 2.0

Definition - Defines the trip threshold for the load software detected overcurrent function. Typical setting is 1.5

PTLOC = desired trip level
(pu dc link current)

The related software variable definition is

LOD OVER CUR E = 1024 * PTLOC

P T L O C C
Units - n/a
Resolution - n/a
Range - 1 to 60

Definition - Controls number of consecutive load overcurrents allowed before a fault is declared. A penalty function is implemented such that 6 good firings will cancel one fault firing. This allows recovery from non-repetitive faults. Typical setting is 3.

PTLOCC = number of consecutive overcurrents

The related software variable definition is

LOD OVER CUR CNT E = 6 * (PTLOCC-1)

P T L P L L C
Units - n/a
Resolution - n/a
Range - 1 to 150

Definition - Controls number of consecutive out-of-tolerance load phase-lock-loop calculations allowed before a fault is declared. A penalty function is implemented such that 6 good calculations will cancel one bad calculation. This allows recovery from non-repetitive faults. Typical setting is 60.

PTLPLLC = # of consecutive out-of tol calcs

The related software variable definition is

LOD PLL CNT E = 6 * (PTLPLLC-1)

P T L P L L T
Units - degrees
Resolution - 1.0
Range - 0 to 60

Definition - Defines the out-of tolerance threshold for the load-side phase-lock-loop calculation. Typical setting is 22 degrees.

PTLPLLT = out-of-tol threshold (degrees)

The related software variable definition is

LOD PLL ERROR TOL E = 1024 * PTLPLLT / 360

P T S O C
Units - pu dc link current
Resolution - 0.01
Range - 0.5 to 2.0

Definition - Defines the trip threshold for the source software detected overcurrent function. Typical setting is 1.5

PTSOC = desired trip level
(pu dc link current)

The related software variable definition is

SRC OVER CUR E = 1024 * PTSOC

P T S O C C
Units - n/a
Resolution - n/a
Range - 1 to 60

Definition - Controls number of consecutive source over- currents allowed before a fault is declared. A penalty function is implemented such that 6 good firings will cancel one fault firing. This allows recovery from non-repetitive faults. Typical setting is 3.

PTSOCC = number of consecutive overcurrents

The related software variable definition is

SRC OVER CUR CNT E = 6 * (PTSOCC-1)

P T S P E T
Units - pu speed (60 hz base)
Resolution - 0.001
Range - 0.0 to 0.5

Definition - Defines the maximum speed error allowed before a 'loss of speed control' alarm is declared. Typical setting is 0.1 pu

PTSPET = max allowed pu speed error
(on 60 hz base)

The related software variable definition is

SPD REG ERR TOL E = 1024 * PTSPET

PTSPLLC
Units - n/a
Resolution - n/a
Range - 1 to 150

Definition - Controls number of consecutive out-or-tolerance source phase-lock-loop calculations allowed before a fault is declared. A penalty function is implemented such that 6 good calculations will cancel one bad calculation. This allows recovery from non-repetitive faults. Typical setting is 60.

PTSPLLC = # of consecutive out-of tol calcs

The related software variable definition is

SRC PLL CNT E = 6 * (PTSPLLC-1)

PTSPLLT
Units - degrees
Resolution - 1.0
Range - 0 to 60

Definition - Defines the out-of tolerance threshold for the source-side phase-lock-loop calculation. Typical setting is 22 degrees.

PTSPLLT = out-of-tol threshold (degrees)

The related software variable definition is

SRC PLL ERROR TOL E = 1024 * PTSPLLT / 360

PTSUVT
Units - seconds
Resolution - 0.1
Range - 0 to 20

Definition - Defines the maximum time a source undervoltage may exist before a source uv fault is declared. If the undervoltage exists for less than this time, then it is considered a bus transfer and the control will restart when the source voltage returns.

PTSUVT = max bus transfer time (sec)

This variable is application dependent and the customer should be consulted to determine its setting. In the absence of other guidlines, set at 7.2 seconds

The related software variable definition is

UV TIME OUT LIM E = 720 * PTSUVT

RRACC0, 1, 2, 3, 4, 5
Units - pu speed per second
Resolution - 0.001
Range - 0.001 to 0.097

Definition - Family of variables controlling the rate-of-change of the drive speed. Selection of a particular speed acceleration rate is controlled in the background. Consult ladder diagram to determine mode.

RRACC? = pu speed change per second

The ralated software variable definition is

SPD RATE ACCL E (?) =279600 * RRACC? * DRMFRQ/DRSFRQ

RRBP0, 1, 2, 3, 4
Units - pu speed
Resolution - 0.01
Range - 0.0 to 1.87

Definition - Selects regions of drive speed where different acceleration/deceleration rates may be selected. 1 pu is defined as motor rated speed.

RRBP? = desired pu speed

The related software variable definition is

SPD REF BREAK POINT E (?) =1024 * RRBP? * DRMFRQ / 60

RRDEC0, 1, 2, 3, 4, 5
Units - pu sppeed per second
Resolution - 0.001
Range - 0.001 to 0.097

Definition - Family of variables controlling the rate-of-change of the drive speed. Selection of a particular speed deceleration rate is controlled in the background. Consult ladder diagram to determine mode.

RRDEC? = pu speed change per second

The related software variable definition is

SPD RATE DECEL E (?) = 279600 * RRDEC? * DRMFRQ/DRSFRQ

RRFMAX
RRFMIN
Units - pu speed
Resolution - 0.001
Range - 0.0 to 1.33

Definition - Variables controlling the maximum and minimum speed reference scaling in speed mode 3. Note : 1 pu speed is motor rataed speed.

RRFMAX = maximum desired pu speed
RRFMIN = minimum desired pu speed

The related software variable definition is

SPD REF MAX/MIN LMT E = 1024 * RRFMAX/MIN * DRMFRQ / 60

RRGMAX
RRGMIN
Units - pu speed
Resolution - 0.001
Range - 0.1 to 1.33

Definition - Define the maximum and minimum speed reference clamps. Note: 1 pu speed is rated motor speed.

RRGMAX = max pu speed ref clamp
RRGMIN = min pu speed ref clamp

The related software variable definition is

SPD REG REF MAX/MIN LMT E = 1024 * RRGMAX/MIN * DRMFRQ / 60

RRMOD
Units -n/a
Resolution -n/a
Range -0 to 7

Definition - Variable selecting the speed reference for normaql operation (not turning gear or estop).

RRMOD	Reference Selected
0	Remote 0-10 volt reference
1	Remote 4-20ma reference
2	Remote -10 to 10 volt reference
3	Local 0-8.2 volt reference
4	Remote contact reference
5	Turning gear reference
6	Emergency speed re quest
7	Special product line reference

The related software variable definition is

SPD REF MODE E = RRMOD

RRPMAX
RRPMIN
Units -pu speed
Resolution -0.0 01
Range -0.0 to 133

Definition - Variables controlling the maximum and minimum speed reference scaling in speed modes 0, 1, 2, and 4.
Note: 1 pu speed is motor rated speed.

RRPMAX = maximum desired pu speed
RRPMIN = minimum desired pu speed

The related software variable definition is

SPD REF PROCESS MAX/MIN E =
1024 * RRPMAXMIN * DRMFRQ / 60

RRTG
Units - pu speed
Resolution - 0.001
Range - 0.1 to 1.33

Definition - Variable setting motor fixed speed when speed set point mode 05 is selected. This mode is normally used for turning-gear function.

RRTG = desired pu speed

Note: 1 pu sped is rated motor speed

The related software variable definition is

SPD REF TURNING GEAR E = 1024 * RRTG * DRMFRQ / 60

SCFLUX
Units - pu motor flux
Resolution - 0.01
Range - 0.10 to 0.90

Definition - Defines the minimum flux level that must be achieved while performing a scan start before the load phase-lock-loop is turned on. Typical setting is 0.25

SCFLUX = desired pu flux

The related software variable definition is

MIN SCAN FLUX E = 1024 * SCFLUX

SCFREQ
Units - hertz
Resolution - 1.0
Range - 30 to 60

Definition - Defines the initial inverter frequency when beginning a scan start. The control scans down from this frequency in an attempt to determine the rotational frequency of the motor. Typical setting is 50.

SCFREQ = desired intitial freq (hz.)

The related software variable definition is

SCAN FREQ E = 6 * SCFREQ

SCRDNR
Units - pu speed per second
Resolution - 0.001
Range - 0.001 to 0.5

Definition - Define the rate at which the inverter fundamental frequency ramps down from the initial frequency (SCFREQ) when the control is attempting to start a motor spinning at an unknown speed (SCAN start). Typical setting is 0.042 pu/sec.

SCRDNR = desired scan rate (pu/sec)

The related software variable definition is

RAMP DN E = 1 / SCRDNR

SCTCOM
Units - pu dc link current
Resolution - 0.01
Range - 0.1 to 1.0

Definition - Defines the level of dc link current while the control is attempting a SCAN start (motor spinning at unknown speed). Typical setting is 0.38 pu.

SCTOM = desired pu current level

The related software variable definition is

SCAN TCOM E = 1024 * SCTCOM

SQBLWR
Units - bit coded
Resolution - n/a
Range - not clamped

Definition - Controls the desired blower alarm/fault configuration. The low order byte sets the alarm configuration, the high order byte sets the fault configuration. If the sum of the running blowers is less than or equal to the number represented in the alarm byte, then a blower alarm will be declared. The blower fault works in the same manner.

The related software variable definition is

BLOWER CNT E = SQBLWR

SQOPT1
Units - bit coded
Resolution - n/a
Range - not clamped

Definition - Individual bits are used to enable/disable various functions or inputs in the application program (background). See the ladder diagram for specific usage. Typically a "1" in the bit position indicated will force the indicated state.

Bit	Function
0	no motor overtemperature
1	no bridge differential pressure alarm
2	mask blower 1 starter energized
3	mask blower 2 starter energized
4	mask blower 3 starter energized
5	mask blower 4 starter energized
6	no user defined alarm
7	no transformer alarm
8	no dc link reactor alarm
9	no pcr alarm
A	no blown bridge filter fuse
B	mask run permissive energized
C	mask start permissive energized
D	blower fault is not system fault
E	select return to normal
FD	reserved

The related software variable definition is

SQ OPT1 E = SQ OPT1

SQOPT2
Units - bit coded
Resolution - n/a
Range - not clamped

Definition - Individual bits are used to enable/disable various functions or inputs in the application program (background). See the ladder diagram for specific usage. Typically a "1" in the bit position indicated will force the indicated state.

0 select RUN(1) / CIC(0) output
1 mask line breaker status in READY, READYLOCAL
2 mask source differential current flt
3 select blower fault test (0=status,1=count)

4 select blower alarm test (0=status,1=count)
5 select reclose logic (1=reclose,0=local)
6 select gate driver alarm(1) / fault(0)
7 select gound fault current(1)/ voltage(0)

8 enable line breaker status in RUNP
9 mask fan door closed
A unused
B unused

C unused
D unused
E unused
F reserved

The related software variable definition is

SQ OPT2 E = SQ OPT2

S Q S P 1, 2, 3, 4
Units - pu speed
Resolution - 0.001
Range - 0.0 to 1.875

Definition - Defines uncommited speed breakpoint thresholds for general use in the application program (background). See the ladder diagram for specific usage. 1 pu is defined as motor rated speed.

SQSP? = desired pu speed

The related software variable definition is

BKGND SPD POINT? E = 1024 * SQSP? * DRMFRQ / 60

S Q T 1, 2, 3, 4, 5, 6, 7
Units - seconds
Resolution - 0.01
Range - 0.0 to 200.0

Definition - Defines time delay durations in uncommitted timers used by the applicaton program (background). See the ladder diagram for specific usage.

SQT? = desired time (seconds)

The related software variable definition is

BKGND TIMER? LIMIT E = SQT? * DRSFRQ

S R E R R F
Units - bit coded
Resolution - n/a
Range - 0,1,2

Definition - Selects the break of the low pass filter applied to the speed error signal. Typically set at 2.

SRRERRF = 0 ->filter is bypassed

= 1 ->double at 17 rad (60hz src)
double at 14 rad (50hz src)

= 2 ->double at 40 rad (60hz src)
double at 28 rad (50hz src)

The related software variable definition is

SPD ERR FILTER E = SRRERRF

S R G A I N
Units - gain
Resolution - 0.1
Range - 0.1 to 20.0

Definition - Speed regulator gain. The relation to system parameters is:

$$SRGAIN = \frac{N^2 * WK^2}{1614308.6 * HP * Tr^2} * \frac{60^2}{DRMFRQ^2}$$

N = rated motor rpm
Tr = regulator response time (1 sec)
WK^2 = total drive WK^2 at motor shaft ($lb\text{-}ft^2$)
HP = drive horse power

The related software variable definition is SPD G2 E = 1024 * SRGAIN / (2*DRSFRQ)

S R L L A D J
Units - bit coded
Resolution - n/a
Range - 0,1,2,3,4,5,6,7,8

Definition - Variable for selecting the speed feedback time constant. This time constant is normally coordinated with the speed regulator crossover so as to give critical damping. Typically set to 7.

SRLLADJ	lag T.C. 60hz	lag T.C. 50hz	speed reg crossover
0	filter is bypassed	filter is bypassed	___
1	0.0125	0.0104	80
2	0.0292	0.0243	34
3	0.0625	0.0521	16
4	0.1292	0.1077	8
5	0.2629	0.2191	4
6	0.5300	0.4417	2
7	1.0640	0.8867	1
8	2.1300	1.7750	0.5

The related software variable definition is LEAD LAG ADJUST E = SRLLADJ

S R S L
Units - pu slip
Resolution - 0.001
Range - 0.0 to 0.1

Definition - Defines full load motor slip.

SRSL = f.l. slip pu

The related software variable definition is

SLIP GAIN E = 1024 * SRSL

S R T C
Units - seconds
Resolution - 0.01
Range - 0.05 to 3.0

Definition - Time constant used to stabilize the speed regulator.

SRTC = desired time constant (seconds)

The related software variable definition is

SPD$G1$E = 8 * SRGAIN * SRTC

S R T C D
Units - pu torque
Resolution - 0.01
Range - 0.0 to 1.0

Definition - Sets the default torque command level when the speed regulator is opened. Typically set to 0.2 pu.

SRTCD = desired pu torque

The related software variable definition is

TCOM DEFAULT E = 1024 * SRTCD

S R T C M A X
Units - pu torque
Resolution - 0.01
Range - 0.5 to 1.5

Definition - Defines the maximum motoring torque command clamp applied to the output of the speed regulator.

SRTCMAX = desired pu torque

The related software variable definition is

TCOM HI CLMP E = 1024 * SRTCMAX

S R T C M I N
Units - pu torque
Resolution - 0.01
Range - 0.0 to 1.0

Definition - Defines the minimum motoring torque command clamp that is calculated in the "Torque Clamp vs Speed" function (see SRTVSG).

SRTCMIN = desired pu torque

The related software variable definition is

TCOM LO CLMP E = 1024 * SRTCMIN

S R T C R E G
Units - pu torque
Resolution - 0.01
Range - 0.5 to 1.5

Definition - Defines the maximum regenerative torque command clamp applied to the output of the speed regulator.

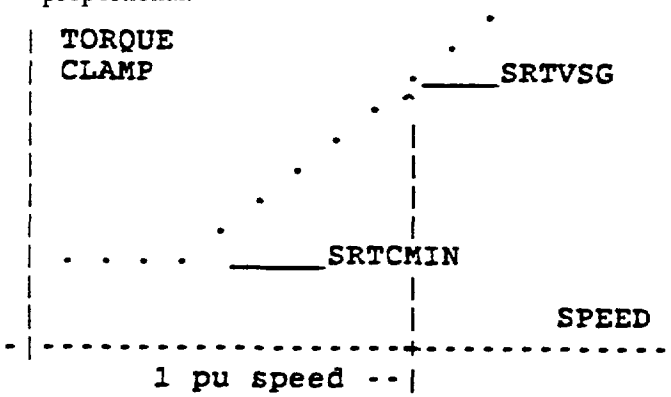
SRTCREG = desired pu torque

The related software variable definition is

TCOM REGEN CLMP E = -1024 * SRTCREG

S R T V S G
Units - pu torque
Resolution - 0.01
Range - 0.1 to 1.87

Definition - Defines the "Torque Clamp vs Speed" function that limits torque as a function of speed. This variable specifies the desired torque limit a 1 pu speed. The torque limit at other speeds will be directly proportional.



SRTVSG = desired pu torque limit

The related software variable definition is

TORQ VS SPD GAIN E = 16 * SRTVSG

S U D S T L
Units - seconds
Resolution - 1.0
Range - 1 to 20

Definition - Defines the maximum length of time that the control will attempt to start the motor from a standstill. Typically set to 7 seconds.

SUDSTL = desired time (seconds)

The related software variable definition is DEAD START TMR LMT E = SUDSTL

S U F T M R
Units - seconds
Resolution - 0.1
Range - 0.0 to 20.0

Definition - Defines the time allowed for the load-side flux integrators to settle before an attempt is made to run the load-side phase-lock-loop. Typically set to 0.7 seconds.

SUFTMR = desired time (seconds)

The related software variable definition is

FLX DELAY TIME E = 360 * SUFTMR

S U K S T L
Units - seconds
Resolution - 1.0
Range - 1.0 to 5.0

Definition - Defines the maximum duration of a known start attempt based on hardware speed. Typically set for 2 seconds.

SUKSTL = desired time (seconds)

The related software variable definition is

KNOWN START TMR LMT E = SUKSTL

S U M F R Q
Units - hertz
Resolution - 0.1
Range - 0.2 to 75.0

Definition - Defines the inverter frequency when the control is in test mode Manual Fix Fire selected in DRVTST. Typically set to 5hz.

SUMFRQ = desired frequency (hertz)

The related software variable definition is

MANUAL FREQ E = 6 * SUMFRQ

S U M I N F L X
Units - pu flux
Resolution - 0.01
Range - 0.1 to 0.75

Definition - Defines the flux magnitude required before the control will start to ramp the inverter frequency up during a dead start attempt. Typically set to 0.4 pu.

SUMINFLX = desired pu flux

The related software variable definition is

MIN START FLUX E = 1024 * SUMINFLX

S U M T C M
Units - pu dc link current
Resolution - 0.01
Range - 0.1 to 1.0

Definition - Defines the current command when the control is in test mode Manual Fix Fire selected in DRVTST. Typically set to 0.25 pu.

SUMTCH = desired pu dc link current

The related software variable definition is

MANUAL TCOM E = 1024 * SUMTCM

S U R E G S P D
Units - hertz
Resolution - 0.1
Range - 1.0 to 30.0

Definition - Defines the inverter frequency above which the speed and flux regulator loops are closed. Typically set to 7hz. This value should always be greater than SUTRNFRQ.

SUREGSPD = desired frequency (hz)

The related software variable definition is

SPD BRK CLOSE REG E = 1024 * SUREGSPD / 60

S U R U P R
Units - pu speed per second
Resolution - 0.001
Range - 0.001 to 0.1

Definition - Defines the rate at which the inverter fundamental frequency ramps up from the initial frequency of SUSFRQ when the control is attempting a dead start. Typical setting is 0.016 pu/sec.

SCRDNR = desired ramp rate (pu/sec)

The related software variable definition is

RAMP UP E = 1 / SCRUPR

S U S F R Q
Units - hertz
Resolution - 0.1
Range - 0.5 to 3.0

Definition - Defines the inverter frequency at which the control initially starts firing in order to establish flux when attempting a dead start. Typically set to 1 hz.

SUSFRQ = desired frequency (hz)

The related software variable definition is

START FREQ E = 6 * SUSFRQ

S U S T C M
Units - pu dc link current
Resolution - 0.01
Range - 0.1 to 1.0

Definition - Defines the level of dc link current command when the control is attempting a dead start. Typically set to 0.75 pu.

SUSTCM = desired pu current

The related software variable definition is

START TCOM E = 1024 * SUSTCM

S U T R N F R Q
Units - hertz
Resolution - 0.1
Range - 1.0 to 15.0

Definition - Defines the minimum frequency at which the control can turn on the phase-lock-loop and transition to closed loop inverter angle control when attempting a dead start. Typically set to 5 hz.

SUTRNFRQ = desired frequency (hz)

The related software variable definition is

MIN TRANS FREQ E = 6 * SUTRNFRQ

S Y B C A
Units - seconds
Resolution - 0.001
Range - 0.01 to 1.0

Definition - Sets the time delay associated with the closing of the bypass contactor. Refer synchronizing Tuneup Instructions for the procedure to determine this value.

SYBCA = measured delay in seconds

The related software variable definition is

SYNC BC ACK TIM E = 2 * DRSFRQ * SYBCA

S Y B C P
Units - seconds
Resolution - 0.001
Range - 0.01 to 1.0

Definition - Defines the time delay associated with the closing of the bypass contactor. Refer Synchronizing Tuneup Instructions for the procedure to determine this value.

SYBCP = Measured delay in seconds

The related software variable definition is

SYNC BC PREDICT TIM E = 2 * DRSFRQ * SYBCP

S Y C N T R L
Units - n/a
Resolution - n/a
Range - 0,1,2,3,4

Definition - Controls selective enabling of various steps involved in the synchronizing/desynchronizing process. Typically set to 3 however the other settings are useful during tune-up and trouble-shooting.

SYCNTRL	function
0	disable all sync/dsync functions
1	allow phase match
2	allow phase/voltage match
3	allow phase/voltage match and allow sync and dsync transfers
4	TEST MODE to open and close bypass contactor in GATE-TEST-MODE

The related software variable definition is

SYNC E value	SYCNTRL value
0	0
1	1
3	2
15	3
32768	4

S Y G A I N
Units - gain
Resolution - 1.0
Range - 5 to 30

Definition - Synchronizing regulator gain. The relation to system parameters is:

SYGAIN = 12.1 * Wc

Wc = regulator crossover (1.24 rad)

The related software variable definition is

SYNC GAIN E = SYGAIN

SYMCNT
Units - seconds
Resolution - 0.001
Range - 0.01 to 1.0

Definition - Sets the length of time that the phase and voltage must be matched during synchronization before the bypass conductor is commanded to close. Typically set to 0.167 seconds.

SYMCNT = desired time in seconds

The related software variable definition is

SYNC MATCH CNT E = 2 * DRSFRQ * SYMCNT

S Y P A D J
Units - degrees
Resolution - 1.0
Range - 0 to 60

Definition - Sets the desired phase angle by which the motor voltage leads the source line voltage during synchronizing. This is typically used to compensate for the dead time experienced by the motor during a transfer and is a function of the pu inertia of the motor/load and the expected loading while trying to synchronize.

SYPADJ = 14 * expected pu load/pu inertia
pu inertia (J) = 2* .31E-6 * wk^2 * RPM^2/Hp

The related software variable definition is

SYNC ADJ E = 1024 * SYPADJ / 360

SYPTOL
Units - degrees
Resolution - 1.0
Range - 0 to 20

Definition - Defines the allowable tolerance on the phase match check during synchronizing. Typically set to 6 degrees

SYPTOL = desired angle tolerance

The related software variabledefinition is

SYNC PHS TOL E = 1024 * SYPTOL / 360

S Y S P D
Units - pu speed
Resolution - 0.001
Range - 0.8 to 1.0

Definition - Defines the motor speed above which bypass is commanded. A typical setting might be 0.96 pu.
Note: pu is based on source line frequency

SYSPD = desired pu speed

The related software variable definition is

SYNC SPD E = 1024 * SYSPD * (DRSFRQ/60)

S Y S T P
Units - pu speed
Resolution - 0.001
Range - 0.995 to 1.005

Definition - Defines the motor speed setpoint while trying to synchronize. Typically set to 1.002 pu.
Note: pu is based on source line frequency

SYSTP = desired pu speed setpoint

The related software vaiable definition is

SYNC SPD STP E = 1024 * SYSTP * (DRSFRQ/60)

S Y VADJ
Units - gain
Resolution - 0.01
Range - 0.6 to 1.4

Definition - Gain to scale the a/d reading of the source flux waves so they can be used as the flux regulator setpoint while trying to synchronize. It is approximately equal to the ratio of rated source voltage to rated motor voltage.

SYVADJ = V11src / V11mot

The related software variable definition is

SYNC FLX SCL E = 27.5 * SYVADJ * (DRMFRQ/DRSFRQ)

S Y V T O L
Units - pu volts
Resolution - 0.01
Range - 0 to 0.20

Definition - Defines the allowable tolerance on the voltage match check during synchronizing. Typically set to 0.05 pu.

SYVTOL = desired voltage tolerance

The related software variable definition is

SYNC VOLT TOLE E = 1024 * SYVTOL

TSMAX
TSMIN
Unit - amps
Resolution - 1.0
Range - no clamps

Definition - Used during the QC FINAL TEST GTO OFF TEST. These values define maximum and minimum values of 1 pu current that the test man is allowed to enter. Typically use 1250 amps max and 100 amps min.

TSMAX = max allowed current (amps)
TSMIN = min allowed current (amps)

The related software variable definition is

QC CUR UPPER LIMIT E = TSMAX
QC CUR LOWER LIMIT E = TSMIN

TSVTOL
Units - %
Resolution - 1
Range - not clamped

Definition - Used during the QC FINAL TEST GTO OFF TEST. This value defines the maximum allowable current in percent. Typically use 5% (5).

TSVTOL = desired tolerance (%)

The related software variable definition is

QC CUR VALUE TOL E = TSVTOL

DESCRIPTION OF PUBLIC VARIABLES

SECTION B

A D V L M T
Equiv Software Name - ADV LMT
Variable Type - INTEGER
Scaling - +16384 counts = +1 pu dc link voltage

Definition - Maximum clamp on the output of the proportional + integral section of the current regulator.

A L A R M
Equiv Software Name - ALARM WORD
Variable Type - WORD
Scaling - BIT CODED

Definition - Variable used to pass alarm states sensed in various software modules to the background (sequencing) programs. The bits of the variable are coded as follows:

Bit	0	unused
	1	unused
	2	I/O addressing problem
	3	Division by zero
	4	Improper interrupt request
	5	Improper running mode for UP
	6	Numerical overflow in calculation
	7	Non-maskable interrupt service
	8	Improper interrupt request on HLCB
	9	System time interrupt running
	A	Interrupt rquest unresolved
	B	Loss of speed control
	C	GTO freeze
	D	spare
	E	Loss of speed reference
	F	spare

The ALARM will remain set until the "RESET" is operated.

A L P H L D
Equiv Software Name - FIRE ANG RL
Variable Type - INTEGER
Scaling - +256 counts -> ALPHLD = 270 degrees
0 counts -> ALPHLD = 180 degrees
-255 counts -> ALPHLD = 90 degrees

Definition - Sets the firing angle for the load side power converter. It is the phase angle result of the vector addition of IQ and IX. During motoring operation, ALPHALD ranges between (90 to 270) degrees . Values between (0 to 90) and (270 to 360) indicate regenerative mode. The load angle control regulator calculates this variable value in RUN FIRE when the phase-lock- loop is locked. In IDLE andFIX FIRE the variable is not calculated but set to 180 degrees.

A L P H S R C
Equiv Software Name - ALPHA
Variable Type - INTEGER
Scaling - +256 counts -> ALPHA = 0 degrees
- 255 counts -> ALPHA = 180 degrees

Definition - Sets the value of the source side power converter firing angle. The variable is calculated in the source current regulator.

BCCNT
Equiv Software Name - BC CNT
Variable Type - WORD
Scaling - 1 count = 1/(2*src freq) seconds

Definition -Variable used to measure the opening and closing times of the bypass contactor. The counter counts up when the bypass contactor status disagrees with the bypass contactor command. The counter counts at twice the source line frequency.

CLSREG
Equiv Software Name - CLOSE FLX REG, CLOSE SPD REG
Variable Type - WORD
Scaling - BIT CODED

Definition - This monitor variable overlays two adjacent byte variable CLOSE FLX REG (low byte) and CLOSE SPD REG (high byte). The least significant bit of each byte commands the respective regulator to close. Thus in normal running CLSREG = 257 = 101H. Do NOT attempt to modify this variable while running the motor!

CPSUV
Equiv Software Name - CP UV FLG
Variable Type - BYTE
Scaling - True/False

Definition - This flag is true to indicate an undervoltage condition on the 400hz. output of the chopper power supply.

C R E R R
Equiv Software Name - RAW ERR
Variable Type - INTEGER
Scaling - 1024 counts = 1 pu dc link current

Definition - Error signal for the current regulator. It is the difference between current command and current feedback (command-feedback) before being clamped.

C R S T I E
Equiv Software Name - CRSSTP
Variable Type - INTEGER
Scaling - 8192 counts = 1 pu motor flux command

Definition - Sets the amount by which the level of commanded motor flux is reduced in order to keep the source power converter firing angle from saturating in advance limit (alpha = 0).

C S T E P
Equiv Software Name - CUR REG STEP
Variable Type - INTEGER
Scaling - 1024 counts -> 1 pu current

Definition - Test input for current regulator. Used to step the current regulator to determine its transient response.

C U R C M P
Equiv Software Name - CURCOMP
Variable Type - INTEGER
Scaling - 16384 counts = 1 pu src link volts

Definition - Source dc link voltage command output from the proportional + integral section of the current regulator. It makes adjustments to the dc link voltage commanded from the feed forward section of the current regulator. Ideally, this variable should vary irregularly around zero whenever the feedforward section is active.

F F M O D E
Equiv Software Name - FIX FIRE MODE
Variable Type - BYTE
Scaling - n/a

Definition - Indicates the specific mode of operation when then control is in inverter open loop frequency control (LMODE=16). The different modes are:

0 - CROWBAR: diametric pairs are fired on then inverter

1 - MANUAL: inverter fundamental frequency and dc link current level are manually controlled

2 - DEADSTART: motor start attempt assuming motor is at standstill

3 - SCANSTART: motor start attempt assuming motor is spinning at unknown speed

4 - KNOWNSTART: motor start attempt assuming motor is spinning at speed indicated by hardware speed signal

---

F I R E A N G
Equiv Software Name - FIRE ANG
Variable Type - INTEGER
Scaling - +256 counts = alpha 270 degrees
0 counts = alpha 180 degrees
-256 counts = alpha 90 degrees

Definition - Unclamped and non-rate-limited version of the inverter firing angle.

F L T W D
Equiv Software Name - FAULT WORD
Variable Type - WORD
Scaling - BIT CODED

Definition - Faults, sensed by the foreground programs, are encoded into FLTWD for transmission to the background. The bits are defined as follows:

Bit	0	spare
	1	Inverter dc link overvoltage
	2	Chopper power supply failure
	3	DC link overcurrent SRC
	4	DC link overcurrent LD
	5	IDC source shoot thru
	6	IDC load shoot thru
	7	Source undervoltage
	8	Source PLL fault
	9	Load PLL fault
	A	Source reverse phase sequence
	B	Stack fault
	C	spare
	D	EEPROM transfer failure
	E	desynchronization failure
	F	spare

F L X D R P
Equiv Software Name - FLX DROOP ADJ
Variable Type - WORD
Scaling - 512 counts = 1 pu flux

Definition - Amount that the flux command is reduced when the drive is operation at light load (the reduction is inversely proportional to torque command).

F L X E R R
Equiv Software Name - FLX ERR FIL
Variable Type - INTEGER
Scaling - 1024 counts = 1 pu flux error

Definition - Flux error is the difference between flux setpoint and flux feedback (setpoint-feedback) after being clamped.

F L X F B K
Equiv Software Name - FLX FBK
Variable Type - INTEGER
Scaling - 1024 counts = 1 pu flux feedback

Definition - Feedback signal for the flux regulator. It is derived from a/d reads of the motor line to line peak flux.

F L X P I
Equiv Software Name - FLX PI SUM
Variable Type - INTEGER
Scaling - 4096 counts = 1 pu flux command

Definition - The output of the proportional + integral section of the flux regulator. This signal is used to adjust the flux setpoint coming from the feedforward section of the flux regulator. These signals are divided by pu motor exciting reactance to obtain required pu motor exciting current.

F L X S T P
Equiv Software Name - FLX STP
Variable Type - INTEGER
Scaling - 1024 counts = 1 pu flux command

Definition - The flux setpoint command into the flux regulator. This signal has been compensated by the cross tie regulator and clamped. This is the non-rate-limited version of FLXSTR.

F L X S T P S Y
Equiv Software Name - SYNC FLX STP
Variable Type - INTEGER
Scaling - 1024 counts = 1 pu flux command

Definition - This is the command to the flux regulator when the control is trying to match motor voltage magnitude to source line voltage magnitude when trying to synchronize. .

F L X S T R
Equiv Software Name - FLX STAR1
Variable Type - INTEGER
Scaling - 1024 counts = 1 pu flux command

Definition - This is the rate-limited version of FLXSTP. The rate limit is in effect only when first turning the flux regulator on and when the reference takes a step change when the synchronizing regulator is turned on or off.

F S T E P
Equiv Software Name - FLX REG STEP
Variable Type - INTEGER
Scaling - 1024 counts = 1 pu flux command

Definition - Variable used to step the flux regulator when tuning the transient response.

I C A P
Equiv Software Name - ICAP FIL
Variable Type - INTEGER
Scaling - 1024 counts = 1 pu dc link current

Definition - Calculated motor capacitor current. PU is based on rated inverter dc link current.

I L D
Equiv Software Name - IDC LOD
Variable Type - INTEGER
Scaling - 1024 counts = 1 pu dc link current

Definition - Current feedback calculated from a/d reads of the inverter cts. PU is based on rated dc link current

---

I N V L I M
Equiv Software Name - INVLIM
Variable Type - WORD
Scaling - BIT CODED

Definition - When any bit of the variable is non-zero, the source and load side converters will be set to fire at inversion limit. The drive will not necessarily fault trip and may recover to normal operation. The bits are coded as follows:

Bit	0	Zero current request
	1	unused
	2	unused
	3	unused
	4	Load idle request
	5	Load dc link overvoltage
	6	Source instantaneous over current
	7	Gto freeze
	8	Drive idle request
	9	Loss of hardware speed
	A	Load instantaneous over current
	B	Source phase lock loop problem
	C	Load phase lock loop problem
	D	Crowbar
	E	unused
	F	Stack overflow

I Q
Equiv Software Name - IQ STAR
Variable Type - INTEGER
Scaling - 1024 counts = 1 pu dc link current

Definition - Command for torque producing component of dc link current. This signal is derived from the speed regulator. It is added vectorally to give dc link current command.

I S R C
Equiv Software Name - IDC SRC
Variable Type - INTEGER
Scaling - 1024 counts = 1 pu dc link current

Definition - Current feedback calculated from a/d reads of the source cts. PU is based on rated dc link current. This signal is calculated by averaging a/d readings of the souce cts at the time of the source firings and 30 degrees later. This averaging tends to eliminate the undesirable affects of the ripple current.

I S T A R
Equiv Software Name - IDC STAR
Variable Type - INTEGER
Scaling - 1024 counts = 1 pu dc link current

Definition - The unclamped and non-rate-limited dc link current command. It is the magnitude result of the vector addition IQ and IX.

I S T A R 2
Equiv Software Name - ISTAR2
Variable Type - INTEGER
Scaling - 1024 counts = 1 pu dc link current

Definition - The clamped and rate-limited version of dc link current command ISTAR.

I X
Equiv Software Name - IX STAR
Variable Type - INTEGER
Scaling - 1024 counts = 1 pu dc link current

Definition - Command for reactive component of dc link current. It is the difference between the reactive current required by the motor and the reactive current supplied by the motor capacitors. It is calculated in the flux regulator. IX is vectorally added to IQ to give dc link current command.

I X M T R
Equiv Software Name - IME STAR
Variable Type - INTEGER
Scaling - 1024 counts = 1 pu inverter current

Definition - Commanded level of motor exciting current. This exciting current is jointly supplied by the inverter and the motor filter capacitors.

L C O M
Equiv Software Name - LINE COM
Variable Type - INTEGER
Scaling - +256 counts = +1 pu dc link voltage
- 256 counts = - 1 pu dc link voltage

Definition - Commanded source dc link voltage. This is the output from the current regulator and is converted to a source firing angle through an arccosine table.

L F R E Q
Equiv Software Name - SPD6
Variable Type - INTEGER
Scaling - 1024 counts = 60 hz

Definition - Fundamental inverter firing frequency based on motor flux crossover intervals. It is calculated in the load side phase lock loop.

L I T C O N D
Equiv Software Name - LOD COND
Variable Type - BYTE
Scaling - bit coded

Definition - Indicates the status of the light load operating condition in which the flux regulator is partially open looped in order to satisfy the demands of the speed regulator.

LITCOND = 0 -> light load cond inactive
LITCOND = 255 -> light load cond active

L M O D E
Equiv Software Name - LAC MODE
Variable Type - WORD
Scaling - BIT CODED

Definition - Indicates the major mode of operation of the inverter control. The different modes are:

00 (00H) - IDLE (idle)
16 (10H) - FIXFIRE (open loop, frequency control)
32 (20H) - RUNFIRE (closed loop, angle control)

L S T A T
Equiv Software Name - LOD STATUS
Variable type - WORD
Scaling - BIT CODED

Definition - Indicates the mode of harmonic elimination. The different modes are:

00 (00H) - idle
16 (10H) - fixfire
32 (20H) - no harmonic elimination
48 (30H) - eliminate 5th
64 (40H) - eliminate 5th and 7th
80 (50H) - eliminate 7th and 11th
96 (60H) - eliminate 11th and 13th

L V L T S
Equiv Software Name - LD FF VDC
Variable Type - INTEGER
Scaling - 256 counts = 1 pu dc link volt command

Definition - Load side dc link voltage feedforward into source current regulator.

M F L A G
Equiv Software Name - MODE FLAG
Variable Type - WORD
Scaling - BIT CODED

Definition - Variable used to interface the background sequencing program to the foreground programs which control the the firing algorithms. The variable is coded as follows:

Bit	0	unused
	1	Run
	2	unused
	3	unused
	4	Crowbar
	5	Switchgear closed
	6	unused
	7	unused
	8	Gate-test-mode
	9	unused
	A	unused
	B	unused
	C	unused
	D	Factory source-side firing
	E	Factory load-side firing
	F	Factory pump back

M T R S E G
Equiv Software Name - MTR SEG
Variable Type - BYTE
Scaling - n/a

Definition - Variable used to represent the identity of the previous motor flux crossover event. There are six distinct flux crossovers per cycle (360 deg).

N P T
Equiv Software Name - NPT
Variable Type - BYTE
Scaling - True/False

Definition - Used in the synchronizing function as a final check that the motor voltage is in fact phase and magnitude matched to the source voltage. Potential transformers on both sides of the bypass contactor are compared. This variable can be thought as NOT Potential Transformer Match. Thus

NPT = True indicates no match
NPT = False indicated match

P 9 0 0 C
Equiv Software Name - PORT 900C
Variable Type - BYTE
Scaling - BIT CODED

Definition - Background sequencing variable representing an input port. The bits represent the following inputs:

Bit	0	Source overvoltage
	1	Source low voltage
	2	Source hardware overcurrent
	3	Source gating undervoltage
	4	Load overvoltage
	5	Load hardware overcurrent
	6	unused
	7	Overspeed

P H L D
Equiv Software Name - PH LD
Variable Type - BYTE
Scaling - n/a

Definition - Variable used to indicate the next load inverter leg to be gated 'on'. The value of the variable corresponds to the leg #.

P H S E R R S Y
Equiv Software Name - SYNC PHS ERR
Variable Type - INTEGER
Scaling - 1023 cnts = 360 degrees

Definition - This is the phase error between the source line voltage and the motor voltage. It is always active but is meaningful only when the drive is trying to synchronize. A positive phase error indicates the motor volatage is leading the source line voltage.

P H S R C
Equiv Software Name - PH SRC
Variable Type - BYTE
Scaling - n/a

Definition - Variable used to indicate the next source converter leg to be gated 'on'. This is a twelve count ph which is used on both 6-pulse and series 12-pulse systems. Each ph's correspond to channel A (master) firings, odd ph's correspond the channel B (slave) firings. Divide the even ph counts by 2 to obtain the next leg # to be fired in channel A bridge. If channel B lags channel A, divide the odd ph count by 2 and round down to obtain the next leg # to be fired in channel B bridge. If channel B leads channel A, divide the odd ph count by 2 and round up to obtain the next leg # to be fired in channel B bridge.

bridge	next leg # to be fired
channel A	even ph/2
channel B lags	(odd ph/2) -0.5
channel B leads	(odd ph/2) + 0.5

P L L L E R R
Equiv Software Name - OLD CORCT ERROR
Variable Type - INTEGER
Scaling - 1024 counts = 360 degrees

Definition - Load phase lock loop regulator error signal (unclamped version).

P L L L L D
Equiv Software Name - PLLC LOAD LD
Variable Type - INTEGER
Scaling - 120 counts = 60 hz

Definition - Value loaded into the load PLL counter to maintain synchronizm with the motor flux frequency. The equation controlling its value is

PLLLLD = 7200 / (frequency)

P L L S E R R
Equiv Software Name - PLL ERROR SRC
Variable Type - INTEGER
Scaling - 1024 counts = 360 degrees

Definition - Source phase lock loop regulator error signal.

P L L S F L G L
Equiv Software Name - PLL SFLAG LOD
Variable Type - INTEGER
Scaling - not applicable

Definition - Variable used as a measure of the load phase-lock-loop reliablility. A calculation where the error is excessive will result in the variable being decremented by 1 count. A good calculation will increment the variable by 1 count.

P L L S F L G S
Equiv Software Name - PLL SFLAG SRC
Variable Type - INTEGER
Scaling - not applicable

Definition - Variable used as a measure of the source phase lock loop reliablility. A calculation where the error is excessive will result in the variable being decrimented by 1 count. A good calculation will incriment the variable by 1 count.

P L L S L D
Equiv Software Name - PLLC LOAD SRC
Variable Type - INTEGER
Scaling - 120 counts = 60 hz

Definition - Value loaded into the source PLL counter to maintain synchronizm with the source line frequency. The equation controlling its value is

PLLSLD = 7200 / (frequency)

Q U A D
Equiv Software Name - QUAD
Variable Type - BYTE
Scaling - bit coded

Definition - Variable indicates the quadrant of the inverter firing angle. The quadrant is generally described in terms of reactive current (IX) and real current (IQ) polarity.

+IX -> supply reactive current
- IX -> absorb reactive current

+IQ -> supply motoring real current
- IQ -> supply regenerating real current

R E T D L S R C
Equiv Software Name - RETARD LINE
Variable Type - INTEGER
Scaling - 256 counts = -1pu dc link voltage

Definition - Sets the negative clamp for the output of the current regulator (dc link voltage command). This is typically referred to as inversion limit. Its value is calculated so as to prevent commutation failures when firing at inversion limit.

R E T L M T
Equiv Software Name - RET LMT
Variable Type - INTEGER
Scaling - -16384 counts = -1 pu dc link voltage

Definition - Minimum clamp on the output of the proportional + integral section of the current regulator.

R F L X S P D O
Equiv Software Name - REG FLX SPD OPEN
Variable Type - BYTE
Scaling - bit coded

Definition - Indicates the status of the speed and flux regulators:

0 - speed and flux reg open
1 - speed and flux reg closed

S F D B K
Equiv Software Name - SPD1
Variable Type - INTEGER
Scaling - 1024 counts = 60 hz

Definition - Motor speed signal. This signal is derived from inverter fundamental electrical frequency which has been compensated to account for motor slip.

S F D B K L
Equiv Software Name - SPD FDBK
Variable Type - INTEGER
Scaling - 1024 counts = 60 hz

Definition - This is the speed feedback signal used by the speed regulator. It is the slip compensated speed signal which has also be passed through a lead-lag filter.

S M O D E
Equiv Software Name - SAC MODE
Variable Type - WORD
Scaling - BIT CODED

Definition - Indicates the major mode of operation of the source control. The different modes are:

00 (00H) - IDLE (idle)
16 (10H) - unused
32 (20H) - RUNFIRE (closed loop, angle control)

S P D E F I L
Equiv Software Name - SPD ERR FIL
Variable Type - INTEGER
Scaling - 1024 counts = 60 hz

Definition - Speed regulator error signal after passing through the speed error low pass filter.

S P D R E G O P
Equiv Software Name - OPEN SPD REG
Variable Type - BYTE
Scaling - n/a

Definition - This variable can be used to force the speed regulator to remain open loop, in which case the level of torque command is set by Tune-Up variable SRTCD.

state	condition
0	speed reg allowed to close
1	speed reg forced open

S P D R P
Equiv Software Name - SPD RAMP
Variable Type - BYTE
Scaling - bit coded

Definition - Specifies the speed accel/decel rates selected by the background program.

S P D S P
Equiv Software Name - SPD SP
Variable Type - INTEGER
Scaling - 2048 counts = 10 volts

Definition - The raw, unscaled a/d reading of the selected speed reference.

S P D S T P
Equiv Software Name - SPD STP
Variable Type - INTEGER
Scaling - 1024 counts = 60 hz

Definition - Speed reference setpoint before the speed ramp function.

S P D S T R
Equiv Software Name - SPD STR
Variable Type - INTEGER
Scaling - 1024 counts = 60 hz

Definition - Speed regulator reference (after speed ramp).

S R C S E G
Equiv Software Name - SRC SEGMENT
Variable Type - BYTE
Scaling - n/a

Definition - Variable used to represent the identity of the previous source flux crossover event. There are six distinct flux crossovers per cycle (360 deg).

S R C U V
Equiv Software Name - UV FLG
Variable Type - BYTE
Scaling - True/False

Definition - This flag is true to indicate an undervoltage condition on the source line voltage.

S R E F M D Equiv Software Name- SPD REF MODEVariable Type- BYTEScaling- n/a

Definition - Variable used to indicate which speed reference is actively being used.

SREFMD	Reference Selected
0	Remote 0-10 volt
1	Remote 4-20 ma
2	Remote -10 to 10 volt
3	Local 0-8.2 volt
4	Remote Contact
5	Turning Gear
6	Emergency Speed Request
7	Special Product Line

S S T E P
Equiv Software Name - SPD REG STEP
Variable Type - INTEGER
Scaling - 1024 counts = 60 hz

Definition - Speed regulator input variable for stepping the regulator and determining its transient response.

S Y M T C H
Equiv Software Name - SYNC MTCH
Variable Type - BYTE
Scaling - True/False

Definition - This variable is true to tell the background that the motor voltage has been synchronized to the source line voltage and that the synchronizing sequencing may proceed.

T C O M
Equiv Software Name - TCOM
Variable Type - INTEGER
Scaling - 1024 counts = 1 pu torque command

Definition - Output from the speed regulator. It is the torque command which get converted into IQ by dividing by flux.

T H C L M P
Equiv Software Name - SPD REG CLMP
Variable Type - INTEGER
Scaling - 1024 counts = 1 pu torque command

Definition - Indicated the maximim torque clamp imposed on the output of the speed regulator.

CHAPTER 5
SECTION A
DIAGNOSTIC MONITOR SPECIFICATIONS

5-1. INTRODUCTION

The diagnostic monitor for the Structured GTO-IMD product provides the user with a variety of functions. Via an RS-232 terminal, the user may display regulator and logic variables and system fault and status information. The monitor also provides the ability to modify system variables and regulator constants, which facilitates troubleshooting and tuning procedures.

Communication with the monitor is handled through the RS232 port on the drive control. Hence, the communication device may be any RS232 compatible terminal, including handheld, hard copy and CRT types. Provisions are made to specially format the monitor output when a handheld terminal is being used.

All communication via the RS232 port are 7-bit data, even parity, with one stop bit. Acceptable baud rates are 300, 1200, 2400, 4800, and 9600 bits per second (baud). Upon power up, the monitor searches for carriage return characters to lock onto the baud rate used by the terminal device.

Monitor functions are accessed by entering a valid command (from the monitor's instruction set) at the diagnostic terminal. Figure 1-1 illustrates the basic command structure for the drive. Many of the commands are called directly from the MONITOR level, '*>' prompt, and control is returned to this level after execution of the requested process. The ENGINEERING and TUNEUP functions, however, are actually extended modes of operation that have their own instruction sets. After execution of one of the ENGINEERING mode commands, for example, control is returned to the ENGINEERING mode, 'E>' prompt. To return to the MONITOR level again, the QUIT command must be invoked -- this is also true for the TUNEUP mode.

In sections 1-3 and 1-4, the purpose and syntax of each monitor command is described in detail. Table 5-1 defines the conventions used in specifying the command formats.

TABLE 5-1 Definitions for Command Syntax Notation

Symbol	Definition
::=	"Is defined to be"
[ ]	Items between brackets are optional
|	Separates items in a list - choose one
{ }	Braces delimit items for the purpose of minimizing ambiguity
( )	Optional characters in a command or parameter are in parentheses

HIGHLITED User entries (Highlited)

<decimal no> ::= <1-5 decimal digits> [T]

<hex no> :: = <1-4 hex digits> [H]

<new value> :: = [+>-]<1-5 decimal digits w/ or w/o dec. pt.>

<pointer> ::= [{CS | DS | <hex no>}:] <hex no>

<register> ::= [CS | DS]

CS ::= Code Segment

DS ::= Data Segment

5-2. OPERATING INSTRUCTIONS

5-2.1. General Information

To initiate communication with the Monitor, the user should enter several "carriage return" characters at the terminal keyboard. The terminal will then respond with a banner message and the monitor prompt character (*>) to indicate that it is ready to accept command inputs. At this point, after passing the PASSCODE. The operator may enter any of the MONITOR level commands or select one of the extended modes of operation (for example TUNEUP) as described in detail in sections 3 and 4.

When finished using the diagnostic monitor, the terminal session may be terminated by entering QUIT at the terminal keyboard (see section 5-3.9). The logoff message is then printed at the terminal, indicating that the session is over, and the monitor will idle, waiting for the carriage return characters that indicate the start of another terminal session.

Terminal Format:

The Diagnostic Monitor is designed to work with either a handheld terminal or a standard keyboard type. The default (at the start of the terminal session) is the handheld format. In this mode of operation, the output messages are sent one line at a time to accommodate the limited display length of the handheld terminal (16 characters max per line). Entering a carriage return causes the next line of output to be sent to the terminal. The TERMINAL command (section 5-3.4) may be used to switch to an 80 column listing format with no pauses for use with a standard CRT or hardcopy terminal.

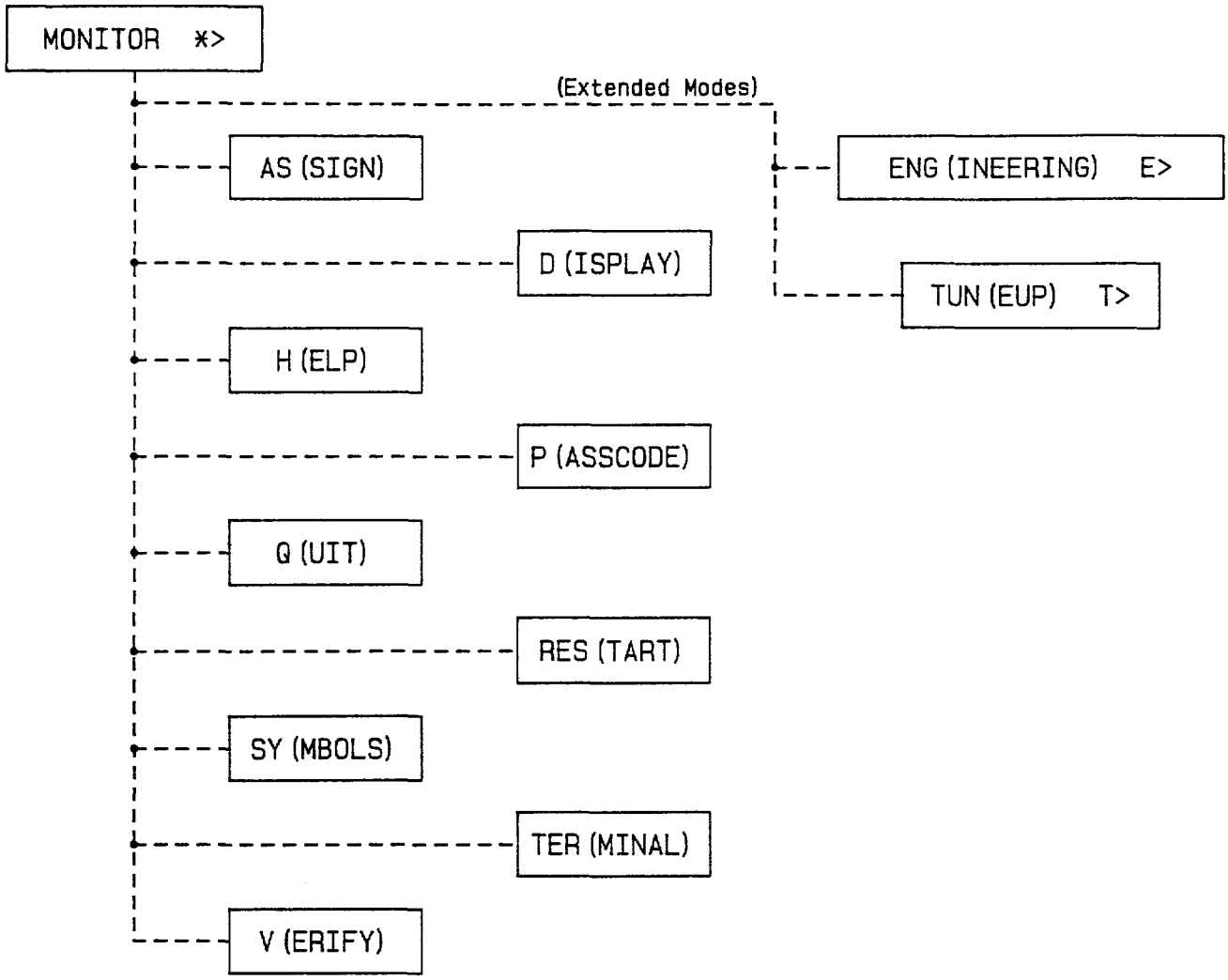


Figure 5A-1. Monitor Command Structure

Power-up Status Messages:

When power is first applied to the drive control, a sequence of self-tests and initialization procedures is performed. Any failure status of these tests is automatically sent to the panel printer. Should the printer fail, diagnostic messages will be rerouted to the diagnostic terminal.

5-2.2. Editing Functions

The diagnostic monitor provides text editing functions for correcting errors entered within a command string. Each function is listed below with a brief description.

NOTE: In the examples below, the symbol *> is the diagnostic monitor prompt (not typed by the user), and the symbol [CUR] represents the cursor position at the end of the sequence shown.

DEL/RUBOUT -

Removes the previous character from the message and echoes the deleted character to the terminal.

Example: Typed - *> HEKP<DEL><DEL>LP

Echoed - *> HEKP /K P LP [CUR]

CONTROL-R

Causes all characters currently in the message to be transmitted to the terminal on a new line. This function is convenient after using DEL/RUBOUT because it shows the state of the message after the editing.

Example:

Typed - *> HEK<DEL>LP<CONTROL-R>

Echoed - *> HEK /K LP
HELP [CUR]

NOTE: Typing subsequent consecutive CONTROL-R keys will invoke the monitor recall function. This feature calls up previous commands that were typed at the terminal, starting with the most recent and progressing to the oldest command remembered. A total of 20 command lines are remembered. Using the recall feature helps the user avoid retyping a command that he wishes to execute repetitively.

CONTROL-X

Deletes the entire message that has been entered so far.

NOTE: This has the same effect as using the DEL/RUBOUT function to remove all the characters from the message.

Example: Typed - *> HEKL<CONTROL-X>HELP
Echoed - *> HEKL
HELP [CUR]

CONTROL-Y

Aborts the current command sequence and returns the monitor to the current mode of operation (e.g., ENGINEERING, TUNEUP or the main level).

NOTE: CONTROL-Y or CONTROL-C may be entered at any time to abort the current process--even during the listing of a message at the terminal.

Example:

Typed - T> HE<CONTROL-Y>

Echoed - T> HE^ Y

T> [CUR]

CONTROL-C

Aborts the current command sequence and returns the monitor to the outermost (main) level of operation. When a CONTROL-C is received, the monitor also displays a message identifying the product as a GTO-IMD Drive as in the login sequence.

Example:

Typed - T> HE<CONTROL-C>
Echoed - T> HE^ C
-- IGTO MONITOR--
*> [CUR]

CONTROL-S

Causes the current output to be halted until it is restarted using the CONTROL-Q switch.

CONTROL-Q

Causes the current output to be restarted after having been stopped using the CONTROL-S switch.

5-2.3. Command Syntax

The general syntax for a monitor command is shown below:

<command> ::= <instruction>
[<delimiter> <parameter> ...] <terminator>

where <instruction> is one of the valid monitor instructions (defined in section 5-3) and <terminator> is a carriage return (<cr>). The <delimiter> and <parameter> entries, if necessary, depend on the particular instruction being called. Each instruction has its own set of valid delimiters and parameters. Also, some commands may require several <delimiter> <parameter> pairs, while some require no parameters at all.

As an aid to the operator, and to facilitate handheld terminal use, all the <delimiter> <parameter> pairs need not be entered on the same line as the instruction -- each needed parameter will be requested by the monitor. The syntax for this type of sequence is:

<command> ::= <instruction> <terminator>
<parameter> <terminator>.

After each <terminator> entry, the monitor displays an appropriate prompt that asks the user for the necessary parameter.

The use of abbreviated commands is another feature designed to aid the handheld user. Rather than typing the complete instruction word, only the first one or two

letters need to be entered (see the syntax definition tables in sections 3 and 4 for the minimum acceptable abbreviations).

5-2.4. Protected Commands

Some monitor commands allow the user to modify system variables and change tuning parameters. These functions, because of their potentially destructive nature, are available only after executing a privilege enabling sequence. This is so the user will not inadvertantly make modifications that could inhibit the proper operation of the drive control.

The privileges are enabled by entering a password at the monitor terminal. Privileges are automatically disabled when: (1) the monitor is logged off (QUIT).

5-2.5. Help

The user may print out a list of available commands by asking for HELP. From either the MONITOR mode, or one of the extended modes, a HELP request will print the list of commands available at that mode only.For example, if in the ENGINEERING mode, only the ENGINEERING mode commands, and not the MONITOR mode commands, will be listed.

5-3. MONITOR COMMANDS

The MONITOR level commands are intended to be the primary tools used in interrogating the drive control and monitoring its status. The instruction set provides display and modify functions, the ability to reset the control, and the power to direct variables to analog outputs for strip chart monitoring, among other things. Table 5-2 shows the complete list of instructions and their correct syntax. The commands are defined in their single line entry format, although the "prompt-through" type sequence (as discussed in section 2.3) is also valid.(See Table 5-2)

TABLE 5-2. Syntax Definitions for MONITOR Level Commands

```
ASSIGN:     A(SSIGN)        {LIST| DA}
DISPLAY:    {D(ISPLAY)  {   <varname> [C] | FAULTS | LIST | RELAYS } |
             DF |
             DL |
             DR   }   <cr>
HELP:       H(ELP) <cr>
PASSCODE:   P(ASSCODE) <cr>

QUIT:       {Q(UIT) |       EX(IT)       <cr>
RESTART:    RES(TART) <cr>
SYMBOLS:    SY(MBOLS) [ <symbolname> ] <cr>
TERMINAL:   TER(MINAL) <cr>
              <current format> CHANGE ? [ Y|       N ] <cr>
VERIFY: V(ERIFY) [ <symbol group name> ] <cr>
```

5-3.1. Assign

Function: To designate that a particular variable be assigned to a circular list and/or D/A converter.

Using the ASSIGN function, the user may modify the configuration of the internal circular lists by changing the number of entries (maximum of 8) to be stored at each sample interval and/or selecting which entries are being captured. A total of 2K bytes of memory are available for storage of the diagnostic list information. This memory is evenly divided among the parameters within the list. As an example, if 4 word parameters are monitored in the circular a total of 256 events will be recorded.

The list may be triggered either by a level trigger or mask trigger. In either case the trigger is compared against the variable selected in list position no. 1.. In a level (>, <, =). When the bit trigger mode is selected, any bit of variable 1 becoming true causes the list to be taken. Bits of variable 1 may be masked using the mask variable option of the list. The list can only be triggered by the level trigger or the bit trigger. When more than one trigger has been activated only the last will be accepted by the list function.

The user may also direct drive variable contents to a D/A converter for observation on an oscilloscope or chart recorder. The analog outputs are available from test points on the front edge of the printed circuit board HLCB. When assigning a variable to a D/A output, the operator selects the two parameters, SCALE and OFFSET. Since the possible range of a variable often exceeds the range of the D/A converter, a function is provided to scale the variable contents before sending data to the D/A converter. The following table gives the possible SCALE parameters; beside each one is the value by which the variable contents are multiplied

SCALE PARAMETERS

SCALE	MULTIPLIER	SCALE	MULTIPLIER
15	1/32768	0	1
14	1/16384	-1	2
13	1/8192	-2	4
12	1/4096	-3	8
11	1/2048	-4	16
10	1/1024	-5	32
9	1/512	-6	64
8	1/256	-7	128
7	1/128	-8	256
6	1/64	-9	512
5	1/32	-10	1024
4	1/16	-11	2048
3	1/8	-12	4096
2	1/4	-13	8192
1	1/2	-14	16384
0	1	-15	32768

before being converted if that SCALE factor is selected. An OFFSET may also be selected during the ASSIGN process -- this value is subtracted from the variable contents before the scaling process. This function may be used when the slight variations around a fixed value need to be examined (see example below).

Examples:

```
*>ASSIGN LIST
List Setup:

0 – quit
1 – samples                    ; selects variables to be sampled
2 – trig level                 ; sets up level triggered list
3 – trig mask                  ; sets up bit triggered list
4 – site                       ; selects site where data is taken

Choice? 1
NO OF ENTRIES=1                ; previously was set to capture1 parameter
NEW VALUE ? 3<cr>              ; set to capture 3 parameters
ENTRY 1 = FLUXFB               ; previously set to capture FLUXFB
NEW ENTRY? SPDREF<cr>          ; speed reference (no. 1)
ENTRY 2 = 0000H                ; null value
NEW ENTRY? SPDFB<cr>           ; speed feedback (no. 2)
ENTRY 3 = 0000H                ; null value
NEW ENTRY? IAFB<cr>            ; armature current feedback (no. 3)

*>ASSIGN LIST

List Setup:
  0 – quit
  1 – samples
  2 – trig level
  3 – trig mask
  4 – site

Choice? 2
TRIGGER LEVEL = 0
NEW VALUE? 200                 ; level of trigger in counts
TRIGGER CONDITION? OFF
NEW VALUE? <                   ; choices are <, >, =
```

```
*>ASSIGN LIST
List Setup:
0 — quit
1 — samples
2 — trig level
3 — trig mask
4 — site

Choice? 4
TRIGGER SOURCE = OFF                ; present data site
CHOICES:                            ; available data sites
0 - OFF
1 - SPLL
2 - SDRV
3 - LPLL
4 - LDRV
5 - FLXR
6 - SPDR
7 - TCMR
8 - SYNR
9 - BKGND
NEW VALUE? 1 <cr>
*>ASSIGN LIST <cr>List Setup:
  0-quit
  1-samples
  2-trig level
  3-trig mask
  4-site

Choice? 3
PRESENT MASK FF                     ;you must turn off level trigger
NEW VALUE? OF                       ;value must be in hexadecimal
*>
*>ASSIGN LIST                       ; go to choice 4 to select site
*>                                  ;because choice 3 turned off list

*> ASSIGN DA1 <cr>                 ; choose DA1 (TP2) on HLCB (slot2E)
ENTRY = SPDREF                      ; the current setup is shown
SCALE = 8
OFFSET = 0
ENTRY = IAFB<cr>                    ; set new parameters -- I feedback
SCALE = 6<cr>                       ; divide by 64 to get in range
OFFSET = <cr>                       ; leave the offset the same (at 0)
*>

*> AS DA2<cr>                       ; choose the DA2 (TP3) on HLCB (slot 2E)
ENTRY = SPDFB                       ; the current setup is shown
SCALE = 8
OFFSET = 0
ENTRY = <cr>                        ; still look at speed feedback
SCALE = -2 <cr>                     ; multiply 4 to accent variations
OFFSET = 1000<cr>                   ; look at fluctuations around 1000
*>
```

The following notes on using the diagnostic list apply.

1. When using the bit trigger option the mask must always be entered in hexadecimal. The bit (s) to trigger the list must be set to a "1" in the appropriate bit location.
2. In either the bit trigger or magnitude trigger mode, entry "1" in the variable selections is compared against either the magnitude or bit trigger word.
3. When using the bit trigger option, the word must be re-entered, wherever selection 3 of "ASSIGN LIST" is accessed, otherwise the mask is reset to "0".
4. The list has completed taking data when the trigger source has returned to "OFF" after entering another selection.
5. List sites must be selected with care. The list may not necessarily fill up if a site is selected which doesn't operate during the event to be captured. For example, LPLL is disabled when the drive "RUN" command is removed.
6. If variable is by address rather than name, use offset only (converter if necessary to data segment DS: [F000] base).

5-3.2. Display

Function: to display the contents of select drive control variables and circular lists.

Several options are allowed in the DISPLAY command:

1. DISPLAY LIST, to display the circular list information.

2. DISPLAY FAULTS, to display the status of the fault and alarm logic.

3. DISPLAY RELAYS, to display a concise list of the drive's background relay status.

The DISPLAY FAULTS command will produce a list of the most recent STATUS messages (stored in a circular "fault log"). This list includes current status indications (shown also with the DISPLAY STATUS command).

Drive variables are displayed in their respective engineering units. Generally they are in per unit (of flux, current, speed, etc...). The general format in which variable contents are displayed is:
<varname> = <value>
A particular variable may be displayed repetitively by simply typing a "C" (for continuous) following the variable name on the command line.

Some DISPLAY command options may be abbreviated to two letters. These abbreviations and their equivalents are listed as follows:

DF ::= DISPLAY FAULTS
DL ::= DISPLAY LIST
DR ::= DISPLAY RELAYS

Examples:
*> **DISP SPDSP<cr>**
SPDSP = 472
*> **D<cr>**
NAME? **SPDSP<cr>**
SPDSP = 472
*>

*> **D RELAY<cr>**
<drive relay status list>
*>

*> **DISPLAY LIST<cr>**
<listing of circular list>
*>

*> **DISPLAY FAULTS<cr>**
<listing of circular fault log>
*>

5-3.3. Help

Function: to display at the terminal a list of the commands available in the current mode of monitor operation.

Examples:
*> **H <cr>**
<list of available commands>
*>

*> **HELP <cr>**
<list of available commands>
*>

5-3.4. Passcode

Function: to enable the monitor's privileged functions.

**** NOTE ****

For security, the characters are not echoed when entering the password at the terminal. The present passcode is:
[GESALEM 9][20:222, 224] "PASS"

Examples:
*> **PASSCODE <cr>**
PASSWORD : <**password**> <cr> ; password not echoed
PRIVILEGES ENABLED
*>
To change the passcode, see SET password under TUNEUP

5-3.5. Quit

Function: to terminiate the current terminal session. Upon execution of the QUIT command, the monitor will print a logout message and then begin idling, waiting for the carriage return inputs that indicate the start of a new terminal session.

Examples:
*> **Q <cr>**
MONITOR OFF

*> **EX <cr>**
MONITOR OFF

5-3.6. Restart

Function: to execute a software reset of the microprocessor.

Executing the RESTART command is essentially a soft reset. The diagnostic checks are run the same as during power-up, and the drive control is re-initialized to restore the hardware (counters,etc.) to its normal running state.

**** NOTE ****

To execute the RESTART command, the drive must be stopped. If the drive is running, the RESTART routines will not run, and a warning will be printed at the terminal.

Examples:

*> **RES <cr>**
<selftest results messages>

*> **RESTART <cr>** <selftest results messages>

5-3.7. Symbols

Function: to display at the diagnostic terminal the symbolic names that may be used to access drive control variables. If no qualifier is given, the complete list will be printed; if a partial name is specified, however, all symbols that match that partial name are listed. See Table 5A-1 for list of symbolic names.

Examples:

*> **SY <cr>**
<complete symbol list>
*>

*> **SYMBOLS TS <cr>**
TSMAX
TSMIN
TSVTOL*>

*> **SYM SFDBK <cr>**
SFDBK
*>

Table 5A-1
List of Available Symbols:

ALARM	ALPHLD	ALPHSRC	
CRGAIN	CRLIMP	CRMARG	CRMAX
CRMIN	CRSTIE	CRVDC	CSTEP
CT	CTADVL	CTGAIN	CTMARG
CTMAX	CTIC	CURCMP	
DRHPPU	DRIPU	DRMC	DRMFRQ
DRMRPM	DRMVPU	DRSFRQ	DRSVPU
DRVCFG	DRVTST	DRZRAT	
FPBRK1	FPBRK2	FPBRK3	FPDMAX
FPDTCOM	FPLEV1	FPLEV2	FPLEV3
FLTWD	FLXFBK	FLXPI	FLXSTP
FLYSTR	FRFETC	FRGAIN	FRICTC
FRLLIH	FRLLI	FRLLS	FRTC
FRX1	FRXM	FSTEP	

List of Available Symbols: (Continued)

ICAP	IFHSO1	IFHSO4	IFHS12
IFHS23	IFHS34	IFSR6	ILD
INVLIM	ISRC		
LCOM	LMODE	LSTAT	LVLTS
MFLAG			
P900C	PLLLERR	PLLLLD	PLLMODE
PLLSERR	PLLSFLGL	PLLSFLGS	PLLSLD
PTLOCC	PTLOC	PTLPLLC	PTLPLLT
PTSOCC	PTSOC	PTSPET	PTSPLLC
PTSPLLT	PTSUVT	PO1	PO2
PO3	PO4	PO5	PO6
POE			
RRACC0	RRACC1	RRACC2	RRACC3
RRACC4	RRACC5	RRBP0	RRBP1
RRBP2	RRBP3	RRBP4	RRDEC0
RRDEC1	RRDEC2	RRDEC3	RRDEC4
RRDEC5	RRFMAX	RRFMIN	RRGMAX
RRGMIN	RRMOD	RRPMIN	RRPMAX
RRTG			
SCFLUX	SCFREQ	SCRDNR	SCTCOM
SFDBK	SFDBKL	SPDEFIL	SPDSTP
SPDSTR	SPEED	SQOPT1	SQOPT2
SQT1	SQT2	SQT3	SQT4
SQT5	SQT6	SRERRF	SRGAIN
SRLLADJ	SRSL	SRTCD	SRTC
SRTCMAX	SRTCMIN	SRTCREG	SRTVSG
SSTEP	SUDSTL	SUFTMR	SUKSTL
SUMFRQ	SUMINFLX	SUMTCH	SURGSPD
SURVPR	SUSFRQ	SUSTCM	SUTRNFRQ
SYADT	SYCNT	SYDBCP	SYFLX
SYGAIN	SYPHS	SYSECP	SYSSPD
SYSTP	SYVOLT		
TSMAX	TSMIN	TSVTOL	TCOM

5-3.8. Terminal

Function: to examine and possibly modify the terminal format. The monitor will display the current format (either "handheld" or"standard") and then ask the operator whether a change is required. Entering Y(ES) will cause the monitor to toggle to the other terminal format. Any other entry will cause no change.

Examples:

*> **TERMINAL <cr>**
HNDHLD-CHANGE? Y <cr>
STANDARD FORMAT
*>

*> **TER <cr>**
STD - CHANGE? <cr>
STANDARD FORMAT
*>

5-3.9. Verify

Function: displays the current state of all drive EEPROM settings and adjustments to allow the user to verify factory settings. Individual groups of variables may be displayed for verification by appending one or more group names to the VERIFY command (see example). The groups will be listed in the order specified. Following are the possible group names:

DR - Drive configuration
CR - Current Regulator
CT - Crosstie Regulator
FP - Flux Profile
FR - Flux Regulator
IF - Inverter Firing
PO - Printer Output
PT - Protective Parameters
RR - Speed Ramp Rate Conditioning
SC - Scan Parameters
SQ - General Sequencing
SR - Speed Regulator
SU - Startup Parameters
SY - Synchronizing Regulator
TS - Test

Examples:

*> **VERIFY <cr>**
<grouped list of all software jumpers and adjustments>
*>

*> **V SR, FR, SQ <cr>**
(list of speed reg. software jumpers and adjustments>
(list of flux reg. software jumpers and adjustments>
(list of software sequencing jumpers>
*>

5-4. EXTENDED MODES

5-4.1. Engineering

The ENGINEERING mode provides primitive functions like examining and modifying memory and input and output ports. Absolute addresses, rather than symbolic names, are used to access memory locations and I/O ports. Table 5-3 summarizes the command syntax for this mode of monitor operation.

Table 5-3 Syntax Definitions for ENGINEERING Level Commands

DISPLAY:	D[W] <start addr>[,<end addr>] <cr>
HELP:	H <cr>
INPUT:	I[W] <port addr>,[, ...] <cr>
PUT:	P[W] <port addr>,[, ...] <cr>
MOVE:	M <start addr>, <end addr>, <dest addr> <cr>
OUTPUT:	O[W] <port addr>, <data>[,<data> ...] <cr>
QUIT:	{Q |EX} <cr>
SUBSTITUTE:	S[W] <addr>,[<new value>, ...] <cr>

<start addr>:: = <pointer>
<port addr >:: = <hex no>
<end addr > :: = <pointer>
<new value>:: = <hex no>
<dest addr> :: = <pointer>
<data> :: = <hex no>
<addr> :: = <pointer>

5-4.2. Display

Function: to display the contents of a memory location or a block of memory. The current value of each byte (or word) is listed in hexadecimal form. Additionally, the ASCII equivalent of the memory shown on a line is displayed in a field to the right of the standard display. Only printable ASCII characters are shown --others are displayed as a period (.). The ASCII display function is suppressed when operating in the handheld terminal format.

Format:

D[W] <start addr>[,<end addr>] <cr>

Examples:

E> **D DS:800 <cr>**
F800:0800 41 A
E>

E> **DW 200, 210 <cr>**
0000:0200 55AA 00FF 1233 5256 4147 4E49
2020 0020 .U..3.VRGAIN .
0000:0210 4243 CB
E>

5-4.3. Help -

Function: to list the available command set for the ENGINEERING mode.

Examples:

E> H <cr>
<list of available commands>
E>

5-4.4. Input

Function: to display a byte or word at an input port.

Format:

I[W] <port addr> <cr>

Examples:

E> **I 9008 <cr>**
7F ,
7F ,
7F <cr>
E>

E> **IW 9008 <cr>**
007F <cr>
E>

5-4.5. Put

Function: to modify the contents of selected memory locations. Typing a comma causes the monitor to remain in the PUT mode and increment to the next sequential memory address. A carriage return at the end of a line executes the current command (if any) and then returns control to the ENGINEERING level. This command is very useful in writing to EEPROM because no verification of the modified information is made.

Format:

P[W] <addr> <cr>

Examples:

E> **P DS:A40 <cr>**
FC00:0A40 A5- ,
FC00:0A41 5A- **AA <cr>**
E>

E> **PW DS:50 <cr>**
FC00:0050 0FFF- **7FF,**
FC00:0051 5A5A- **<cr>**
E>

5-4.6. Move

Function: to move a block of data from one area of memory to another. Memory contents are transferred sequentially from low memory to high.

Format:

M <start addr>, <end addr>, <dest addr> <cr>CB

Examples:

E> **M DS:50,59,CS:00 <cr>**
E>

E> **M 800,900,DS:42 <cr>**
E>

5-4.7. Output

Function: to output a byte or word to an output port.

Format:

O[W] <port addr>, <data><cr>[<data>, ... <cr>]

Examples:

E> **O 8010, FF <cr>**
- **F8,**
- **F0 <cr>**
E>

5-4.8. Quit

Function: to return from the ENGINEERING mode to the MONITOR mode.

Examples:

E> **Q <cr>**
*>
E> **EX <cr>**
*>

5-4.9. Substitute

Function: to exam and modify the contents of selected memory locations. Typing a comma causes the monitor to remain in the SUBSTITUTE mode and increment to the next sequential memory address. A carriage return at the end of a line executes the current command (if any) and then returns control to the ENGINEERING level. This command is very useful in writing to EEPROM because no verification of the modified information is made.

Format:

S[W] <addr><cr>

Examples:

E> S DS:A40 <cr>
FC00:0A40 A5- ,

FC00:0A41 5A- **AA** <cr>
E>

E> **SW DS:50** <cr>
FC00:0050 0FFF- **7FF,**
FC00:0051 5A5A- <cr>
E>

5-5. TUNEUP

In the TUNEUP mode of monitor operation, the user is able to adjust the performance of the drive by modifying regulator parameters and configurations. A step-by-step tuneup procedure is contained in Chapter 6 . Following are descriptions of the commands that are used as part of this procedure. Table 5-4 summarizes the command syntax for the TUNEUP mode.

For on-line tuneup of any regulator, the background sequencing RUN logic must be satisfied. Monitor commands alone will not allow the drive to run. This mode is intended to be used primarily when the drive is initially installed.

Table 5-4 Syntax Definitions for TUNEUP Level Commands

APPLY:	AP(PLY) <cr>
ASSIGN: *	AS(SIGN) {LIST | DA}
DISPLAY: *	{D(ISPLAY) { <varname> [C]| STATUS | LIST <listname> | } | DS | DL <listname>} <cr>
HELP:	H <cr>
MODIFY:	M(ODIFY) <varname> <cr> <old value message> <new value prompt> [I | D | <new value>] <cr>
MODIFY REFERENCE:	M(ODIFY) REF(ERENCE) <cr> or MREF
QUIT:	{Q | EX } <cr>
REMOVE:	REM(OVE) <cr>
RESTART	RES(TART) <cr>
SAVE:	SAV(E) <cr> <verify query> [Y | N] <cr>
SAVE D/A	SDA <cr> <verify query>[Y | N]<cr>
SET PASSCODE	Set (PASSCODE) <cr>
SYMBOLS: *	SY(MBOLS) [<symbol name qualifier>] <cr>
TERMINAL: *	TER(MINAL) <cr> <current format> CHANGE ? [Y | N] <cr>
TUNEUP SYMBOLS:	TSY(MBOLS) <cr>
TUNEUP:	TUN(EUP) [symbol group name] <cr>
VERIFY: *	V(ERIFY) [<symbol group name>] <cr>

* - Also available in MONITOR mode -- see section 5-3 for descriptions.

5-5.1. Apply -

Function: to apply the test generator reference to the regulator being tuned up.

Examples:

T> **AP** <cr>
T>

5-5.2. Help-

Function: to list the available command set for the TUNEUP mode.

Examples:

T> **H** <cr>
<list of available commands>
T>

5-5.3. Modify -

Function: to allow the operator to modify the contents of select drive control variables. A new value may be entered, or the current value may be incremented or decremented repeatedly. Also,the increment (or decrement) interval may be selected by placing a new interval after the I or D entry (see example below). If none is specified, the increment interval will be one unit of the least significant digit. Values are displayed and entered in appropriate engineering units.

**** NOTE ****

Because modifying certain memory locations could have adverse effects on the drive control operation, the MODIFY command may only be executed after completing the privilege enabling sequence.

Examples:

```
T> MOD VRGAIN <cr>
VRGAIN = 1.1
NEW VALUE ? 15 <cr>
VRGAIN = 15.0
NEW VALUE ? <cr>
T>
T> M VRGAIN <cr>
VRGAIN = 1.1
NEW VALUE ? I<cr>
VRGAIN = 1.2
NEW VALUE ? I .3<cr>
VRGAIN = 1.5
NEW VALUE ? I<cr>
VRGAIN = 1.8
NEW VALUE ? D<cr>
VRGAIN = 1.5
NEW VALUE ? <cr>
T>
```

**** NOTE ****

When a tuneup case is selected (section 5-4.20),a numbered list of symbols associated with that case is displayed at the terminal. While running that particular test, a symbol's NUMBER (from the list) rather than its name may be used as the argument in the DISPLAY and MODIFY commands. This eliminates the need to laboriously type out long symbol names at the terminal during a tuneup session.

Examples:

```
T> MOD 3 <cr>
FRKGAIN = 1.1
NEW VALUE ? 1.3 <cr>;change gain from 1.1 to 1.3
FRKGAIN = 1.3
NEW VALUE ? <cr>
T>
```

5-5.4. Modify Reference -

Function: to modify (customize) an internal reference generator. When a tuneup case is selected, its corresponding reference generator is automatically selected (see Section 5-4.20). The user then has the option to modify the characteristics of the reference generator as shown in the following examples. A reference generator is a series of two steps, each with selectable amplitude and duration. Step durations are entered in seconds (i.e., .05 for 50 milliseconds) and reference levels are specified in per unit levels (i.e., for a current test, the level is in per unit current and for a speed test in per unit speed, etc.).

After the steps are fully defined and the test generator is enabled by the APPLY command, the regulator specified by the TUNEUP command will be excited with a periodic signal corresponding to the test reference levels defined. A fixed constant level may be obtained by setting the amplitudes of both steps to the same value.The REMOVE command disables the test generator and removes all excitation to the specified regulator.

Examples:

(Fixed Reference)

```
T> MOD REF <cr>
STEP      LEVEL(PU)      TIME(SEC)
  1         0.100           1.00
  2         0.000           1.00

STEP 1 LEVEL = <cr>
STEP 1 DURATION = <cr>
STEP 2 LEVEL = 0.10 <cr>  ;make levels agree
STEP 2 DURATION = <cr>
T>
```

(Stepped Reference)

```
T> MOD REF <cr>
STEP      LEVEL(PU)      TIME(SEC)
  1         0.100           1.00
  2         0.100           1.00

STEP 1 LEVEL = <cr>
STEP 1 DURATION = <cr>
STEP 2 LEVEL = 0.00 <cr>  ;set for 10% step
STEP 2 DURATION = <cr>
T>
```

5-5.5. Quit -

Function: to return from the TUNEUP mode to the MONITOR mode. On leaving the TUNEUP mode, the monitor asks the user whether hewould like to have any new constant values transferred into the non-volatile memory (i.e., execute a SAVE command). This is so the operator will not inadvertantly forget to save the new drive status after finishing the TUNEUP sequence.

Examples:

T> **Q** <cr>
DO YOU WANT TO SAVE NEW CONSTANTS?
? **Y** <cr>
CONSTANTS SAVED
*>

T> **EX** <cr>
DO YOU WANT TO SAVE NEW CONSTANTS ?
? **N** <cr>
CONSTANTS NOT SAVED
*>

5-5.6. Remove -

Function: to remove the test generator signal from the regulator being tuned up.

Examples:

T> **REM** <cr>
T>

5-5.7. Set Passcode -

Function: to to allow the user to put in his own password — up to 6 characters long. For this to be a permanent change it must be saved.

** NOTE **

For security, passcodes are not
echoed
:Password address

Example:

OLD PASSCODE =: <**OLD PASSCODE**> <cr>
; password not echoed
NEW PASSCODE =: <**NEW PASSCODE**> <cr>
; password not echoes
NEW PASSCODE =: <**NEW PASSCODE**> <cr>
; verify to enable
T > ; can now save if want permanent change.

5-5.8. Save -

Function: to store the present values of the regulator variables in the system's nonvolatile memory. After modifying system parameters (for example, gains and time constants during a tuneup session), the new values may be transferred into permanent storageusing the SAVE command. The previously stored values, of course, are lost during this process; hence, the monitor will prompt the user for confirmation before actually executing the command.

NOTE: Privileges must be enabled to execute the SAVE command. Also, the hardware write enable switches on the DMPB cards must be in the "store enable" position before values may be stored in the nonvolatile memory.

Examples:

*> **SAVE** <cr>
ARE YOU SURE? **Y**<cr>
CONSTANTS
SAVED
saved if DMPB switch on
*>

*> **SAVE** <cr>
ARE YOU SURE? <cr>
CONSTANTS NOT SAVED
*>

5-5.9. Save D to A (SDA)

Function: To store the present values of the D to A assignment variables in the system's nonvolotile memory. During tuneup mode, the D to A converters (1 and 2) are continually reassigned according to the default values required for the regulator being turned. If the D to A values were included in the global "SAVE" command, the intended setup values would be lost. A separate "SAVE" command is therefore included.

When it is required to modify, the D to A assignments permanently, the "ASSIGN DA" command used first be used to setup the required parameters.

The hardware write enable switches on the DMPB card must be in the "write enable" position before values may be stored in the nonvolatile memory.

Examples:

*> **SDA** <cr>
ARE YOU SURE? Y <cr>
CONSTANTS SAVED
* >

*>**SDA** <cr>
ARE YOU SURE? <cr>
CONSTANTS NOT SAVED
*>

5-5.10. Tuneup Symbols -

Function: to list all tuneup variables associated with the current tuneup case.

Examples:

T> **TSY** <cr>
TUNEUP VARIABLES ARE:
<numbered symbol list> or NONE
T>

5-5.11. Tuneup -

Function: to initiate a particular regulator tuneup sequence. The user must specify which of several regulator is to be tuned. He is then allowed to select the desired tuneup case. Examples of valid commands are:

CR - Current Regulator
FR - Flux Regulator
SR - Speed Regulator

Example:

(Current Regulator Tuneup)
T>**TUNE CR**<cr>

1 - CRGAIN = 338
2 - CRTC = 0.020
3 - CRVDC = 0.96
4 - CRMIN = 0.20
5 - CRMAX = 1.00
6 - CRLIMP = 0.12
7 - CRMARG = 17
T>

5-5.12. Monitor Commands -

The following main-level monitor commands are also available in the TUNEUP mode:

ASSIGN
DISPLAY
SYMBOLS
TERMINAL
VERIFY

The syntax definitions and functional descriptions for these commands are contained in section 5-3.

5-6. LOADING DISPLAY MESSAGES HEADERS

Customized display messages can be loaded into EEPROM to identify the user, drive, personnel to contact, actions to be taken, etc. Five (5) different messages can be loadeed - starting at the locations below. Each message can be up to 64 characters long (minus 1 for end of message = 63).

Starting Address	Message type
3000:250	Header - message printed at power-up
3000:290	Repetition - message printed at beginning of "data logging"
3000:2D0	Requisition - message printed when "Status" P.B. used
3000:310	Fault - message printed at beginning of "Fault" messages
3000:350	Alarm - message printed at beginning of "Alarm" messages

Procedure (see example 5B-1):

1. Write the message out on Form 5B-1 (remember to leave 2 spaces) for <cr> <LF> (by word) if you don't need to force a new line.
2. Convert the characters to their ASCII equivalent using Table 5B-1. Add after the last conversion 00 to signal end-of-message.
3. Using the monitor Put command, load the ASCII equivalents.

**** NOTE ****

You can use a "," instead of a <cr> after each ASCII code to increment to the next location - rather than reloading the address (faster).

Suggestions
Always include user's name and the drive/channel #. For the fault, alarm, and repetition messages, a reminder such as "Log items in maint book' may be useful. The default messages should be okay . It should only be necessary to add the user's name and drive I.D. to the beginning (if sufficient space) or to the end (will need to move "00" of the default message.

MONITOR OFF <cr> ; Form 5B-1 or 5B-2 should be filled out.

IGTO MONITOR
*>PASSWORD <cr>
PASSWORD ?**PASS** <cr>

PRIV. ENABLED
*> **ENG**<cr>

E>**PUT 3000:200**<cr> ; Enable EEPROM write switch
3000:200 ?? **53,**
3000;201 ?? **43,** ; Continue till all characters done
:
:
3000:2?? ?? **00**<cr>

Example 5B-1 : MESSAGE HEADERS

Message Type_____

Starting Address - 3000:_____

Message Character	ASCII (HEX)	Message Character	ASCII (HEX)
01___	___	33___	___
02___	___	34___	___
03___	___	35___	___
04___	___	36___	___
05___	___	37___	___
06___	___	38___	___
07___	___	39___	___
08___	___	40___	___
09___	___	41___	___
10___	___	42___	___
11___	___	43___	___
12___	___	44___	___
13___	___	45___	___
14___	___	46___	___
15___	___	47___	___
16___	___	48___	___
17___	___	49___	___
18___	___	50___	___
19___	___	51___	___
20___	___	52___	___
21___	___	53___	___
22___	___	54___	___
23___	___	55___	___
24___	___	56___	___
25___	___	57___	___
26___	___	58___	___
27___	___	59___	___
28___	___	60___	___
29___	___	61___	___
30___	___	62___	___
31___	___	63___	___
32___	___	64 < EOM >	00

FORM 5B-1: For Header Message

Message Type_____

Starting Address - 3000:_____

Message Character	ASCII (HEX)	Message Character	ASCII (HEX)
01____	20_	33_d_	64_
02____	20_	34____	20_
03____	20_	35_i_	69_
04____	20_	36_t_	74_
05____	20_	37_e_	65_
06____	20_	38_m_	6D_
07____	20_	39____	20_
08____	20_	40_i_	61_
09____	20_	41_n_	6E_
10____	20_	42____	20_
11____	20_	43_L_	4C_
12____	20_	44_O_	4F_
13____	20_	45_G_	47_
14____	20_	46____	47_
15____	20_	47_b_	20_
16____	20_	48_o_	6F_
17____	20_	49_o_	6F_
18____	20_	50_k_	6B_
19____	20_	51	00_
20____	20_	52____	____
21____	20_	53____	____
22____	20_	54____	____
23____	20_	55____	____
24____	20_	56____	____
25____	20_	57____	____
26 < cr >	0D_	58____	____
27 < 1f >	0A_	59____	____
28_R_	52_	60____	____
29_e_	65_	61____	____
30_c_	63_	62____	____
31_o_	6F_	63____	____
32_r_	72_	64 < EOM >	00

FORM 5B-2: Default for Header Message

Character	ASCII	Character	ASCII
A	41	a	61
B	42	b	62
C	43	c	63
D	44	d	64
E	45	e	65
F	46	f	66
G	47	g	67
H	48	h	68
I	49	i	69
J	4A	j	6A
K	4B	k	6B
L	4C	l	6C
M	4D	m	6D
N	4E	n	6E
O	4F	o	6F
P	50	p	70
Q	51	q	71
R	52	r	72
S	53	s	73
T	54	t	74
U	55	u	75
V	56	v	76
W	57	w	77
X	58	x	78
Y	59	y	79
Z	5A	z	7A
0	30	<space>	20
1	31	<cr>	0D
3	32	<1f>	0A
3	33	#	23
4	34	&	26
5	35	'	2C
6	36	-	2D
7	37	.	2E
8	38	:	3A
9	39	@	40

TABLE 5B-1: ASCII Character Set

Message Type_____ANY_____

Starting Address - 3000:___ANY

Message Character	ASCII (HEX)	Message Character	ASCII (HEX)
01_S_	53_	33_d_	64_
02_C_	43_	34____	20_
03_O_	4F_	35_i_	69_
04_T_	54_	36_t_	74_
05_T_	54_	37_e_	65_
06____	20_	38_m_	6D_
07_P_	50_	39____	20_
08_A_	41_	40_i_	61_
09_P_	50_	41_n_	6E_
10_E_	45_	42____	20_
11_R_	52_	43_L_	4C_
12<cr>	OD_	44_O_	4F_
13	OA_	45_G_	47_
14_D_	44_	46____	20_
15_r_	72_	47_b_	62_
16_i_	69_	48_o_	6F_
17_v_	76_	49_o_	6F_
18_e_	65_	50_k_	6B_
19_#_	23_	51	00_
20_1_	31_	52____	____
21____	20_	53____	____
22_C_	43_	54____	____
23_H_	48_	55____	____
24_._	2E_	56____	____
25_A_	41_	57____	____
26<cr>	0D_	58____	____
27<1f>	0A_	59____	____
28_R_	52_	60____	____
29_e_	65_	61____	____
30_c_	63_	62____	____
31_o_	6F_	63____	____
32_r_	72_	64<EOM>	00

FORM 5B-2: Default for Header Message

EXAMPLE 5B-1: MESSAGE HEADERS

CHAPTER 6
INSTALLATION AND STARTUP GUIDE (OUTLINE)

I. Keep detailed installation log for each drive.

II. Install Equipment
(Align motor, setup coupling to run uncoupled from driven device. Setup motor protectives - lubrication, thermocouples, RTDs, vibration detectors. Check-out transformers and reactors - fault devices, alarm devices; source and load breakers - close and trip circuits, overcurrent devices.).

III. Interconnect equipment

IV. Wire check

V. Check-out source and load breakers

VI. Megger

VII. Run motor in "Bypass".

VIII. Check Berg jumpers, daughter board components.

IX. Apply control power
Check starters and blowers
Check power supplies
Check relays, lights

X. Operate in gate-test-mode
Check source and load firing
Check process inputs and outputs
Set current feedback

XI. Apply source voltage
Check attenuator signals
Set scaling and trip levels

XII. Simulate load voltage
Check attenuator signals
Set scaling and trip levels

XIII. Perform crowbar test

XIV. Perform "locked" rotor test

XV. Run motor uncoupled
Check bearings, vibration
Check acceleration, deceleration, regeneration

XVI.Check transfer schemes

XVII. Run motor coupled
Tune control
Check loaded transfer schemes
Evaluate control

XVIII. Clean-Up
Send back to factory corrected prints
Make 3 lists of EEPROM Default Values
(Customer, A&ES, Factory)

6-1. DETAILED INSTALLATION INSTRUCTIONS

It is assumed that all interconnecting wire and cables are in place and are connected. The erection and alignment of the motors is not described here, nor is the routine checkout of the circuit interrupters. It is advised that a detailed installation log be maintained for each drive.

6-2. MEGGER TEST

a. Short all power buses together in the IMD panel. L1[A/B], L2[A/B], L3[A/B], P2, N1, N2, M1, M2, M3)
b. Pull the ribbon cables that connect to the front edge of the HPTK cell firing cards.
c. Disconnect ribbon cable to each HPTK card in the Leg module of the bridge compartment. Also remove the ground (GND) stab on this card and on the XPTN.
d. Also remove the XPTP ground (GND) stab. This card is located in the motor capacitor bank compartment.
e. Remove NATL ground (GND) JG stab and ribbon connector JV. The NATL board is mounted on the wall of the chopper power supply compartment. Disconnect capacitor grounding resistor.
f. Confirm that both the source side isolator switch and the load side motor isolator are open.
g. Perform the 1000V megger test and record the test values. Indicated resistance should exceed 100 megohms.
h. Remove jumpers and reconnect the ribbons and stab connections.
i. Jumper the load side motor isolator load side terminals (T1, T2, T3) together and megger to ground. Indicated resistance should exceed 100 megohms.
j. Remove jumpers. Megger test is now completed.

6-3. RUN MOTOR UNCOUPLED ON BYPASS CONTACTOR

CONSIDERATIONS:
Bearing monitoring protection should be functioning at this point to protect the motor. The motor bearing temperature rise, vibration, and oil rings (slingers) should be checked.

a. Stop the drive and rack-out and lock-out the drives load circuit interrupter. Rack-in the bypass interrupter. Source interrupter

should be closed. (Drive's isolation switch should be open).

b. Place INVERTER/BYPASS switch in BY-PASS. Start motor with start switch. Check motor direction of rotation - matches drive. Check motor vibration, magnetic center, etc. Stop motor.

6-4. PRELIMINARY CONSIDERATIONS

a. Verify the variable component values in the SEM per sheet EA of the drive elementary.
b. Verify Berg jumper locations as shown on drive elementary sheet EA.
c. Inspect all compartments for loose hardware/debris removing any such material prior to applying power.
d. Inspect all stab connections in both SCR and GTO for tightness.
e. Open all circuit breakers; refer to drive elementary.
f. Open or turn off all switches and power supplies.
g. Partially extract all PWB's from the SEM.
h. Check each SRC, diodes, and GTO's for shorts.

6-5. CHECK SCR AND GTO CONTINUITY

a. Using a multimeter, check across each SCR in the source bridge. Measurements should indicate approximately 10K ohms. If greater, check the equalizing resistors (R1) in the snubber boxes. If less than 100 ohms check for defective snubber capacitor (C1) Reference Figure 6.1.

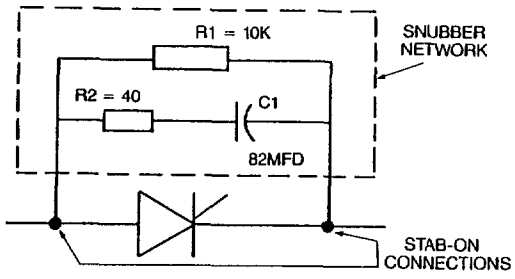


Figure 6.1

b. Using a multimeter, check across each diode and GT0 in load side bridge. See Figure 6.2. Parts 1-2 should read approximately the same reading for each pair of diodes and GTO. Parts 2-3 should read approximately the same reading for each pair of diodes and GTO.

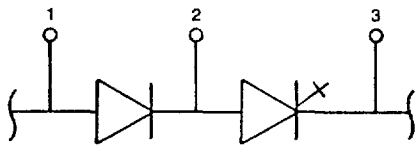


Figure 6.2

6-6. AUXILIARY POWER

a. Open the power supplies circuit breakers.
b. Apply the 480Vac and/or 120Vdc control power.
c. Check bridge cooling fans for proper direction of rotation, air should enter the front of each of the bridge compartments and exit the top of the blowers when the doors are closed.

6-7. POWER SUPPLIES

1. PSFF

   a. Turn on control power supply, PSFF.
   b. Verify presence of P5, P15, N15 and P28 at the electronics module backplane.
   c. Insert the PWBs and verify that the power supply remains constant.
   d. With all PWBs in, adjust the e power supply to obtain the following voltages at the MEM.

- Adjust R1 on NPPB until P5 = + 5.0 +/- 0.1 volts.
- Adjust R6 on NPSK until P15 = +15 +/- 0.2 volts.
- Adjust R7 on NPSK until N15 = - 15.0 +/- 0.2 volts.

2. CPS

   a. Turn on chopper power supply CPS.
   b. Monitor the voltage across the dc filter cap.
   c. Adjust R20 and see that voltage across the cap changes.
   d. Set voltage at 162Vdc.
   e. At XDCE, verify that hot side of R154 (12KΩ) is at ≈ 115 Vac to chassis. This is the side nearest the molex JY connector. If it is at ground potential, swap 400 Hz connection @ the chopper power supply.

6-8. ELECTRONICS MODULE INITIALIZATION

Null all meters.

6-8.1. Set-up Panel Printer

Consult the panel printer's manual to enable the battery backup (clock). Then set the time and date. Also, set the

interval time greater than 60 minutes.

6-8.2. Run IMD Selftest:

Turn-on (up) NLCB test switch. Reset HMPG by jumpering TP1(RST) to TP2(DCOM). Follow the indicated instruction and investigate any fault messages. Repeat till unit passes all tests.

6-8.3. Operating Set-up Adjustments and EEPROM Memory Initialization:

Measure and set the voltages for all the OV, SUPP, RTD, UVR, OSP test rings as tabulated in the system elementary sheet 1EC.

Jumper testpoint TINO on HLCB-2E to DCOM, turn-on(up) WREN switch on DMPB(HMPG), WREN LED on the DMPB card should come on. Reset the HMPG. Remove the jumper and turn off WREN switch. Reset the HMPG again. Dump EEPROM memory, using VERIFY, on a printing terminal and compare to values shown on system elementary (sheet 1EE - 1EH).

6-8.4. Auxiliary Outputs

If one or more of the auxiliary 4-20 mA outputs are via DAHI1 or DAHI2 (see elementary sheet CL), (From the D to A port), then use the "ASSIGN DA"command to assign the correct variable. Then use "SDA" command to save the variable in the nonvoltile memory. see elementary sheet BF.

6-9. BRIDGE CURRENT FEEDBACK CALIBRATION (MVB) :

Verify resistors and Berg jumpers on the XDCE, against system elementary (sheet EA). Remove CT lead plugs JU and JX from the XDCE. Measure the resistance across each of the burden resistors on the XDCE; it should equal the "R BURDEN SETTING" value per the elementary. Adjust the corresponding pot to fine tune. Re-connect plugs JU and JX.

6-10. GATE TEST MODE OPERATION

Refer to the paragraph entitled GATING TESTS in Chapter 7 for a description of procedure.

Verify that the source-side circuit interrupt and load-side circuit interrupter are open.

Follow the procedure described in GATING TESTS.

6-11. CHECK PROCESS INPUTS

Verify that the source-side circuit interrupter and load side circuit interrupter are open and locked out.

The purpose of the tests in this section is to operate each external input shown on system elementary sheets BD, BE into the SEM. Operate each of these inputs by the most convenient means and confirm that the correct LED lights with a closed circuit in each case.

6-12. CHECK PROCESS OUTPUTS

Verify that the source contactor and load isolator are open and locked out.

Connect the hand terminal (or equivalent) to the HMPG and MODIFY EEPROM variable (DRVTST) to "1" to temporarily disable the interrupts. (See Chapter 5, section MONITOR - section EXTENDED MODES for procedure). Save the modification and reset the HMPG.

Using the OUTPUT command, operate the output relays shown on system elementary sheets BG, BJ in accordance with the table 6-1. Confirm that the relays operate the proper device(s) per the system elementary.

Relays may be operated in any desired combination by adding the appropriate DATA values from table 6-1.

Re-MODIFY DRVTST to original value.

TABLE 6-1
SLOT 1A

ADDRESS	DATA	LED ON
8040	01	0
8040	02	1
8040	04	2
8040	08	3
8040	10	4
8040	20	5
8040	40	6
8040	80	7

ADDRESS	DATA	LED ON
8044	01	8
8044	02	9
8044	04	A
8044	08	B
8044	10	C
8044	20	D
8044	40	E
8044	80	F

6-13. NO-LOAD FULL-VOLTAGE TEST

The tests in this section require that the source power be connected and available. Open and lock-out the load side circuit interrupter. (If no interrupter is available disconnect leads to motor.)

Apply the control power, reset the SEM and check that all the IMOK LEDs are on. Check for any fault indications. Investigate and clear any indicated faults. Null all meters.

6-13.1. DC Offset Check

This procedure should be performed whenever the NLIB card is replaced by a new one.

a. With source power off and the control not in GATE-TEST-MODE, measure TP4 (FACR), TP5 (FCBR), and TP6 (FACE) on NLIB's in slots 1C and 1D with a DVM in the millivolts dc range. Adjust dc offset pots R1, R2 and R3, respectively on the NLIB MOTHER BOARD so that TP4, TP5, and TP6 read 0.0 millivolts dc (±5).

6-13.2. VOLTAGE PHASING AND BALANCING

a. Close source side circuit interrupter. Observe that all source side bridge neon lamps are illuminated.

b. Connect an oscilloscope to simultaneously observe the waveforms on testpoints TP7 (VBA), TP8 (VCB), and TP9 (VAC) on the NSFC-2C. Confirm that the phase sequence is 1-2-3 and that all these are appromimately equal. (V_{BA} leads V_{CB} leads V_{AC})

The actual system voltage = (ATTN RATIO) X VAC

where ATTN RATIO is given on the system elementary sheet 1EG.

Using a DVM, measure the DC voltage offset at TP7, TP8 and TP9. This value should be less than 5 millivolts.

NOTE: Must measure between TP and ACOM testpoint on the card.

c. Connect an RMS-measuring voltmeter to TP1 (FAB) on the NLIB-1C. Adjust R1(FLXAB) to obtain voltage as follows:

$$\text{FAB} = \frac{\text{ACTUAL SYSTEM VOLTAGE}}{\text{NOMINAL SYSTEM VOLTAGE}} \times 4.75 \text{ VAC RMS}$$

Nominal system voltage is given on the system elem. Sh. 1EG.
Now connect the voltmeter to TP2 (FBC) and adjust R2(FLXBC) for the same rms voltage as above. Repeat for TP3 (FCA) and R3(FLXCA). Balance signals as accurately as possible (+5 millivolts). Check for dc offset at TP4, TP5 and TP6 of NLIB (<10 millivolts). TP4 TP5 abd TP6 should read the same voltage as TP1, TP2, and TP3 respectively.

d. Check for presence of zero crossing pulses at TP8 (SFT1) on the HLCB-2E. These are narrow 2.5v pulses evenly spaced at 2.78 millisecond intervals.

e. Connect the voltmeter to measure TP13 (VABS) on the NLIB-1C and check that the dc voltage level is approximately 2.229 X VAC at TP9 of NSFC-2C.

6-13.3. Set Overvoltage:

Connect the voltmeter to TP18 (OVR) on the NLIB-1C. Turn R8 (OV) slowly counterclockwise until the "OV" LED comes on. The source side circuit interrupter should trip.

Observe the dc voltage on TP-18 and re-adjust R8 so that the voltage is [per unit source overvoltage] times the trip value. (If incoming line is high or low, compensate by the ratio of RATED LINE VOLTS/ACTUAL LINE VOLTS.) See elementary sheet EC for per unit source overvoltage.

6-13.4. Set Undervoltage.

Clear any faults and close the line contactor. Connect voltmeter to TP19 (UVR) of NLIB-1C.

Connect an oscilloscope to backplane pin 55-1C; this should be a logic zero. Turn R9 (UV) slowly cw until pin 55 just transitions high (may alternate high and low).

Observe TP19-1C and adjust UV(R9)-1C so that the measured value is per unit source undervoltage X trip level. (If incoming line is high or low, compensate by the ratio of RATED LINE VOLTS/ACTUAL LINE VOLTS.) See system elementary sheet 1EC for per unit source undervoltage.

6-13.5. Slave Bridge:

Using an oscilloscope, observe the waveforms on NSFC (2A) VBA (TP7), VCB (TP8), and VAC (TP9). Confirm that the phase sequence is TP7, TP8, TP9.

Use a DVM to measure the voltage - record each voltage.

TP7 = _____ TP8 = _____ TP9 =_____

The actual L-L system voltage = (ATTN RATIO) X V1-n avg.

where ATTN RATIO is given in system elementary (sheet EA)

6-13.6. Check Phasing Between master and Slave:

Using an oscilloscope, observe TP7 (VBA) - NSFC(2C) and TP7 (VBA) - NSFC(2A). Verify master(2C) leads slave(2A) by 30 degrees. If not, MODIFY DRVCFG.

The actual system voltage = (ATTN RATIO) X VAC

where ATTN RATIO is given on the system elementary sheet 1EG.

Using a DVM, measure the dc voltage offset at TP7, TP8 and TP9. This value should be less than 5 millivolts.

** NOTE **

Must measure between TP and ACOM testpoint on the card.
Connect an RMS-measuring voltmeter to TP1 (FAB) on the NLIB-1C. Adjust R1(FLXAB) to obtain voltage as follows:

$$FAB = \frac{\text{ACTUAL SYSTEM VOLTAGE}}{\text{NOMINAL SYSTEM VOLTAGE}} \times 4.75 \text{ VAC RMS}$$

Nominal system voltage is given on the system elem. Sh. 1EG.
Now connect the voltmeter to TP2 (FBC) and adjust R2(FLXBC) for the same rms voltage as above. Repeat for TP3 (FCA) and R3(FLXCA). Balance signals as accurately as possible (+5 millivolts). Check for DC offset at TP4, TP5 and TP6 of NLIB (< 10 millivolts). TP4, TP5 and TP6 should read the same voltage as TP1, TP2, and TP3 respectively.

Check for presence of zero crossing pulses at TP8 (SFT1) on the HLCB-2E. These are narrow 2.5v pulses evenly spaced at 2.78 millisecond intervals.

Connect the voltmeter to measure TP13 (VABS) on the NLIB-1C and check that the DC voltage level is approximately 2.229 X VAC at TP9 of NSFC-2C.

6-14. LOAD SIDE BRIDGE VOLTAGE TESTS WITH SIGNALS FROM SOURCE SIDE BRIDGE

Trip source circuit interrupter. Disconnect ribbon cable JA from NSFC-2B and move the ribbon cable JA from NSFC-2C to NSFC-2B. Connect a test jumper from 1C-55 to 1C-1 (disables source undervoltage fault).

6-14.1. Voltage Signal Phasing and Balancing

Close source circuit interrupter.

a. Connect oscilloscope to simultaneously observe the waveforms on testpoints TP7 (VBA), TP8 (VCB), and TP9 (VAC) on the NSFC-2B. Confirm that the phase sequence is 1-2-3 and all balanced. The testpoints should measure the same as for the source-side. Using a DVM, measure the DC voltage offset at TP2, TP3 and TP4. This value should be less than 5 millivolts.
b. Connect an RMS-measuring voltmeter to TP1 (FAB) on the NLIB-1D. Adjust R1 (FLXAB) to obtain voltage as follows:

$$FAB = \frac{\text{ACTUAL VOLTAGE}}{\text{NOMINAL MOTOR VOLTAGE}} \times 4.75 \text{VAC RMS} \times \frac{\text{Rated Freq.}}{60} \times \frac{\text{LOAD ATTN RATIO}}{\text{SRC ATTN RATIO}}$$

Rated frequency and nominal motor volts from elementary sheet 1EG.
Now connect the voltmeter to TP2 (FBC) and adjust R2 (FLXBC) for the same rms voltage as above. Repeat for TP3 (FCA) and R3 (FLXCA). Balance signals as accurately as possible (+5 millivolts). Check for DC offset at TP4, TP5 and TP6 of NLIB (< 10 millivolts).

** NOTE **

Temporarily unplug NCCD card at 1F slot, TP4, TP5 and TP6 at NLIB -1D. It should read the same voltage as TP1, TP2 and TP3 respectively.

c. Check for the presence of zero-crossing pulses at TP9 (SFT2) on the HLCB-2E. These are narrow 5 vdc pulses evenly spaced at 2.78 millisecond intervals.
d. Connect the voltmeter to measure TP13 (VABS) on the NLIB-1D and check that the DC voltage level is the same as for the source-side 2.229 X VAC at TP9 of NSFC-2B

6-14.2. Set Overvoltage

Connect voltmeter to TP18 (OVR) on the NLIB-1D. Turn R8 (OV) slowly ccw until the "OVA" LED comes on. The motor contactor should trip. Observe the DC voltage on TP18 and re-adjust R8 so that the voltage is as follows:

$$\text{"OVA" volts} = 1.2 \times \text{trip value} \times \frac{\text{RATED MOTOR VOLTS}}{\text{ACTUAL LINE VOLTS}}$$

6-14.3. Set Speed and Voltage Indicators

Since speed is derived from the motor voltage frequency, the source 60 Hz can be used to set the overspeed.

Close-in line contactor. Connect a voltmeter to TP1-1E (SPD). Adjust R1 (SPD) until TP1 = 10.0 Vdc @ 60 Hz .

Motor Voltage Indication:

Adjust VM (R7) -1D on DLIB so that the motor voltmeter reads actual line voltage.

6-14.4. Set Overspeed:

Set OSP -1E to +11Vdc. Connect voltmeter to testpoint OSP (TP6)-1E and slowly turn OSP (R2)-1E ccw until

the OSP LED comes on. The voltage on TP6 should be +10 Vdc at the trip point. See elementary sheet DC for per unit overspeed.

Adjust OSP until TP6 = [per unit overspeed] X trip level.

Trip the source circuit interrupter.

Restore NSFC ribbon cables to correct configuration.

Remove jumpers.

6-14.5. Capacitor Current Calculation

1. Measure NSFC (2B) -Vac(TP9) and drive frequency(freq).
2. Calculate capacitor current Icap

$$Icap = \frac{Vln}{Xc} = \frac{Vac*ATTNRATIO*2*PI*FREQ*C}{1.732}$$

where ATTNRATIO and C are given on system elementary sh. 1EA.

3. Check that MIFB(TP11)-NCCC(1F) =

$$\frac{Icap * 3.887}{\text{Rated Inverter Current}}$$

where Rated Inverter Current is listed on system elementary sh. 1EA.
Reduce (increase) RO8 & RO9 on DCCD-1F if MIFB is too high (low).

**** NOTE ****

Individual phase (motor) current is scaled 3.322 Vacrms on inverter base.

4. Adjust GAIN-NLIB(1D) so that ammeter reads Icap.

MONITOR OFF <cr>
*> PASSWORD
PASSWORD? **PASS**<cr>

PRIV. ENABLED
*>**TUN**<cr>

T>**MOD DRVTST**<cr>

DRVTST = 0
VALUE?**2**<cr>

T>**MOD CRMIN**<cr>
CRMIN = 0.13
VALUE? **.09**<cr>
CRMIN = 0.09
VALUE?<cr>

T>**MOD SUMTCH**<cr>
SUMTCH = 0.75
VALUE?**.15**<cr>

SUMTCH = -.75	Start drive and check IFB
VALUE?**I.05**<cr>	;Increase by 0.05
SUMTCH = 0.20	;check IFB level & if continous
VALUE?**I**<cr>	;Increase till IFB = .25 PU
SUMTCH = 0.25	check CTs, Vdc(master & slave)
VALUE?**I**<cr>	;adjust SUPP to min and
	;increase IFB till trip.
	; approximately 0.9 pu

Example 6-2

6-15. CROWBAR TEST

The crowbar test permits controllable dc link current to be circulated without involving the motor windings. On the load-side, the dc current circulates though one or more diametric cell pairs bypassing the motor. The signals to be observed on the oscilloscope will indicate whether the source-side CTs are phased correctly.

**** NOTE ****

Bypass must be open or STATUS disconnected to fine cells.

a. Reset SEM and apply primary power. Clear all existing faults.
b. Connect hand terminal to source-side HMPG.
c. MODIFY the following EEPROM variables (Example 6-2):
   CRMIN to .10
   DRVTST to 2
   SUMTCH to .15
d. Connect channel one of the scope to NSFC-2C SIG1 (TP1).
   Connect channel two of the oscilloscope to NLIB (1C) "OIA" (TP11). Trigger on channel one.
e. Initiate a run sequence; i.e., sequence the drive to pick-up the software logic ladder's "RUN". The current regulator should now be active but regulating min current. Check that word MFLAG = 50H.(See elem. Sh. 1FF for software logic "RUN").
f. Using the MODIFY command, increase (I) SUMTCH to .25.
g. Verify that test point "IFB" on NLIB (1C) = 1.25 Vdc (+/- 25%) (current should be continous - not going to zero).
h. Connect channel one to NLIB (1C), 0IA (TP11). Channel 2 to NSFC (2C), SIG1 (TP1). Compare the display to Figure 6-8.

i. Connect channel one to NLIB (1C), 0IB (TP10). Channel 2 to NSFC (2C), SIG3 (TP3). Compare the display to Figure 6-8.
j. Connect channel one to NLIB (1C), 0IC (TP9). Channel 2 to NSFC (2C), SIG5 (TP5). Compare the display to Figure 6-8.
k. Check NSFC (2C) - Vdc - (TP11) for sawtooth (360 Hz) wave form.
l. If there is no slave source bridge skip 6-16.1. step f. to test for hardware suppression.

6-15.1. Check Slave Bridge

a. Verify that testpoint IFB2(TP4) - NCCD(1F) = 1.25 Vdc (+/- 30%).
b. Connect channel one to NCCD (1F), 0IA2 (TP1). Channel 2 to NSFD (2C), SIG1 (TP1). Compare the display to Figure 6-3.
c. Connect channel one to NCCD (1F), 0IB2 (TP2). Channel 2 to NSFC (2B), SIG3 (TP3). Compare the display to Figure 6-3.
d. Connect channel one to NCCD (1F), 0IC2 (TP3). Channel 2 to NSFC (2B), SIG5 (TP5). Compare the display to Figure 6-3.
e. Check NSFC(2A) - Vdc - (TP11) for sawtooth (360 Hz) wave form.
f. Adjust suppression (R6) 1C on NLIB to minimum (fully clockwise). Testpoint SUPPR should read approximately 3.0 volts.
g. Increase SUMTCH to 0.9 so that suppression LED on NLIB-1C comes on. The source breaker should trip.
h. Readjust suppression (R6) 1C for values as shown on system elementary Sheet 1EC.
i. Re-MODIFY CRMIN, DRVTST, and SUMTCH to original values.

6-16. "LOCK ROTOR"

The following test is designed to check the phasing of the load-side CTs. The motor may start and turn slowly,

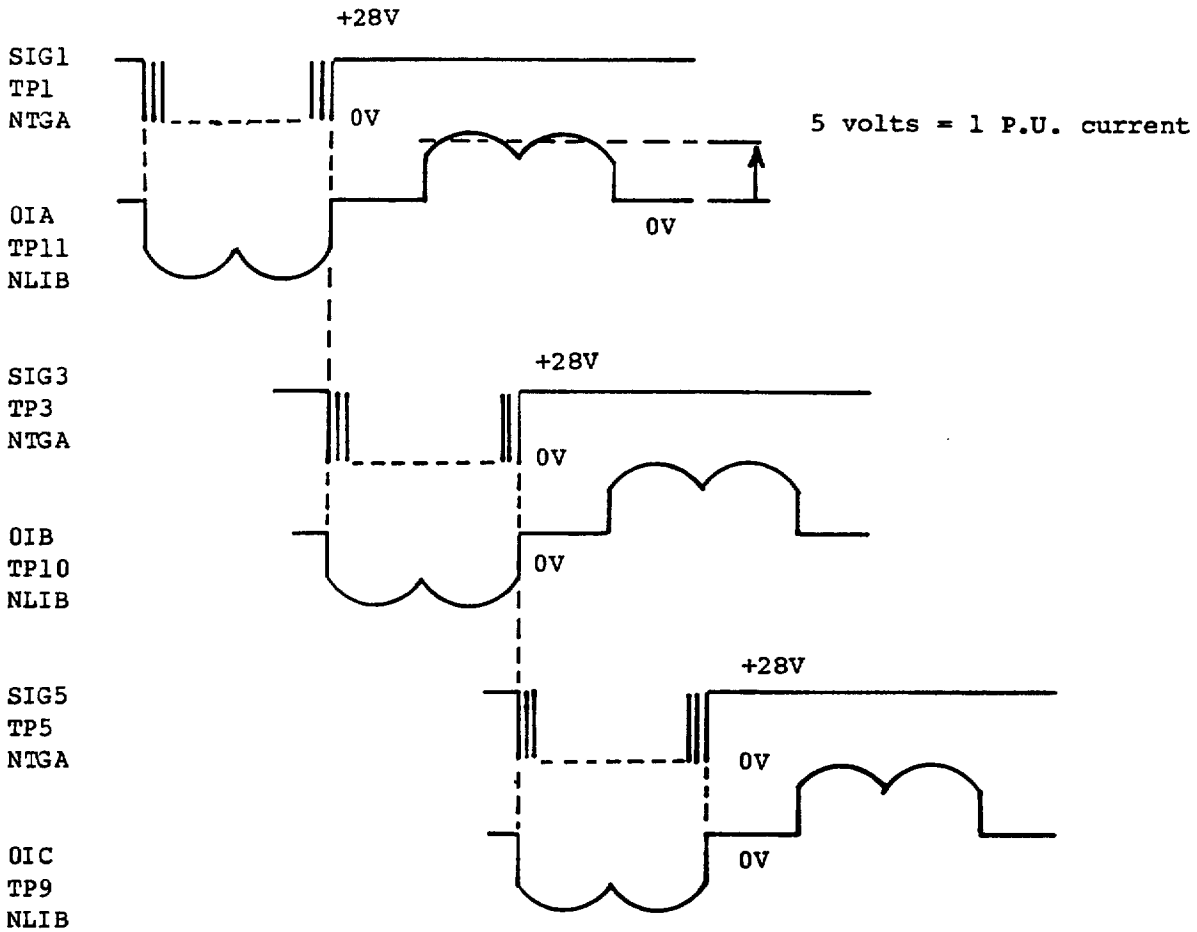


Figure 6-3. Firing Signals and Current Signals

therefore, the lubrication system should be operating at this time.

a. Reset the SEM and apply primary power.
b. MODIFY the following EEPROM variables:
   DRVTST to 3
   SUMFRQ to 3
   SUMTCH to .09
c. Connect a brush recorder to the following signals; channel A to 0IA-1D (TP11) and channel B to SIG1-2B (TP1) Start brush recorder and then drive. Move brush to 0IB-1D (TP10) and SIG3-2B (TP3). Move brush to 0IC-1D (TP9) and SIG5-2B (TP5). Verify traces against Figure 6-3.
d. After a few seconds move channel B of the brush to OIB-1D (TP10). After a few seconds move channel B to OIA-1D (TP11). Compare current brush traces against Figure 6-4.
e. Verify IFB-1C = 0.65 Vdc (+/- 12.5%). Stop drive.
f. Return DRVTST, SUMFRQ, and to original values.

6-17. RUN MOTOR UNCOUPLED CONSIDERATIONS:

Bearing monitoring protection should be functioning at this point to protect the motor. The motor should be run to 50% speed once started and bearing temperature rise, vibration, and oil rings(slingers) checked.

6-17.1. Test Connections

1. Make the following connections to a six channel chart recorder:

CH1 - DA1(TP1) - 2E
CH2 - OIA(TP11) - 1D
CH3 - FCBR(TP5) - 1D
CH4 - OIC(TP9) - 1D
CH5 - IFB(TP7) - 1C
CH6 - SPD(TP10) - 1E2.

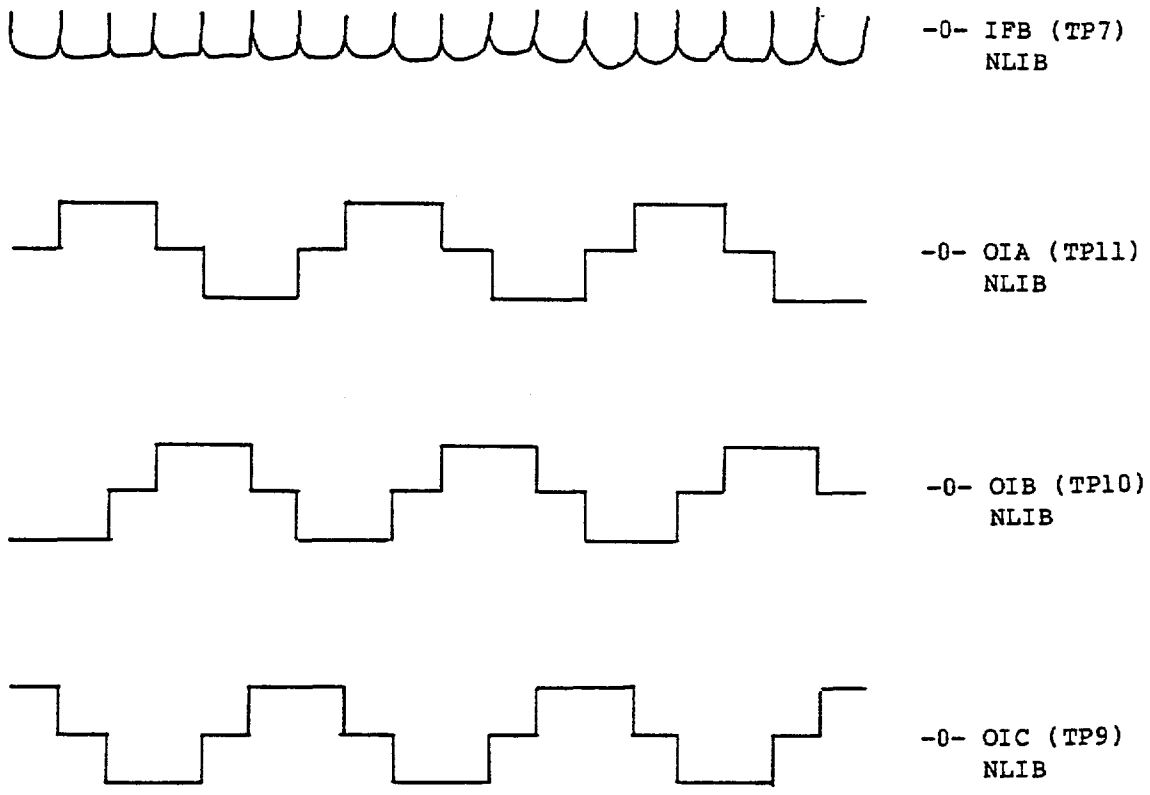


Figure 6-4

2. MODIFY the following EEPROM variables:

SRGAIN	to	1.0
SRTC	to	1.0
SUSTCM	to	.25
RRGMAX	to	.5
RRGMIN	to	.15
CRMIN	to	.09

3. ASSIGN LSTAT(MODE) to D/A #1
   (scale = 0)
   (offset = 0)

6-17.2. Start and Run Motor

1. Check that auxiliary and control power is applied.
2. Clear all alarms and faults.
3. Close source circuit interrupter and rack-in load circuit interrupter (bypass remains racked-out).
4. Select local speed reference and set it to minimum.
5. Initiate run.
   Monitoring LSTAT(MODE) changes on DA1-2E: starts from mode LOOK START (idle) 0 Vdc
   when run initiated
   switches to mode FIX FIRE (open loop) 1.25 Vdc
   at approx. 5Hz
   switches to mode RUNFIRE (closed-loop) 2.5 Vdc
   at approx. 10 Hz.
   switches to mode CHOPA (reduces 11th & 13th harmonics) 7.5 Vdc
   at approx. 17Hz
   switches to mode CHOPB (reduces 7th and 11th harmonic) 6.3 Vdc
   at approx. 23Hz
   switches to mode CHOPC (reduces 5th and 7th harmonics) 5.0 Vdc
   at approx 34Hz
   (when motor coupled up)switches to mode CHOPD (reduces 5th harmonic) 3.8 Vdc
   at approx . 45Hz
   (when motor coupled up)switches to no harmonic elimination 2.5 Vdc
6. If motor fails to start, then decrease SUSFRQ by steps of 0.5 (minimum of 1) until motor starts, and/or increase system.
7. Check motor rotation. If motor rotates backward, check IMD to motor wiring; swap two motor leads if necessary.
8. Initiate stop and motor should coast stop.

6-17.3. Speed Range Test

Start drive and gradually increase the drive speed to 50% speed . Check magnetic center, vibration, and bearing temperature rise. Verify panel speed meter reads rated rpm.

6-17.4. Speed Ramp Test

While monitoring speed with the brush recorder, decelerate the drive as fast as possible by changing the speed reference to minimum as quickly as possible.
Then check acceleration by increasing reference as fast as possible.

6-17.5. Remote Operation

Place drive control in remote and verify drive follows remote reference and remote stop/start commands.

6-18. SYNCHRONIZING/DE-SYNCHRONIZING TUNE-UP PROCEDURES (OPTION)

6-18.1. General Considerations

1. When instructed to temporarily modify tune-up variables, record the original setting and restore the variable to this original value when the testing is completed.

6-18.2. Verify Phasing of Source/ Bypass/Inverter Contactors

1. Turn drive channel select switch to "off".
2. Select "bypass" on inverter/bypass selector switch.
3. Rack in motor and inverter limit-amps.
4. Close source no-load isolation switch and user's switchgear. High voltage should be on source bridge, check phase sequence and magnitude. Look at public variable NPT to verify pt signal into control. It should be True.
5. Push start to run motor in bypass (only bypass contactor should close). verify pt signal match (check at XDCE and NCCD). Look at public variable NPT to verify pt signal into control. It should be False.
   Verify correct motor rotation.
6. Temporarily close inverter contactor by jumpering close circuit (may have to open trip circuit

also). High voltage should be on load bridge, check phase sequence and magnitude.
7. Verify source to load voltage phases, match (VBAs-VBA1, VCBs-VCB1, VACs-VAC1) on NSFC-2C and NSFC-2B.
8. Remove jumper to open (trip) inverter contactor and stop motor.
9. Rack out bypass contactor.
10. Turn drive channel select switch to "on".
11. Select "inverter" on inverter/bypass selector switch.
12. Run motor with drive and verify correct motor rotation (same as in bypass operation).

6-18.3. Test Bypass Contactor Opening/ Closing Times

1. Confirm source breaker is open and source converter is "dead".
2. Confirm the primary of the bypass contactor is "dead"
3. Remove MISOLS interlock (TB4-TB5). This makes the control think that the motor isolator is open and allows gate-test-mode to function even if the inverter contactor is really closed.
4. It is necesssary to be in "remote" when synchronizing so select "remote" on the local/remote selector switch.
5. Select "inverter" on the inverter/bypass selector switch.
6. Rack in inverter contactor.
7. Rack in bypass contactor.
8. Lower local speed pot to minimum.
9. Set up the diagnostic list:

# variables	= 1
variable #1	=BCCNT B
trigger level	= 0
trigger condition	= >
sample site	= SYNR (8)

10. Enter gate-test-mode (move NLCB-1E test switch to up position). Verify test oscillator is running.
11. Push START, the inverter contactor should close. Verify test oscillator is still running.
12. Set bit in tune-up variable DRVCFG (bit 7) to enable synchronizing.
13. Set tune-up variable SYCNTRL to 4. The bypass contactor is should close and the list should trigger. Record maximum count reached lby variable 1 in the list (variable 1 starts counting up at the middle list sample,. start display of the list near the middle with command DL 520).
14. Re-arm the diagnostic list by setting sample site to SYNR.
15. Set tune-up variable SYCNTRL to 3H. The bypass contactor should open and the list should trigger. Record maximum count reached by variable 1 in the list (variable 1 starts counting up at the middle list sample, start display of the list near the middle with command DL 520).
16. Repeat closing and opening cycle several times to get average closing and opening counts for the bypass contactor.
17. Convert opening and closing counts to opening and closing times:

$$\text{close time (sec)} = \frac{\text{bypass closing counts}}{2 * (\text{test oscillator frequency})}$$

$$\text{open time (sec)} = \frac{\text{bypass opening counts}}{2 * (\text{test oscillator frequency})}$$

where test oscillator frequency is typically around 54Hz. (Measure and record test oscillator frequency).
18. Assign tune-up variables:
SYBCP (predict time = close time - 40ms
SYBCA (acknowledge time) = 70ms
19. Push test STOP to open inverter and bypass contactor.
20. Set tune-up variable SYCNTRL to 0.
21. Excuse Tune Save to permanently save these new tune-up values.

6-19. OPERATE WITH MOTOR COUPLED

6-19.1. Test Connections

Make the following connections to a six channel chart recorder.

CH1 - DA1	TP1-2E
CH2 - 0IA	TP11-1D
CH3 - FCBR	TP5-1D
CH4 - 0IC	TP9-1D
CH5 - IFB	TP7-1C
CH6 - SPD	TP10-1E

6-19.2. Starting Tune-up

The start-up of the IMD should be optimized by varying EEPROM variables SUSFREQ and SUSTCM
The procedure is as follows:

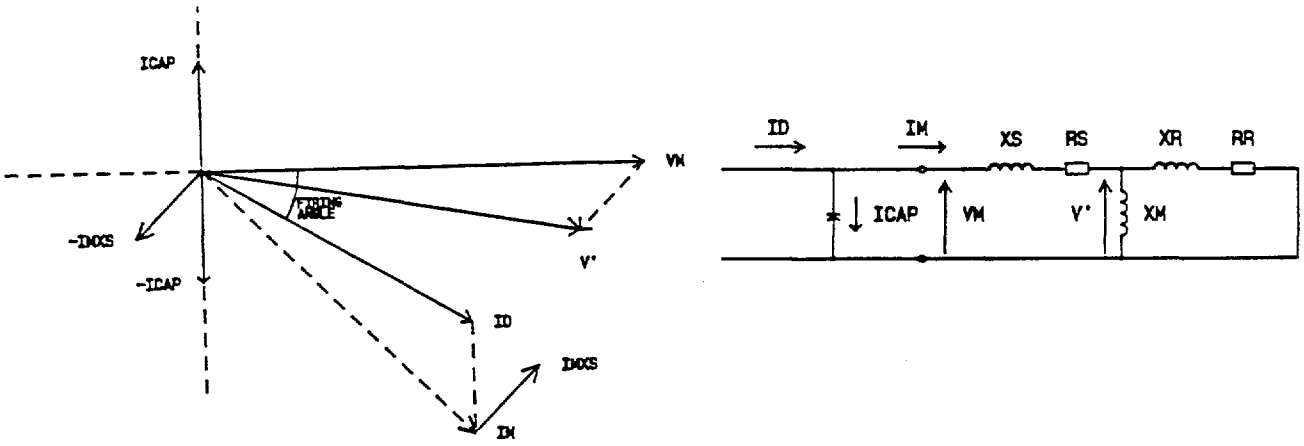


Figure 6-4A. Per Phase Equivalent Circuit

1. Attempt to start drive and monitor chart recorder.
2. If you fail to get load flux, start decreasing SUSFREQ until system starts(starts getting flux) (to minimum 6H).
3. If IMD still fails to start, return SUSFREQ to original value, increase SUSTCM, and repeat step 2.

6-19.3. Motor Flux Check

1. Start the drive and run up to some speed between 50% and 75%. Connect oscilloscope to testpoint "VAC" (TP7-2B). The waveform should appear as shown in Figure 6-5:
   Use DVM to measure V_{AC} RMS

   $$f = \frac{1}{T}$$

   Calculate motor volts/Hz (flux).

   $$\overline{V/Hz} = \frac{(\text{attn ratio})(V_{AC}\text{RMS})/f}{(\text{rated motor volts})/60\text{Hz}}$$

   The flux integrators on 1D should then have RMS voltages of:

   $Vf = (\overline{V/Hz}) * 4.75 =$ _____ V_{rms}

2. Check for accurate reconstruction. See Figure 6-10 for illustration

6-19.4. Capacitor Current

1. Measure NSFC(2B)- V_{AC}(TP9) and drive frequency(freq)
2. Calculate capacitor current I_{cap}

   $$I_{cap} = \frac{V_{LN}}{X_C} = \frac{V_{AC} * \text{ATTNRATIO} * 2 * \text{PI} * \text{FREQ} * C}{1.732}$$

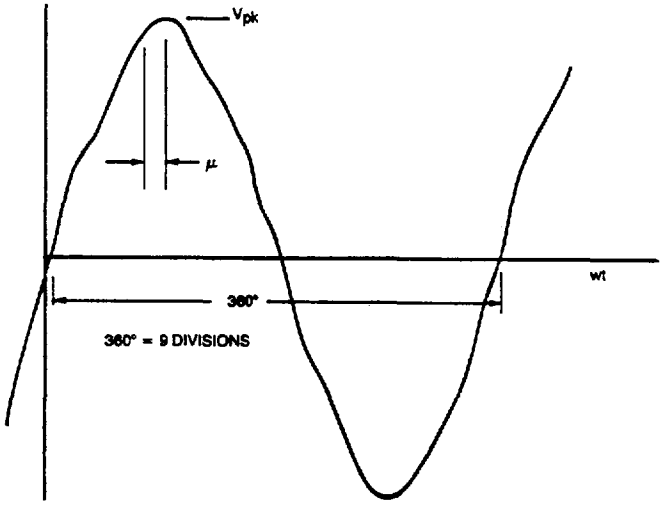


Figure 6-5. Waveform for Motor Flux

where ATTNRATIO and C are given on system elementary sheet 1EA.

3. MODIFY EEPROM variable DRMC so that RAM variable ICAP

$$ICAP = \frac{I_{CAP} * 1024}{\text{Rated Drive Current}} (\pm 5\%)$$

4. Save DRMC if changed.

6-19.5. Current Regulator

The current regulator for the IMD drive is calculated for a 180 (±25) radian response. The parameters involved in this calculation are fairly well known and measurement of the response is impractical.
The stability of the regulator can be checked by the following procedure:

Run drive to the middle of the operating (between the speed limits) range. The current, IFB-1C, should be greater than 25% (1.25 Vdc).

** NOTE **

In "TUN CR" DA1 has been temporarily assigned to current feedback (ISRG , divided by 8) so that the step reponse can be examined. DA2 has been temporarily assigned to CURCMP (divided by 64) so that feed-forward can be adjusted.

6-19.6. Check Feedforward (VDCGAIN)

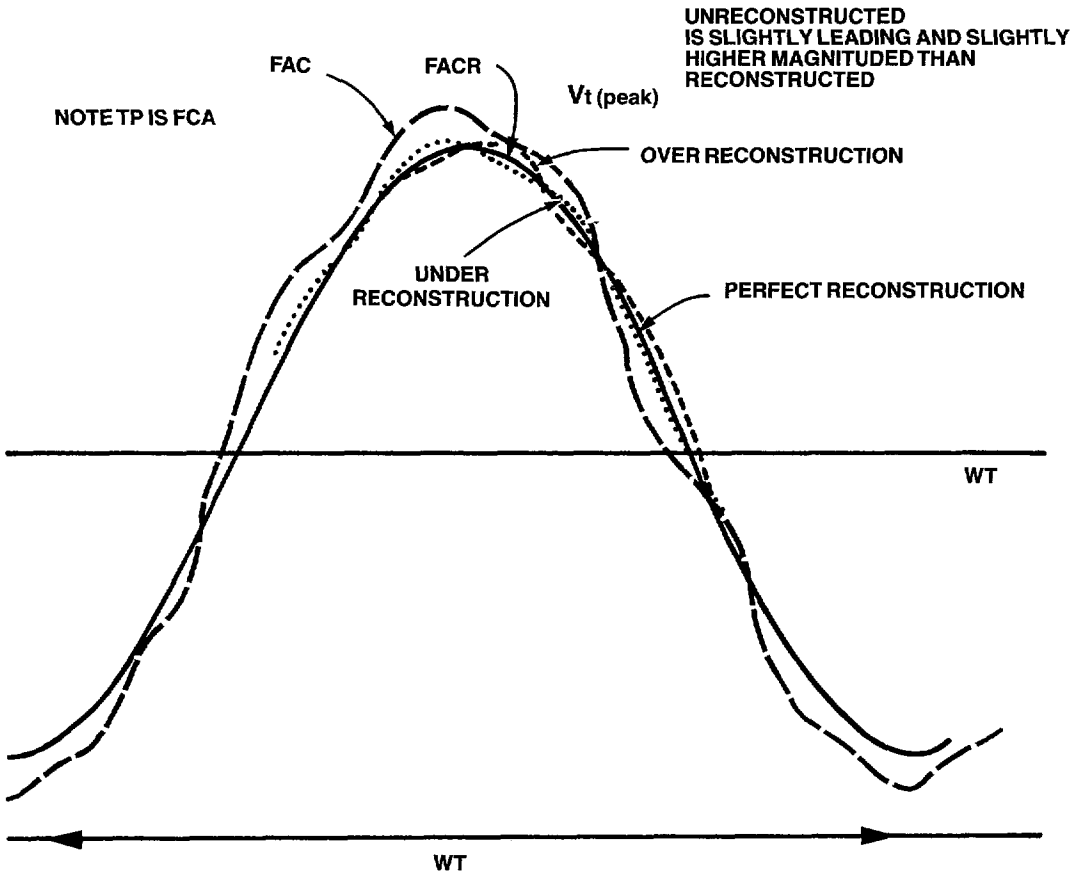
Using the monitor commands - MODIFY (see example 6-3), MODIFY CRVDC till the output of DA2-HLCB (TP2-2E) is approximately zero (but positive).

SAVE CRVDC if changed.

Apply a 2-5% step for 0.1-0.5 seconds to the current gulator using the monitor commands - MODIFY REFERENCE, APPLY, and REMOVE (TUNEUP MODE - CURRENT REGULATOR). See Example 6-3.

The brush recorder should be running at its fastest speed.

Verify IFB-1C does not overshoot or oscillate on both



The voltage at testpoints FBAR, FCBR, and FACR should appear as follows:
It is important that these be balanced as closely as possible.

the up-step to a higher current and speed, and the return step to the lower current and speed.

```
*>TUN <cr>
T >TUN CR <cr>
Available Variables are:
(List of Variables)
T >MOD REF <cr>

Step        Level(PU)          Time(sec)
1
2
Step 1.     Level. =           0.02 <cr>
Step 1      Time   =           .1 <cr>
Step 2      Level  =           0 <cr>
Step 2      Time   =           .1 <cr>
T > APPLY <cr>                 ; take brush recording of
                                 up +down response
T > REMOVE <cr>                ; remove step function
```

Example 6-3

6-19.7. Flux Regulator

The response of the flux regulator should be critically damped at 10 (+/-2) radian.

Connect TP1(DA1) -2E and TP7(IFB) -1C to chart recorder. Run drive to the middle of the operating (between the speed limits) range. The current, IFB-1C, should be greater than 25% (1.25 Vdc).

In "TUN FR" DA1 has been temporarily assigned to flux error (FLXERR, divided by 32) so that the step response can be examined. DA2 has been temporarily assigned to FLXPI (divided by 32) so that the flux regulator feed-forward can be adjusted.

6-19.8. Checking Flux Feedforward (XMGAP)

Using the monitor, in the TUNEUP MODE - FLUX REGULATOR case and MODIFY (see example 6-4), MODIFY FRXM till the output of DA2-HLCB (TP2-2E) is approximately zero (but positive).

SAVE FRXM if changed.

a. Apply a 5-10% step for 3-5 seconds to the flux regulator using the monitor commands - MODIFY REFERENCE, APPLY, and REMOVE (TUNEUP MODE - FLUX REGULATOR) . See Example 6-3.
b. Observe the response on the chart recorder. Response should be critically damped and non-oscillating. Measure and record response.
c. If the response is too fast (or to slow), decrease (increase) EEPROM variable FRTC until the nearest desired response is obtained.
d. If the response is overdamped or underdamped, increase or decrease FRGAIN to obtaincritical damping. If damping is now correct but response is not, return to step c.
e. SAVE EEPROM variables if changed.
f. Verify the NSFC(2B)-VDC(-) TP11) for sawtooth (6Xload frequency) waveform.

```
*TUN <cr>
T> TUN <cr>

Available Variables Are:
(List of Variables)

T> MOD REF <cr>     ;MODIFY FRXM as
                      needed

Step        Level(PU)          Time(sec)
1
2
Step 1.     Level(pu) =        0.02 <cr>
Step 1      Time   =           5 <cr>
Step 2      Level  =           0 <cr>
Step 2      Time   =           5 <cr>
t> APPLY <cr>      :Take brush of up + down
                      response
T> REMOVE-<cr>     ;Remove step function
```

Example 6-4

6-19.9. Speed Regulator

Run the drive speed up to middle of operating speed range. IFB-1C should be greater than 1.25 Vdc. MODIFY DRVCFG to disable torque vs speed limit.

TCOM is greater than CURLL and that CSTIE = 0. Apply a step size (2-5% forr 5 seconds) to obtain as large a step as possible without hitting current limit (observe IFB on chart recorder). (Using TUNEUP mode-SPDREG: commands MODIFY REFERENCE, APPLY, REMOVE) See Example 6-5.

**** NOTE ****

It may be necessary to buck-out the speed signal with a battery voltage to obtain enough amplification. If step down response is different from step, use step down response.

Observe the step response and adjust SRGAIN and SRTC for a 1 radian critically damped response as shown in Figure 6-8. It

should only be necessary to modify SRGAIN at all times.
Save the values for SRGAIN, SRTC, into EEPROM.
Record

**** NOTE ****

DA1 has been temporarily assigned to speed error, so that the step response could be examined. DA2 has been temporarily assigned to load firing angle (IFANG), divided by 2.

List of Variables are:
***> TUN <cr>**
T> TUN FR <cr>

T> MOD REF <cr>

Step	Level(PU)	Time(sec)
1		
2		
Step 1	Level(pu) =	**.05 <cr>**
Step 1	Time =	**2 <cr>**
Step 2	Level =	**0 <cr>**
Step 2	Time =	**2 <cr>**

t> **APPLY <cr>** :Take brush of up + down response
T> **REMOVE <cr>** ;Remove step function

Example 6-5

6-19.10. Source-side Line Impedance

Normally the source-side operates in the rectifying quadrant. However, when the drive is forced to slow down, the source converter may have to regenerate (invert). To avoid commutation failures, the angle of retard must be programmed as a function of current. Since a regenerating condition is difficult to hold steady-state, a measurement of system impedance will be made and this value will be programmed into EEPROM.

Run the drive to some convenient speed (the higher the better). Connect channel 1 of the oscilloscope to testpoint "VAC"-2C. Connect channel 2 to testpoint "0IA"-1C. The signal should be similar to those shown in Figure 6-10:

$\overline{V1\text{-}1}$ = (attn ratio)*(Vpk/1.414)/Vnom where Vnom = transformer nominal voltage

$\overline{Id} = I/5$

$\overline{Xc} = (\overline{V1\text{-}1}\ \overline{Id})\ [\cos\gamma - \cos(\gamma + \mu)]$

$CRLIMP = (\overline{Xc})$

MODIFY and SAVE into EEPROM.
Take drive to top spd, then decelerate as fast as possible. If drive trips on overcurrent then increase CRLIMP.

6-19.11. Horsepower Meter

1. Run drive to middle of operaing range and measure the drive horsepower (using chapter 6 - Section 6 - APPENDIX -A.2.II and B.3)
2. Adjust NLCB(1E) -HPM so that horsepower meter reads measured horsepower.

6-19.12. Ammeter Check

1. Measure motor current (IMM) by means of clamp-on ammeter at motor lead if possible and check that MIFB(TP) -NCCC(IF) =

$$\frac{IMM * 3.887}{\text{Rated Inverter Current}}$$

where rated inverter current is listed on system elementary sheet 1EP

**** NOTE ****

Individual phase (motor) current is scaled 3.322 Vacrms on inverter base.

2. Adjust GAIN-NLIB(1D) so that ammeter reads correct motor amps

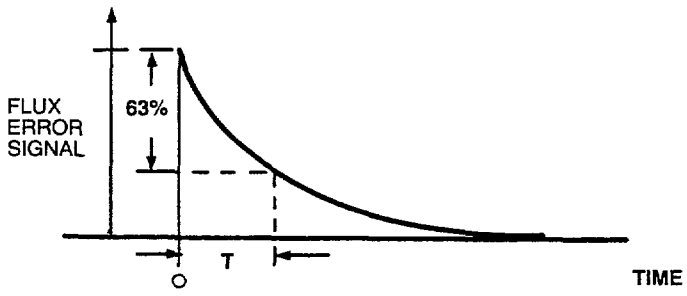


Figure 6-6. Flux Error Signal

6-19.13. Speedmeter

1. Run motor to full speed and measure motor rpm and speed reference input variable SPD_STR.
2. MODIFY EEPROM variable SRSL so that motor rpm = requested rpm

$$RPM = \frac{SPD_STR_*}{1024} \text{ (Rated RPM @ 60 Hz)}$$

3. Adjust gain-NLIB(IC) so that speedmeter reads correct motor rpm

6-19.14. Re-start on fly

It is possible to catch the motor while it is still spinning and to return to requested speed. Monitor the following on a brush recorder:

1. SPD (TP1)-1E
2. FCBR (TP5)-1D
3. IFB (TP7)-1C
4. Run drive to full speed, stop and immediately re-start (before flux dies) drive should 'catch' and control the motor
5. Run drive to full speed stop, wait 15 secs and restart (flux will be zero and speed signal will be zero). After 8-15 sec spd signal will go to ≈ 9-10 volts then ramp down. As speed ramps down flux will peak. After flux peaks drive should be "caught" if not, decrease SCFLUX by .1 (min of 0.10)

6-20. SYNCHRONIZING/DE-SYNCHRONIZING TUNE-UP PROCEDURES

6-20.1. General Considerations

1. When instructed to temporarily modify tune-up variables, record the original setting and restore the variable to this original value when the testing is completed.

6-21. SYNCHRONIZING CHECKS (OPTION)

6-21.1. Test Synchronizing/Bypass Logic

1. Confirm source breaker is open and source converter is "dead".
2. Confirm the primary of the bypass contactor is "dead".
3. Remove_MISOLS interlock (TB4-TB5). This makes the control think that the motor isolator is open and allows gate-test-mode to function even if the inverter contactor is really closed.
4. It is necesssary to be in "remote" when synchronizing so select "remote" on the local/remote selector switch.
5. Select "inverter" on the inverter/bypass selector switch.
6. Rack in inverter contactor.
7. Rack in bypass contactor.
8. Lower local speed pot to minimum.
9. Set bit in tune-up variable DRVCFG (bit 7) to enable synchronizing.
10. Set tune-up Variables:

RRMOD	to	3	select local speed test
SYSPD	to	0.89pu	lower sync speed threshold
SYCNTRL	to	3	enable all sync logic
SYPADJ	to	0	don't try to phase, motor advanced

11. Enter gate-test-mode (move NLCB-1E test switch to up position). Verify test oscillator is running.
12. Push START, the inverter contactor should close. Verify test oscillator is still running.
13. Push STOP, inverter contactor should open.

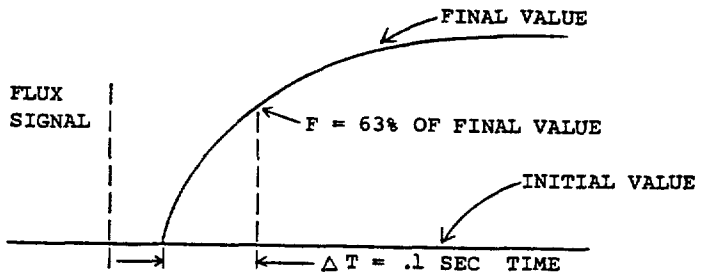


Figure 6-7.

14. Look at the public word variable LFREQ using display continuous command and record the most common value.
15. Set tune-up variable SYSTP to the most common value of LFREQ/1024
16. Set public word variable CLSREG to 257. This enables the speed and flux regulators. Expect to get a loss of speed control alarm at this point.
17. Push START, the inverter contactor should close.
18. Using the local speed pot, raise the speed setpoint to the maximum. When the rate limited speed setpoint has reached maximum, then the forward transfer sequence should occur (bypass close and inverter open).

** NOTE **

If synchronizing is controlled by contact closure rather that speed reference then use contact.

19. Lower speed setpoint below synchronizing speed threshold (or open sync contact).
20. Set public word variable CLSREG to 257 to close the flux and speed reqs. Return speed setpoint to maximum, the forward transfer sequence lshould occur (bypass close and inverter open).
21. Lower the speed setpoint and the reverse transfer sequence should occur (bypass open and inverter close).
22. Verify the the STOP will open the bypass contactor when in a bypassed condition.
23. Set tune-up variable SYCNTRL to 0.
24. Clear the excessive speed error alarm.
25. Exit gate-test-mode.

** NOTE **

For drives that allow across-the-line starts, do the following

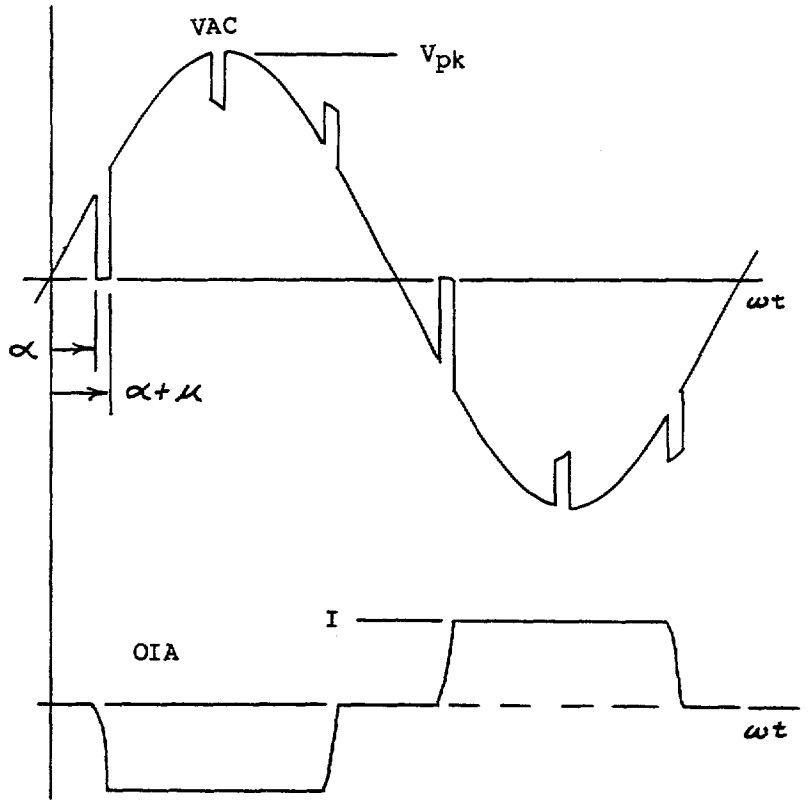


Figure 6-8.

26. Verify that a transfer to bypass will occur if a drive fault occurs when in a normal run condition.

6-21.2. Preliminaries for Actual Synchronizing

1. Initially run the drive with tune-up variable SYCNTRL set to 0H.
2. Verify all regulators are properly tuned and the drive is stable up to 60Hz.
3. With the drive at top speed (60Hz.), set tune-up variable SYCNTRL to 1H. This enables the phase synchronizing regulator. Verify that the control tries to match phase between source and load voltage.
4. Look at public integer FLXSTPSY. This variable should be in the range of 819 to 1228.
5. If FLXSTPSY is in the appropriate range, then enable the voltage matching regulator by setting SYCNTRL to 2. Motor flux should raise or lower to agree with FLXSTPSY.
6. With a DVM, compare source ac volts to motor ac volts (use NSFC testpoints). They should agree to within approximately 5%. Adjust tune-up variable SYVADJ to obtain agreement (raise SYVADJ to araise motor voltage).
7. The motor should now be phase and amplitude matched to the ac line. The regulators should be stable.
8. Lower the speed setpoint below the synchronous speed threshold and the drive should respond to speed setpoint.
9. Raise the speed setpoint above the synchronous speed threshold (or close sync contact) and the drive should match motor voltage phase and magnitude to the ac line. Monitor the public byte variable SYMTCH when attempting to synchronize. Phase and voltage should be matched and stablized before SYNTCH goes true. If not, increase tune-up variable SYMCNT.
10. With the motor synchronized to the line, allow the transfer to occur by setting tune-up variable SYCNTRL to 3. Monitor motor current and voltage. Verify successful traansfer (bypass closed and inverter open).
11. Lower the speed setpoint below the synchronous speed (or open the sync contact) and the drive will de-synchronize. Monitor drive current and voltage. Verify successful transfer (bypass open and inverter closed).
12. Execute TUNE SAVE command to permanently store all new tune-up values.

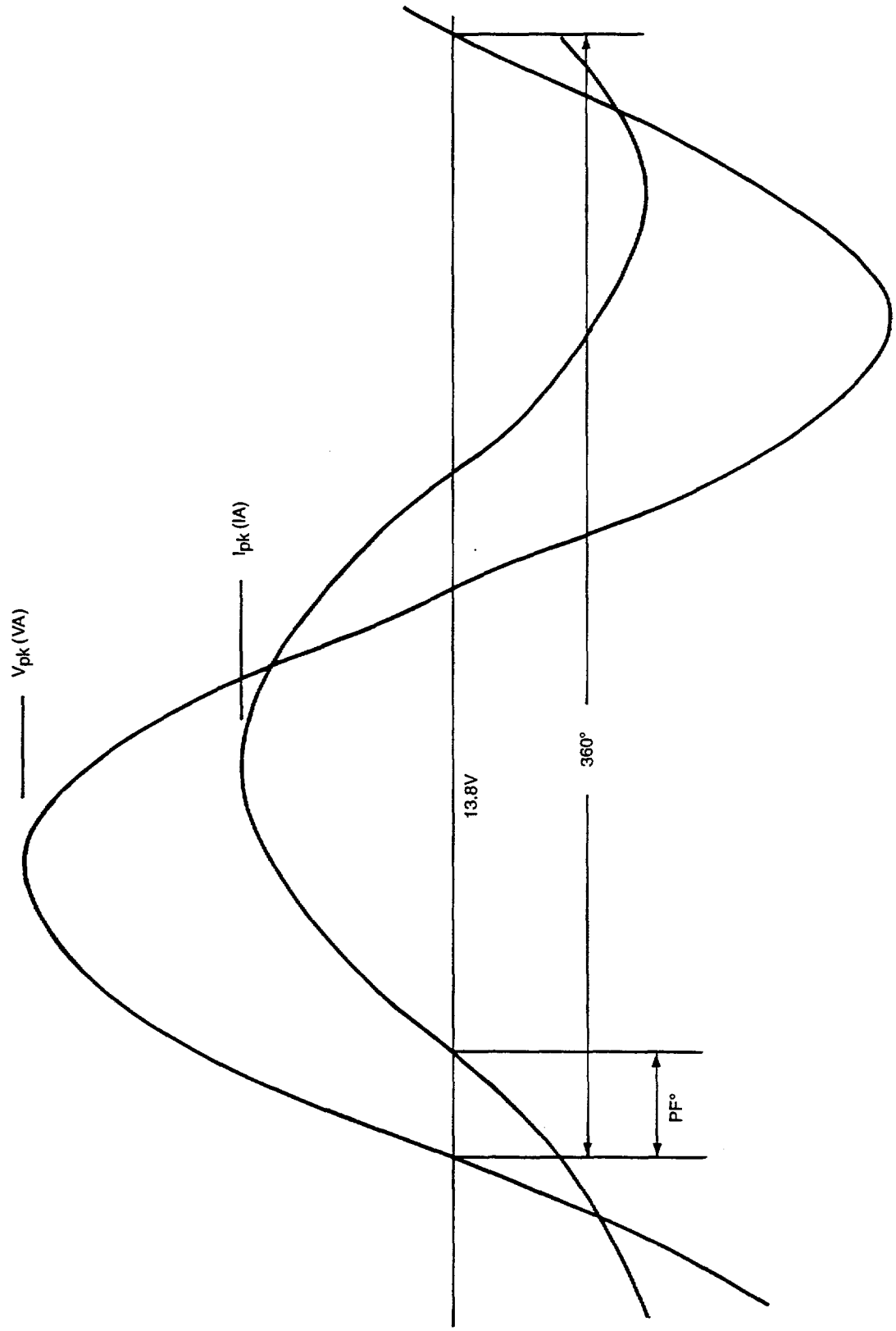


Figure 6-9.

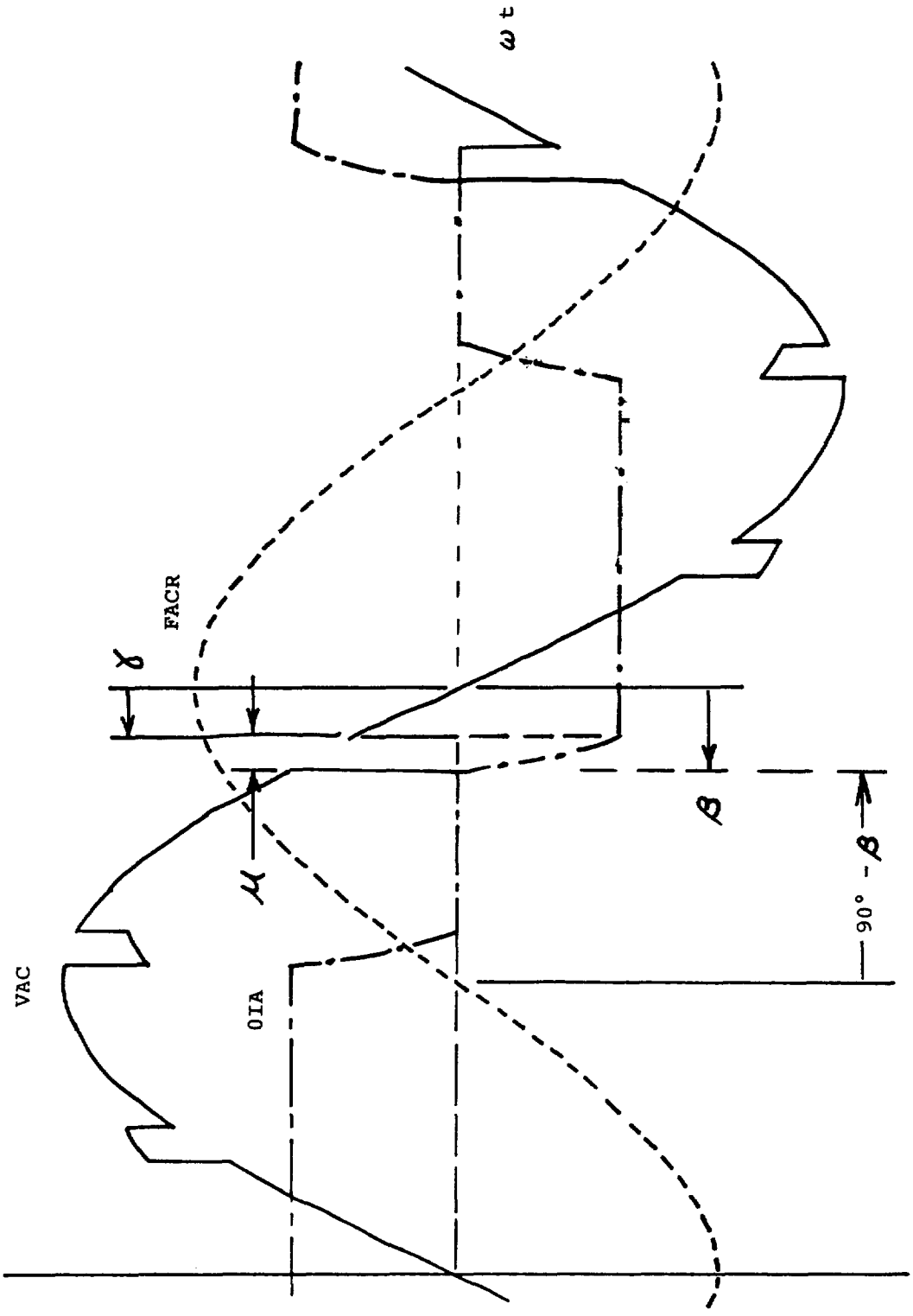


Figure 6-10

ADDITIONAL INFORMATION
POWER READINGS

A. PCM input power factor and power readings (per channel) use Figure 6-10.

1. Power factor (PF)

   a. Measure μ
   b. Measure γ

$$\text{Power factor} = \frac{\cos(\gamma - \mu) + \cos\gamma}{2}$$

2.I. Kilowatts (KW)

   a. Measure power factor (PF).
   b. Measure OIA (peak average) (OIA).
   c. Measure peak voltage of VAC (Vacpk).

$$\text{Kilowatts} = \frac{\text{Vacpk} \bullet \text{attn ratio}}{1000 \bullet 2} * \frac{\text{OIA}}{5\text{V/unit}}$$

* rated bridge ac current • PF.

2.II Alternative to 2 (eliminates P.F reading)

   a. If VDC not being used, turn to min (full CCW).
   b. Measure VDC
   c. Measure IFB

$$\text{Kilowatts} = \frac{\text{IFB}}{5\text{V/volt}} \bullet \text{rated bridge ac current}$$

$$\bullet\ 1.2825 * \frac{\text{VDC} \bullet \text{ATTN RATIO}}{1000}$$

B. Motor power factor and power readings - use Figure 6-9.

1. Power factor (PF)
   With an oscilloscope, observe the potential transformer (PT) signal -VA and the current transformer (CT) signal - IA (with a current probe).
   Measure the power factor
   PF = cos (PF°)
   Kilowatts (KW)
   a. Measure power factor (PF)
   b. Measure voltage peak (V_{pk})
   c. Measure current peak - (I_{pk})

$$\text{Kilowatts} = \left[\frac{I_{pk}}{\sqrt{2}} \bullet \text{CT Ratio}\right] \bullet \left[\frac{V_{pk}}{\sqrt{2}} \bullet \text{PT ratio}\right] \bullet \left[\frac{1}{1000}\right] \bullet \text{PF}$$

2. Horsepower (HP)

   a. Measure Kilowatts (KW)
   Horsepower = • KW • Eff./7457
   [Eff - efficiency can be estimated at 0.94 for large motors (see motor data sheets for accurate data)].

3. Torque (τ)
   a. Measure horsepower (HP)
   b. Measure SPD (RPM)
   Put oscilloscope's time base on calibration for this entire measurement. Use as much as possible at the screen to display a half-cycle. Measure the time of this half cycle (T).
   SPD = [(rated frequency) . rated RPM at rated frequency)/(2-t)]

$$\text{Torque} = \frac{5256 \bullet \text{HP}}{\text{SPD}}$$

d.

ADDITIONAL INFORMATION FOR ALTERNATE AMMETER CHECKS

Alternate method to calculate motor current.

a. Calculate capacitor current
   ICAP = Vac * Att. ratio * 2 * PI * freq * C/1.732
b. Calculate drive current
   ID = (IFB * Drive rated current)/(1.2825 8 5)
c. Calculate load firing angle
   use display command to display
   Variable FIREANGCL

Firing angle = FIREANGCL * .3516 degree
Where Vac is measured at TP 'Vac' 2B
Frequency is the motor stator frequency
IFB is measured at TP 'IFB' 1C
ratio is given on Sh. 1EA
C is given on Sh. 1EA
Drive rated current is given on Sh. 1EP

Calculate motor current

$$IM = [ID * Coso)^2 + (ID * SIN\ o - ICAP)^2]^{1/2}$$

Where o is the firing angle in degree

CHAPTER 7
DIAGNOSTICS AND TROUBLESHOOTING

7-1. INTRODUCTION

Included in the drive power and electronic circuitry is a multi-layered diagnostic and troubleshooting capability. The main elements of this capability are the following:

1. Self testing of critical microcomputer system elements every time the control is reset or power applied.
2. A memory type fault monitor which displays the cause of shutdown in a printed message at the printer.
3. Lights and test points on front of cards provide the following information:
   a. Attenuated system bus voltages and currents on test points allow a check of magnitude and phasing.
   b. Gate pulse signals to converter bridge available at test points.
   c. Neon cell voltage indicating lights across each thyristor in power converters for indication of shorted cells.
   d. Overcurrent, overvoltage, and overspeed indicating lights in electronics module.
   e. Card front lights indicate status of all switching inputs and outputs to/from electronics module.
4. Gate-test-mode allows checkout of analog signal conditioning circuitry, microprocessor firing control, and cell gate signal integrity while off-line.
5. Serial monitor software which allows examination of system memory, inputs, and outputs.(Chapter 5)
6. Capability to store diagnostic lists of system variables preceding and following fault conditions.(Chapter 4)

7-2. DIAGNOSTICS INITIALIZATION SELFTEST

SECTION A - SELFTEST

7-3. BASIC STRUCTURE

The selftest routines are PLM86 procedures which may be executed by the HMPG resident in the Structured control Small Electronics Module (SEM) to test individual PWB's and functions. These routines are executed upon power-up if the TEST switch on the NLCB is in the up position. These routines may also be run when the drive is in a non-RUN condition by operator commands received via a terminal connected to the HMPG's RS-232 port or by depressing a test switch. The function of the selftest is to test most of the control panel for the Structured GTO-IMD, with great emphasis placed on the testing of the PWB's in the SEM.

Testing of the panel begins by performing tests of the hardware elements within a neighborhood of the HMPG's microprocessor and continues to tests of increasingly remote hardware. All of the tests incorporated within the selftest may be classed into 3 groups:
1. PWB-Specific tests
2. Paired PWB tests
3. Panel-level tests

7-4. PWB-SPECIFIC TESTS:

A great amount of effort has been placed on performing tests that will pinpoint problems on specific PWB's first. In this manner, if any easily diagnosed problem exists, it will be clearly spelled out to the operator, instead of vaguely defined by a list of possible solutions. Tests such as these comprise the PWB-Specific test group.

7-5. PAIRED PWB TESTS:

Tests which depend on the operation of several PWB's are classed within the Paired-PWB group. Diagnostic messages for failures of these tests will indicate a functional problem and also a list of possible solutions which may be general in nature.

CAUTION

Selftest exercises the output relays; therefore, the output terminal boards should be disconnected from the I/O module, and/or source and load breakers should be racked-out and drive stopped.

7-6. PANEL-LEVEL TESTS:

Panel-level tests also require correct operation of several PWB's or modules, however, these tests are segregated to allow the user to perform either a module-level test (involving only the control SEM) or a complete panel-level test which would include this last category of procedures. From a practical point of view, the sequence of testing within the control SEM progresses in the following manner:

1. HMPG
2. HLCB
3. HAIA
4. NSFC (source-master)
5. NSFC (load)
6. NSFC(source-slave)

7. NLCB
8. NLIB (source)
9. NLIB (load)
10. HISA
11. HRDA

Further tests which include PWB's outside the SEM, progress in the following sequence:

1. Cables leading from the SEM.
2. the I/O Module.

7-7. EXECUTING THE TESTS

The self test routine may be initiated by resetting (toggle TP1 (RESET) to TP2 (DCOM) the HMPG with the TEST switch (on the NLCB) in the UP position. The entire test is completed in approx 20 seconds. At its completion, status information will be printed out at the door-mounted printer. In the present product, the entire self test routine is executed; however, provisions for executing small sections of the code are in place.

Upon completion of the self-test, the drive will operate in the gate-test-mode. To return to normal operation, the NLCB's test switch must be returned to the normal position and the HMPG must be reset (Toggle TP1 (RESET) to TP2 (DCOM)).

CAUTION

Selftest exercises the output relays; therefore, the output terminal boards should be disconnected from the I/O module, and/or source and load breakers should be racked-out and drive stopped .

** NOTE **

If the VBol setting is greater than 10.5 Vdc then set down to 10.5 Vdc before executing selftest. Return to original setting when selftest is done.

7-8. DIAGNOSING FAILURES

As mentioned above, failure of any of the selftest procedures will produce a diagnostic message on the door-mounted printer. The format of the diagnostic message is:

Usually, a failure can easily be diagnosed by the problem statement. At least one solution will be suggested in the print out. It is suggested that troubleshooting be performed at the PWB-level (Replace the PWB not its component). Although the diagnostic messages may indicate more specific failures, it is not recommended that PWB components be replaced or repaired (always write or attach the diagnosis to the card)

The TEST NUMBER and SUB-TEST NUMBER uniquely define a failure within the selftest procedure. The TEST NUMBER is incremented from 1 to 31 (hex) as the tests progress. For different test numbers, the SUB-TEST parameter will vary from 1 up to the number of sub-tests within that major test. The Description of Tests section that follows details the complete selftest procedure according to the TEST NUMBER and SUB-TEST NUMBER parameters.

7-8.1. Description of Tests

The following lists all tests available with sub-test information:

TEST	DESCRIPTION
0	HMPGROMTST: Reads all of ROM memory (F000:0 to F000:FFFF). Possible failures are:
.0	BusHog error
.1	- Read error
1	HMPGRAMTST: Reads all of RAM memory (0:0 to 0:3FFF).Possible failures are:
.0	- BusHog error
.1	- Read error
.2	- Write error
2	HMPGINTRTST
.1	- Verifies control and data lines to HMPG's 8259
3	HMPG_TMR
.1	- Verifies addr lines & data bits 6 & 7
.2	- Verifies addr lines & data bits 6 & 7
.3	- Verifies addr lines & data bits 6 & 7
.4	- Verifies data bits 5-0 with walking 1 pattern
.5	- Verifies that all outputs are free to toggle.
.6	- FOR TMR0: Checks for null count = 1
.7	- FOR TMR0: Loads TMR; checks for null count = 0
.8	- FOR TMR0: Verify TMR time out
.9	- FOR TMR0: Verify TMR reloads
.A	- FOR TMR1: Checks for null count = 1
.B	- FOR TMR1: Loads TMR; checks for null count = 0
.C	- FOR TMR1: Verify TMR time out
.D	- FOR TMR1: Verify TMR reloads
.E	- FOR TMR2: Checks for null count = 1
.F	- FOR TMR2: Loads TMR; checks for null count = 0
.10	- FOR TMR2: Verify TMR time out
.11	- FOR TMR2: Verify TMR reloads
4	HMPG_SIO_PRES
.1	- Check that USART trans and rec buffers are empty
5	HMPG_STALL
.1	- Verifies software reset of Stall circuit.
.2	- Initiate stall timer; checks for no STALL after 100
.3	- Verifies STALL asserted after 1.5 seconds.

TEST	DESCRIPTION
6	HMPG_INTR
.1	- Verify operation of BusHog interrupt.
.2	- Verify that BusHog doesn't occur for legitimate read.
.3	- Verify operation of Time Tic interrupt.
7	DMPB
.1	- Checks that Write Enable Switch is not asserted.
.2	- Checks for no BusHog during EEPROM reads.
.3	- Checks for expected pattern in EEPROM.
8	HLCB_PRES
.1	- Enables IMOK; checks for 0IMOK=0.
.2	- Disables IMOK; checks for 0IMOK=1.
9	HLCB_DAK
.1	- Checks for no BusHog during reads of HLCB.
A	HLCB_TMR_U22
.1	- Verifies addr lines & data bits 6 & 7
.2	- Verifies addr lines & data bits 6 & 7
.3	- Verifies addr lines & data bits 6 & 7
.4	- Verifies data bits 5-0 with walking 1 pattern
.5	- Verifies that all outputs are free to toggle.
.6	- FOR TMR0: Checks for null count = 1
.7	- FOR TMR0: Loads TMR; checks for null count = 0
.8	- FOR TMR0: Verify TMR time out
.9	- FOR TMR0: Verify TMR reloads
.A	- FOR TMR1: Checks for null count = 1
.B	- FOR TMR1: Loads TMR; checks for null count = 0
.C	- FOR TMR1: Verify TMR time out
.D	- FOR TMR1: Verify TMR reloads
.E	- FOR TMR2: Checks for null count = 1
.F	- FOR TMR2: Loads TMR; checks for null count = 0
.10	- FOR TMR2: Verify TMR time out
.11	- FOR TMR2: Verify TMR reloads

TEST	DESCRIPTION
B	HLCB_TMRU23
.1	- Verifies addr lines & data bits 6 & 7
.2	- Verifies addr lines & data bits 6 & 7
.3	- Verifies addr lines & data bits 6 & 7
.4	- Verifies data bits 5-0 with walking 1 pattern
.5	- Verifies that all outputs are free to toggle.
.6	- FOR TMR0: Checks for null count = 1
.7	- FOR TMR0: Loads TMR; checks for null count = 0
.8	- FOR TMR0: Verify TMR time out
.9	- FOR TMR0: Verify TMR reloads
.A	- FOR TMR1: Checks for null count = 1
.B	- FOR TMR1: Loads TMR; checks for null count = 0
.C	- FOR TMR1: Verify TMR time out
.D	- FOR TMR1: Verify TMR reloads
.E	- FOR TMR2: Checks for null count = 1
.F	- FOR TMR2: Loads TMR; checks for null count = 0
.10	- FOR TMR2: Verify TMR time out
.11	- FOR TMR2: Verify TMR reloads
.12	- FOR TMR2: Initialize to Mode 0; verify status.
.13	- FOR TMR2: Load with 10 ms delay; verify status.
.14	- FOR TMR2: After 4 ms, verify status.
.15	- FOR TMR2: After 15 ms, verify time out.
.16	- FOR TMR2: Check that count = 0.
C	HLCB_INTR_PRES
.1	- Verify addr and data lines to HLCB's 8259.
D	HLCB_SIO
.1	- Initialize and check status of HLCB's 8251.
.2	- Verify that single byte of data is send, by sensing status.
.3	- Verify transfer of data pattern.
E	HAIA_DAK
.1	- Checks for no BusHog during reads of HAIA.
.2	- Checks for no BusHog during writes to HAIA.
F	HAIA_PRES
.1	- Checks that HAIA is present by reading single bit.

TEST	DESCRIPTION
10	HAIA_TMR
.1	- Verifies addr lines & data bits 6 & 7
.2	- Verifies addr lines & data bits 6 & 7
.3	- Verifies addr lines & data bits 6 & 7
.4	- Verifies data bits 5-0 with walking 1 pattern
.5	- Verifies that all outputs are free to toggle.
.6	- FOR TMR0: Checks for null count = 1
.7	- FOR TMR0: Loads TMR; checks for null count = 0
.8	- FOR TMR0: Verify TMR time out
.9	- FOR TMR0: Verify TMR reloads
.A	- FOR TMR1: Checks for null count = 1
.B	- FOR TMR1: Loads TMR; checks for null count = 0
.C	- FOR TMR1: Verify TMR time out
.D	- FOR TMR1: Verify TMR reloads
.E	- FOR TMR2: Checks for null count = 1
.F	- FOR TMR2: Loads TMR; checks for null count = 0
.10	- FOR TMR2: Verify TMR time out
.11	- FOR TMR2: Verify TMR reloads
11	HAIA_CONVERSION
.1	- Start A/D conversion; check that BUSY=1.
.2	- Check that 1DAVAL = 0.
.3	- Wait for conversion to complete; check that BUSY=0.
.4	- Check that 1DAVAL = 1.
12	HAIA_PSUPPLY
.1	- Check that N15 supply is read correctly.
.2	- Check that P15 supply is read correctly.
13	NSFC_S_PRES
.1	- Set IMOK true; read back IMOK signal.
.2	- Set IMOK false; read back IMOK signal.
14	NSFC_S_TMR_PRES
.1	- Verifies addr lines & data bits 6 & 7
.2	- Verifies addr lines & data bits 6 & 7
.3	- Verifies addr lines & data bits 6 & 7
.4	- Verifies data bits 5-0 with walking 1 pattern
.5	- Verifies that all outputs are free to toggle.

TEST	DESCRIPTION
15	NSFC_S_TMODE
.1	- FOR TMR0: Checks for null count = 1
.2	- FOR TMR0: Loads TMR; checks for null count = 0
.3	- FOR TMR0: Verify TMR time out
.4	- FOR TMR0: Verify TMR reloads
.5	- FOR TMR1: Checks for null count = 1
.6	- FOR TMR1: Loads TMR; checks for null count = 0
.7	- FOR TMR1: Verify TMR time out
.8	- FOR TMR1: Verify TMR reloads
.9	- Perform operations to set pulse train to 0; verify.
.A	- In TEST Mode, write all on pattern, expect echo of same
.B	- In TEST Mode, walk 1 thru primary & firing latch.
.C	- In TEST Mode, see if pulse train is stuck low.
.D	- In TEST Mode, see if pulse train is stuck high.
.E	- In TEST Mode, check that pulse train is toggling.
16	NSFC_L_PRES
.1	- Set IMOK true; read back IMOK signal.
.2	- Set IMOK false; read back IMOK signal.
17	NSFC_L_TMR_PRES
.1	- Verifies addr lines & data bits 6 & 7
.2	- Verifies addr lines & data bits 6 & 7
.3	- Verifies addr lines & data bits 6 & 7
.4	- Verifies data bits 5-0 with walking 1 pattern
.5	- Verifies that all outputs are free to toggle.
18	NSFC_L_TMODE
.1	- FOR TMR0: Checks for null count = 1
.2	- FOR TMR0: Loads TMR; checks for null count = 0
.3	- FOR TMR0: Verify TMR time out
.4	- FOR TMR0: Verify TMR reloads
.5	- FOR TMR1: Checks for null count = 1
.6	- FOR TMR1: Loads TMR; checks for null count = 0
.7	- FOR TMR1: Verify TMR time out
.8	- FOR TMR1: Verify TMR reloads
.9	- Perform operations to set pulse train to 0; verify.
.A	- In TEST Mode, write all on pattern, expect echo of same.
.B	- In TEST Mode, walk 1 thru primary & firing latch.
.C	- In TEST Mode, see if pulse train is stuck low.
.D	- In TEST Mode, see if pulse train is stuck high.
.E	- In TEST Mode, check that pulse train is toggling.
19	NSFC_SRC_SLAVE_PRES
.1	- Set IMOK true; read back IMOK signal.
.2	- Set IMOK false; read back IMOK signal.
1A	NSFC_SRC_SLAVE_TMR_PRES
.1	- Verifies addr lines & data bits 6 & 7
.2	- Verifies addr lines & data bits 6 & 7
.3	- Verifies addr lines & data bits 6 & 7
.4	- Verifies data bits 5-0 with walking 1 pattern
.5	- Verifies that all outputs are free to toggle.
1B	NSFC_SRC_SLAVE_TMODE
.1	- FOR TMR0: Checks for null count = 1
.2	- FOR TMR0: Loads TMR; checks for null count = 0
.3	- FOR TMR0: Verify TMR time out
.4	- FOR TMR0: Verify TMR reloads
.5	- FOR TMR1: Checks for null count = 1
.6	- FOR TMR1: Loads TMR; checks for null count = 0
.7	- FOR TMR1: Verify TMR time out
.8	- FOR TMR1: Verify TMR reloads
.9	- Perform operations to set pulse train to 0; verify.
.A	- In TEST Mode, write all on pattern, expect echo of same.
.B	- In TEST Mode, walk 1 thru primary & firing latch.

TEST	DESCRIPTION
.C	- In TEST Mode, see if pulse train is stuck low.
.D	- In TEST Mode, see if pulse train is stuck high.
.E	- In TEST Mode, check that pulse train is toggling.
1C	NSFC_S_PLL
.1	- FOR TMR2: Enables PLL; Checks for null count = 1
.2	- FOR TMR2: Loads TMR; checks for null count = 0
.3	- FOR TMR2: Verify TMR time out
.4	- FOR TMR2: Verify TMR reloads
1D	NSFC_L_PLL
.1	- FOR TMR2: Enables PLL; Checks for null count = 1
.2	- FOR TMR2: Loads TMR; checks for null count = 0
.3	- FOR TMR2: Verify TMR time out
.4	- FOR TMR2: Verify TMR reloads
1E	NSFC_SRC_SLAVE_PLL
.1	- FOR TMR2: Enables PLL; Checks for null count = 1
.2	- FOR TMR2: Loads TMR; checks for null count = 0
.3	- FOR TMR2: Verify TMR time out
.4	- FOR TMR2: Verify TMR reloads
1F	ROLL_CALL
.1	- Test for ribbon cable connecting HAIA(JA) to SEM(JG).
.2	- Enable NLCB IMOK; check for IMOK true.
.3	- Disable NLCB IMOK; check for IMOK false.
.4	- Enable NLIBS IMOK; check for IMOK true.
.5	- Disable NLIBS IMOK; check for IMOK false.
.6	- Enable NLIBL IMOK; check for IMOK true.
.7	- Disable NLIBL IMOK; check for IMOK false.

TEST	DESCRIPTION
20	NLCB_DIFF_AMPS
.1	- Test for 0 volts at REFO.
.2	- Test for 0 volts at TREFO.
.3	- Test for 0 volts at 12PREFO.
.4	- Inject current in REFO circuit; test for change in output.
.5	- Inject current in TREFO circuit; test for change in output.
.6	- Inject current in 12PREFO circuit; test for change in output.
.7	- Test for 0 volts at SP (speed).
.8	- Reset existing faults; test for 1OSP (overspeed) = 0.
21	HLCB_SQ_WAVES
.1	Check for absence of square waves with oscillator off.
.2	Check for any square waves with oscillator on.
.3	Check for source VA, VB, VC square waves. Check for any source flux waves.
.4	Check for SRC VA flux wave.
.5	Check for SRC VB flux wave.
.6	Check for SRC VC flux wave.
.7	Check SRC phase sequence
.8	Check for load VA, VB, VC square waves. Check for any source flux waves.
.9	Check for load VA flux wave.
.A	Check for load VB flux wave.
.B	Check for load VC flux wave
.C	Check load phase sequence.
22	NLIB_S_ANALOG
.1	- Reset existing faults; verify UV & not OV condition.
.2	- Verify not SUPPression condition.
.3	- Verify 1UV70=0.
.4	- Test C17 for time delay. (Overvoltage circuit)
.5	- Test C17 for time delay. (Overvoltage circuit)
.6	- Test for 0 volts at IFBS.
.7	- Test for 0 volts at FLUX BAS.
.8	- Inject current in source current condition circuit. Detect change in IFBS.
.9	- Detect change in FLUX BAS.

TEST	DESCRIPTION
.A	- For FLUX BAS: Check for 0 volts dc offset.
.B	- For FLUX BAS: Check level of positive peak.
.C	- For FLUX BAS: Check level of negative peak.
23	NLIB_L_ANALOG
.1	- Reset existing faults; verify UV & not OV condition.
.2	- Verify not SUPPression condition.
.3	- Verify 1UV70=0.
.4	- Test C17 for time delay. (Overvoltage circuit)
.5	- Test C17 for time delay. (Overvoltage circuit)
.6	- Test for 0 volts at IFBL.
.7	- Test for 0 volts at FLUX BAL.
.8	- Inject current in source current condition circuit. Detect change in IFBL.
.9	- Detect change in FLUX BAL.
.A	- For FLUX BAS: Check for 0 volts dc offset.
.B	- For FLUX BAS: Check level of positive peak.
.C	- For FLUX BAS: Check level of negative peak.
.D	- For FLUX CBS: Check for 0 volts dc offset.
.E	- For FLUX CBS: Check level of positive peak.
.F	- For FLUX CBS: Check level of negative peak.
.10	- For FLUX ACS: Check for 0 volts dc offset.
.11	- For FLUX ACS: Check level of positive peak.
.12	- For FLUX ACS: Check level of negative peak.
.13	- Turn off TST2 signal, verify 10YLTCH is 1 (latched).
.14	- Verify 1KILLLD = 1.
.15	- Reset faults, verify 10VLTCH is 0.
.16	- Verify 1KILLLD = 0.
24	NCCD_ANALOG
.1	-Verify 1GNDF2 = 0
.2	-Verify 0IFLT = 1
.3	-Verify10VLTCB = 0
.4	-Verify 1KILLD = 0
.5	-Verify 1CAPFLT = 0
.6	-Verify 0CIBERR = 1
.7	-Verify IFB(-) is Approximately 0.0 VDC
.8	-Take OTST1 low, verify OIFLT is still 1
.9	-Turn on source current signal. Verify 0IFLT is 0

TEST	DESCRIPTION
.A	Verify 1GNDF2 is 0
.B	-Verify 0CIBERR is 0
.C	-Turn off source current, verify 0IFLT is 1
.D	-Verify 1GNDF2 is 0
.E	-Verify 0CIBERR is 1
.F	-Turn on load current signal, verify 0IFLT is 0
.10	-Turn off load current, verify 0IFLT is 1. Also set 0TST1 high
.11	-Turn off TST2 signal, verify 10VLTCH is 1
.12	-Verify 1KILLLD = 1
.13	-Turn off TST2 signal, verify 10VLTCH is 1 (Latched)
.14	-Verify 1KILLLD = 1
.15	-Reset faults, verify 10VLTCH is 0
.16	-Verify 1KILLLD = 0
25	HMPG_INTR_
.1	- Enable test oscillator; verify occurance of source zero crossing interrupt (slave intr 0).
.2	- Verify frequency of source zero crossing intr.
.3	- Enable test oscillator; verify occurance of load zero crossing interrupt (slave intr 1).
.4	- Verify frequency of load zero crossing intr.
.5	- Verify occurance of source firing interrupt.
.6	- Verify occurance of load firing interrupt.
.7	- Verify occurance of field firing interrupt.
26	NLCB_SPEED
.1	- Enable test oscillator; verify speed circuits reads 55 Hz.
.2	- Verify no overspeed condition.
27	HISA_PRES
.1	- Disable IMOK; verify by reading back bit.
.2	- Enable IMOK; verify by reading back bit.
28	CABLE_CONT
.1	- Tests for continuity in ribbon cable from SEM,JA to I/O Module.
.2	- Tests for continuity in ribbon cable from SEM, JK to I/O Module.
.3	- Tests for continuity in ribbon cable from SEM, JH to I/O Module.
.4	- Tests for continuity in ribbon cable from SEM, JB to Door Module.

TEST DESCRIPTION

29 HISA_IO
- .1 - Set all inputs from I/O Module high; verify by reading back inputs to HISA.
- .2 - Set all inputs from I/O Module low; verify by reading back inputs to HISA

2A NCCD_XDCD_FDBK_OK
- .1 -Check #DDCDS in XDCD against EEPROM variable XDCD configuration E
- .2 -Check CPS is OK - have output voltage
- .3 -Check that GTO gate drivers are in "off" position

2B HMPG INTERRUPT 7
- .1 Check master interrupt 7

2C Load NSFC Pulse Train
- .1 Check load side NSFC pulse train

2D P105_SUPPLY
- .1 - Check that 105 volt dc power supply is OK.

2E HROA_1A_IO: In sequence, pick-up single relays on the HROA in slot 1A of the I/O module. Expect feedback signal to indicate change.
- .1 -Force all HROA(1A) relays to drop-out; verify by feedback.
- .2 -Force 105 Vdc relays to pick-up; verify by feedback.
- .3 -Drop out 105 Vdc relays, wait 40 ms, verify by feedback.
- .4 -Pick-up single 28 V relay; verify by feedback.
- .5 -Drop-out all relays; verify by feedback.
- .6 -Pick-up single 28 V relay; verify by feedback.
- .7 -Drop-out all relays; verify by feedback.
- .8 -Pick-up single 28 V relay; verify by feedback.
- .9 -Drop-out all relays; verify by feedback.
- .A -Pick-up single 28 V relay; verify by feedback.
- .B -Drop-out all relays; verify by feedback.

2F HROA_1B_IO: In sequence, pick-up single relays on the HROA in slot 1B of the I/O module. Expect feedback signal to indicate change.
- .1 -Force all HROA(1A) relays to drop-out; verify by feedback.
- .2 -Force 105 Vdc relays to pick-up; verify by feedback.
- .3 -Drop out 105 Vdc relays, wait 40 ms, verify by feedback.
- .4 -Pick-up single 28 V relay; verify by feedback.
- .5 -Drop-out all relays; verify by feedback.
- .6 -Pick-up single 28 V relay; verify by feedback.
- .7 -Drop-out all relays; verify by feedback.
- .8 -Pick-up single 28 V relay; verify by feedback.
- .9 -Drop-out all relays; verify by feedback.
- .A -Pick-up single 28 V relay; verify by feedback.
- .B -Drop-out all relays; verify by feedback.
- .C -Pick-up single 28 V relay; verify by feedback.
- .D -Drop-out all relays; verify by feedback.
- .E -Pick-up single 28 V relay; verify by feedback.
- .F -Drop-out all relays; verify by feedback.

30 NLCB_R_DVR
- .1 - Verify presence of HROA in slot 1B of I/O Module.
- .2 - Verify presence of HROA in slot 1A of I/O Module.
- .3 - Force relay drivers on NLCB on; verify response from I/O Module via feedback.
- .3 - Force relay drivers on NLCB on; verify response from I/O Module via feedback.

31 DOOR_MOD_PRES
- .1 - Test for CTS asserted.
- .2 - Test for DSR asserted.

After the microcomputers have successfuly powered up and/or self-test operation have been finished successfuly, the printer can then display faults/alarms as described below.

7-9. FAULT/ALARM MONITOR

Built into the electronics software module are fault/alarm monitors.

7-10. FAULT DETECTION

In the event of a fault condition, the IMD will shut down and a fault indication will appear on the SEM. The cause of the shutdown will be displayed on the printer terminal located on the control panel door. If more than one fault has occurred, the fault finder will store and print the subsequent faults (up to 10 total).

To reset the control after a fault condition, it is necessary to press the reset switch on the HLCB card in the SEM on the reset pushbutton on the control panel door. But first record the fault/alarm/relay status, and I/O

states. If the fault condition is clear, the fault light will go out after a door reset, and stay out when the reset is completed.
HMPG INTERRUPT 7_ .1 Check Master Interrupt 7

Consult the system elementary ladder diagram for a complete list of faults. Most of the faults are standard and are defined in the following paragraphs, along with suggested corrective action.

LOSS OF COOLING FAN- This fault indicates a loss of bridge cooling fan(s). In units having redundant fans, this fault occurs only after both the normal and backup fans have failed.
Possible action to be taken:

1. Replace fan motor fuses
2. Replace defective fan motor (and/or centrifugal switch)
3. Replace HRDA boards
4. Replace I/O module

SOURCE, LOAD, MASTER, SLAVE HARDWARE OVERCURRENT - This fault indicates that the source or load side current has exceeded an acceptable level for more than 26 milliseconds. Under normal circumstances, the electronic control can reduce current to zero in less than 26 milliseconds. Actuation of the hardware overcurrent generally indicates a severe problem from which the control cannot recover.

Possible action to be taken:

1. Check converter bridges for shorted cells
2. Replace NLIB card

MICROPROCESSOR STALL - This microprocessor has failed to execute its program in a normal amount of time.

Possible action to be taken:

1. Replace HMPG card
2. Replace NLCB and NLIB cards
3. Replace HLCB card

LOAD, SOURCE, MASTER, SLAVE SOFTWARE OVERCURRENT - Source and load current are compared in software to an adjustable overcurrent threshold. If the current exceeds this threshold, both source and load converters will act to reduce the current to zero. If it is not possible to reduce current to zero and regain control, a fault will be declared and the drive will be tripped.

Possible action to be taken:

1. Go into gate-test-mode and look for missing pulse
2. Replace in order NLCB, NLIB, HAIA cards
3. Reset and try again

SOURCE, LOAD OVERVOLTAGE - Indicates an excessive level of voltage on the source or load ac buses.

Possible action to be taken:

1. If source overvoltage, check incoming voltage from switchgear.
2. Replace HLCB, NLIB cards

OVERSPEED - Indicates excessive motor speed has occurred.

Possible action to be taken:

1. Replace NLCB PWB
2. Check attenuators

GROUND FAULT - Indicates a ground exists in the converter bridges, transformer, reactor, or motor. The ground detector will also actuate if the load converter ac line filter circuits are unbalanced due to an open fuse or defective capacitor.

Possible action to be taken:

1. Check ac and dc filter fuses and capacitors in bridges
2. Check for grounds in motor, transformer, reactor (megger)
3. Replace NLCB PWB

For other possible faults and solution patterns see Appendix C.

7-11. ALARM DETECTION

Alarms are conditions which are cause for concern but which do not warrant immediate shutdown of the drive but should be investigated as soon as possible. A composite alarm signal is generally supplied to operate a light on the control panel door, a remote annunciator supplied by the user, or both.

Some of the alarms will vary from one installation to the next, so the software ladder diagram section of the system elementary must be consulted for a complete list of. The following alarms are standard (if applicable) and so will be defined here.

LOSS OF SPEED REFERENCE - When the drive is controlled from a 4 - 20 milliamp speed reference, a check is made to determine if this signal has gone to zero (Less than 4 mA). If so, an alarm is declared and the drive speed is locked at its present value.

The remaining alarms to be defined deal with unused or unexpected interrupts. In case one of these interrupts occurs, software is present to record data, declare an alarm, and then return to normal processing. The data saved upon execution of one of these interrupts is stored in a stack in RAM memory. There is also a counter for each of these interrupts so that the number of occurrences can be logged. The stack variable and event counter names are given in Table 5-2. The definition of the information contained in the stack is given in Table 5-3. It is not expected that the user would be able to fully utilize this information, but it is given here in the interest of completeness.

Rapid repetition of these interrupts can result in shutdown of the system but it has been shown that the system may recover from these if they are infrequent. The cause of these interrupts is generally due to noise or other unusual conditions stated in the following definitions.

I/O ADDRESSING - Whenever any memory or I/O device is addressed by the microprocessor, it is expected that that device will respond with an acknowledge signal. If an acknowledge is not received, the microprocessor will wait for up to 80 microseconds, generate a bus hog interrupt, and then proceed to the next task. Failure to receive this acknowledge indicates that some memory or I/O device is "hogging" the data bus, and an alarm is declared. Actually, addressing an undefined address with the terminal will invoke a bus hog alarm. The bus hog alarm may be tested in this manner by executing the following command:

DW500:500 CR

Continued receipt of the bus hog alarm usually indicates that a digital card is defective or not plugged in properly, or a wrong address was requested.

The following are combined to form one composite alarm - MICROPROCESSOR ALARM - DUMMY INTERRUPT (DUM INT) - If any of the masked (disabled) interrupts should occur, a dummy interrupt alarm will occur.

DIVISION BY ZERO INTERRUPT (DIV ZERO) - This is a non-maskable interrupt that is invoked whenever the microprocessor attempts a division by zero.

POWER DOWN INTERRUPT (POW DN) - This is a non-maskable interrupt that places the microprocessor in a special mode where all instructions and data must be one byte in length. This interrupt is generated whenever the result of a calculation exceeds the available register size.

SINGLE STEP INTERRUPT (SING STP) - A single step alarm is an alarm declared whenever the non-maskable interrupt used to single step the microprocessor execution is invoked. Single step mode is not used in the LCI control.

SYSTEM TIMING INTERRUPT (SYS TIM) - This is a non-maskable interrupt from an external timer that is ordinarily used to give a "heartbeat" interrupt for overall system timing functions. This interrupt is not used on the LCI.

USART RECEIVE, TRANSMIT (USART R, USART T) - USART receive, transmit is an interrupt that is generated whenever the USART fails to indicate that it is ready to receive, transmit data. These interrupts are not used in the LCI control.

Table 7-2.

INTERRUPT ALARM COUNTER AND STACK VARIABLES

ALARM	:STACK ARRAY VARIABLE	: TRAP CNTR BH
DUM INT	: SAVE STACK DUM	: TRAP CNTR DUM
DIV ZERO	: SAVE STACK 0	: TRAP CNTR 0
POW DN	: SAVE STACK 2	: TRAP CNTR 2
ONE B	: SAVE STACK 3	: TRAP CNTR 3
OV FL	: SAVE STACK 4	: TRAP CNTR 4
SING STP	: SAVE STACK 1	: TRAP CNTR 1
SYS TIM	: SAVE STACK SYS TIM	:TRAP CNTR SYS TIM
USART R	: SAVE STACK US R	: TRAP CNTR US R
USART T	: SAVE STACK US T	: TRAP CNTR US T

NOTE: In order to access information in the stacks, it is necessary to obtain the address of the desired SAVE STACK variable from a set of memory maps. Then the data from SAVE STACK ADDRESS to SAVE STACK ADDRESS + 1AH must be printed. To do this, use the following command structure:

<DW SAVE STACK ADDRESS , SAVE STACK ADDRESS + 1AH CR>

NOTE: See paragraph entitled "Monitor Command Structure" for an explanation of < >.

To obtain the trap counter value, the following must be done:

<DW TRAP CNTR ADDRESS CR>

7-12. SECTION B - GATE-TEST-MODE

7-12.1. Gate-Test-Mode

The gate test mode allows off-line testing of much of the analog signal conditioning, microprocessor firing control, and gate pulse generation circuitry. In the gate-test-mode, a local test oscillator supplies 3 phase, approximately 60 Hz signals in place of the attenuated bus signals from the source and load bridges. The electronic control can then lock onto these signals and

Table 7-3.

MAP FOR INTERRUPT SAVE STACKS

STACK OFFSET	REGISTER	DEFINITION
0	BP	
2	ISR	CALL RETURN (OFFSET)
4	ISR	CALL RETURN (SEGMENT)
6	DI	
8	DI	
A	BX	
C	DX	
E	CX	COUNT REGISTER
10	AX	16BIT ACCUMULATOR
12	DS	DATA SEGMENT
14	ES	EXTRA SEGMENT
16	IR	INTERRUPTED ROUTING (OFFSET)
18	IR	INTERRUPTED ROUTING (SEGMENT)
1A	FLAG	FLAG REGISTER

generate signals to "dry fire" the thyristors in both bridges. Thyristor gate pulses may then be examined for proper magnitude and phasing. Firing may be adjusted between advance and retard limit for each bridge. In the gate-test-mode, the control produces pulse train firing, thus it is necessary to run the converter ventilating fans for this test.

To perform the following tests, a dual channel oscilloscope with voltage probes and a current probe is required.

7-12.2. Getting Into Gate-Test-Mode

a. The drive should be stopped, primary switchgear and/or transformer isolation switch (if any), and load isolation device (if any) should be open. Check the software ladder diagram in the system elementary for any other conditions that must be met.
b. Put the NLCB (1E) "TEST" switch (SW1) in the test position.
c. Reset the electronics module by pressing the "RESET" (SW1) switch on the HLCB (2E or 2J).

Note: Always use dc coupling on the oscilloscope channels.

7-12.3. Gate-Test-Mode Checkout

a. Connect channel one of the oscilloscope to NLIB (1C) "FCBR" (TP5) and channel two of HLCB (2E) "VA1" (TP12). Trigger the scope on channel one and verify the waveforms look like figure 7B-1.
b. Connect channel one of the oscilloscope to NLIB (1D) "FCBR" (TP5) and channel two to HLCB (2E) (TP13). Trigger the scope on channel one and verify the waveforms look like figure 7B-1.
c. Connect channel one of the oscilloscope to HLCB (2E) "VA1" (TP12) and trigger on this signal. Adjust the variable time/division scale so that one period equals six major divisions. (i.e, 60 degrees per major division or 12 degrees per minor division).
d. One at a time, connect channel two to SIG1 thru SIG6 on the source side "A" NSFC (2C) (TP1 thru TP6) and verify that the gate pulses are positioned as in figure 7B-2A. Note: The firing angle can be adjusted by varying the speed demand reference (place in "LOCAL" and adjust "LOCAL REF").
e. One at a tine, connect channel two to SIG1 thru SIG5 on the source side NSFC (2A) (TP1 thru TP6) and verify that the gate pulses are positioned as in figure 5-2B. Note: The slave firing lags the master firing by 30°.
f. Connect channel one of the oscilloscope to HLCB (2E) "VA2" (TP13) and trigger on this signal. Adjust the variable time/division scale so that one period equals six major divisions. (i.e., 60 degrees per major division or 12 degrees per minor division).
g. One at a time, connect channel two to SIG1 thru SIG6 on the load side NSFC (2B) (TP1 thru TP6) and verify that the gate pulses are positioned as in Figure 7B-2A.
h. Verify that each of the red gate pulse LEDS on the HPTN cards in the leg cell modules is illuminated indicating that the cell is getting gate pulses.
i. With a current probe, look at the gate pulse of each thyristor while simultaneously monitoring the corresponding gate pulses on the NSFC card. The two signals should line up with each other as shown in Figure 7B-4A.
j. Connect a current probe to the scope and trigger on this channel. Hook the current probe to each of the gate leads (white wire) of the SCRs with the arrow on the current probe pointing away from the gate lead termination point. Verify that the waveform has the same characteristics as Figure 7B-4B. If the pulse fails the criterion, the HPTN card in that leg cell module is most likely the

cause, or the NTDA connections may be reversed.

k. Verify proper fiber optic terminations in the GTO bridge by following this procedure:
   1. Reference elementary sheets for the Signal Distribution XDCD Board (Sht. IDC) and for the GTO Load Converter (Sht. 1AC)
   2. Energize the 400Hz Power Supply. Follow the procedure in Chapter 7 to get into Gate-Test-Mode. The amber GTO "Gating" LED's should now be illuminated.
   3. Pull out the gray "Send" fiber optic links at the XDCD board one at a time and insure the expected "Gating" LED goes out at each GTO driver module in the bridge.
   Example: Pull out 10N1, Z0N1, 30N1, and 161, 162 and 163 respectfully. The should go out in the bridge. Follow this procedure to check out all fiber optic links.

l. Put the NLCB (2E) "TEST" switch (SW1) in the OFF position and reset the electronics module by pressing the "RESET" (SW1) switch on the HLCB (2E).

7-12.4. SECTION C - SOFTWARE REGULATOR BLOCK DIAGRAMS

Following are symbolic block diagrams of various regulators and functions found in the restructured LCI software. Note - most of the regulator diagrams are presented in both "S" domain (analog) and Z Domain (digital). Also, most of the variables listed (in "S" domain)

are available by name through the monitor (Review Chapter 5) either as EEPROM variable (Chapter 4, Section A) or public variables (Chapter 4, Section B).

These diagrams should be useful for understanding the overall control of the drive and for troubleshooting by identifying variable that could be checked or used on the diagnostic list or D/A function.

7-13. SECTION D - TERMINALS

7-13.1. Serial Monitor and Hand Terminal (or other Diagnostic Terminal)

Integrated into the firmware for both source and load electronics is a monitor program which supports manipulation of system memory and I/O. The serial monitor drives an RS-232 port on the front of each microcomputer card. Communication with the microcomputer can be established by connecting the hand-held terminal supplied with the drive to RS-232 port. This connection must be made using the HNMB null modem (supplied) with the electronics.

Table 2
Master Bridge

CORRESPONDING TP ON NSFC	GATE LEAD OF EACH SCR
SIG1(TP1, 2B)	LTM1
SIG1(TP2, 2B)	LTM2
SIG1(TP3, 2B)	LTM3
SIG1(TP4, 2B)	LTM4
SIG1(TP5, 2B)	LTM5
SIG1(TP6, 2B)	LTM6

Table 2A
Master Bridge

CORRESPONDING TP ON NSFC	GATE LEAD OF EACH SCR
SIG1(TP1, 2C)	STM1A
SIG1(TP2, 2C)	STM2A
SIG1(TP3, 2C)	STM3A
SIG1(TP4, 2C)	STM4A
SIG1(TP5, 2C)	STM5A
SIG1(TP6, 2C)	STM6A

Table 2B
Master Bridge

CORRESPONDING TP ON NSFC	GATE LEAD OF EACH SCR
SIG1(TP1, 2A)	STM1B
SIG1(TP2, 2A)	STM2B
SIG1(TP3, 2A)	STM3B
SIG1(TP4, 2A)	STM4B
SIG1(TP5, 2A)	STM5B
SIG1(TP6, 2A)	STM6B

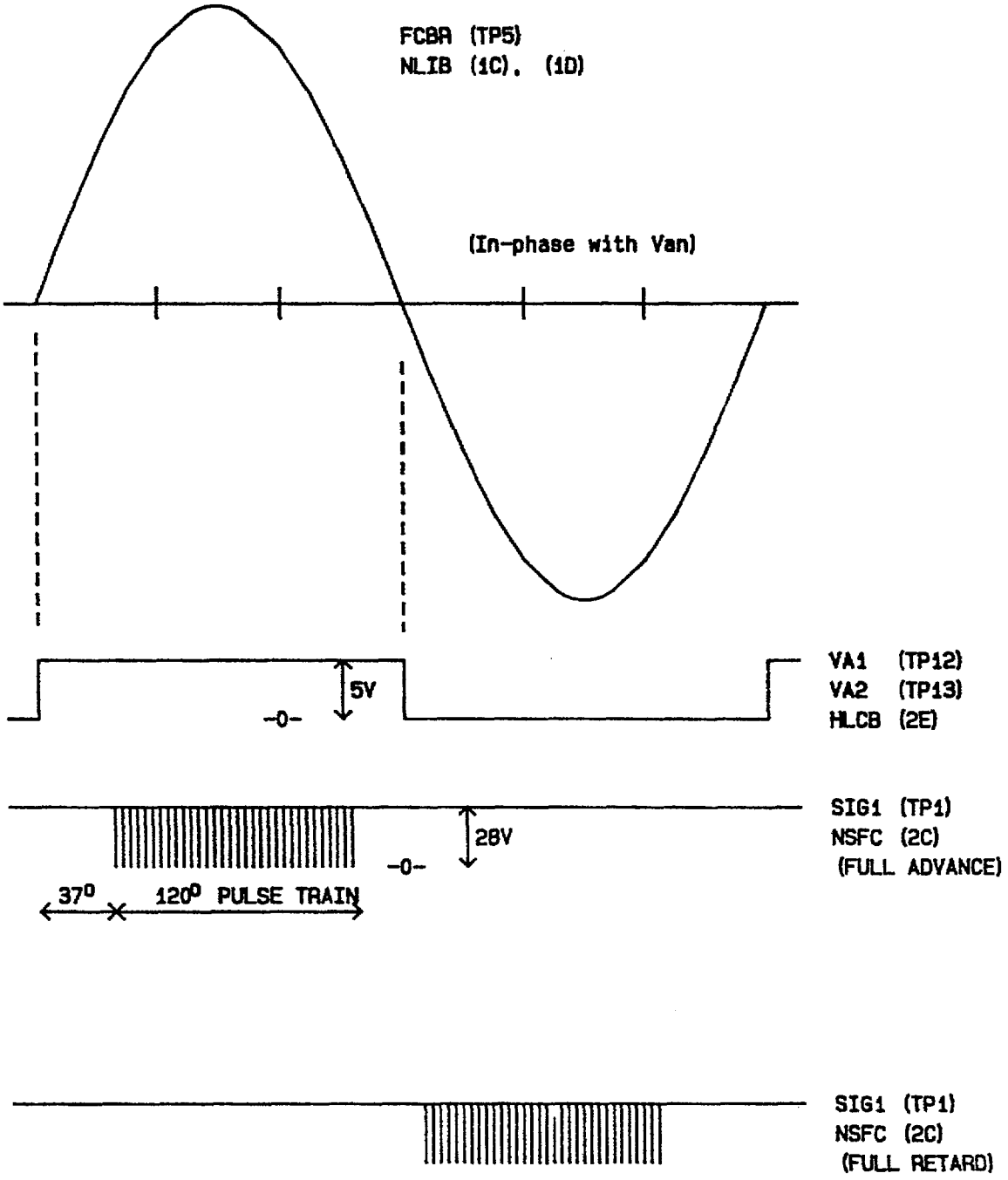


Figure 7B-1. Firing Angle Limits

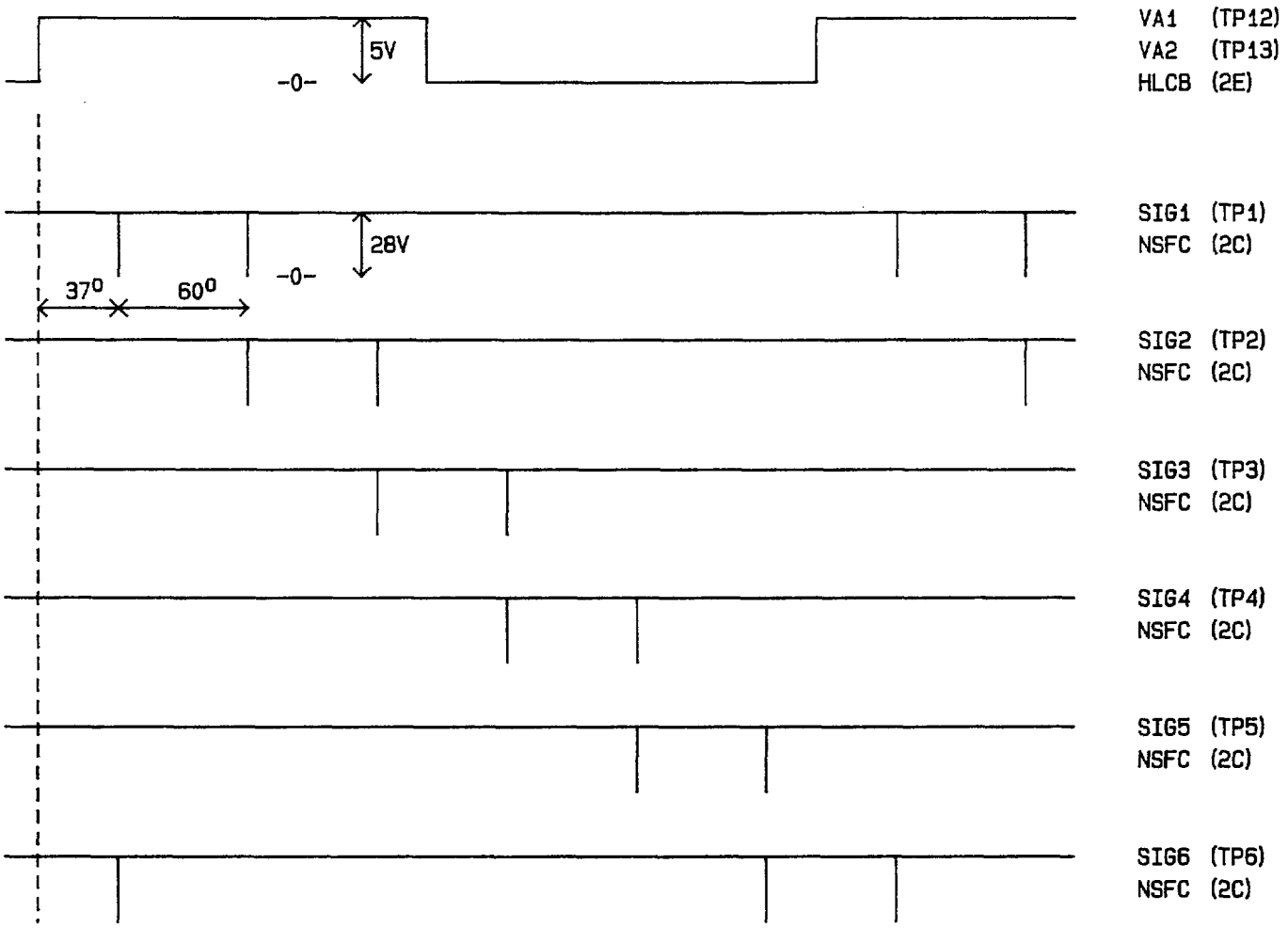


Figure 7B-2A. Source Firing (Master) - Full Advance

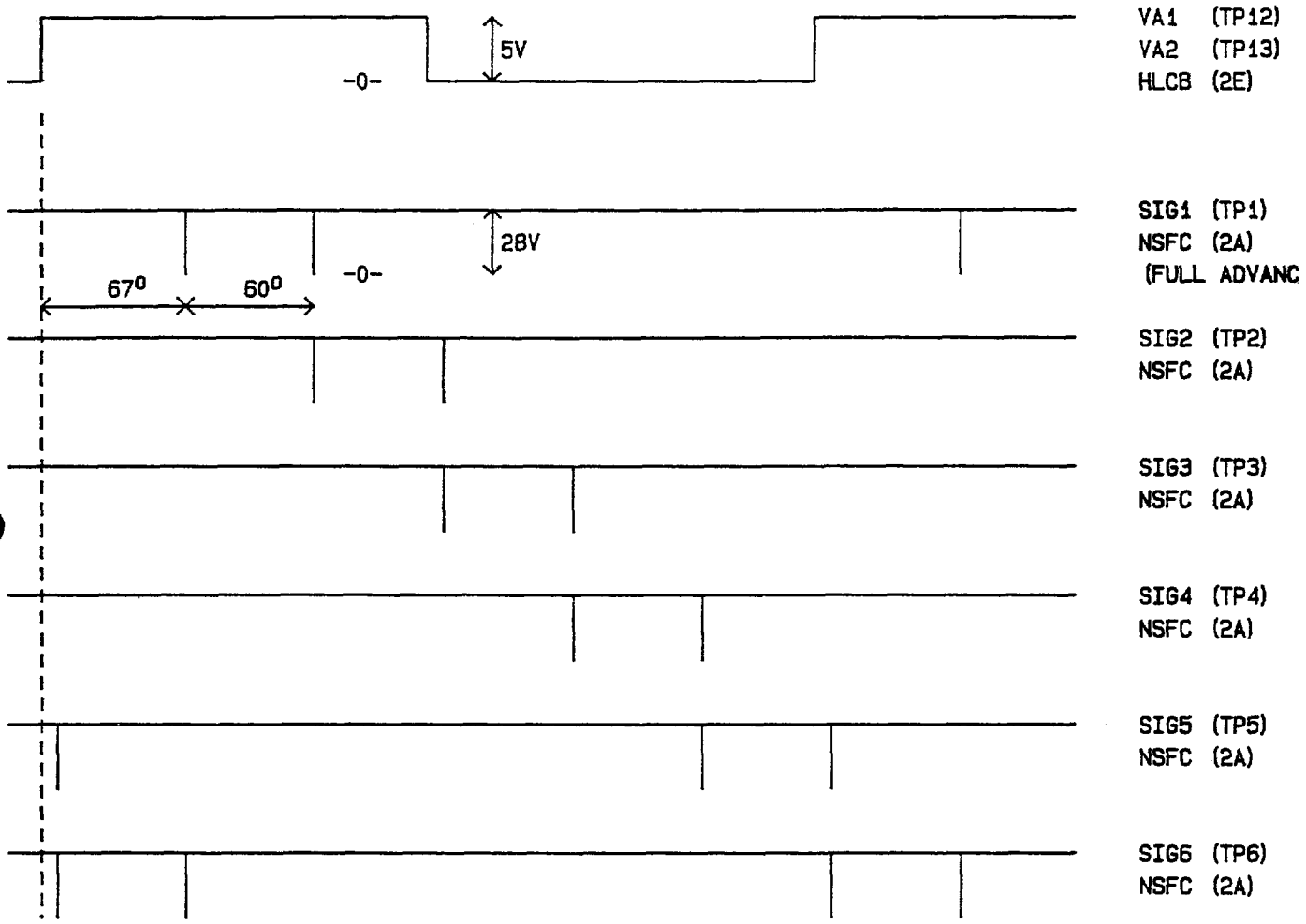


Figure 7B-2B. Source Firing (Slave) - Full Advance

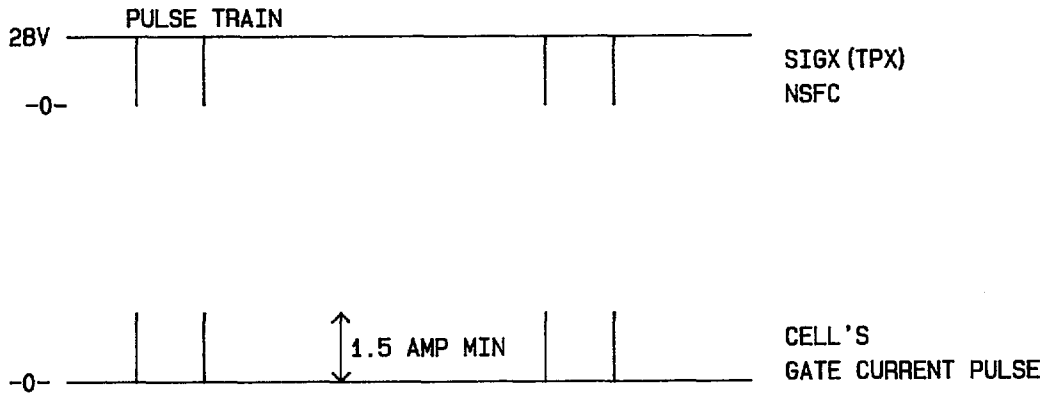


Figure 7-4A.

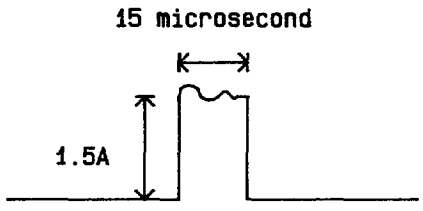


Figure 7-4B. Typical Firing Pulse (Cell Gate Current Pulse)

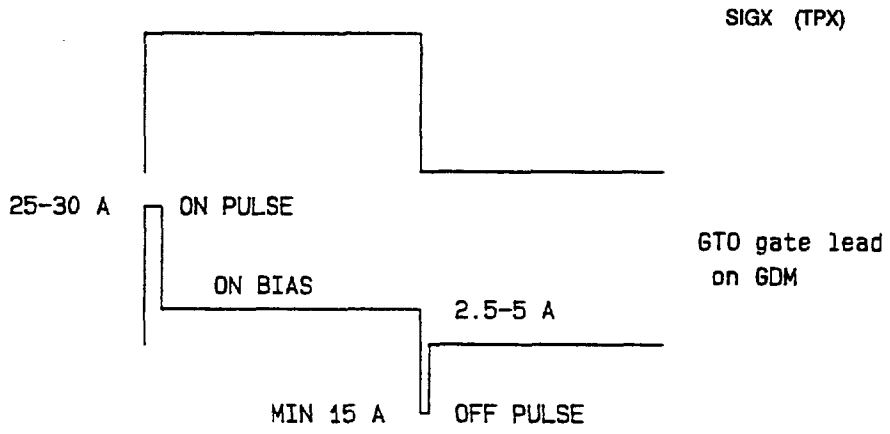


Figure 7B-5B. Typical GTO Gating Signal

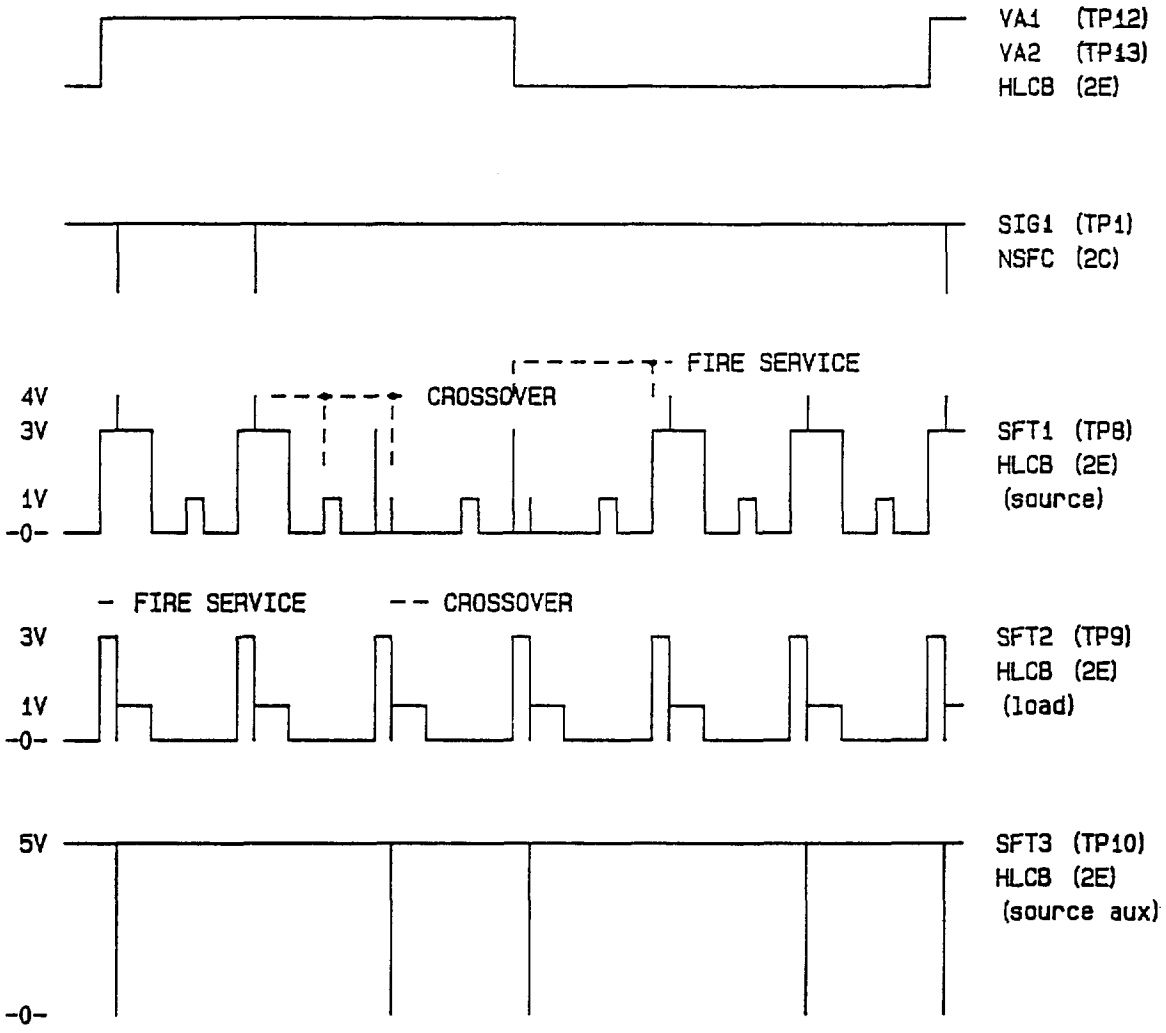


Figure 7B-6. Software Interrupts - Full Advance (Source)

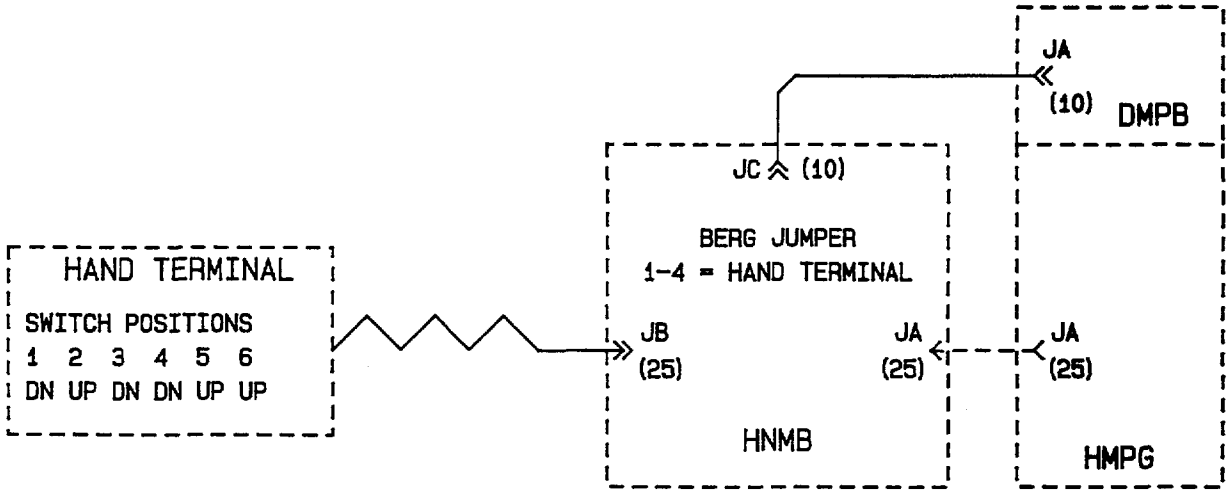


Figure 7-D1. Hand Terminal Connection

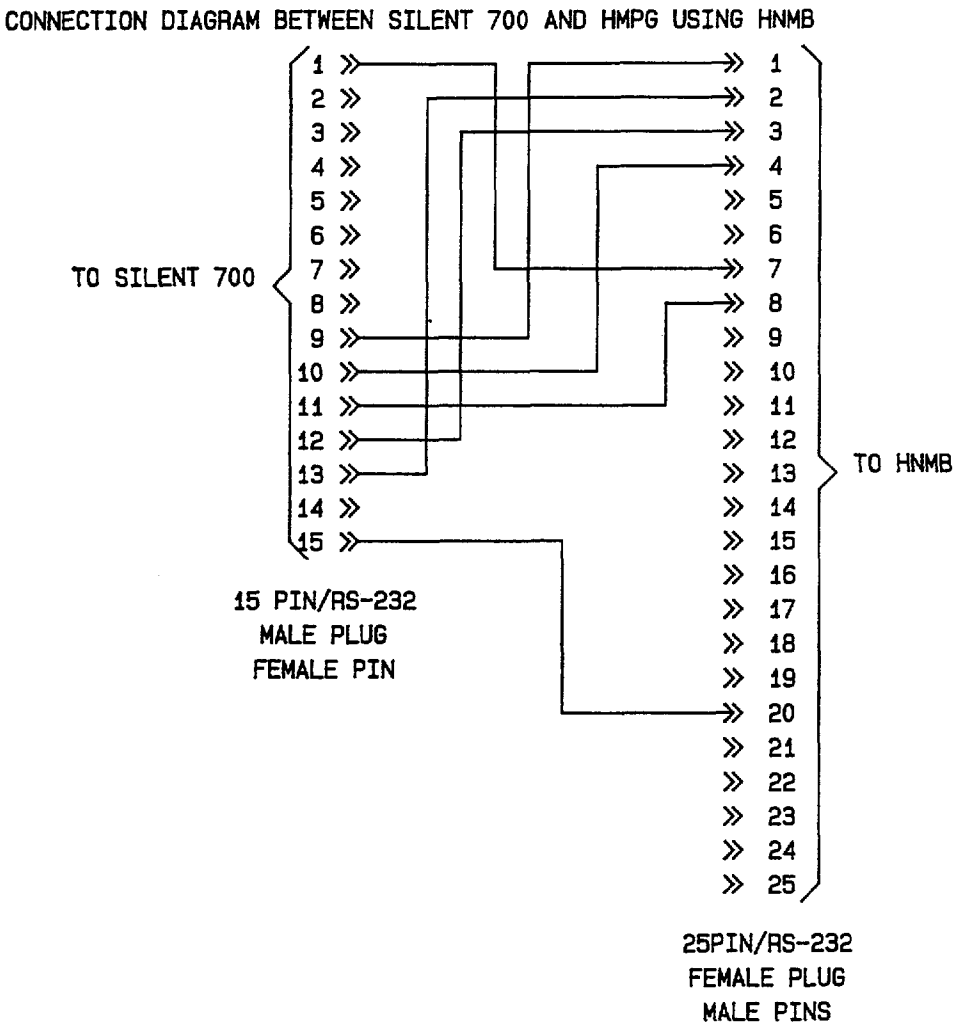

CONNECTION DIAGRAM BETWEEN SILENT 700 AND HMPG USING HNMB
TO SILENT 700
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
15 PIN/RS-232
MALE PLUG
FEMALE PIN
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25
TO HNMB
25PIN/RS-232
FEMALE PLUG
MALE PINS

Figure 7D-2.

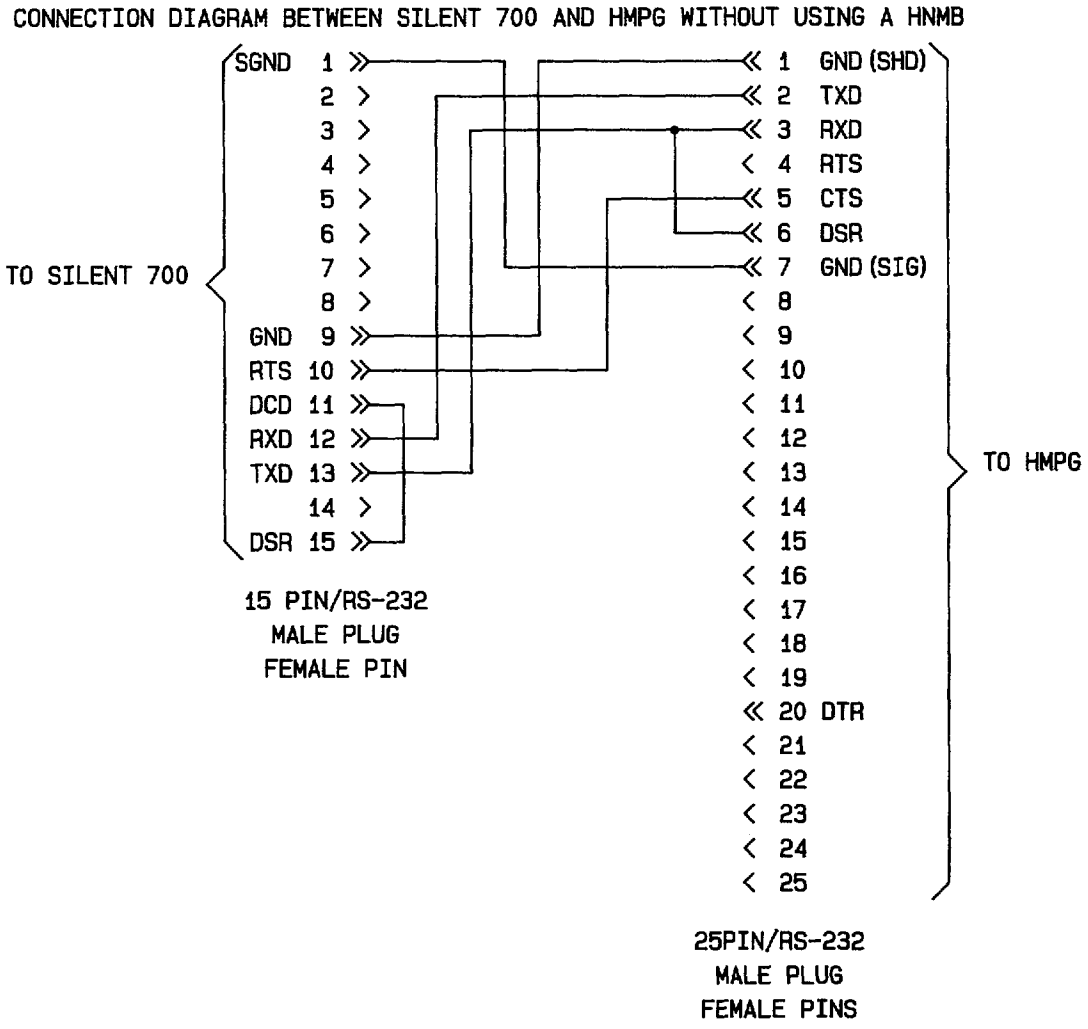

CONNECTION DIAGRAM BETWEEN SILENT 700 AND HMPG WITHOUT USING A HNMB
TO SILENT 700
SGND 1
2
3
4
5
6
7
8
GND 9
RTS 10
DCD 11
RXD 12
TXD 13
14
DSR 15
15 PIN/RS-232
MALE PLUG
FEMALE PIN
1 GND (SHD)
2 TXD
3 RXD
4 RTS
5 CTS
6 DSR
7 GND (SIG)
8
9
10
11
12
13
14
15
16
17
18
19
20 DTR
21
22
23
24
25
TO HMPG
25PIN/RS-232
MALE PLUG
FEMALE PINS

Figure 7D-3.

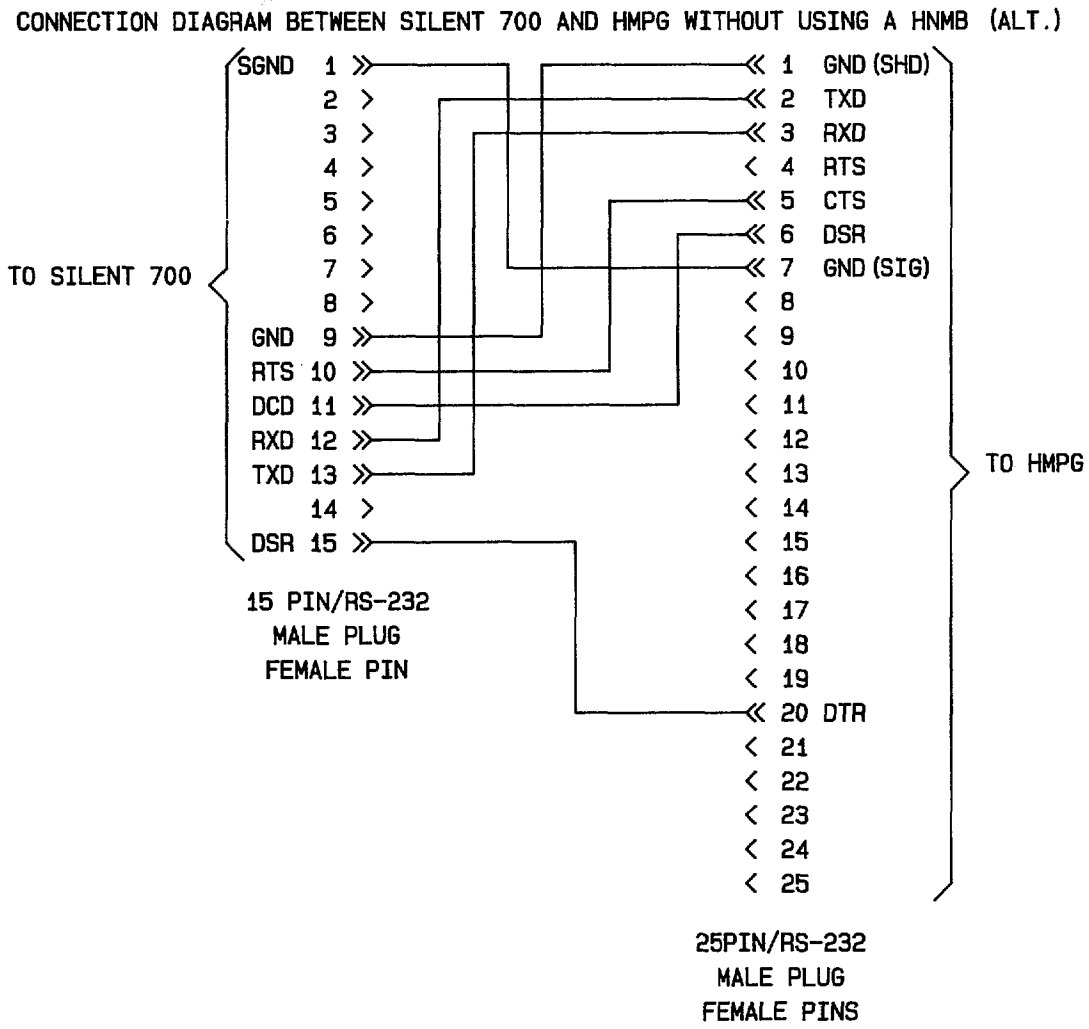
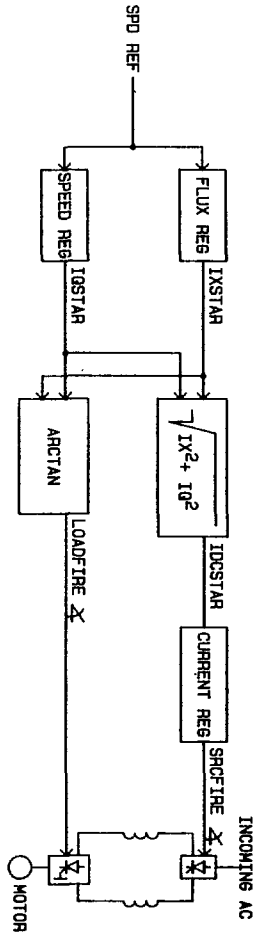
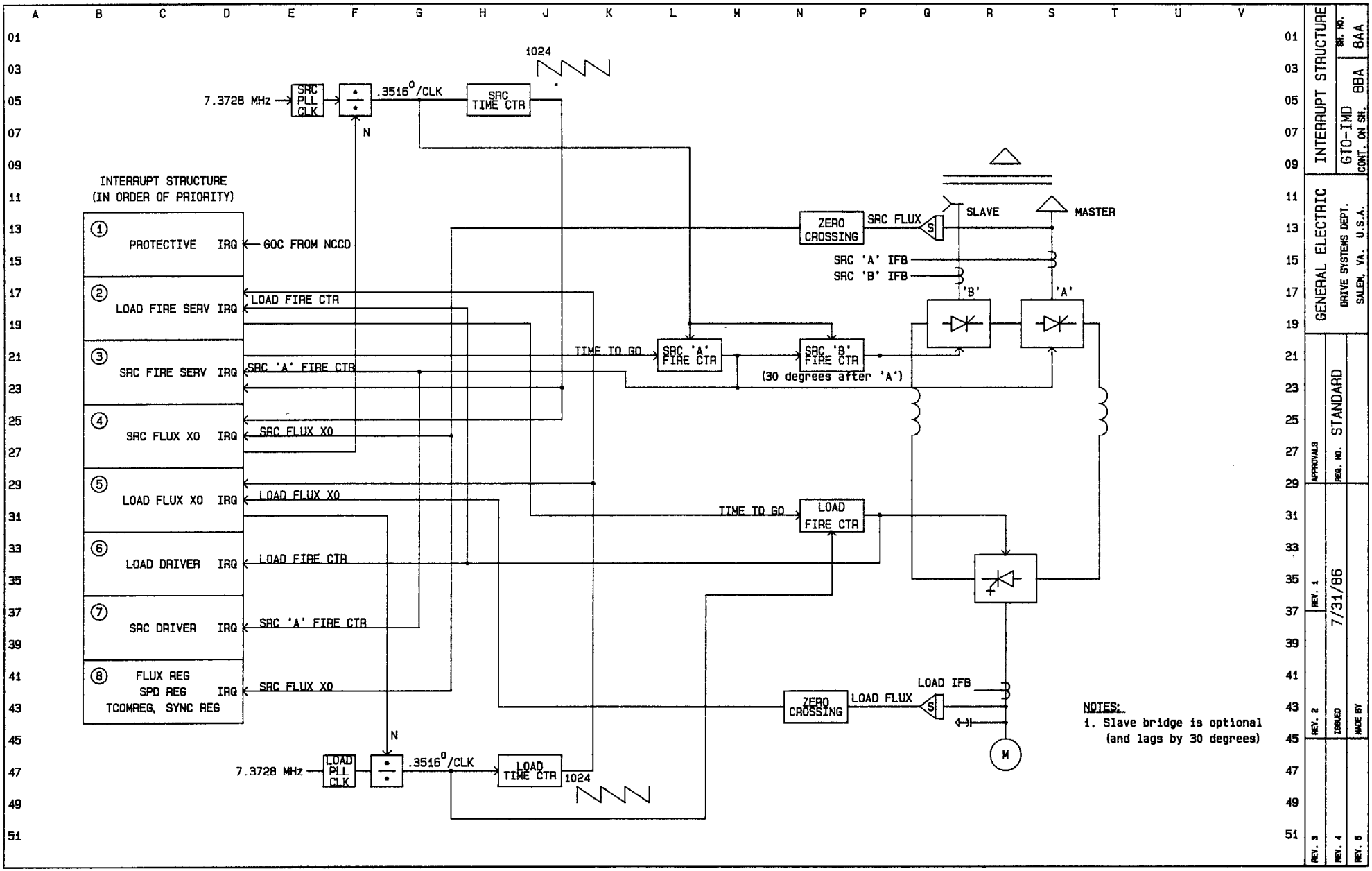
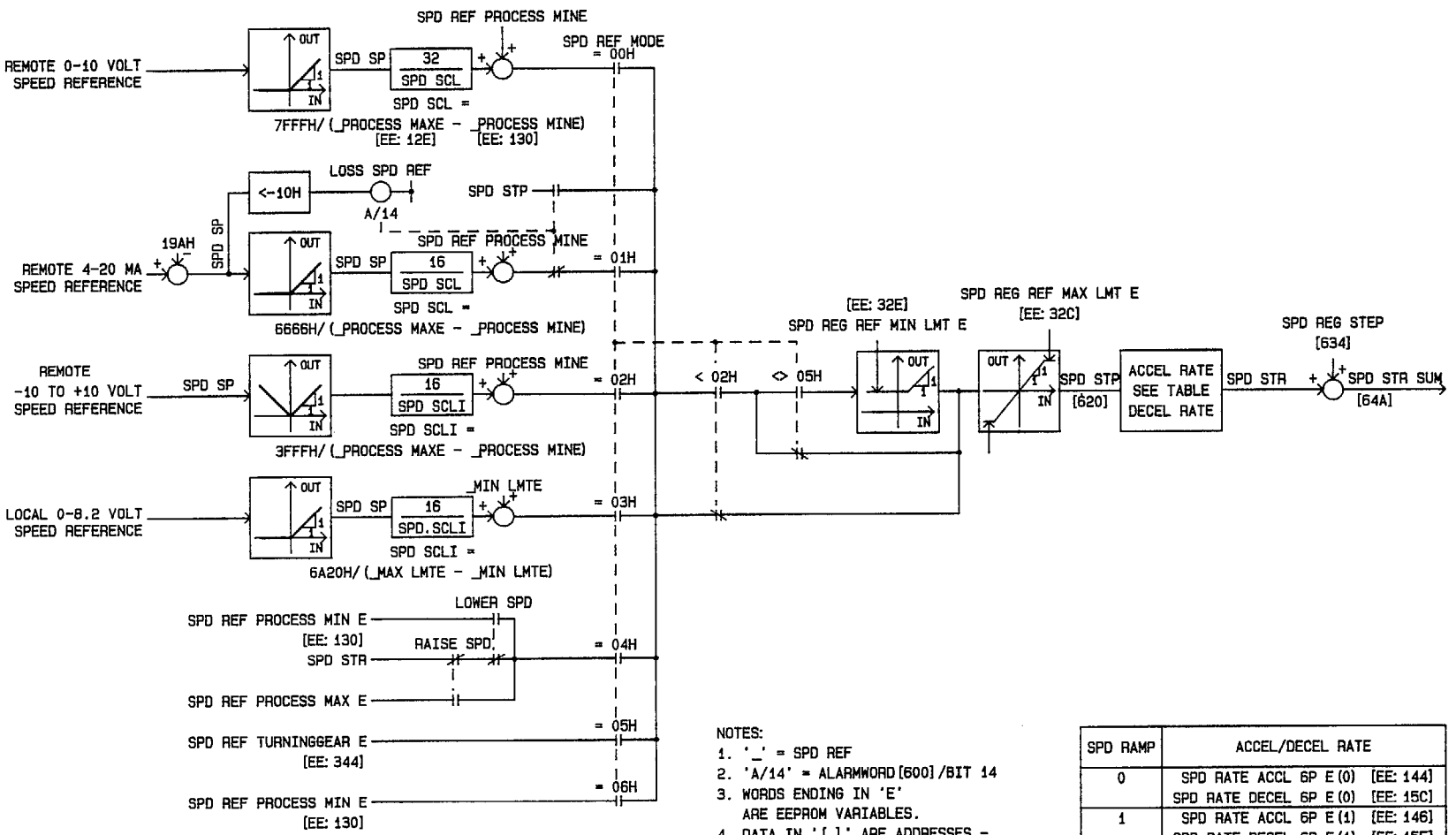

CONNECTION DIAGRAM BETWEEN SILENT 700 AND HMPG WITHOUT USING A HNMB (ALT.)
TO SILENT 700
SGND 1
2
3
4
5
6
7
8
GND 9
RTS 10
DCD 11
RXD 12
TXD 13
14
DSR 15
15 PIN/RS-232
MALE PLUG
FEMALE PIN
1 GND (SHD)
2 TXD
3 RXD
4 RTS
5 CTS
6 DSR
7 GND (SIG)
8
9
10
11
12
13
14
15
16
17
18
19
20 DTR
21
22
23
24
25
TO HMPG
25PIN/RS-232
MALE PLUG
FEMALE PINS

Figure 7D-4



REV. 3	REV. 2	REV. 1	APPROVALS	GENERAL ELECTRIC	ONELINE
REV. 4	ISSUED	7/31/86	REG. NO. STANDARD	DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.	GTO-IMD CONT. ON SH. 8AA / SH. NO. 0AF
REV. 5	MADE BY		PCAD\GTOIMD\SH.0AF		


INTERRUPT STRUCTURE
(IN ORDER OF PRIORITY)
① PROTECTIVE IRQ
② LOAD FIRE SERV IRQ
③ SRC FIRE SERV IRQ
④ SRC FLUX XO IRQ
⑤ LOAD FLUX XO IRQ
⑥ LOAD DRIVER IRQ
⑦ SRC DRIVER IRQ
⑧ FLUX REG SPD REG TCOMREG, SYNC REG IRQ
GOC FROM NCCD
LOAD FIRE CTR
SRC 'A' FIRE CTR
SRC FLUX XO
LOAD FLUX XO
LOAD FIRE CTR
SRC 'A' FIRE CTR
SRC FLUX XO
7.3728 MHz
SRC PLL CLK
.3516° /CLK
SRC TIME CTR
1024
N
7.3728 MHz
LOAD PLL CLK
.3516° /CLK
LOAD TIME CTR
1024
N
TIME TO GO
SRC 'A' FIRE CTR
SRC 'B' FIRE CTR
(30 degrees after 'A')
ZERO CROSSING
SRC FLUX
S
SLAVE
MASTER
SRC 'A' IFB
SRC 'B' IFB
'B'
'A'
TIME TO GO
LOAD FIRE CTR
ZERO CROSSING
LOAD FLUX
S
LOAD IFB
M
NOTES:
1. Slave bridge is optional
(and lags by 30 degrees)
INTERRUPT STRUCTURE
GTO-IMD
CONT. ON SH. 8BA
SH. NO. 8AA
GENERAL ELECTRIC
DRIVE SYSTEMS DEPT.
SALEM, VA. U.S.A.
APPROVALS
REV. NO. STANDARD
REV. 1
REV. 2
ISSUED 7/31/86
MADE BY
REV. 3
REV. 4
REV. 5

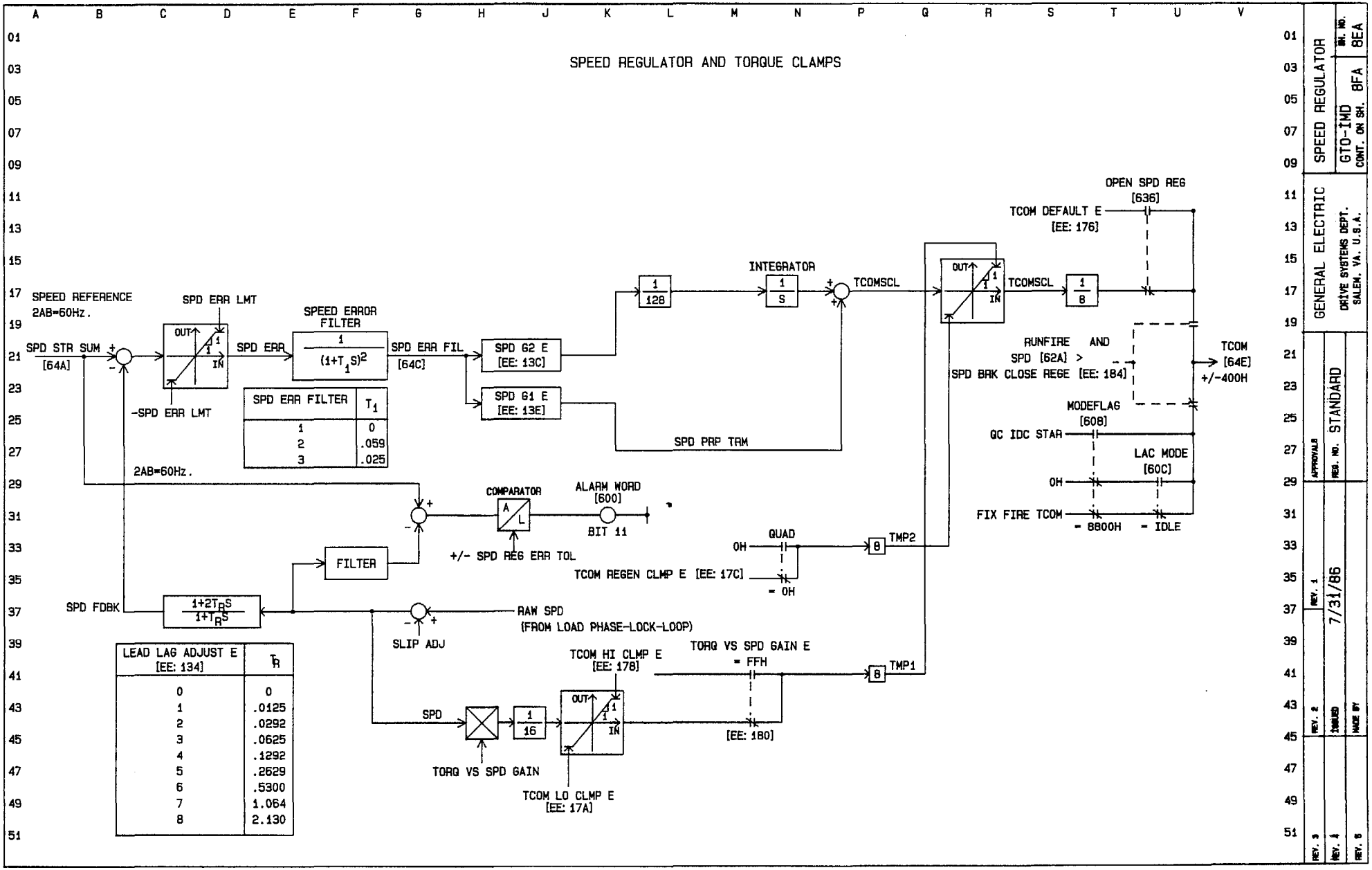


NOTES:
1. '_' = SPD REF
2. 'A/14' = ALARMWORD[600]/BIT 14
3. WORDS ENDING IN 'E' ARE EEPROM VARIABLES.
4. DATA IN '[ ]' ARE ADDRESSES - WITH 'EE' BEING THE EEPROM OFFSET.

SPD RAMP	ACCEL/DECEL RATE
0	SPD RATE ACCL 6P E(0) [EE: 144] SPD RATE DECEL 6P E(0) [EE: 15C]
1	SPD RATE ACCL 6P E(1) [EE: 146] SPD RATE DECEL 6P E(1) [EE: 15E]
2	SPD RATE ACCL 6P E(2) [EE: 148] SPD RATE DECEL 6P E(2) [EE: 160]
3	SPD RATE ACCL 6P E(3) [EE: 14A] SPD RATE DECEL 6P E(3) [EE: 162]
4	SPD RATE ACCL 6P E(4) [EE: 14C] SPD RATE DECEL 6P E(4) [EE: 164]
5	SPD RATE ACCL 6P E(5) [EE: 14E] SPD RATE DECEL 6P E(5) [EE: 166]

SPEED REFERENCE
GTO-IMD 8EA
DWG. NO. 8BA
CONT. ON SH.
GENERAL ELECTRIC
DRIVE SYSTEMS DEPT.
SALEM, VA. U.S.A.
APPROVALS
REQ. NO. STANDARD
7/31/86
REV. 1
REV. 2
REV. 3
REV. 4
REV. 5
ISSUED
MADE BY

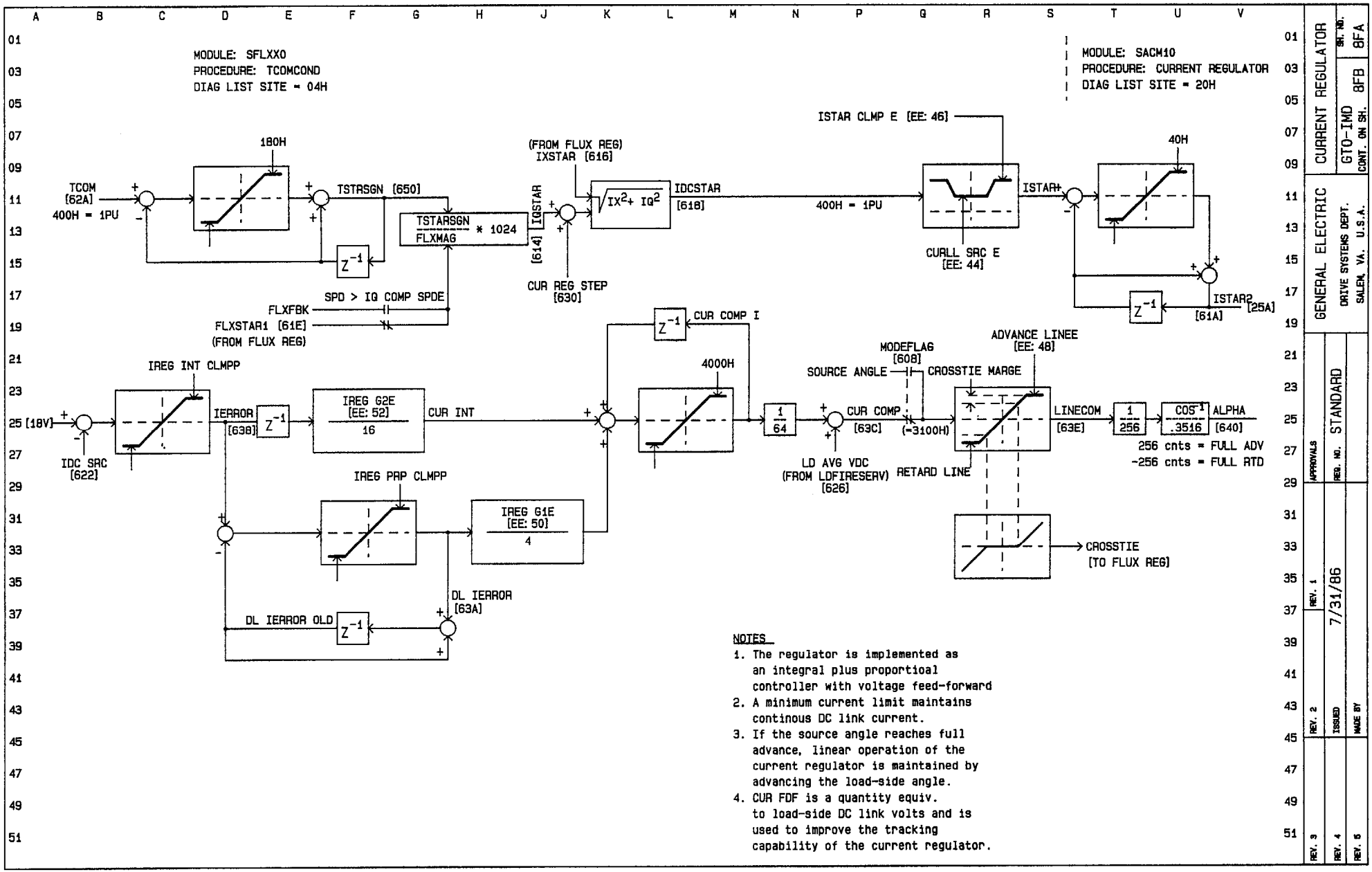
SPEED REGULATOR AND TORQUE CLAMPS



GENERAL ELECTRIC
DRIVE SYSTEMS DEPT.
SALEM, VA. U.S.A.

SPEED REGULATOR
GTO-IMD
CONT. ON SH. 8FA
DW. NO. 8EA

APPROVALS
REQ. NO. STANDARD
REV. 1
7/31/86
REV. 2
ISSUED
MADE BY
REV. 3
REV. 4
REV. 5



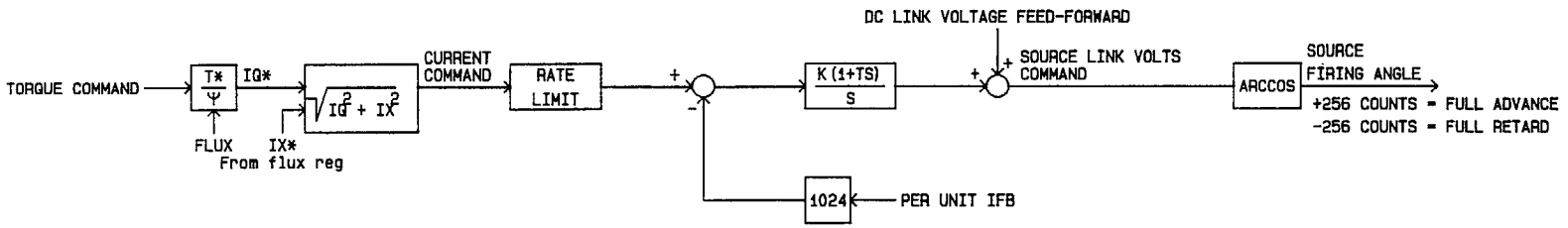
NOTES

1. The regulator is implemented as an integral plus proportioal controller with voltage feed-forward
2. A minimum current limit maintains continous DC link current.
3. If the source angle reaches full advance, linear operation of the current regulator is maintained by advancing the load-side angle.
4. CUR FDF is a quantity equiv. to load-side DC link volts and is used to improve the tracking capability of the current regulator.

GENERAL ELECTRIC
DRIVE SYSTEMS DEPT.
SALEM, VA. U.S.A.

CURRENT REGULATOR
GTO-IMD
CONT. ON SH. 8FB
SH. NO. 8FA

APPROVALS
REV. NO. STANDARD
REV. 1
REV. 2 ISSUED 7/31/86
REV. 3
REV. 4
REV. 5 MADE BY



GENERAL ELECTRIC
DRIVE SYSTEMS DEPT.
SALEM, VA. U.S.A.

CURRENT REGULATOR
GTO-IMD CONT. ON SH. 8GA
SH. NO. 8FB

APPROVALS
REA. NO. STANDARD
REV. 1
REV. 2 ISSUED 7/31/86
REV. 3
REV. 4 MADE BY
REV. 5

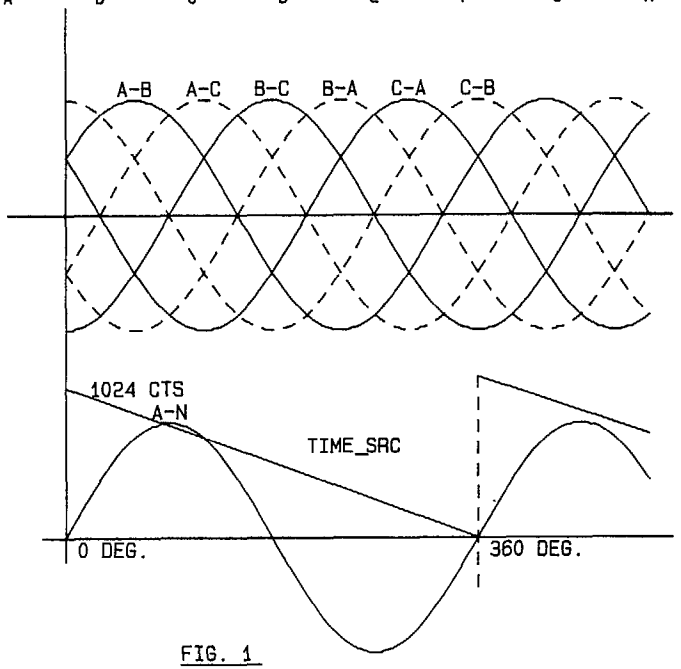


FIG. 1

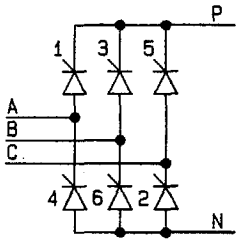


FIG. 2

PH TABLE

L-L	PH	OFFSET (CTS)
A-B	1	3ABH
A-C	2	300H
B-C	3	255H
B-A	4	1ABH
C-A	5	100H
C-B	6	055H

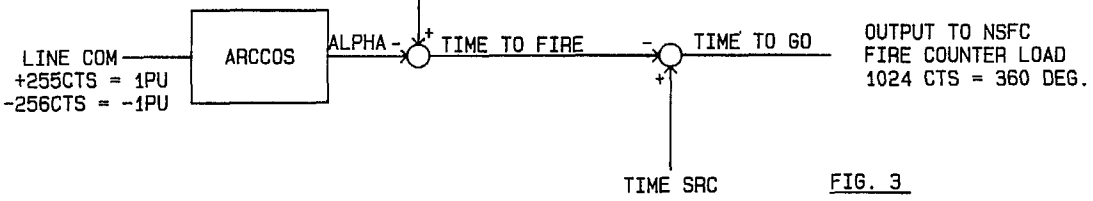


FIG. 3

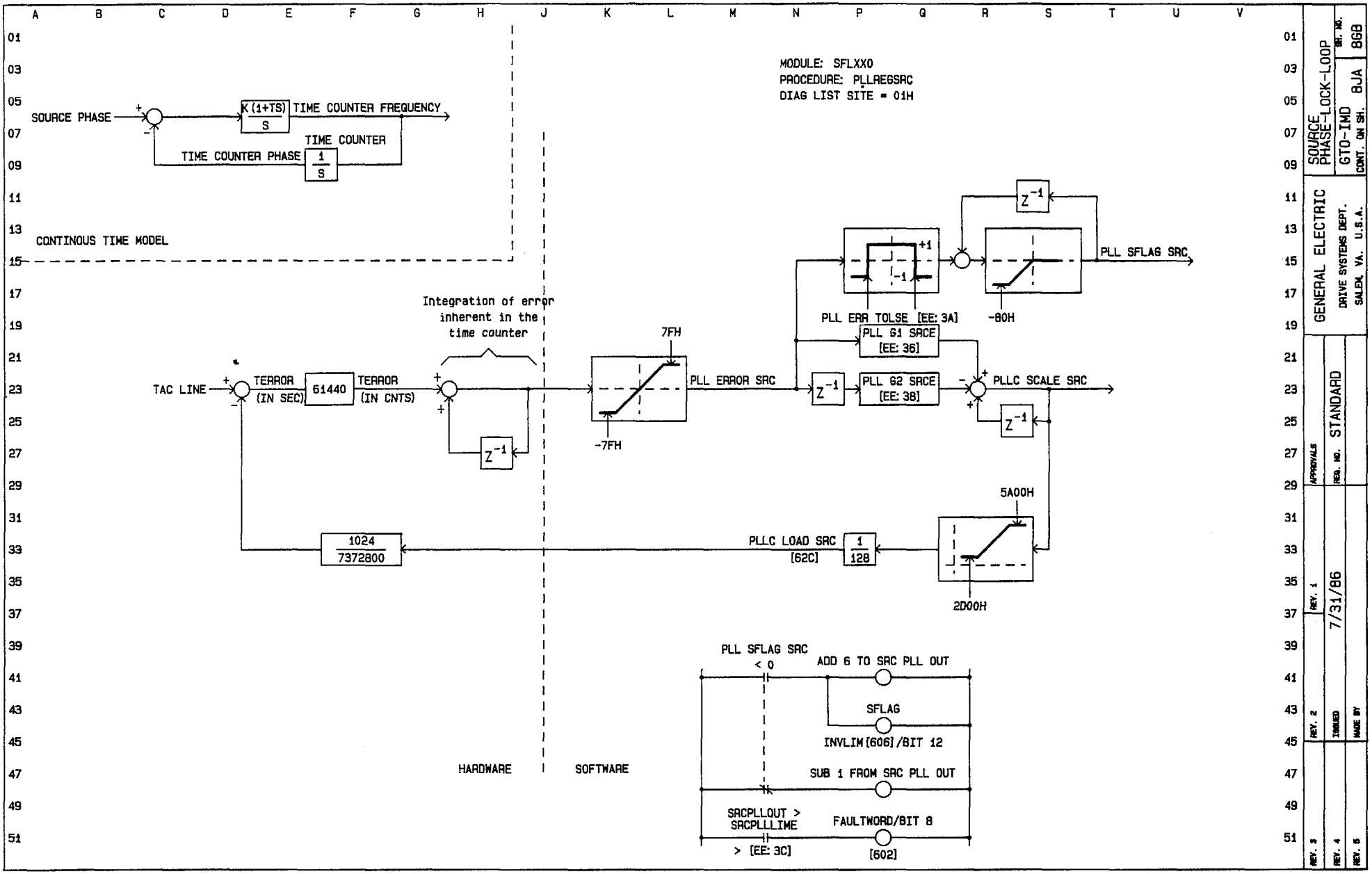
TIME TO FIRE CALCULATION

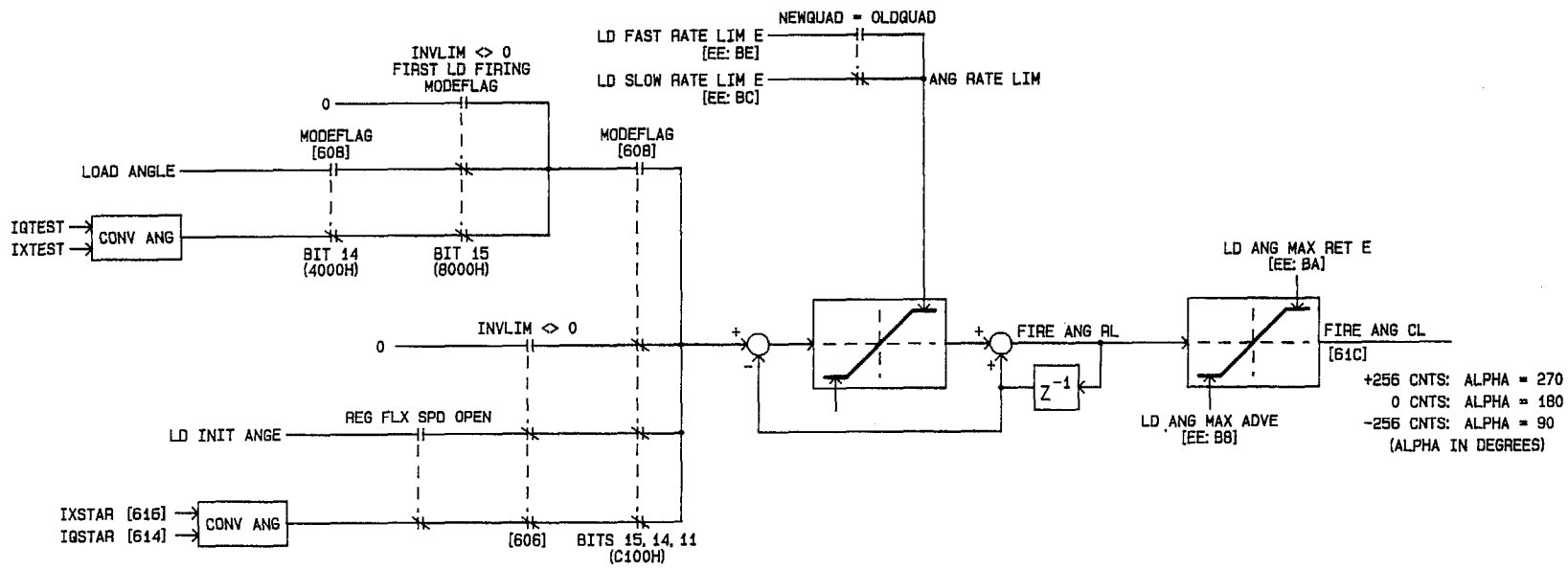
1. THE TIMING FOR CELL FIRING IS ESTABLISHED BY THE SOURCE PHASE LOCK LOOP. IT ACTS TO MAINTAIN SYNCHRONIZM WITH THE BRIDGE VOLTAGE SUCH THAT 360 DEGREES IS DIVIDED INTO 1024 COUNTS (SEE FIG. 1). NOTE THAT THE REFERENCE PHASE IS A-N.
2. EACH CELL PAIR WHICH CONDUCTS ON A GIVEN L-L VOLTAGE IS ASSIGNED A VALUE CALLED "PH". FOR EXAMPLE, CELLS ASSOCIATED WITH A -B ARE CELLS NUMBERED 1 AND 6 (SEE FIG. 2). FROM THE PH TABLE THIS IS A PH OF 1. NOTE THAT THE ASSOCIATED OFFSET IS 55H COUNTS. THIS CORRESPONDS TO 30 DEGREES. THE DEFINITION OF OFFSET CORRESPONDS TO ALPHA = 0 DEGREES (FULL ADVANCE) FOR A GIVEN PH.
3. THE VALUE OF LINECOM CALCULATED BY THE CURRENT REGULATOR IS CONVERTED TO A FIRING ANGLE ACCORDING TO THE EQUATION ALPHA = ARCCOS (LINECOM). THIS IS IMPLEMENTED IN THE SOFTWARE WITH AN INDEXED LOOKUP TABLE.
4. THE VALUE OF ALPHA IS SUB FROM THE OFFSET FOR THE PARTICULAR PH BEING CALCULATED.
   THE RESULTANT IS TIME_TO_FIRE.
5. TIME TO GO (TIME SRC - TIME TO FIRE) IS THE DIFFERENCE BETWEEN PRESENT TIME AND TIME TO FIRE. THE SFTWRE ALLOWS FOR ROLLOVER (ANGLES > 360 DEGREES) AND TOO LATE TO FIRE.
6. THE RESULTING TIME TO GO (IN COUNTS) IS LOADED INTO THE FIRE COUNTER FOR THE CORRESPONDING NSFC. WHEN THE FIRE COUNTER IS DECREMENTED TO ZERO, THE FIRING INTERRUPT OCCURS AND THE NEW CELL MASK IS TRANSMITTED TO THE NSFC.

GENERAL ELECTRIC
DRIVE SYSTEMS DEPT.
SALEM, VA. U.S.A.

SOURCE PHASE-LOCK-LOOP
GTO-IMD 8GB
CONT. ON SH. 8GA

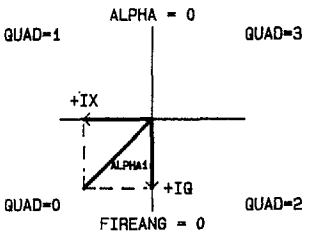
APPROVALS
REQ. NO. STANDARD
REV. 1 7/31/86
REV. 2 ISSUED
REV. 3 MADE BY
REV. 4
REV. 5

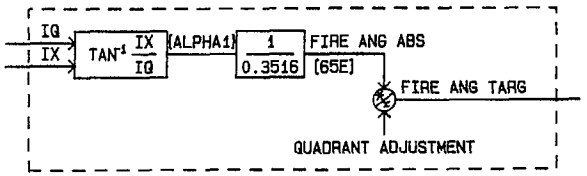

MODULE: SFLXX0
PROCEDURE: PLLREGSRC
DIAG LIST SITE = 01H
SOURCE PHASE
K(1+TS)/S
TIME COUNTER FREQUENCY
TIME COUNTER
TIME COUNTER PHASE
1/S
CONTINOUS TIME MODEL
Integration of error inherent in the time counter
TAC LINE
TERROR (IN SEC)
61440
TERROR (IN CNTS)
7FH
-7FH
PLL ERROR SRC
PLL ERR TOLSE [EE: 3A]
+1
-1
-80H
PLL SFLAG SRC
PLL G1 SRCE [EE: 36]
PLL G2 SRCE [EE: 38]
PLLC SCALE SRC
5A00H
2D00H
PLLC LOAD SRC [62C]
1/128
1024/7372800
HARDWARE
SOFTWARE
PLL SFLAG SRC < 0
ADD 6 TO SRC PLL OUT
SFLAG
INVLIM [606]/BIT 12
SUB 1 FROM SRC PLL OUT
SRCPLLOUT > SRCPLLLIME
> [EE: 3C]
FAULTWORD/BIT 8
[602]
GENERAL ELECTRIC
DRIVE SYSTEMS DEPT.
SALEM, VA. U.S.A.
SOURCE PHASE-LOCK-LOOP
GTO-IMD
CONT. ON SH.
8JA
8GB
APPROVALS
REQ. NO. STANDARD
REV. 1
7/31/86
REV. 2
ISSUED
REV. 3
REV. 4
REV. 5
MADE BY



CONV ANG

IX	IQ	QUAD	QUADRANT ADJUSTMENT
+	+	0	(*1)
+	-	1	(*-1, +/- 200H)
-	+	2	(*-1)
-	-	3	(-200H)





LOAD ANGLE REGULATOR
GTO-IMD 8KA
CONT. ON SH.
SH. NO. 8JA

GENERAL ELECTRIC
DRIVE SYSTEMS DEPT.
SALEM, VA. U.S.A.

APPROVALS
REV. 1
REV. 2
REV. 3
REV. 4
REV. 5

REQ. NO. STANDARD
7/31/86
ISSUED
MADE BY

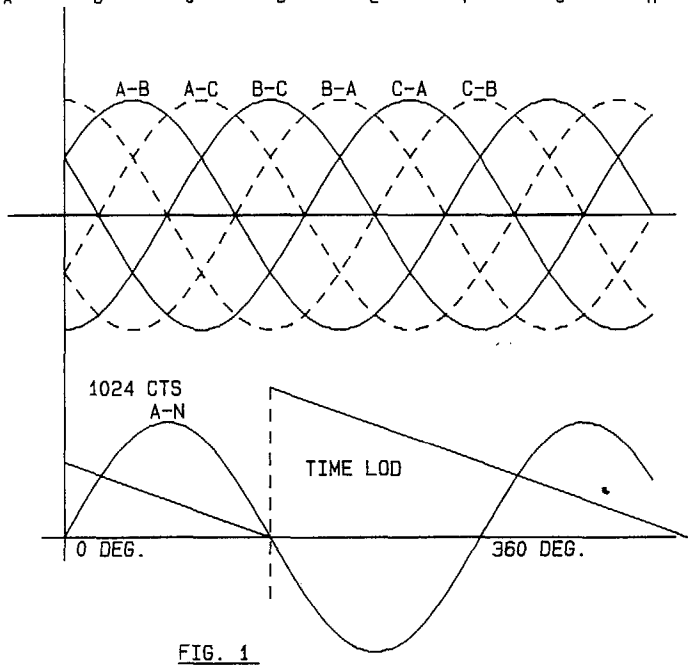


FIG. 1

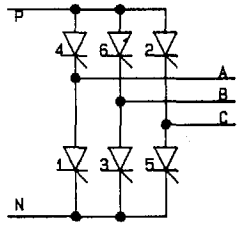
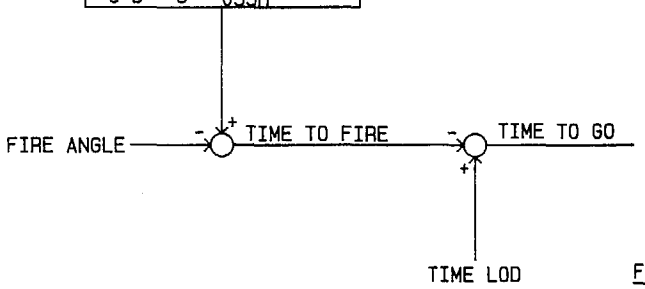


FIG. 2

PH TABLE

L-L	PH	OFFSET (CTS)
A-B	1	3ABH
A-C	2	300H
B-C	3	255H
B-A	4	1ABH
C-A	5	100H
C-B	6	055H



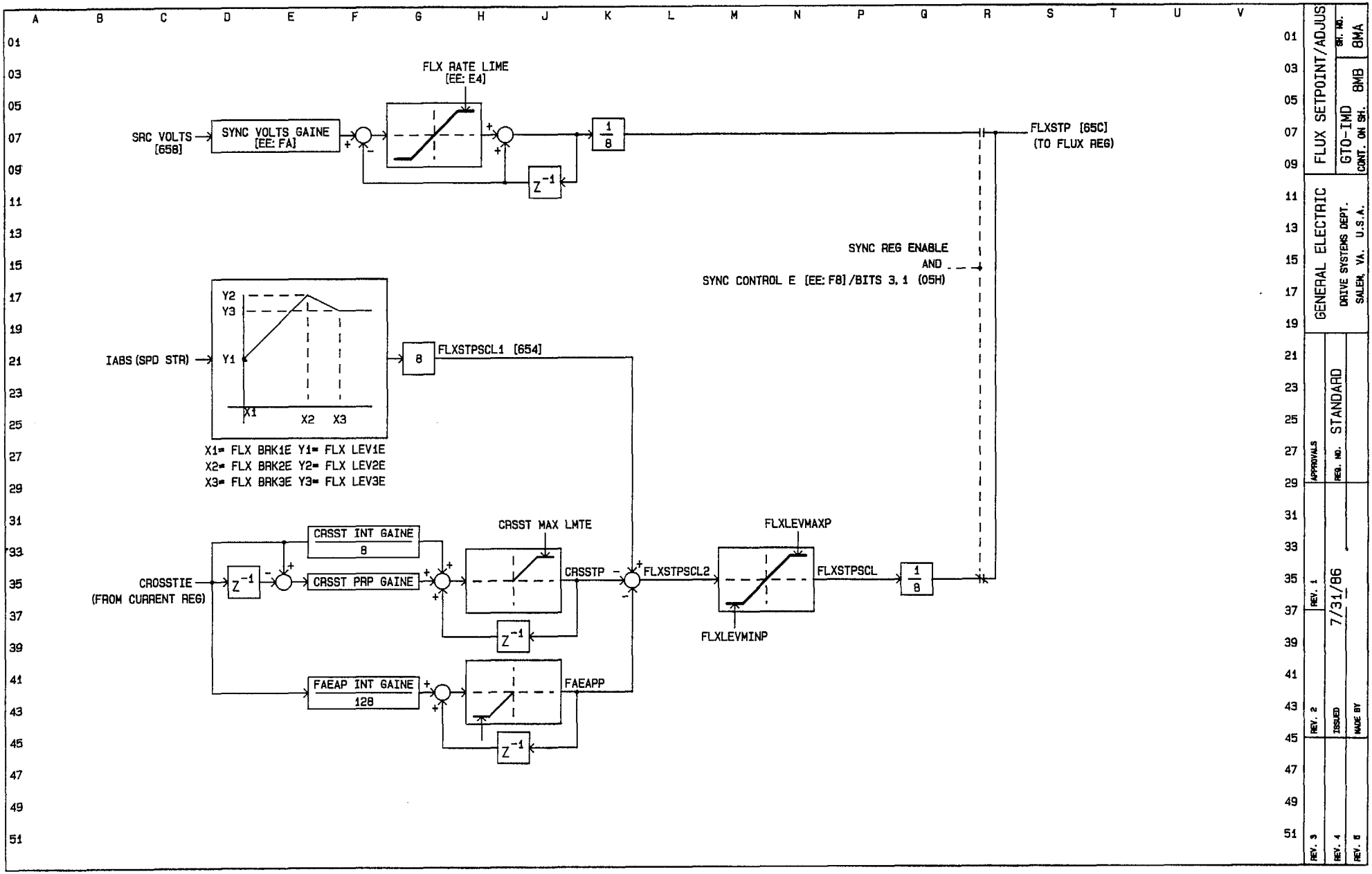
OUTPUT TO NSFC
FIRE COUNTER LOAD
1024 CTS = 360 DEG.

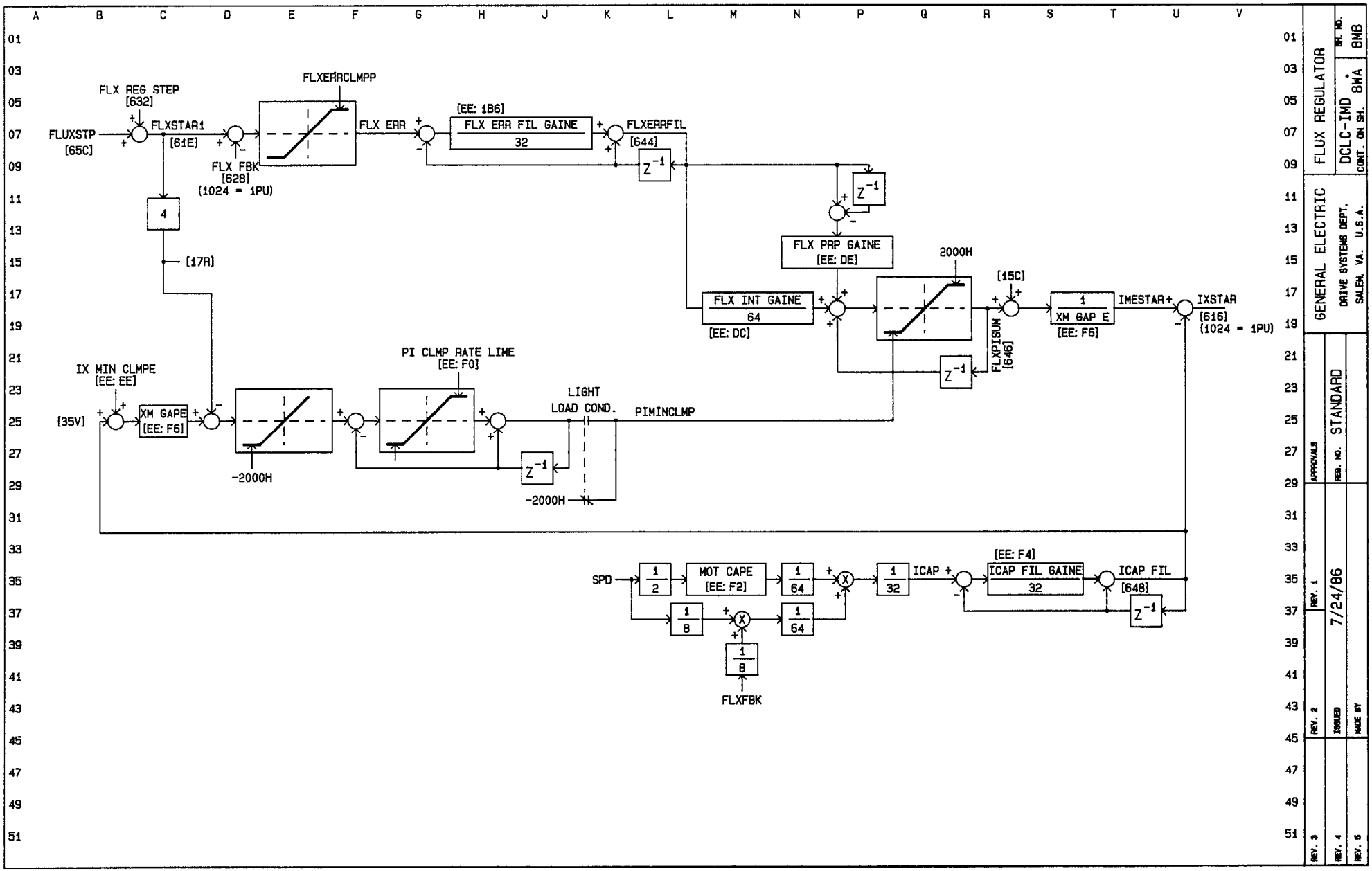
FIG. 3

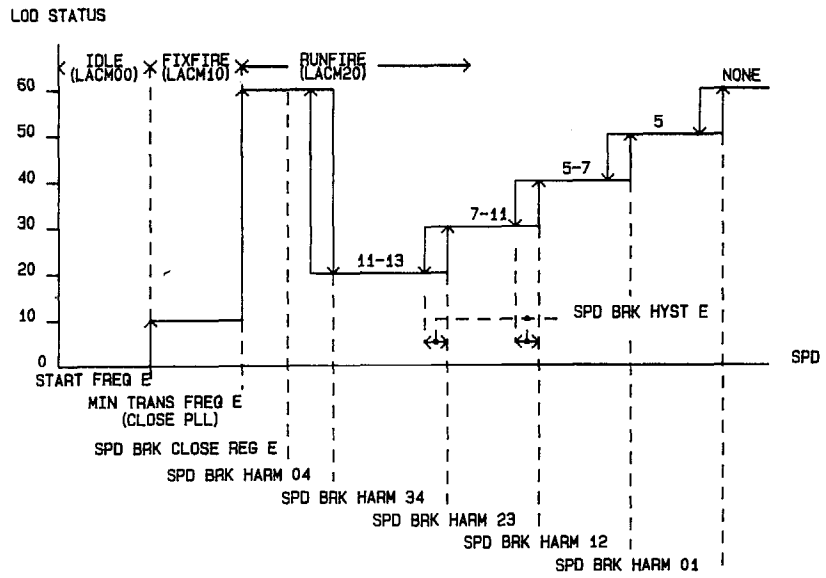
TIME TO FIRE CALCULATION

1. THE TIMING FOR CELL FIRING IS ESTABLISHED BY THE LOAD PHASE LOCK LOOP. IT ACTS TO MAINTAIN SYNCHRONIZM WITH THE MOTOR VOLTAGE SUCH THAT 360 DEGREES IS DIVIDED INTO 1024 COUNTS (SEE FIG. 1). NOTE THAT THE REFERENCE PHASE IS A-N.
2. EACH CELL PAIR WHICH CONDUCTS ON A GIVEN L-L VOLTAGE IS ASSIGNED A VALUE CALLED "PH". FOR EXAMPLE, CELLS ASSOCIATED WITH A -B ARE CELLS NUMBERED 1 AND 6 (SEE FIG. 2). FROM THE PH TABLE THIS IS A PH OF 1. NOTE THAT THE ASSOCIATED OFFSET IS 55H COUNTS. THIS CORRESPONDS TO 30 DEGREES. THE DEFINITION OF OFFSET CORRESPONDS TO ALPHA = 180 DEGREES (FULL MOTOR) FOR A GIVEN PH
4. THE VALUE OF FIRE ANGLE IS SUB FROM THE OFFSET FOR THE PARTICULAR PH BEING CALCULATED
THE RESULTANT IS TIME_TO_FIRE.
5. TIME TO GO (TIME LOD - TIME TO FIRE) IS THE DIFFERENCE BETWEEN PRESENT TIME AND TIME TO FIRE. THE SFTWRE ALLOWS FOR ROLLOVER (ANGLES > 360 DEGREES) AND TOO LATE TO FIRE.
6. THE RESULTING TIME TO GO (IN COUNTS) IS LOADED INTO THE FIRE COUNTER FOR THE CORRESPONDING NSFC. WHEN THE FIRE COUNTER IS DECREMENTED TO ZERO, THE FIRING INTERRUPT OCCURS AND THE NEW CELL MASK IS TRANSMITTED TO THE NSFC.

LOAD PHASE-LOCK-LOOP
GTO-IMD
CONT. DW SH. 8MA
DR. NO. 8KA
GENERAL ELECTRIC
DRIVE SYSTEMS DEPT.
SALEM, VA. U.S.A.
APPROVALS
REV. 1
REV. 2
REV. 3
REV. 4
REV. 5
REQ. NO. STANDARD
7/31/86
ISSUED
MADE BY


FLX RATE LIME
[EE: E4]
SRC VOLTS
[658]
SYNC VOLTS GAINE
[EE: FA]
1/8
FLXSTP [65C]
(TO FLUX REG)
SYNC REG ENABLE
AND
SYNC CONTROL E [EE: F8] /BITS 3, 1 (05H)
Y2
Y3
Y1
X1
X2
X3
IABS (SPD STR)
8
FLXSTPSCL1 [654]
X1= FLX BRK1E Y1= FLX LEV1E
X2= FLX BRK2E Y2= FLX LEV2E
X3= FLX BRK3E Y3= FLX LEV3E
CRSST INT GAINE
8
CRSST MAX LMTE
FLXLEVMAXP
CROSSTIE
(FROM CURRENT REG)
CRSST PRP GAINE
CRSSTP
FLXSTPSCL2
FLXSTPSCL
FLXLEVMINP
FAEAP INT GAINE
128
FAEAPP
GENERAL ELECTRIC
DRIVE SYSTEMS DEPT.
SALEM, VA. U.S.A.
FLUX SETPOINT/ADJUS
GTO-IMD
CONT. ON SH. 8MB
SH. NO. 8MA
APPROVALS
REV. 1
REV. 2
REV. 3
REV. 4
REV. 5
REQ. NO. STANDARD
7/31/86
ISSUED
MADE BY


FLX REG STEP
[632]
FLUXSTP
[65C]
FLXSTAR1
[61E]
FLX FBK
[628]
(1024 = 1PU)
FLXERRCLMPP
FLX ERR
[EE: 1B6]
FLX ERR FIL GAINE
32
FLXERRFIL
[644]
Z-1
4
[17R]
FLX PRP GAINE
[EE: DE]
2000H
[15C]
FLX INT GAINE
64
[EE: DC]
FLXPISUM
[646]
1
XM GAP E
[EE: F6]
IMESTAR+
IXSTAR
[616]
(1024 = 1PU)
IX MIN CLMPE
[EE: EE]
[35V]
XM GAPE
[EE: F6]
-2000H
PI CLMP RATE LIME
[EE: F0]
LIGHT
LOAD COND.
PIMINCLMP
-2000H
SPD
1
2
MOT CAPE
[EE: F2]
1
64
1
32
ICAP
[EE: F4]
ICAP FIL GAINE
32
ICAP FIL
[648]
1
8
FLXFBK
GENERAL ELECTRIC
FLUX REGULATOR
DCLC-IMD 8WA
DRIVE SYSTEMS DEPT.
SALEM, VA. U.S.A.
APPROVALS
REV. 1
REV. 2
REV. 3
REV. 4
REV. 5
REG. NO. STANDARD
7/24/86
ISSUED
MADE BY



GTO-IMD STARTUP/HARMONIC ELIMINATION SCHEME

GENERAL ELECTRIC
DRIVE SYSTEMS DEPT.
SALEM, VA. U.S.A.

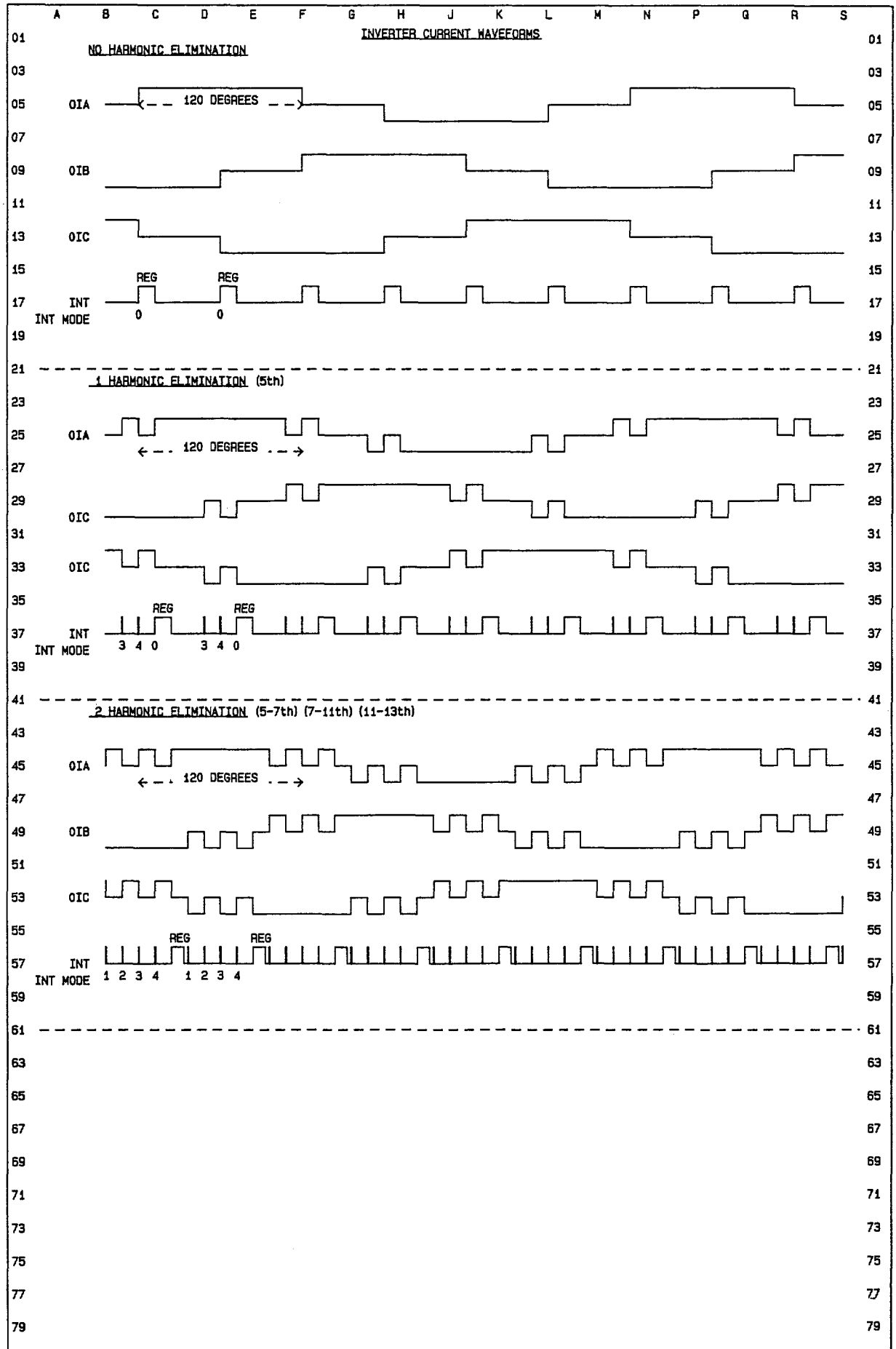
STARTUP SEQUENCE & FIRING REGIONS

GTO-IMD

DR. NO. 8WA
8WB
CONT. ON SH.

APPROVALS
REV. 1
REV. 2
REV. 3
REV. 4
REV. 5

REG. NO. STANDARD
7/31/86
ISSUED
MADE BY



REV. 3	REV. 2 / REV. 1	APPROVALS	GENERAL ELECTRIC	COMMUTATION MODES	
REV. 4	ISSUED 7/31/86	REG. NO. STANDARD	DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.	GTO-IMD CONT. ON SH. FNL	SH. NO. 8WB
REV. 5	MADE BY				



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Salem, Virginia 24153
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